What Power-Network Models Preserve

A book and knowledge base

Draft

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Part I

Start: problem and counterexample

Chapter 1

Home

Page status: reader-facing overview and navigation map.

Read the searchable HTML book here, or download the long-form PDF for the argument-shaped monograph.

Graphs, reductions, and decision boundaries

This book starts with a failure mode: a familiar power-network representation can remain numerically plausible while silently discarding the identity, grounding, limits, controls, decisions, measurements, or provenance needed by the study. The question is therefore not "which graph is correct?" but:

> What does this representation preserve for this observation, constraint, or decision—and what does it forget?

The general multiconductor model is the baseline for making that question visible: explicit terminals and neutral/grounding, coupled line factors, switching states, multiwinding transformers, limits, controls, and recoverable decisions. Balanced transmission models are derived specializations, not the starting ontology.

The central thesis is:

> A graph transformation is meaningful only relative to declared observations, constraints, and decisions. Simpler graphs should be derived, traceable views with preservation and recovery contracts.

Read the numbered spine from 1 to 5 across the first row, then continue down to 6 and follow the second row from right to left through 9. The arrows show reading order, not logical implication or a hierarchy of claims.

The argument route

Follow How to use this book for the full route. If you want to ask questions through ChatGPT, start with Use this resource with ChatGPT, which documents the supported reader and developer access paths. The companion page Use this resource with Claude documents the same paths for Claude, including the Model Context Protocol route into this repository. The compact argument is One network, many graphs → A first failure: heterogeneous parallel branches → Scope and thesis → Formal representation frameworks → From source data to a canonical network model → When the general model collapses → Translation traps → Preservation contracts → Transformation semantics and register → the guarded transformations and worked decision cases.

For the complete retrieval surface, use the generated knowledge-base index, which lists claims, artifacts, chapters, open items, and their evidence. The generated cross-community vocabulary indexes translate in

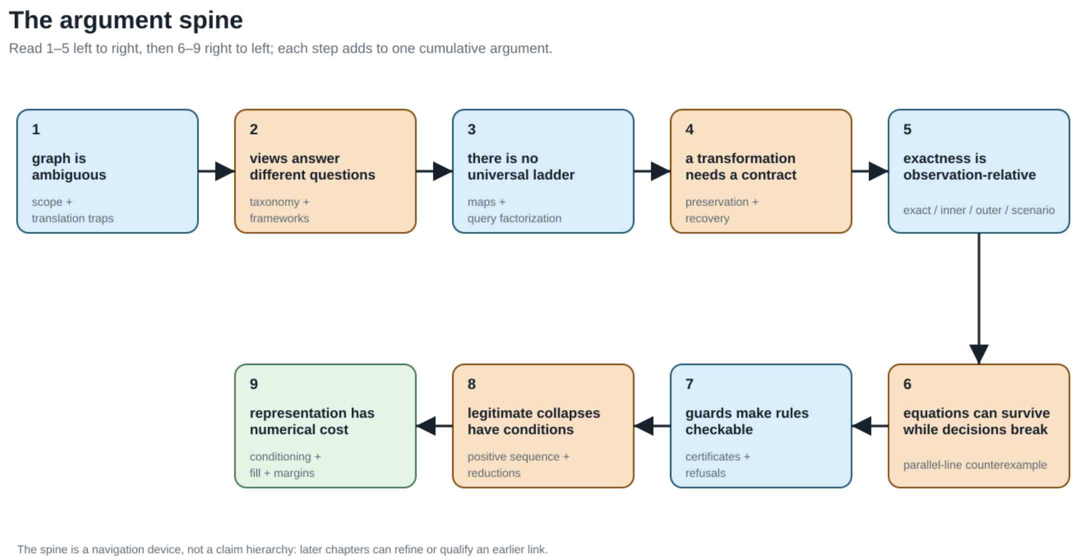

Figure 1.1: The argument spine of the book.

both directions between familiar community phrases and the book's maintained terms. The chapter-status table records the scope and verification boundary of each page.

The long-form PDF is the argument-shaped monograph. The HTML route is both that argument and the exhaustive knowledge base: use its generated indexes when the question is retrieval rather than sequence.

The audience map is a route selector, not another taxonomy. Power engineers, software and data experts, mathematical modellers, graph theorists, and graph machine-learning experts enter through different questions, but every route returns to the same preservation-contract and evidence language.

What to read by question

- **I recognize the words but not their use here.** Read One network, five languages, then use the maintained Terminology page for compact definitions or the cross-community vocabulary indexes for bidirectional lookup.
- **I know simple graphs or balanced transmission models.** Start with the graph and transmission reading guide, then follow the route for your background into the shared four-wire and preservation examples.
- **Which graph is this?** Read Representation frameworks, Representation taxonomy, and Maps between representation frameworks.
- **What does an arrow, cycle, parallel line, or radial feeder mean?** Read Translation traps, Orientation, terminal quantities, and power transfer, and Cycles, parallelism, and radial structure.
- **How did this graph get made?** Read From source data to a canonical network model, then From conductor geometry to impedance fidelity.
- **Why is a Y-bus not always enough?** Read Circuit formulations and the lowering boundary, then Two topology levels and the nodal projection.

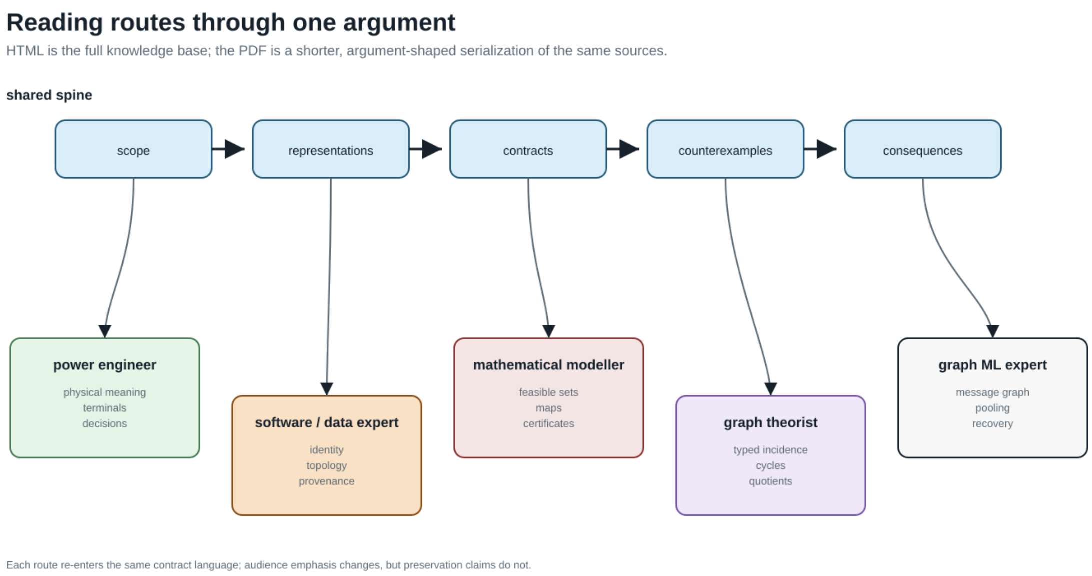

Figure 1.2: Audience routes through the shared HTML/PDF argument.

- **Why can the same graph produce different decisions?** Read Load models and decision dependence, followed by the parallel-line decision cases.
- **Can I reduce or compile a model?** Read Preservation contracts, Projection, compilation, and reduction, and

Kron, Ward, and optimized network equivalents, then follow the four-wire impedance-model ladder.

- **What does a transformation preserve, and what does it erase?** Read Transformation semantics and register, then Degree-two series elimination and BIM/BFM parallel lines: an expressiveness audit.
- **What survives a decision problem?** Read the parallel-line cases, the transformer-tap case, Certificate schema and composition, and Guarded normalization rules.
- **What changes numerically?** Read Numerical consequences of representation and reduction.

Scope and conventions

The book uses the BMOPFTools-style index discipline: ℓij denotes the stored reference orientation and terminal order for line ℓ; it does not assert the operating direction of power. An element-intrinsic impedance is $\mathbf{Z}_\ell$. Multiwinding devices retain device and winding indices until an explicit compilation creates two-terminal arcs.

The target is stronger than matching voltages. A valid transformation may need to preserve or map conductor and winding limits, continuous and discrete controls, switching and outage choices, feasible sets, objectives, active constraints, eliminated quantities, and source provenance.

The current text is a research draft. Claims, generated witnesses, citations, and unresolved boundaries are tracked explicitly in the generated indexes.

Chapter 2

How to use this book

Page status: reading guide for the long-form monograph and the exhaustive HTML knowledge base.

This project has two complementary reading surfaces. They share Markdown sources, claims, citations, generated artifacts, and evidence boundaries, but they answer different reader needs.

For access through ChatGPT, see Use this resource with ChatGPT. It explains the Project, custom GPT, and developer-service routes, and how to distinguish a book-grounded answer from an ordinary model response. For access through Claude, see Use this resource with Claude, which adds a Model Context Protocol route that reads this repository's generated corpus directly.

Follow the monograph argument

The long-form route is organized around a problem rather than a catalogue of graph types:

1. **Problem and counterexample:** see why a plausible simplification can change the question being answered, beginning with One network, many graphs and A first failure: heterogeneous parallel branches.
2. **Representation obligations:** identify which physical objects, equations, observations, constraints, decisions, and provenance a study requires.
3. **Canonical model:** establish the source data and semantic projection before deriving a graph or matrix view.
4. **Collapses and failure modes:** study when balanced, nodal, radial, series, parallel, or reduced views are valid and when they discard needed meaning.
5. **Preservation contracts:** state the exact object being preserved, the guards, the recovery map, and the evidence boundary.
6. **Transformations and recovery:** read the executable rules and certificates only after the contract is clear.
7. **Cases and consequences:** compare feasible sets, active limits, objectives, decisions, and recovery—not only state or terminal-equation error.

The curated PDF route follows this sequence. It remains a long-form draft and retains detailed reference and case material; the route is refactored, not shortened.

Use the HTML knowledge base for retrieval

Use the generated knowledge-base indexes to find claims by type, chapter, verification state, unresolved issue, or generated artifact. Use the chapter-status table to see what each page establishes and what remains open. Use the evidence map for coverage gaps, the vocabulary indexes for cross-community terminology, and the references page for the bibliography.

The knowledge base is intentionally more exhaustive than the monograph. Its presence does not promote a proposal into a theorem, a local numerical witness into a global result, or an independent numerical reimplementation into external peer review.

Read the evidence labels correctly

- **Definition / proposal:** the book's vocabulary or architecture; not a claim that the field has adopted it.
- **Theorem / established result:** a mathematical or literature claim whose scope and assumptions must be read with its citation or derivation.
- **Empirical witness:** a result recorded for a declared fixture, solver, state, and tolerance.
- **Independently implemented:** a separate numerical path for the declared case; it may still share source fixtures, matrices, or model assembly.
- **Externally reviewed:** human review recorded with reviewer, date, scope, and response. The current ledger has no claims in this category.

For every transformation, ask four questions: what is the source model, what is the target model, what is preserved for the declared observation or decision, and how are forgotten quantities and constraints recovered or bounded?

Choose a route by background

- Power engineers: begin with One network, five languages, the reading guide, and the parallel or grounding cases.
- Optimization researchers: begin with Load models and decision dependence, Preservation contracts, and the decision certificates.
- Software and data practitioners: begin with From source data to a canonical network model, Data-model crosswalk, and the generated indexes.
- Graph and formal-methods readers: begin with Formal representation frameworks, Maps between representation frameworks, and Transformation semantics.

Chapter 3

Use this resource with ChatGPT

Page status: reader-facing access and grounding guide. The repository's developer/operator details remain in `llm/README.md`.

This resource can be used through ChatGPT at three levels. The first two are convenient for readers; the third is the supported integration route for a software application. In every case, treat the book's release identity and evidence boundary as part of the answer, not as implementation detail.

Quickest route: a ChatGPT Project

Create a Project, upload the released book PDF or the chapters relevant to the question, and add the following project instructions:

> Use the attached resource as the primary authority. Answer in the reader's domain language, but preserve the resource's qualifications. State the source representation, derived representation or formulation, operating state, and exactness or preservation object. Distinguish physical assets and behavioural factors from graph and matrix views. Explain the tempting short answer and why it can fail. Cite the relevant chapter, claim, or source anchor. If the resource does not establish the answer, say **unsupported by this resource** and identify what evidence would be needed.

Projects keep related files, chats, and instructions together. They are useful for a reader who wants to ask follow-up questions over several sessions. See Projects in ChatGPT for the current product instructions and limits.

Begin with a question that declares the audience and task, for example:

> I am a power engineer. Using only this resource, explain whether a constant-admittance load belongs in the graph, in the nodal admittance matrix, or both. Separate source-model identity from formulation placement and cite the relevant sections.

Uploading a book to a conversation or Project gives ChatGPT a snapshot of the files supplied for that workspace. It is not a live connection to this repository, so check the release identity when the answer matters.

Reusable route: a custom GPT

A custom GPT can package the same approach for a class, team, or public reader experience. Configure it with:

- the curated release bundle as Knowledge;
- the answer contract above as Instructions;
- conversation starters for students, power engineers, software engineers, and mathematical modellers;
- a requirement to name the release identity and cite chapter or claim anchors; and
- an explicit `unsupported` response when no qualified evidence is found.

Custom GPTs use uploaded Knowledge files and configured Instructions inside ChatGPT. They are not a live synchronization mechanism: update the Knowledge files whenever a new book release changes the supported content. See GPTs in ChatGPT and Creating and editing GPTs for current configuration and sharing details.

Creating or publishing a new GPT depends on the current account and workspace permissions. If GPT creation is unavailable, use a Project with the same files and instructions; an existing shared GPT can still be used when the reader has access to it.

Do not describe a custom GPT as an independent authority. It is a reader interface over a particular resource release. Its instructions should require the same distinctions and abstention behaviour as the book.

Developer route: the grounded access service

The repository contains a model-independent corpus and a deterministic local access service. From the repository root:

```
python3 scripts/serve_llm_access.py --port 8787
```

The service provides:

- `GET /healthz` for availability;
- `GET /v1/manifest` for the corpus release identity and hashes;
- `GET|POST /v1/search` for ranked source records; and
- `GET|POST /v1/context` for an answer-oriented context packet.

The context packet is the preferred interface for an application. It carries the supported answer basis, representation and scope, qualifications, counterexamples, source anchors, evidence status, and an explicit `unsupported` or `under_retrieved` result when the corpus cannot support a qualified answer. The downstream model should render that packet into the user's domain language; it should not replace the packet with uncited model memory.

Compatible local clients can use the MCP adapter:

```
python3 scripts/mcp_llm_server.py
```

The local service and MCP adapter are developer interfaces. A normal ChatGPT conversation cannot call a service running on a private computer unless a separate, trusted integration makes it available. For a hosted application, load the generated corpus into a retrieval service or API vector store and retain the corpus manifest identifier and source metadata.

Recommended answer contract

Every grounded answer should contain, as applicable:

1. a direct answer supported by the resource;
2. the named representation, formulation, state, and exactness object;
3. assumptions and load-bearing qualifications;
4. the tempting shortcut and its failure consequence;
5. a translation into the reader's domain language without changing meaning;
6. chapter, claim, or source anchors and their evidence status; and
7. an unresolved boundary or an explicit `unsupported` status.

The answer should not call a nodal matrix “the network” without naming the source model and formulation. It should not call a graph self-loop a shunt, or assume that a load or generator has one fixed graph membership across all studies. These are precisely the shortcuts for which the retrieval layer keeps misconception contracts.

Keeping ChatGPT context synchronized

The canonical sources are the Markdown chapters, claims ledger, vocabulary registry, evidence artifacts, and release manifest. The generated corpus is a derived access artifact. Do not edit it by hand.

At each release:

1. regenerate the corpus from canonical sources;
2. run the LLM accessibility and retrieval checks;
3. publish the corpus manifest with its source commit and hashes;
4. rebuild any hosted retrieval index or custom GPT Knowledge files; and
5. test the audience prompts and high-risk misconception cases against the new release before presenting it as current.

A Project or custom GPT uploaded before step 4 is a stale snapshot. Its answers must not be described as representing the latest book release.

Choosing the route

For scientific or engineering decisions, use a grounded route and inspect the source anchors. The model's fluency is not evidence that the resource supports the conclusion.

Need	Recommended route	Main limitation
One reader asking questions	ChatGPT Project with uploaded chapters	File snapshot; not automatically synchronized
Class or team reader experience	Custom GPT with a pinned release bundle	Knowledge must be rebuilt when the book changes
Software or service integration	`/v1/context`, MCP, or hosted retrieval/API	Requires deployment and integration work
Unstructured ordinary ChatGPT question	Use only for orientation	The answer is not necessarily book-grounded

Chapter 4

Use this resource with Claude

Page status: reader-facing access and grounding guide. The repository's developer/operator details remain in `llm/README.md`.

This resource can be used through Claude at four levels. The first two are convenient for readers; the third connects Claude to a live, versioned view of this repository; the fourth is the supported integration route for a software application. In every case, treat the book's release identity and evidence boundary as part of the answer, not as implementation detail.

The companion page Use this resource with ChatGPT documents the same obligations for that assistant. The two pages differ in one substantive way: Claude can connect directly to this repository's local access service through the Model Context Protocol, so a grounded route is available to a reader without building an application.

Quickest route: a Claude Project

Create a Project, add the released book PDF or the chapters relevant to the question as project knowledge, and set the following custom instructions:

> Use the attached resource as the primary authority. Answer in the reader's domain language, but preserve the resource's qualifications. State the source representation, derived representation or formulation, operating state, and exactness or preservation object. Distinguish physical assets and behavioural factors from graph and matrix views. Explain the tempting short answer and why it can fail. Cite the relevant chapter, claim, or source anchor. If the resource does not establish the answer, say **unsupported by this resource** and identify what evidence would be needed.

Projects keep related files, chats, and instructions together, so a reader can ask follow-up questions over several sessions without restating the contract. See Projects in Claude for the current product instructions and limits.

Begin with a question that declares the audience and task, for example:

> I am a power engineer. Using only this resource, explain whether a constant-admittance load belongs in the graph, in the nodal admittance matrix, or both. Separate source-model identity from formulation placement and cite the relevant sections.

Adding files to a Project gives Claude a snapshot of what was uploaded to that workspace. It is not a live connection to this repository, so check the release identity when the answer matters.

Reusable route: a shared Project

A Project shared with a class or team packages the same approach for repeated use. Configure it with:

- the curated release bundle as project knowledge;
- the answer contract above as custom instructions;
- starter questions for students, power engineers, software engineers, and mathematical modellers;
- a requirement to name the release identity and cite chapter or claim anchors; and
- an explicit `unsupported` response when no qualified evidence is found.

Project sharing depends on the current plan and workspace permissions. If sharing is unavailable, each reader can build the same Project from the same release bundle and instructions.

A shared Project is not a synchronization mechanism: update its knowledge files whenever a new book release changes the supported content. Do not describe such a Project as an independent authority. It is a reader interface over a particular resource release, and its instructions should require the same distinctions and abstention behaviour as the book.

Connected route: the MCP server

This repository ships an adapter for the Model Context Protocol, so Claude can query the generated corpus directly instead of reading an uploaded snapshot. From the repository root the adapter runs as a local stdio server:

```
python3 scripts/mcp_llm_server.py
```

It advertises the server name `multi-graph-book` and exposes two tools and one resource:

- `book_search` returns ranked, stable, source-linked corpus records. It takes `query`, and optionally `limit` and a `method` of `lexical`, `char_tfidf`, `hybrid`, or `graph`.
- `book_context` returns an answer-oriented context packet. It takes the same arguments plus an optional `audience` of `student`, `software_engineer`, or `power_engineer`.
- `book://multi-graph-book/manifest` exposes the corpus release identity and record count as a resource.

To use it from Claude Code, register the server once:

```
claude mcp add multi-graph-book -- python3 /absolute/path/to/scripts/mcp_llm_server.py
```

To use it from the Claude desktop app, add the same command to the app's MCP server configuration:

```
{
  "mcpServers": {
    "multi-graph-book": {
      "command": "python3",
      "args": ["/absolute/path/to/scripts/mcp_llm_server.py"]
    }
  }
}
```

See the Claude Code MCP documentation and the desktop app guide for current configuration details.

This route differs from an uploaded Project in a way that matters for correctness. The adapter reads the generated corpus in the working tree, so the answers follow whatever release is checked out, and `book_context` supplies the qualification structure rather than leaving Claude to reconstruct it from prose. The adapter is local and read-only over generated artifacts; it does not publish the repository, and it cannot be reached by a Claude conversation on another machine.

The corpus is a derived access artifact. If it has not been regenerated for the current sources, this route reports a stale release identity rather than a silently different answer, which is why the manifest resource is worth reading before trusting a result.

Developer route: the grounded access service

For a hosted application, the same service is available over HTTP:

```
python3 scripts/serve_llm_access.py --port 8787
```

The service provides:

- `GET /healthz` for availability;
- `GET /v1/manifest` for the corpus release identity and hashes;
- `GET|POST /v1/search` for ranked source records; and
- `GET|POST /v1/context` for an answer-oriented context packet.

The context packet is the preferred interface for an application. It carries the supported answer basis, representation and scope, qualifications, counterexamples, source anchors, evidence status, and an explicit `unsupported` or `under_retrieved` result when the corpus cannot support a qualified answer. The downstream model should render that packet into the user's domain language; it should not replace the packet with uncited model memory.

An application built on the Claude API should expose `book_search` and `book_context` as tools rather than pasting corpus text into a prompt, so that the abstention status and source anchors survive into the answer. See the tool use documentation for the current request shape. Record the model identifier alongside the corpus manifest identifier: an answer is reproducible only when both are known.

Recommended answer contract

Every grounded answer should contain, as applicable:

1. a direct answer supported by the resource;
2. the named representation, formulation, state, and exactness object;
3. assumptions and load-bearing qualifications;
4. the tempting shortcut and its failure consequence;
5. a translation into the reader's domain language without changing meaning;
6. chapter, claim, or source anchors and their evidence status; and

7. an unresolved boundary or an explicit `unsupported` status.

The answer should not call a nodal matrix “the network” without naming the source model and formulation. It should not call a graph self-loop a shunt, or assume that a load or generator has one fixed graph membership across all studies. These are precisely the shortcuts for which the retrieval layer keeps misconception contracts.

Keeping Claude context synchronized

The canonical sources are the Markdown chapters, claims ledger, vocabulary registry, evidence artifacts, and release manifest. The generated corpus is a derived access artifact. Do not edit it by hand.

At each release:

1. regenerate the corpus from canonical sources;
2. run the LLM accessibility and retrieval checks;
3. publish the corpus manifest with its source commit and hashes;
4. rebuild any hosted retrieval index or Project knowledge files; and
5. test the audience prompts and high-risk misconception cases against the new release before presenting it as current.

The MCP and HTTP routes pick up step 1 automatically, because they read the generated corpus rather than a copy of it. A Project uploaded before step 4 is a stale snapshot, and its answers must not be described as representing the latest book release.

Choosing the route

Need	Recommended route	Main limitation
One reader asking questions	Claude Project with uploaded chapters	File snapshot; not automatically synchronized
Class or team reader experience	Shared Project with a pinned release bundle	Knowledge must be rebuilt when the book changes
Reader who has the repository checked out	MCP server via Claude Code or the desktop app	Local process; requires a checkout and a regenerated corpus
Software or service integration	`/v1/context`, MCP, or hosted retrieval/API	Requires deployment and integration work
Unstructured ordinary Claude question	Use only for orientation	The answer is not necessarily book-grounded

For scientific or engineering decisions, use a grounded route and inspect the source anchors. The model's fluency is not evidence that the resource supports the conclusion.

Chapter 5

One network, many graphs

Page status: explanatory synthesis introducing the representation landscape.

Calling something *the network graph* hides the modelling decision that produced it. The same physical power network can yield several non-isomorphic structures, each correct for some questions and inadequate for others.

Power-system shorthand

Treat *the network graph* as an omitted noun phrase. Ask whether the sentence means an asset graph, active topology, terminal-connectivity graph, bus–branch multigraph, factor incidence graph, or equation/sparsity graph.

A deliberately difficult network

The running network for this book contains ordered conductor terminals, an explicit neutral and grounding impedance, heterogeneous parallel lines, switchgear, and a genuinely multiwinding transformer. It also contains limits and controls used in a decision problem. Its complete semantic specification is given in The running multiconductor network. Its first numerical realization and the six illustrated views described below are given in the Executable running network.

The representation-scoped meanings of cycles, parallelism, bridges, leaves, and radial ends are developed in Cycles, parallelism, and radial structure.

The first surprise is not a new device; it is a change of graph. At the bus level the feeder can be radial, while dense multiconductor stamps create cliques—and therefore cycles—in the scalar support graph used by a matrix algorithm:

The resolving phrase is **which graph?** Bus-level radiality is a statement about equipment connectivity. Support-graph cycles are a statement about algebraic coupling. The latter can still be chordal and admit useful leaf-clique elimination; it is not evidence of an additional physical loop.

Nothing exotic is required to create representational disagreement. Parallel lines already show the issue. If two branches ℓ_1 and ℓ_2 connect buses i and j, a multigraph retains both triples

$$\ell_1 ij,\ \ell_2 ij \in \mathcal{T}^{L\rightarrow}.$$

Here i and j are bus labels, not necessarily row and column numbers in an array. If a software implementation enumerates the bus set, its stored entry is written $[\mathbf{Y}]_{\kappa(i),\kappa(j)}$; the semantic relation remains $\mathbf{Y}_{ij}$. This is the same label-before-coordinate rule used in the equation-reading bridge.

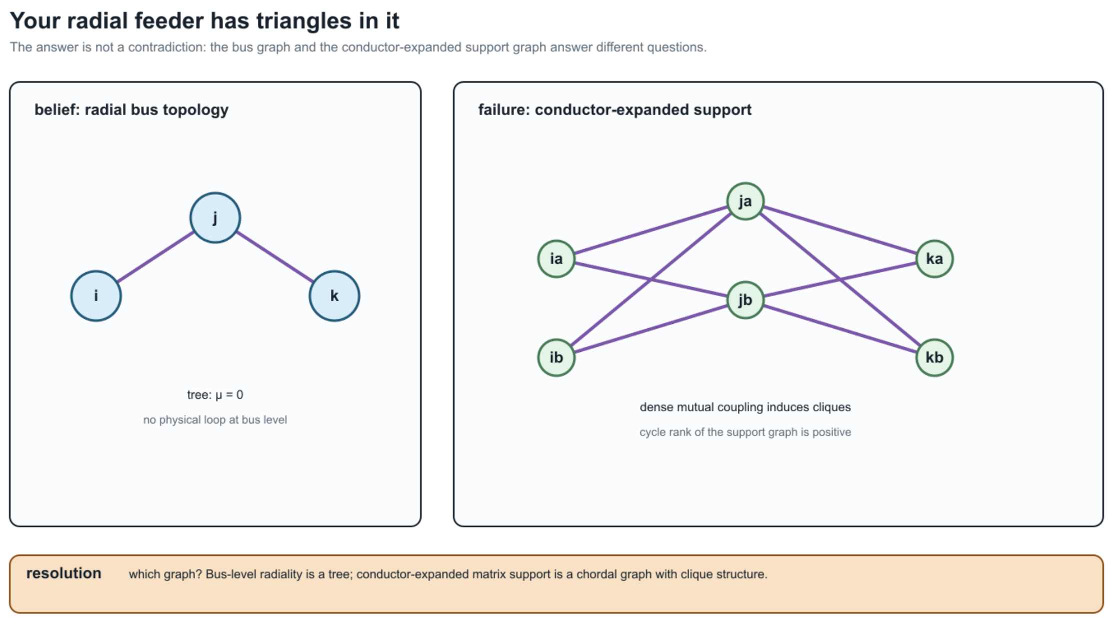

Figure 5.1: A radial bus-level feeder can have cycles in its conductor-expanded support graph.

A simple graph retains only the adjacency $i \sim j$. A weighted simple graph might store

$$\mathbf{Y}_{ij}^{\mathrm{eq}} = \mathbf{Y}_{\ell_1} + \mathbf{Y}_{\ell_2},$$

but this does not by itself retain individual current limits, outages, maintenance states, ownership, or investment choices.

Six useful views

Asset view

The asset graph distinguishes each line, winding, switch, grounding device, measurement, and owner. It supports questions such as *which circuit is unavailable?* and *which construction record produced this impedance?* It need not contain the virtual buses introduced by an OPF formulation.

Terminal-connectivity view

This view records ordered bus terminals and the maps by which element conductors attach to them. It can distinguish phase a from a neutral and can represent a conductor permutation between the ends of a line. It is the natural place to resolve switchgear and grounding connectivity.

Bus–branch multigraph

Buses are vertices and identified two-terminal elements are edges. Parallel circuits remain distinct. This view is effective when the device vocabulary is genuinely two-terminal or when multi-terminal devices have been compiled into an equivalent network with explicit provenance.

Port–factor view

Ports carry terminal variables and factors impose constitutive, limit, control, or measurement relations. A multiwinding transformer can remain one factor with one port bundle per winding. The number of ports is not forced to two.

Equation or optimization view

Variables and constraints form a bipartite graph, or blocks form a computational dependency graph. This view exposes separability, coupling, and decomposition opportunities. An auxiliary variable created for numerical convenience becomes a graph vertex even though it is not a physical object.

Matrix sparsity view

The nonzero pattern of an admittance, Jacobian, KKT, or Schur-complement matrix defines another graph. Elimination may reduce the number of variables while making this graph denser. A sparsity edge means algebraic coupling, not necessarily a physical branch.

Loads, generators, and the graph boundary

Loads and generators expose why an element inventory must be declared before the word *graph* is used. In an asset or bus–branch graph they are usually devices attached to a bus, not ordinary line edges. In a port–factor graph they are explicit one-terminal or multi-terminal factors. In a nodal-support graph they appear only when a declared formulation stamps a linear or linearized part of their relation into the nodal operator. In an equation or optimization graph they also appear through injections, controls, limits, and decision variables.

The same device can therefore be absent from one graph and present in another without contradiction:

View	Device role	Typical question
asset or bus–branch graph	attached equipment or a bus-side factor; not necessarily an edge	which device is switchable, owned, or removed?
port–factor graph	one-terminal or multi-terminal constitutive factor with a terminal map	what relation, limit, or control does the device impose?
nodal-support graph	a diagonal shunt, off-diagonal block, or no direct stamp, depending on the declared model	which retained voltage coordinates are coupled by this formulation?
solver/Jacobian graph	injection, residual, derivative, control, or constraint block	which variables and equations determine the next iterate or decision?

A constant-admittance load can therefore be stamped into a diagonal nodal block without becoming a physical network edge. A Norton generator or a multi-terminal device can contribute a different block pattern. The stamping choice is a property of the declared study formulation, not a definition of the source asset graph. The circuit formulation boundary and load-model chapter make this split explicit.

Different questions select different views

The views are not arranged in a universal hierarchy. The asset graph may know more about ownership and less about electrical variables than a compiled optimization graph. Expressiveness is relative to the declared question.

Question	Required retained meaning	A useful view
Are two assets independently switchable?	member identity and switch state	asset graph or multigraph
Which conductors share a junction?	ordered terminals and terminal maps	terminal-connectivity model
What is the boundary current response?	constitutive relation at retained ports	port–factor or admittance model
Which constraints determine the OPF optimum?	feasible set, controls, objective	optimization model
Which variables should be eliminated first?	numerical nonzero structure	sparsity graph
Can a result be mapped to the source data?	provenance and recovery	linked source and generated views

The first preservation test

Suppose each parallel line obeys

$$\mathbf{I}^{\mathrm{s}}_{\ell ij} = \mathbf{Y}_\ell\big(\mathbf{U}_i[\mathbf{N}_{\ell i}] - \mathbf{U}_j[\mathbf{N}_{\ell j}]\big)$$

and has a conductor-current feasible set $\mathcal{C}_\ell$. Aggregating admittance preserves the sum of terminal series currents, but the source feasible voltage differences satisfy

$$\{\Delta\mathbf{U} : \ \mathbf{Y}_\ell \Delta\mathbf{U} \in \mathcal{C}_\ell \quad \forall \ell\}\,.$$

A single conventional edge limit need not reproduce this intersection. The transformation can be exact for one observation and wrong for a decision problem. This is why the book asks for a Preservation contracts rather than calling a smaller graph simply *equivalent*. A first failure: heterogeneous parallel branches gives an analytic witness and executable certificate.

The route through the book

Before the formal taxonomy, One network, five languages translates the recurring terms used by power engineers, software and data experts, mathematical modellers, graph theorists, and graph-machine-learning readers. The Representation taxonomy separates the major model families. Notation and modelling conventions fixes the element, arc, terminal, and winding indices. Representation implementation record records the linked architecture's executable implementation status. The transformation parts then ask which views can be derived, under what guards, and with what consequences for feasible decisions. Five buses through a multi-port lowering is the compact bridge from these view names to an explicit three-winding compilation and loss ledger.

Chapter 6

How to read diagrams and equations

Page status: introductory translation bridge; the examples are deliberately small, while the terminology is used throughout the multiconductor chapters.

A power-network diagram is not a circuit equation, and a circuit equation is not automatically a graph. Before interpreting an arrow, edge, node, or matrix entry, ask three questions:

1. **What object is drawn?** An asset, a terminal, a port, a factor, a retained voltage coordinate, or an algebraic support relation?
2. **What quantity is attached to it?** A voltage, current, complex power, impedance, admittance, limit, state, or provenance identifier?
3. **What has already been compiled away?** Parallel-member identity, conductor coordinates, an internal transformer node, a neutral, or a decision state?

The same physical feeder can therefore have several faithful diagrams. The diagram is a view with a declared semantic level, not an unqualified picture of the network.

A visual legend before the equations

The book uses familiar electrotechnical symbols where possible, but the symbol does not carry the entire model. IEC 60617 is a symbol library and IEC 61082 is a document-presentation standard; neither one determines the semantics of a book-specific lowering or reduction. Read each maintained figure with this legend:

Single-line diagrams answer "which equipment and terminal-level connections are present?" Multi-line diagrams answer "which conductors, neutral paths, and terminal connections are present?" A factor or equation view answers "which constitutive relation is being assembled?" A compiled graph answers "what does this particular algorithm receive?" These are related views, not successive truth values.

When a figure expands a transformer, line, regulator, or switch, look for the identity fibre and the omitted-semantics note before interpreting a new edge. For example, a complete graph of pairwise transformer leakage factors may be a faithful equation decomposition while still being the wrong graph for asset outages or ownership. The same warning applies to a nodal-support graph whose edges record algebraic coupling rather than physical lines.

The high-risk cases are collected in the special semantic overlays plate in the foundations section: neutral grounding, nominal-π shunts, phase-selective switching, and n-winding leakage factors. When one of these appears compressed into a single-line symbol, look for the state scope and edge provenance in the caption or map certificate.

Mark	House meaning	Do not infer
solid electrical connector	declared attachment in the displayed view	that the attachment is a physical conductor in every other view
double or bundled stroke	several ordered conductors	that phases or neutral may be discarded
dashed arrow	refinement, quotient, decomposition, or compilation map	physical power-flow direction
separate control arrow	tap, switch, protection, or decision relation	an extra series branch
grounding branch	explicit neutral/earth/reference connection	that the grounding can be absorbed without a guard
x_1 or ℓij label	persistent source identity	an array coordinate or a flow sign
pole/state label such as σ_a	per-conductor or per-pole operating state	one scalar open/closed state for the whole asset
λ_{ij} factor label	computational coupling with declared provenance	a physical line, outage asset, or ownership edge

The scalar circuit in one minute

For a scalar branch stored from i to j, write the voltage difference

$$\Delta v_{\ell ij} = v_i - v_j.$$

Ohm's law, or the branch constitutive relation, is

$$i_{\ell ij} = y_\ell \Delta v_{\ell ij}, \qquad y_\ell = z_\ell^{-1}.$$

This equation says how the branch responds to its terminal voltages. It does not say that current must flow from i to j in operation: the computed complex current may have either sign or phase. The stored order ℓij is an index convention, consistent with the BMOPFTools-style notation used in this book.

Kirchhoff's current law (KCL) is a balance at a node. With currents defined as entering the node, one possible convention is

$$\sum_{e \in \delta(i)} i_{e \to i} + i_i^{\mathrm{inj}} = 0.$$

The signs change if the convention changes; the physical balance does not. Kirchhoff's voltage law (KVL) says that the oriented voltage differences sum to zero around a compatible closed circuit path. In a nodal formulation this is usually enforced indirectly by assigning one voltage to each retained node and deriving every branch drop from endpoint voltages. A cycle in a matrix support graph is not automatically such a physical circuit path.

These three statements have different jobs:

Statement	Role	What it does not define
Ohm/constitutive law	element behaviour	asset identity or operating direction
KCL	balance at a junction	which graph view produced the junction
KVL	compatible voltage differences around a circuit loop	a cycle in every derived support graph

Labels are not coordinates

The subscripts in a power-network equation are semantic labels first and array positions second. A bus may be named $source$, $load$, or i_{north}, and a line may be identified by ℓ_{main}; neither name is required to be a consecutive integer. Software normally enumerates those labels so that it can store an ordinary array. That enumeration is a coordinate chart, not a change in the network.

For example, begin with the labelled bus set

$$\mathcal{B} = \{\text{source}, \text{load}, \text{neutral}\}, \qquad \kappa_{\mathcal{B}} : \mathcal{B} \xrightarrow{\sim} \{1, 2, 3\}.$$

The semantic nodal blocks are $\mathbf{Y}^{\mathrm{N}}_{ij}$, with $i, j \in \mathcal{B}$. After choosing $\kappa_{\mathcal{B}}$, a software array stores the same block as

$$\left[\mathbf{Y}^{\mathrm{N}}\right]_{\kappa_{\mathcal{B}}(i), \kappa_{\mathcal{B}}(j)} = \mathbf{Y}^{\mathrm{N}}_{ij}.$$

The familiar integer-indexed matrix is therefore a realization of a label-indexed operator family. Reordering the array changes positions, not the buses or the physical relation. The same rule applies to the signed edge–cycle matrix: its semantic entries are $A_{i\ell}$ and $C_{\ell\gamma}$ for $i \in \mathcal{B}$, $\ell \in \mathcal{L}$, and $\gamma \in \Gamma$; an implementation may enumerate all three sets before performing ordinary matrix multiplication.

This is why $\mathbf{Y}_\ell$ (an element-intrinsic matrix), $\mathbf{Y}^{\mathrm{N}}_{ij}$ (a bus-to-bus block), and $\left[\mathbf{Y}^{\mathrm{N}}\right]_{\kappa(i),\kappa(j)}$ (an array position) should not be read as interchangeable notation. In the scalar case a block may be one number; in the multi-conductor case it is generally a matrix.

The same line with multiple conductors

For a four-wire line, the scalar voltage and current become ordered vectors:

$$\mathbf{U}_i = [U_{i,a}, U_{i,b}, U_{i,c}, U_{i,n}]^{\intercal}, \qquad \mathbf{I}_{\ell ij} = \mathbf{Y}_\ell \left(\mathbf{U}_i[\mathbf{N}_{\ell i}] - \mathbf{U}_j[\mathbf{N}_{\ell j}]\right).$$

The matrix $\mathbf{Y}_\ell$ may be dense. Its off-diagonal entries represent mutual coupling between conductor coordinates; they are not extra physical lines. A vector-valued edge is therefore a useful bridge phrase for a two-terminal multiconductor factor, but it is not a new canonical graph class. The precise source object remains a typed attributed multigraph edge with vector terminal spaces and matrix-valued constitutive data.

When a device has three or more ports, the bridge phrase stops being adequate: retain a typed port–factor relation. A multiwinding transformer is one factor with several port bundles until a guarded compilation explicitly creates a two-terminal realization.

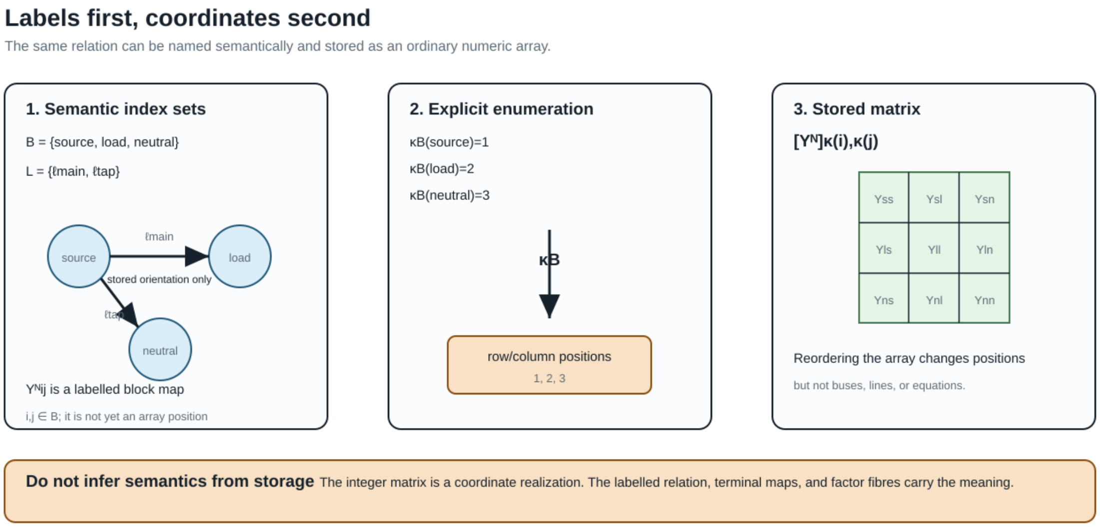

Figure 6.1: Semantic network labels are enumerated into storage coordinates before an ordinary matrix is formed.

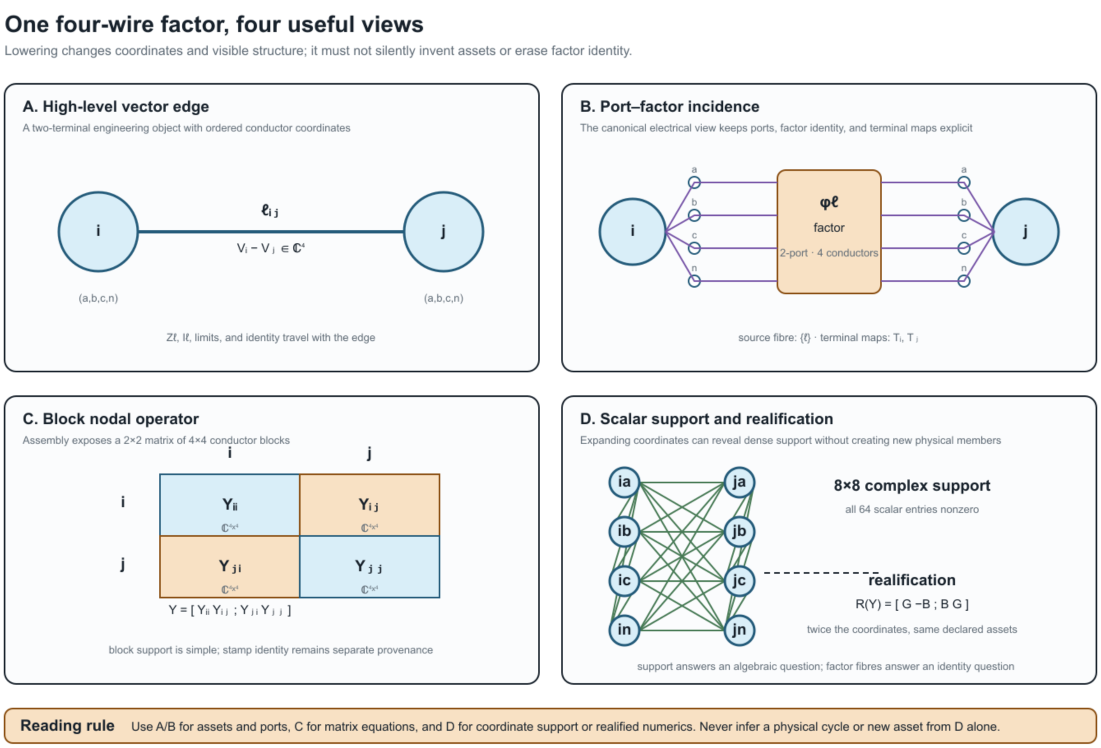

Figure 6.2: One four-wire factor across a vector edge, a port–factor incidence view, a block nodal operator, and scalar or realified coordinate support.

The four panels are deliberately paired. Panels A and B are the useful places to ask *which asset or factor is present*; panel C is the natural place to write the assembled block equation; panel D is a coordinate-level support or solver view. The dense lines in D are algebraic coupling, not a claim that the source contains that many physical branches. This scoped correspondence is checked by the executable witness registered as `ARCH-BLOCK-001`.

What a lossy edge looks like

Power engineers often draw an edge and imagine power flowing through it as one quantity. A circuit model normally computes terminal currents first, then complex power at each terminal, for example

$$S_{\ell ij} = \mathrm{diag}(\mathbf{U}_i[\mathbf{N}_{\ell i}])\mathbf{I}^*_{\ell ij}.$$

Series impedance, endpoint shunts, mutual coupling, and grounding can all make the two terminal powers differ. Even the terminal currents need not be simple negatives when shunts are present. A lossy edge is not a violation of graph theory; it is a constitutive factor with nonzero dissipation or local exchange. If a diagram shows one line with a π model, decide whether its shunts are inside the factor or drawn as separate factors before comparing currents or limits.

Reading scalar, vector, and block notation

The following translation is safe only when the qualifiers are retained:

Diagram or phrase	Mathematical reading	Main qualifier
scalar edge $i - -j$	one voltage coordinate at each endpoint	positive-sequence or single-conductor scope
vector-valued edge	$\mathbf{Z}_\ell$ or $\mathbf{Y}_\ell$ acts on ordered conductor coordinates	still a two-terminal factor
vector-valued node	bus i owns an ordered terminal space $\mathbf{U}_i$	terminals may be missing, permuted, or grounded
block nodal matrix	$\mathbf{Y}^{\mathrm{N}}_{ij}$ maps one bus-terminal block to another	block support is not an asset multigraph
scalar-expanded support graph	each conductor coordinate is a vertex	cycles mean algebraic coupling, not necessarily physical loops
realified model	real and imaginary coordinates are stacked	doubled coordinates are not doubled assets

The report by Geth, Claeys, and Heidari gives a practical four-wire example of this translation: series impedances and shunts are matrices, current-injection methods are centered on a nodal-admittance representation, and radial backward–forward sweep uses a different impedance-oriented computational view [1]. Those are alternative formulations of the same declared circuit model, not competing definitions of what a bus or line is.

A diagram-reading checklist

Before using a graph in a proof, algorithm, or optimisation model, annotate it with:

- **level:** asset, equipment/terminal, port–factor, equation, block support, or scalar support;
- **coordinates:** scalar, phase/neutral vector, sequence, rectangular real, or another declared basis;

- **orientation:** stored terminal order, sign convention, or actual control direction;
- **constitutive scope:** series only, nominal-π, transformer, shunt, grounding, nonlinear load, or an n-port relation;
- **preservation:** identities, KCL/KVL/terminal behaviour, limits, decisions, and provenance that remain available;
- **loss:** what has been summed, eliminated, projected, realified, or made implicit.

This checklist is the short route into the longer two topology levels and the nodal projection chapter. There the same distinctions are stated as maps between typed objects and block operators rather than as visual intuition alone.

Chapter 7

One network, five languages

Page status: explanatory cross-community vocabulary bridge; not a standards crosswalk.

The communities that work on power networks often use the same word for different objects, and different words for nearly the same object. This is more than a stylistic inconvenience. If *edge*, *state*, *flow*, or *equivalent* changes meaning halfway through an argument, a correct statement about one representation can become a false statement about another.

This book therefore uses a **bridge vocabulary**. It does not ask readers to abandon familiar language. It asks them to qualify a familiar term when that term carries mathematical, physical, software, or decision meaning.

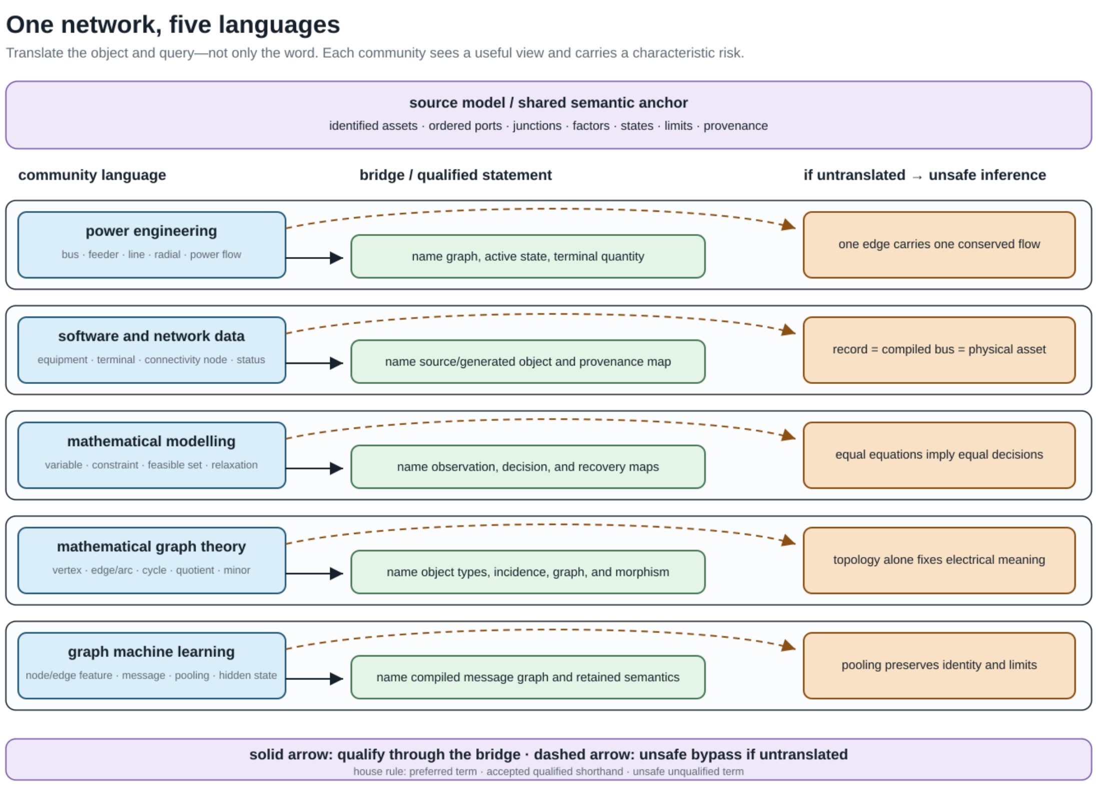

Figure 7.1: Five communities approach the book's shared semantic anchor through different vocabularies; a dashed bypass shows the unsafe inference made when the bridge is skipped.

The running network in five dialects

Consider the same multiconductor network with parallel circuits, explicit neutral and grounding, phase-selective switches, limits, and a multiwinding transformer. Each community has a useful description of it:

- **Power engineering:** buses, feeders, lines, transformers, radial operation, and power flow support physical interpretation and operating practice. The graph, terminal sign, device boundary, and active state may remain implicit.
- **Power-system software and data:** equipment records, terminals, connectivity nodes, statuses, profiles, and compiled buses support exchange, topology processing, and provenance. A record may represent a physical asset, generated object, or equation object.
- **Mathematical modelling and optimization:** variables, constraints, parameters, feasible sets, objectives, and relaxations support decision semantics and formal comparison. Source identity and physical recovery may be outside the formulation.
- **Mathematical graph theory:** vertices, edges, arcs, multigraphs, incidence, cycles, quotients, and minors support structural theorems and algorithms. The physical referent and load-bearing attributes must be added.
- **Graph machine learning:** heterogeneous nodes and edges, features, message passing, pooling, embeddings, and hidden states support learned computation. Parallel identity, n-port structure, limits, and physical state survive only when the compiled graph and feature maps retain them.

None of these descriptions is the universal one. The bridge is the typed map from a community's phrase to the object and query meant in this book.

Circuit theory is not counted as a sixth target community in this route. Its language of nodes, branches, ports, multiports, incidence, tableau equations, and equivalents is a shared technical inheritance of power engineering and mathematical modelling, and part of the precise target vocabulary used here. Its own translation failures remain important; they are developed in Translation traps and Circuit formulations and the lowering boundary.

Translation is not word substitution

For each community term, the relation to the book's term should be classified before it is reused:

- **Exact alias:** interchangeable under the declared scope. For example, Ybus can be an alias for the declared nodal operator after its coordinate order and model class are fixed. The alias is to that operator, not to the source network or factor decomposition that assembled it.
- **Scoped alias:** conventional shorthand that is safe only with a qualifier. The sentence *branch* ℓ *has stored reference orientation* ℓij is safe; *branch* ℓ *is directed from* i *to* j is not safe if it can be read as an operating-flow or one-way-admissibility claim.
- **Broader or narrower term:** one term contains distinctions omitted by the other. A physical line can contain several homogeneous model sections.
- **Representation-dependent term:** its referent changes with the selected view. *Node*, *edge*, *cycle*, and *radial* are the leading examples.
- **False friend:** the words coincide but the concepts do not. Electrical loss is not an ML loss; a GNN message is not a conserved power flow.

These labels are deliberately asymmetric. A term can be acceptable when reading a source and still be too weak to use in a preservation claim.

The collision set to learn first

Objects

Bus, node, vertex, junction, terminal, and port must not be collapsed into one noun. A bus may mean a busbar section, a connectivity node, a state-dependent topological node, a group of nodal variables, or a reporting aggregate. A graph vertex is whatever the declared graph makes a vertex. A port is a typed component interface; a junction supplies interconnection and conservation semantics.

Likewise, **line, branch, edge, arc, relation, and matrix nonzero** do not share one identity. A line ℓ owns intrinsic equipment or model data. The oriented triple ℓij orders its terminals. An edge in a support graph records algebraic coupling and may have no one-to-one physical asset.

An especially important collision is **factor**. Here it means a typed constitutive, control, measurement, limit, or decision relation over ports. It does not mean power factor or matrix factorization, although the same factor may become a node in a factor graph.

Electrical coordinates and reference

Ground, earth, neutral, grounding impedance, and voltage reference are not names for one zero-voltage node. Earth can be a physical return medium; a neutral is a conductor with a state and current; a grounding impedance is a factor between declared terminals; and a voltage reference removes gauge freedom. Eliminating a neutral coordinate does not eliminate its recovered current or a limit on that current.

Phase, conductor, terminal coordinate, and sequence also name different structures. A conductor is a physical or modelled path, a terminal coordinate is one ordered component of an interface vector, and a sequence component is a transformed coordinate. *Phase* a may refer to an asset label, a bus terminal, a voltage coordinate, or a phase-domain component; the terminal and coordinate maps decide whether those uses coincide.

Structure and state

Graph, topology, adjacency, parallel, cycle, tree, and radial require a named representation and usually an active state. A simple bus projection, an identified asset multigraph, a conductor-coordinate support graph, and a GNN message graph can give different answers for the same source network.

State is also overloaded. It can mean continuous electrical state variables, discrete equipment status, an operating scenario, an estimator state, or an ML hidden representation. The book uses a modifier whenever two of these meanings are in scope.

Quantities and computation

Direction may mean stored orientation, terminal-current sign, observed power transfer, rooted-tree order, causal dependence, one-way admissibility, or message direction. None implies the others.

Flow may mean a conserved commodity variable, internal series current, terminal current, terminal complex power, an operating transfer, or a learned message. A lossy AC device generally exposes a tuple of terminal powers rather than one antisymmetric edge-flow scalar.

Limit, rating, and constraint occupy different layers. A rating is an equipment or operational datum; a mathematical constraint is a particular encoding of admissibility; whether that constraint binds belongs to an operating point or solution.

Transformations and evidence

Projection, compilation, lowering, elimination, aggregation, reduction, coarsening, and pooling are not interchangeable ways of saying *make the graph smaller*. Compilation can make a graph larger. Elimination can create fill. Pooling can erase member identities required by a decision problem.

Equivalent, exact, and structure preserving are incomplete until their object is named. The relevant claim may concern an algebraic identity, boundary behaviour, topology, feasible decisions, objective value, limits, measurements, numerical sparsity, or provenance.

Two cross-community false friends deserve permanent warnings:

- **normalization** can mean per-unit conversion, conductor-coordinate canonicalization, schema normalization, or ML feature scaling;
- **loss** can mean electrical dissipation, information discarded by a map, or an ML training objective.

House policy: familiar words with explicit qualifiers

The five relation classes above classify the **map between vocabularies**. The three statuses below classify **how a term may be used in this book**. They are orthogonal: a false friend is often unsafe, but a representation-dependent term can be accepted shorthand once its representation is named.

The book uses three vocabulary statuses (`VOCAB-BRIDGE-001`):

1. A **preferred house term** is used in definitions, claims, contracts, and executable artifacts.
2. An **accepted qualified shorthand** is retained when it helps a community read naturally and its scope is stated nearby.
3. An **unsafe unqualified term** is accompanied by the missing representation, quantity, state, or preservation object.

For example, *radial feeder* remains useful prose. A load-bearing statement is written as *the active simple bus projection is a tree in state* σ. Similarly, *power flow on line* ℓ is replaced in an equation or limit claim by the relevant terminal observation $\mathbf{S}_{\ell ij}$ or $\mathbf{S}_{\ell ji}$ and its sign convention.

The recurring **Vocabulary bridge** callout belongs to this policy. It marks a local crossing between community language and the house vocabulary; it must name the missing object or qualifier and the inference that would otherwise be unsafe.

Three worked translations

> **Community phrase:** Power flows downstream on each edge of the radial feeder.

A testable translation names (i) the active bus-level graph, state, and root that define the parent relation; (ii) the oriented terminal at which active power is measured; and (iii) the fact that its sign is an operating result. It does not infer member-radiality, losslessness, or permanent upstream and downstream asset labels.

> **Community phrase:** Pool the parallel edges and run message passing on the network graph.

A testable translation names the computational graph and pooling map, then asks whether line identity, outage state, member limits, conductor coordinates, and recovery are inputs to the downstream query. If they are, the pooled graph needs side information or is not a sufficient representation for that task.

Community phrase: Ground the neutral, then remove the neutral node.

A testable translation separates the neutral conductor, its connections, each grounding impedance or earth-return factor, and the gauge reference. It then names whether *remove* means a topological projection, a fixed linear Kron elimination, or omission from the source model. If the neutral is eliminated, the recovery map must still evaluate every retained neutral-current, grounding, protection, and decision constraint in the declared domain.

Where to go next

The Translation traps chapter develops the most dangerous false inferences. Representation taxonomy defines the graph families, Notation and modelling conventions fixes semantic ownership and indices, and the maintained Terminology page provides the compact lookup vocabulary. The generated cross-community vocabulary indexes support both community-to-book and book-to-community lookup. The later preservation-contract chapters turn these linguistic qualifications into mathematical obligations.

Chapter 8

Reading guide: graph and transmission readers

Page status: audience-specific route map; the formal definitions and executable evidence remain in the linked foundation and case chapters.

This book has two common starting points. A graph reader brings vertices, edges, incidence, cycles, and sparsity. A power-system reader brings a balanced bus–branch model, often in positive sequence. Both are useful specializations. Neither is the full multiconductor power-network model.

Choose a route below, then join the shared path at the four-wire relation and the first decision counterexample.

Two routes into one language

Starting point	Familiar model	First qualification	Shared question
Simple graph theory	vertices, edges, paths, cycles, incidence	an electrical factor can be vector-valued, and a support cycle need not be a physical loop	which graph, coordinates, and identities are retained?
Balanced transmission modelling	one complex relation per bus pair, often positive sequence	phase/neutral coordinates, coupling, shunts, grounding, ports, and limits may have been collapsed	which assumptions and decision constraints make the specialization valid?

If you come from simple graph theory

Follow this short sequence:

1. **Edge → terminal relation.** A scalar branch may satisfy $i_{\ell ij} = y_\ell(v_i - v_j)$. A four-wire factor instead maps ordered terminal spaces: $\mathbf{I}_{\ell ij} = \mathbf{Y}_\ell(\mathbf{U}_i[\mathbf{N}_{\ell i}] - \mathbf{U}_j[\mathbf{N}_{\ell j}])$. The identity ℓ and attachment triple ℓij are not matrix coordinates.
2. **Vertex → terminal space.** A bus can own phase and neutral terminals, so one graph vertex may represent a block of variables. A dense block records coupling; it does not create one asset per nonzero entry.
3. **Simple projection → identified multigraph.** Parallel members retain separate ratings, states, owners, and outage decisions even when their unconstrained admittances add.

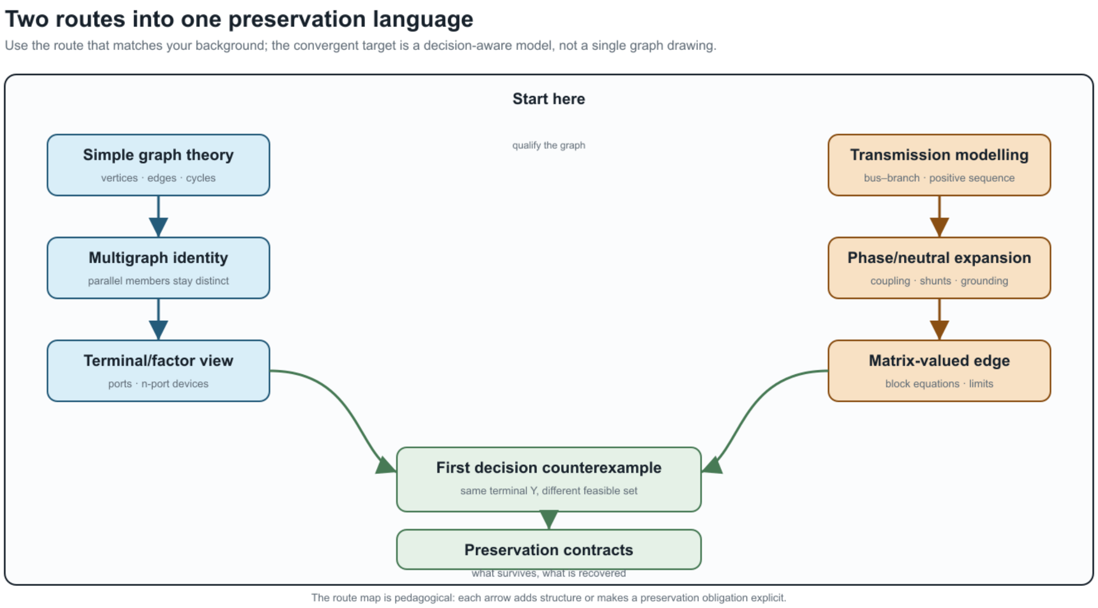

Figure 8.1: Two onboarding routes converge on preservation contracts.

4. **Cycle → qualified cycle.** Member cycles, simple-quotient cycles, port–factor cycles, and scalar support cycles answer different questions.

5. **Edge → factor.** A three-winding transformer is one typed factor with three port bundles. A star or clique is a derived lowering target, not an automatic replacement.

Then read Five-bus cycle spaces, Five buses through a multi-port lowering, Two topology levels and the nodal projection, and Circuit formulations and the lowering boundary.

If you come from balanced transmission modelling

Use this sequence to unpack the familiar model without discarding it:

1. **Positive sequence is a specialization.** Ask which balance, transposition, grounding, and frequency assumptions justify it for the study.

2. **Expand one edge first.** Move from scalar edge to four-wire matrix edge, then to a port–factor relation and a block nodal operator. At each step record added coordinates, coupling, shunts, terminal maps, and decisions.

3. **Requalify flow.** A lossy AC factor has terminal currents and powers; endpoint shunts and grounding can make the two terminal observations different. Stored orientation fixes signs, not operating direction.

4. **Keep eliminated quantities accountable.** Kron reduction can preserve a boundary voltage relation while removing neutral current, grounding, or protection constraints from the reduced problem unless they are recovered.

5. **Promote transformers to factors.** Multiwinding devices, connection maps, internal grounding, and controls may require a typed factor or tableau/MNA target rather than an ordinary branch.

6. **Treat radial language as conditional.** Upstream/downstream belongs to a selected active rooted tree; switching and meshing can invalidate it.

Then read When the general model collapses, From conductor geometry to impedance fidelity, A first failure: heterogeneous parallel branches, and Kron, Ward, and optimized network equivalents.

The shared convergence point

Both routes meet at one preservation question:

> What does this representation preserve for the observation, constraint, and decision I care about?

The deliberately small counterexample is two identified parallel branches: their terminal admittances can sum exactly while their individual current limits produce a different feasible set. Continue with Preservation contracts and Translation traps whenever *edge*, *flow*, *radial*, or *equivalent* appears without a qualifier.

The practical checklist is short: name the objects and index meanings; state the coordinates and graph view; distinguish stored orientation from operating transfer; and write the preservation target (behaviour, limits, decisions, objectives, recovery, or provenance). This is enough to let both communities share notation without treating either starting graph as the universal ontology.

Chapter 9

A five-bus multigraph: identities, cycles, and tree coordinates

Page status: generated worked graph example with hash-bound figures, BMOPFTools cross-checks, and companion conductor-terminal and multi-port lowering witnesses.

Purpose and status

This worked example separates four operations that are often drawn as if they formed one reduction chain: forgetting parallel identity, constructing an electrical aggregate, eliminating internal variables, and selecting a spanning tree. They do not have the same source, target, or preservation contract.

The general definitions of simple cycles, line-identity cycles, parallel fibres, bridges, leaves, and adjacency-versus member-radiality are given in Cycles, parallelism, and radial structure. This chapter supplies the running numerical witness for those definitions.

The graph identities below are standard finite-dimensional results. The line-indexed calculations, nodal-admittance comparison, decision witness, and BMOPFTools topology cross-check are executable. The figures are generated from that same analysis record rather than maintained as independent sketches. The scalar values are chosen to make the distinctions visible; the final section lifts the construction to the book's multiconductor and multi-terminal scope.

Source bus–branch multigraph

Let the buses and identified lines be

$$\mathcal{B} = \{i, j, k, l, m\}, \qquad \mathcal{L} = \{q, r, s, t, u, v, w\}.$$

Using the book's element–endpoint convention, declare

$$\mathcal{T}^{L\to} = \{qji,\ rij,\ sjk,\ tki,\ vlj,\ wkl,\ ulm\}.$$

Thus q and r are distinct parallel lines even though their declared orientations are opposite. Reversing either arrow changes signs in oriented coordinates but does not change the underlying undirected multigraph.

With bus order (i, j, k, l, m) and line order (q, r, s, t, v, w, u), take -1 at the declared from bus and $+1$ at the declared to bus. The incidence matrix is

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 & 1 & 0 & 0 & 0 \\ -1 & 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The connected graph has $\operatorname{rank}(\mathbf{A}) = 4$. Its real cycle space therefore has dimension

$$\mu = \dim\ker\mathbf{A} = |\mathcal{L}| - \operatorname{rank}(\mathbf{A}) = 7 - 4 = 3.$$

More generally, a loopless undirected multigraph with c connected components has

$$\mu = |\mathcal{L}| - |\mathcal{B}| + c.$$

The displayed integer matrix below is only one coordinate realization. Its semantic rows are indexed by buses $i \in \mathcal{B}$ and its columns by identified lines $\ell \in \mathcal{L}$. If $\kappa_{\mathcal{B}}$ and $\kappa_{\mathcal{L}}$ enumerate those sets, then the stored array satisfies

$$[\mathbf{A}]_{\kappa_{\mathcal{B}}(i),\kappa_{\mathcal{L}}(\ell)} = A_{i\ell}.$$

This distinction matters even in this small example: the line labels q and r are not row or column numbers, and the cycle labels $\mathcal{C}_q, \mathcal{C}_t, \mathcal{C}_v$ are not required to be integers. The integer ordering is useful for computation; the labelled incidence is what retains the identity of parallel members.

This count uses identified lines. Deduplicating endpoint pairs before counting cycles gives a different graph and can give a different answer.

A line-indexed fundamental cycle basis

Choose the spanning tree

$$\mathcal{T}_{\text{tree}} = \{r, s, w, u\}.$$

Its chords are q, t, and v. Adding each chord to the tree gives

$$\mathcal{C}_q = \{q, r\}, \qquad \mathcal{C}_t = \{r, s, t\}, \qquad \mathcal{C}_v = \{s, v, w\}.$$

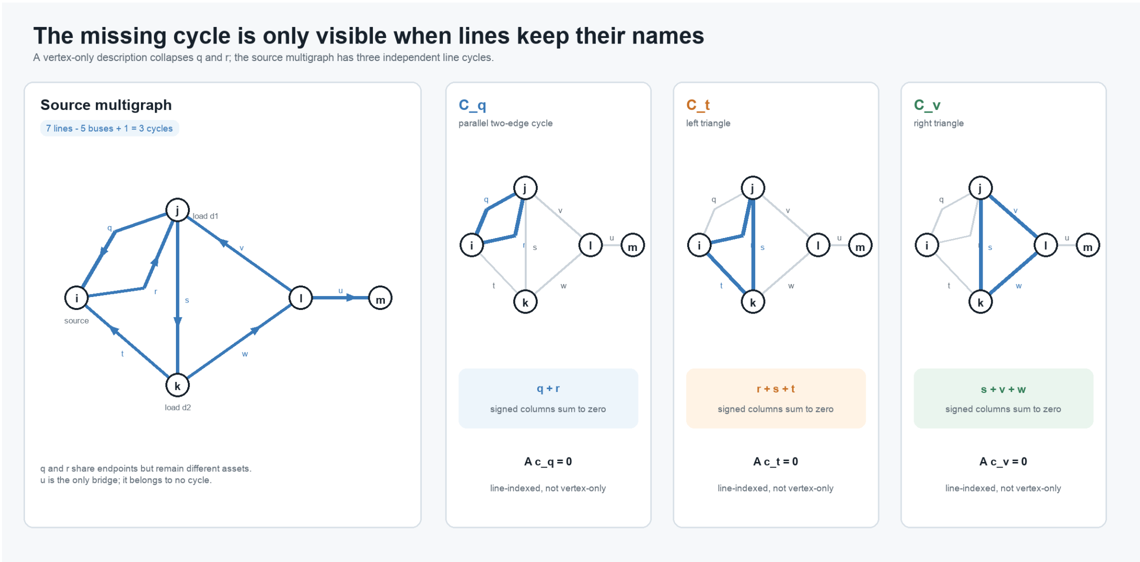

Figure 9.1: The source multigraph and the three line-indexed fundamental cycles.

The small multiples make the missing dimension visible. The cycle $\mathcal{C}_q$ has only two buses but two distinct line columns; it disappears as soon as q and r are represented by one adjacency.

The first cycle is the two-edge cycle made by the parallel pair. With the declared orientations, the signed edge–cycle matrix is

$$
\mathbf{C} = \begin{array}{c|ccc}
 & \mathcal{C}_q & \mathcal{C}_t & \mathcal{C}_v \\
\hline
q & 1 & 0 & 0 \\
r & 1 & 1 & 0 \\
s & 0 & 1 & 1 \\
t & 0 & 1 & 0 \\
v & 0 & 0 & 1 \\
w & 0 & 0 & 1 \\
u & 0 & 0 & 0
\end{array}, \qquad \mathbf{AC} = \mathbf{0}.
$$

Equivalently, write the relation without choosing storage coordinates as $C_{\ell\gamma}$ for $\ell \in \mathcal{L}$ and $\gamma \in \Gamma$. The equation $A_{i\ell}C_{\ell\gamma}$ is then a finite labelled sum over the shared line index ℓ. This is the same label-before-coordinate principle used for nodal blocks and terminal maps.

Cycle bases are not unique. A *minimum* cycle basis additionally requires nonnegative edge weights and minimizes total basis weight [2]. A list of vertex tuples alone is insufficient for the source multigraph: replacing q by r produces a different line cycle, and the two-edge cycle cannot be represented without member identity.

Line u is the only **bridge**: removing it increases the number of connected components. Calling u a radial edge is less precise, and calling m a radial bus does not generalize well. A cycle-core bus can also be the attachment point of a bridge and a pendant tree. Bridges, the graph 2-core, and block–cut structure express those roles without assigning radiality to an individual vertex.

Projection is not electrical aggregation

Let π forget line identity and retain only a canonical unordered endpoint pair. Then

$$\pi(q) = \pi(r) = e_{ij}.$$

The resulting simple graph has six edges and cycle rank

$$6 - 5 + 1 = 2.$$

This is a topology projection. It loses the q–r two-cycle, member orientations, ratings, states, ownership, and provenance. Under the standard definition it also has neither parallel edges nor self-loops.

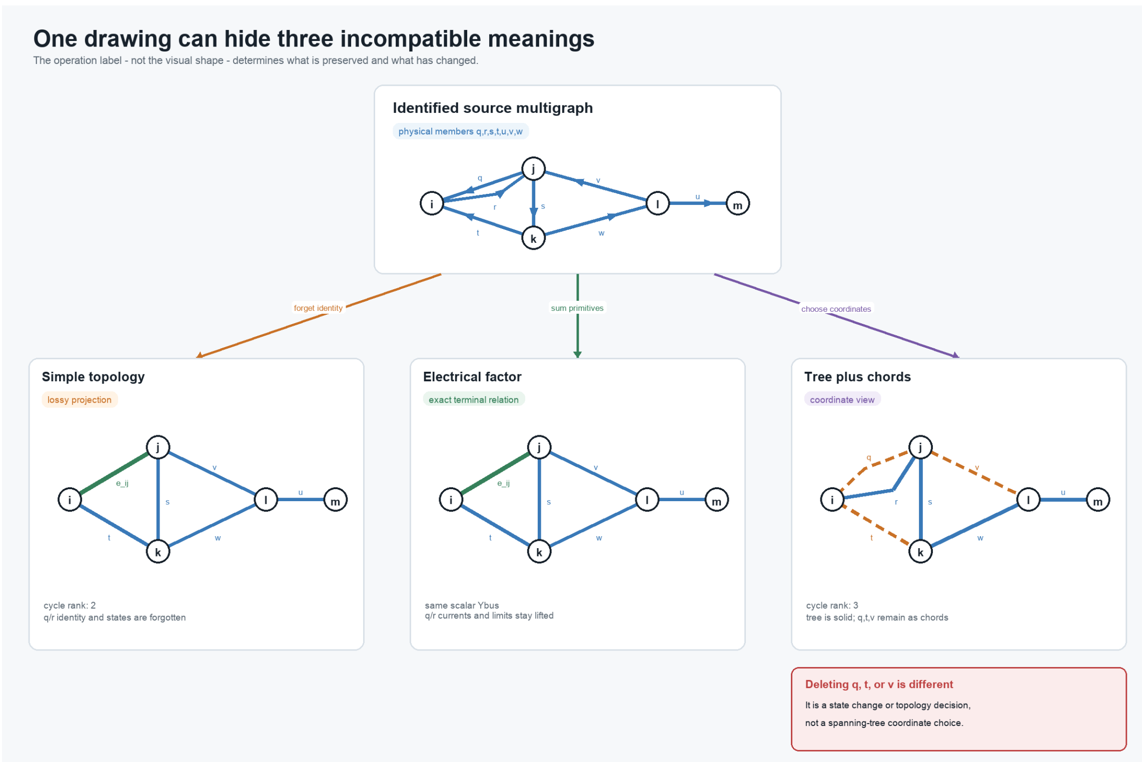

Figure 9.2: Three typed constructions that can otherwise look like one graph-simplification chain.

The first two lower panels intentionally have the same graph shape. One means only that member identity was forgotten. The other carries a summed electrical factor while the source-member recovery maps and limits remain available. The third panel has not removed anything: it merely draws the selected tree solid and the three source chords dashed.

An electrical aggregate is a separate construction. For reciprocal scalar series branches on common endpoint coordinates,

$$I_{ij}^{\text{total}} = (Y_q + Y_r)\,(U_i - U_j), \qquad Y_{ij}^{\text{eq}} = Y_q + Y_r.$$

If both impedances are invertible, this means

$$Z_{ij}^{\mathrm{eq}} = \left(Z_q^{-1} + Z_r^{-1}\right)^{-1},$$

not $Z_q + Z_r$. With the oriented incidence matrix and diagonal series admittance matrix,

$$\mathbf{Y}^{\mathrm{N}} = \mathbf{A}\,\mathrm{diag}(Y_\ell)\mathbf{A}^\mathsf{T}.$$

The executable example verifies that stamping the seven source members and stamping the six-edge weighted projection produce identical $\mathbf{Y}^{\mathrm{N}}$. In an explicitly expanded scalar matrix, the m, m entry is $+Y_x$; the off-diagonal l, m entries are $-Y_x$.

That equality is only a terminal-current statement. In the recorded witness, $Y_q = 10$ S, $Y_r = 1$ S, and each member has a 100 A limit. At $|U_i - U_j| = 15$ V,

$$|I_q| = 150\ \mathrm{A}, \qquad |I_r| = 15\ \mathrm{A}.$$

The source is infeasible, while the summed check

$$|I_q + I_r| = 165\ \mathrm{A} \le 200\ \mathrm{A}$$

passes.

The same source-versus-aggregate feasible-set geometry is shown in the complex-plane figure in A first failure: heterogeneous parallel branches. This five-bus chapter keeps the network topology and recovery discussion local rather than repeating the scalar card.

The weighted simple graph can still be used exactly if the member recovery maps and nonredundant source constraints remain lifted. This is the same decision-preservation mechanism developed in A first failure: heterogeneous parallel branches.

A spanning tree is a coordinate choice

The chosen tree gives a unique parent–child relation after selecting a root. It does so even though the source graph is meshed. What is nonunique is the choice of tree, not the possibility of constructing a hierarchy.

Let the vector of oriented branch-voltage drops be

$$\mathbf{u}_L = \mathbf{A}^\mathsf{T}\mathbf{U}.$$

Then every source cycle satisfies

$$\mathbf{C}^\mathsf{T}\mathbf{u}_L = \mathbf{C}^\mathsf{T}\mathbf{A}^\mathsf{T}\mathbf{U} = \mathbf{0}.$$

Tree paths can parameterize voltage differences, while chord equations impose the remaining cycle consistency. Similarly, the nullspace term in a branch-current description is indexed by cycle coordinates. Lines q, t, and v remain physical members with constitutive equations, limits, states, and provenance.

Deleting a chord is therefore not a graph-coordinate operation. It changes the network unless the line is already open, a decision selects it open, or a separate certificate proves the relevant relation and constraints redundant. For a radial-topology decision, the active identified lines must form a tree or forest under the declared connectivity contract. Counting only simple adjacencies would miss the two-edge cycle created when both q and r are active.

Why the apparent reductions are guarded

In the projected simple graph, bus i has degree two, but it is also the source/injection bus. The degree-two series rule must reject its elimination: topological degree alone does not establish a common series current. The exact guards are given in Degree-two series elimination.

Line u is a dangling bridge, but it cannot be removed merely because it is outside every cycle. If bus m carries a load, generator, measurement, bound, contingency, or retained voltage, pruning u changes the study. Pendant-tree removal is valid only under an observation and decision contract that either excludes that subtree or replaces it with a certified boundary model.

These distinctions classify the useful constructions as follows:

Construction	Classification	What must remain explicit
identified multigraph to unweighted simple graph	projection	forgotten member data if recovery is required
parallel primitive summation	exact terminal normalization under guards	member recovery and nonredundant constraints
guarded junction elimination	exact behavioral reduction	recovery, residual constraints, and provenance
spanning-tree selection	coordinate or algorithmic view	every chord and its source equations
chord opening	fixed state or topology decision	switch/line identity and connectivity contract

The same distinctions survive when scalar branches are expanded into conductor-terminal relations and multi-terminal factors.

Multiconductor and multi-terminal lift

For a multiconductor line, the scalar relation becomes

$$\mathbf{I}^{\mathrm{s}}_{\ell ij} = \mathbf{Y}_{\ell} \left(\mathbf{U}_i[\mathbf{N}_{\ell i}] - \mathbf{U}_j[\mathbf{N}_{\ell j}] \right).$$

Parallel primitives can be added only after orientations, ordered conductor coordinates, terminal maps, units, and base quantities are aligned. For a nominal-π or general fixed linear two-terminal member, the object to stamp is its complete terminal primitive, not only $\mathbf{Y}_{\ell}$. Component, grouped, and both-end limits remain member-indexed unless a separate implication certificate removes them.

The asset-level cycle space still records line identity, but it need not equal a conductor-resolved cycle space. Missing phases, explicit neutrals, grounding, and conductor permutations can give different connectivity by terminal. A cycle-based formulation must say whether its incidence describes buses, conductor terminals, galvanic zones, factors, or algebraic variables.

A genuinely multiwinding transformer has no canonical representation as one ordinary bus–branch edge. Its natural cycle participation belongs to the port–factor or terminal-incidence view. Any cycle basis of a compiled two-terminal realization is relative to that compilation and must retain a map back to the source transformer and windings.

The companion Five buses through a multi-port lowering keeps this chapter's line-induced graph unchanged and adds a symbolic three-port transformer extension. It compares factor incidence, a generated star, and the terminal clique produced by internal elimination, with a separate cycle rank for every declared graph.

Executable evidence

The package-independent implementation constructs $\mathbf{A}$, the fundamental basis, bridges, simple projection, and both nodal admittances. BMOPFTools independently analyzes the same seven identified line records.

The recorded checks give seven source lines on five buses, source incidence rank four, source cycle rank three, and simple-projection cycle rank two. They also give

$$\|\mathbf{AC}\|_\infty = 0, \qquad \|\mathbf{Y}^{\mathrm{N}}_{\mathrm{source}} - \mathbf{Y}^{\mathrm{N}}_{\mathrm{projected}}\|_\infty = 0.$$

BMOPFTools reports three extra physical edges, and the independently computed bridge set is $\{u\}$.

Chapter 10

A first failure: heterogeneous parallel branches

Page status: introductory counterexample with executable scalar and multiconductor evidence; independent mathematical review remains open.

The tempting replacement

Here is the decision-level version of the same trap. Two models can have the same assembled nodal admittance and still answer a dispatch question differently:

The resolving phrase is **which observations and constraints?** The equality of the unconstrained terminal map is real; it simply does not include member ratings, outage states, or independent controls.

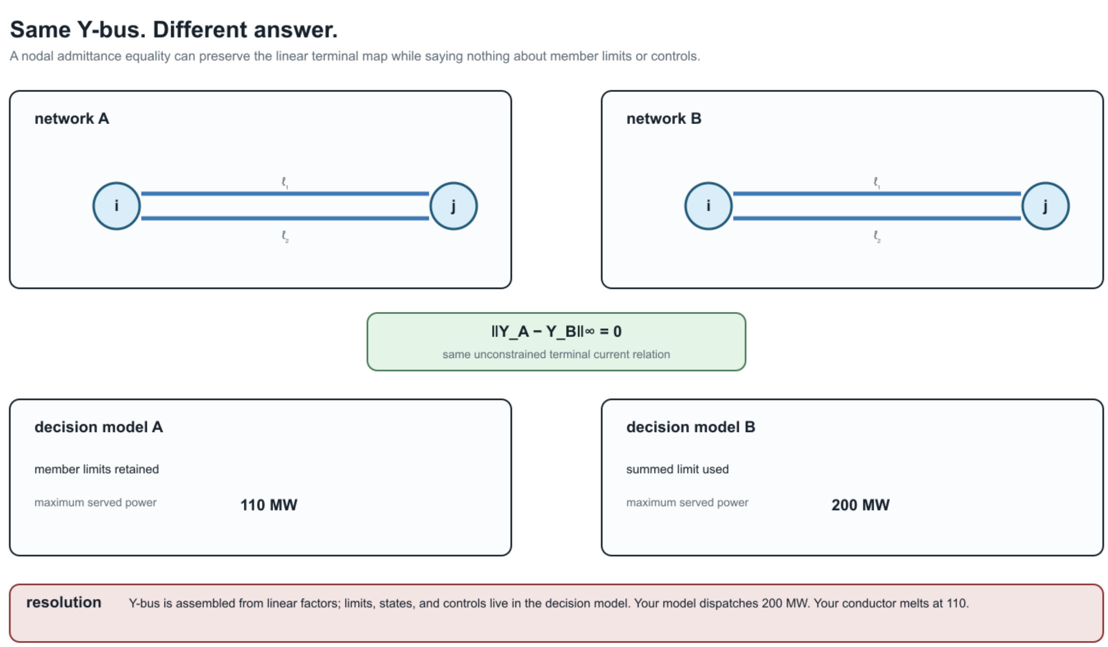

Figure 10.1: Same nodal admittance, different dispatch because member limits were lost.

Consider two scalar resistive branches $\ell_1 ij$ and $\ell_2 ij$ with intrinsic impedances

$$Z_{\ell_1} = 0.1\ \Omega, \qquad Z_{\ell_2} = 1.0\ \Omega,$$

and member current limits

$$I_{\ell_1}^{\max} = I_{\ell_2}^{\max} = 100\ \mathrm{A}.$$

Writing $\Delta U = U_i - U_j$ and $Y_\ell = Z_\ell^{-1}$, the terminal currents are

$$I_{\ell_k ij} = Y_{\ell_k} \Delta U, \qquad k \in \{1, 2\}.$$

Here k is a local member label for the two identified branches. It is not an instruction to use the first and second rows of a stored array; the member identity is ℓ_k and remains meaningful under any reordering.

Replacing the pair by one branch with

$$Y_{\mathrm{eq}} = Y_{\ell_1} + Y_{\ell_2} = 11\ \mathrm{S}$$

is exact for the unconstrained total terminal-current relation. It does not follow that the replacement is exact for a decision problem with member limits.

The feasible sets differ

The source model requires both inequalities

$$|Y_{\ell_1} \Delta U| \le I_{\ell_1}^{\max}, \qquad |Y_{\ell_2} \Delta U| \le I_{\ell_2}^{\max}.$$

Hence its admissible voltage-drop magnitude is

$$|\Delta U| \le \min\left\{ \frac{I_{\ell_1}^{\max}}{|Y_{\ell_1}|}, \frac{I_{\ell_2}^{\max}}{|Y_{\ell_2}|} \right\} = 10\ \mathrm{V}.$$

A common aggregate construction assigns the equivalent branch the summed rating, $I_{\mathrm{eq}}^{\max} = 200\ \mathrm{A}$. It admits

$$|\Delta U| \le \frac{200}{11} = 18.18\ \mathrm{V}.$$

At the witness $\Delta U = 15\ \mathrm{V}$, the equivalent branch carries $165\ \mathrm{A}$ and satisfies its aggregate limit. Recovery gives

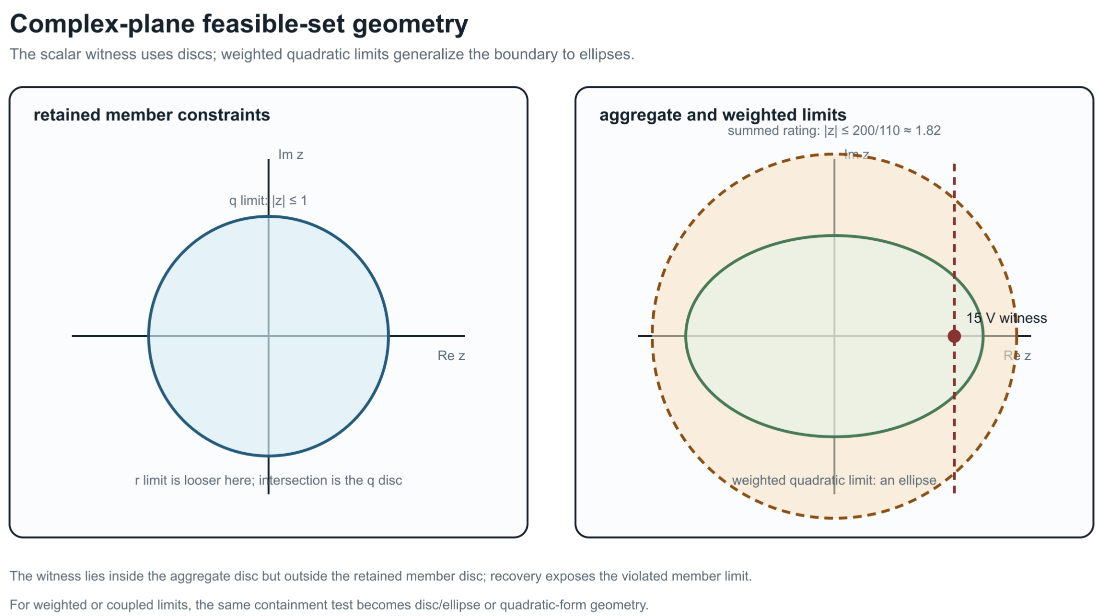

Figure 10.2: Complex-plane feasible-set geometry for retained member limits, aggregate limits, and weighted quadratic limits.

$$I_{\ell_1 ij} = 150\ \mathrm{A}, \qquad I_{\ell_2 ij} = 15\ \mathrm{A},$$

so the source violates the ℓ_1 limit. The target feasible set is therefore an outer relaxation of the source feasible set.

The figure is generated by `experiments/render_parallel_feasible_set_card.py`. For the scalar witness, the retained and aggregate regions are discs in the complex ΔU plane; the ellipse is the corresponding visual reminder for weighted or coupled quadratic limits.

Preservation contract

Contract field	Value
source	two identified parallel branches with individual limits
target	one equivalent branch with summed admittance and rating
preconditions	common endpoints and voltage coordinates; linear branch laws
preserves	unconstrained total terminal current as a function of ΔU
does not preserve	the member-constrained feasible set
forgets	member identity and independent outage, maintenance, or investment state
recovery	$I_{\ell_k ij} = Y_{\ell_k} \Delta U$ if member admittances remain available
classification	outer/relaxed for this rating construction

This counterexample does not say that parallel aggregation is never useful. It says that an aggregate rating needs its own derivation relative to the intended observation or decision set. Choosing a tighter equivalent limit can reproduce this one scalar voltage-drop bound, but it still does not recreate independent member states or arbitrary member-wise constraints.

Certified redundancy is different from aggregation

Some member limits can be removed exactly without replacing the physical members. For a fixed scalar AC π-line, write the endpoint-voltage state as

$$x = \begin{bmatrix} \Re(U_i) & \Re(U_j) & \Im(U_i) & \Im(U_j) \end{bmatrix}^\mathsf{T}.$$

After normalization by the applicable current or apparent-power rating, a flow limit at the ij end has a positive-semidefinite quadratic feasible set

$$\mathcal{E}_{\ell ij} = \{x : x^\mathsf{T} M_{\ell ij} x \le 1\}.$$

If $\mathcal{E}_{\ell_k ij} \subseteq \mathcal{E}_{\ell_r ij}$, satisfying member ℓ_k's limit implies member ℓ_r's limit at that end. Molzahn gives an eigenvalue-based containment test that also handles singular quadratic forms, whose feasible sets are cylinders rather than bounded ellipsoids [3]. A complete line-limit removal requires the implication at both ij and ji ends.

This is exact constraint pruning, not asset aggregation: both line laws, parameters, identities, and recovered flows remain in the model. Failure to identify redundancy is also not proof that a limit is essential; the test is a sufficient certificate. Its stated scope is fixed scalar AC transmission π-models, including fixed complex transformer ratios and shunts. It does not by itself cover arbitrary multiconductor constraint sets, switching, outages, investment states, or other state-dependent parameters. Claim `LIT-PAR-001` records that boundary.

Multiconductor form

For multiconductor branches,

$$\mathbf{I}_{\ell_k ij} = \mathbf{Y}_{\ell_k} \Delta \mathbf{U},$$

and the source feasible set is the intersection of the inverse images of every member constraint set:

$$\mathcal{D}_{\mathrm{src}} = \bigcap_k \{\Delta \mathbf{U} : \mathbf{Y}_{\ell_k} \Delta \mathbf{U} \in \mathcal{C}_{\ell_k}\}.$$

The summed terminal map $\mathbf{Y}_{\mathrm{eq}} = \sum_k \mathbf{Y}_{\ell_k}$ does not, by itself, encode that intersection. Mutual coupling and per-conductor limits make a single conventional rating still less likely to be an exact representation.

The generation script emits the numerical witness and its machine-readable contract as claim `TR-PAR-001`.

A decision problem exposes the gap

The same mechanism changes an optimum in a two-bus maximum-served-load model. Let the parallel members have

$$(b_{\ell_1}, b_{\ell_2}) = (1000, 100)\ \mathrm{MW/rad}, \qquad (F^{\max}_{\ell_1}, F^{\max}_{\ell_2}) = (100, 100)\ \mathrm{MW},$$

with $F_{\ell ij} = b_\ell \delta_{ij}$, nonnegative δ_{ij}, and served power equal to total flow. The source model keeps both member laws and both limits. The naïve target uses $b_{\mathrm{eq}} = 1100$ MW/rad and $F^{\max}_{\mathrm{eq}} = 200$ MW.

Formulation	Maximum served power	δ_{ij}	Active restriction
source members	110 MW	0.1 rad	$F_{\ell_1 ij} \leq 100$ MW
naïve aggregate	200 MW	$2/11$ rad	summed 200 MW rating
aggregate relation with exact lifted member constraints	110 MW	0.1 rad	recovered $F_{\ell_1 ij} \leq 100$ MW

The displayed exact lifted formulation retains every member law and both member limits:

$$F_{\ell ij} = b_\ell \delta_{ij}, \qquad F_{\ell ij} \leq F^{\max}_\ell \quad \text{for every } \ell,$$

alongside the aggregate terminal relation. It therefore reproduces the source feasible set and optimum without pretending that a summed scalar rating is exact. This computed comparison is claim `TR-PAR-003`. JuMP and Ipopt produce the machine-readable result in `experiments/generated/parallel-opf-comparison.json`; the analytic values above also provide a solver-independent check.

Here $b_{\ell_2} = 0.1 b_{\ell_1}$ and the ratings are equal, so the ℓ_2 limit is actually implied by the ℓ_1 limit and can be certifiably pruned. Keeping both in the table isolates the lifting argument from presolve; the next case implements the pruned formulation explicitly.

The Multiconductor parallel AC decision case extends this comparison to coupled complex conductor equations, phase-to-neutral constant power, and voltage-magnitude constraints.

Chapter 11

Five buses through a multi-port lowering

Page status: generated structural lowering example composed from the stable five-bus topology fixture and the checked three-winding transformer contract; it is not a complete transformer power-flow model.

The preceding five-bus multigraph deliberately uses only identified two-terminal lines. That restriction makes its incidence matrix, cycle space, simple projection, and spanning-tree coordinates unambiguous. This chapter keeps that line graph unchanged and adds a second, orthogonal question:

> What happens when one genuinely three-port transformer must be presented to representations and algorithms with different device vocabularies?

The answer is not one longer reduction ladder. The source asset, its electrical factor, an optional ordinary-edge realization, an assembled operator, and the operator's support graph are different objects. Each has a different interface, and the valid targets branch according to the intended study.

The unchanged topology kernel

Retain the line-induced multigraph

$$G_L = (\mathcal{B}, \mathcal{L}, \partial), \qquad \mathcal{B} = \{i, j, k, l, m\}, \qquad \mathcal{L} = \{q, r, s, t, u, v, w\}.$$

Its member cycle rank remains $\mu_L = 3$ and its simple projection remains at $\mu_{L,s} = 2$. Adding another asset to the source inventory does not alter those statements because their domain is explicitly the line-induced graph G_L.

Now introduce transformer x_1 with winding set

$$\mathcal{K}_{x_1} = \{1, 2, 3\}, \qquad \beta_{x_1}(1) = j, \quad \beta_{x_1}(2) = l, \quad \beta_{x_1}(3) = m.$$

The labels $1, 2, 3$ here identify windings in this worked example; they are not claims that winding identities are intrinsically integer-valued. A source format may name the same ports HV, LV, and $tertiary$. The terminal ordering and any storage enumeration are separate declarations.

These attachments form a pedagogical structural extension. The electrical data and winding-interface semantics are inherited from the checked running transformer contract; this page does not pretend that moving that factor onto the five-bus drawing creates a new validated power-flow case.

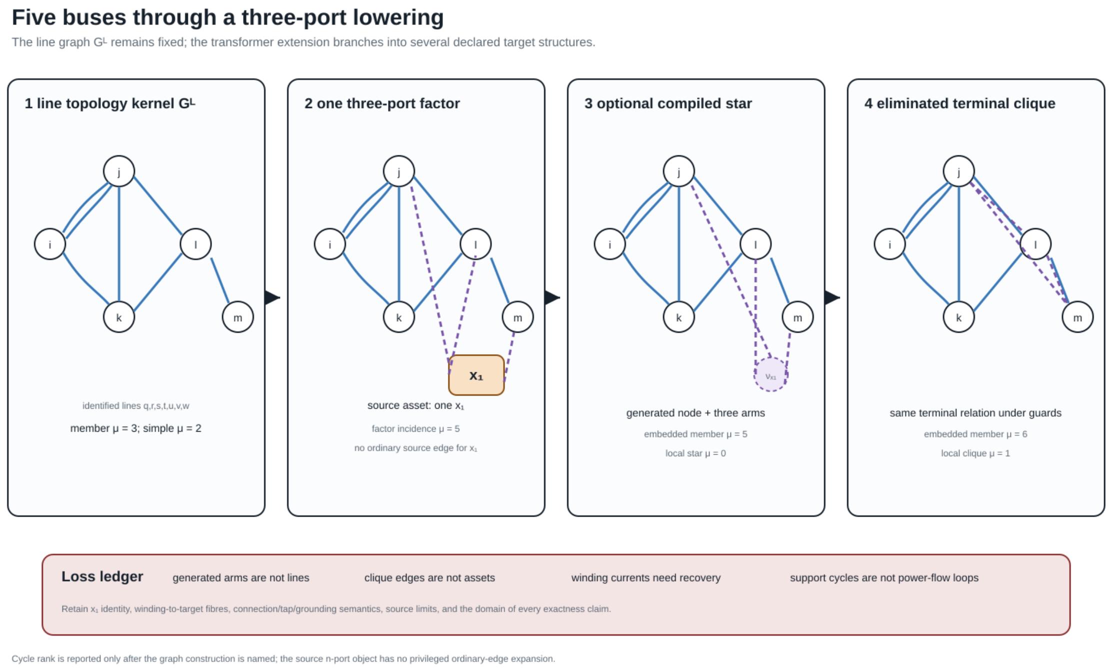

Figure 11.1: The stable five-bus line graph extended by one three-port transformer and two guarded target constructions.

The first panel is still G_L and shows both members q and r of its parallel fibre. The second adds one factor of arity three, not three source lines. The third and fourth are generated targets. Some ranks coincide and others differ; neither fact establishes semantic equivalence because the edge and vertex types are different.

One transformer, several legitimate graphs

Four constructions must be separated.

Identified asset and three-port factor

At the source level, x_1 is one asset with three identified windings. At the canonical electrical level it is one factor ϕ_{x_1} with three ordered port bundles

$$\mathcal{Q}_{x_1} = \{p_{x_1 1}, p_{x_1 2}, p_{x_1 3}\}.$$

Each port carries its own voltage/current space, terminal order, connection map, limit observations, and winding identity. WYE and DELTA ports need not have the same terminal dimension. Factor arity is therefore not the number of scalar conductor terminals.

Pair-test data graph

Complete three-winding short-circuit input supplies the three pair-indexed quantities z^{sc}_{12}, z^{sc}_{13}, and z^{sc}_{23}. Drawing these as the edges of K_3 is a useful **data-incidence graph**. The pair labels 12, 13, and 23 identify tests between winding pairs; they are not entries of a three-by-three matrix. Its edges index tests; they are not three independent two-winding transformers and do not acquire independent outage states.

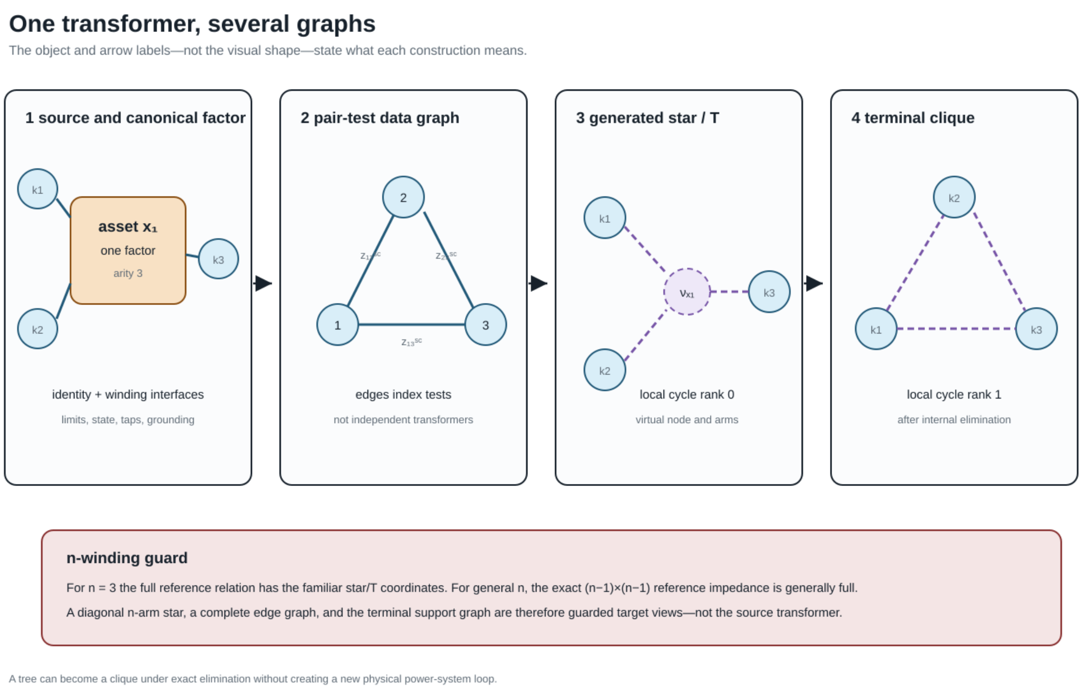

Figure 11.2: One transformer can induce a test-data graph, a generated star, and a terminal clique without becoming three independent line assets.

Generated star

For three windings, the familiar referred leakage coordinates may be drawn as three generated arms meeting at a virtual point ν_{x_1}. Locally, this ordinary graph has four vertices, three edges, and cycle rank zero. Its arms are coordinate objects owned by x_1. Treating them as physical lines would invent asset identities, states, and permissible decisions.

Terminal clique after elimination

Eliminating ν_{x_1} generically produces direct pairwise terminal coupling. For generic pairwise leakage data all three off-diagonal blocks are structurally nonzero, so the support graph is K_3 and has cycle rank one. A proper subgraph requires structural decoupling or an exceptional numerical cancellation and must be justified from the coefficients. The cycle is an algebraic coupling cycle, not evidence that a new circulating power-system route appeared inside the transformer.

> **Graph-theory trap**
>
> *The transformer is a tree* and *the transformer contains a cycle* can both describe exact target structures. Neither sentence is meaningful until it names the star realization, factor-incidence graph, terminal support graph, or another declared construction.

A general cycle-count statement

Let an n-port factor have distinct boundary attachments. Its local factor-incidence or star expansion has $n+1$ vertices and n edges, hence

$$\mu_{\text{star}} = n - (n+1) + 1 = 0.$$

If elimination yields a structurally complete terminal support graph, its local clique has n vertices and $\binom{n}{2}$ edges, hence

$$\mu_{\text{clique}} = \binom{n}{2} - n + 1 = \frac{(n-1)(n-2)}{2}.$$

For $n = 3$, the two local counts are zero and one. The complete clique is the generic terminal support; if a coupling block is structurally absent or cancels numerically, the exceptional subgraph and its count must be declared. This is a statement about graph constructions, not a claim that elimination changes the physical asset.

Embedding the transformer into an already connected graph makes the same point more forcefully. Direct factor stamping adds no ordinary bus–branch object, so the line graph has $\Delta\mu = 0$. A factor vertex joined to n existing attachment vertices and a generated star both add n member edges and one vertex, so each has $\Delta\mu = n - 1$. Their equal count does **not** make their vertices equivalent: one is an equation/factor object and the other is a generated electrical coordinate. A complete generated clique instead adds $\binom{n}{2}$ identified member edges, hence $\Delta\mu = \binom{n}{2}$. For three ports these increments are respectively 0, 2, 2, and 3. They can all encode the same declared terminal behaviour under their guards. A simple projection may add fewer adjacencies where the source already contains an edge between two attachment buses. Therefore the phrase *the transformer adds two cycles* is no safer than *the network is radial* without a representation qualifier.

The recorded five-bus extension gives:

Declared graph	Member cycle rank	Simple cycle rank	Interpretation
line-induced G_L	3	2	seven identified source lines
subdivided line factors plus one three-port factor	5	5	bipartite factor-incidence graph
line members plus generated star	5	4	optional edge target with ν_{x_1}
line members plus generated terminal clique	6	3	eliminated terminal-coupling target

The numerical values are useful diagnostics only when the row label travels with them.

Interfaces along the lowering branches

The source-to-target construction uses five stages, but the arrows need not visit all five:

$$\mathcal{M}_{\text{asset}} \xrightarrow{C_{x_1}} \mathfrak{P} \begin{cases} \xrightarrow{A_{x_1}} \mathcal{E}, & \text{direct factor stamping,} \\ \xrightarrow{L_{x_1}} G_{\text{edge}} \xrightarrow{R_{\text{internal}}} \mathcal{E}, & \text{guarded ordinary-edge branch,} \end{cases} \qquad \mathcal{E} \xrightarrow{S} G_{\text{support}}.$$

Direct stamping is the default. The edge branch exists for a target algorithm that genuinely requires ordinary incidence. It is not an obligatory intermediate representation.

An implementation should serialize this interface record beside the generated objects. The important question is not merely whether a reverse graph map exists, but whether the source quantities needed by the study can be evaluated from the target solution.

Stage	Declared interface
source asset/property	transformer and winding identity, attachment relation, state, ratings, ownership, provenance
canonical port–factor	ordered winding voltage/current ports, constitutive relation, limits, controls, observations
ordinary-edge realization	boundary buses, generated IDs, source and winding fibres, current/constraint recovery
equation or operator	variable coordinates, residuals, constraint ownership, feasible set, recovery operator
support graph	block ordering and numerical-zero policy; no automatic asset or decision meaning

Where power-system structure is lost

The representation can remain electrically exact while becoming structurally unsafe. The following omissions are especially important:

Boundary	Structure at risk if it is not carried separately
asset to factor	ownership, maintenance, common-mode failure, source nameplate meaning
factor to generated edges	one-device identity, winding identity, connection and tap semantics, excitation, grounding, winding limits
generated edges to assembled operator	internal current recovery, virtual-object provenance, source constraint ownership
operator to support graph	coefficients, signs, constitutive meaning, feasible set, decisions and objectives

Decision-model consequence

An exact terminal leakage relation does not authorize independent switching, outage, investment, or rating decisions on generated star or clique edges. Those target objects remain in the provenance fibre of x_1. Winding and coil limits remain source constraints evaluated through a recovery map.

Grounding and shunts are a particularly dangerous case. Magnetizing branches, core loss, neutral grounding, and connection-specific shunts may attach at a particular winding or internal coordinate. A target line model that permits only identical from/to shunts cannot silently absorb these objects into symmetrical generated edges.

Three windings do not define the general case

For $n_x = 3$, complete pairwise leakage data have the familiar star/T coordinates

$$z_1 = \tfrac{1}{2}(z_{12} + z_{13} - z_{23}), \quad z_2 = \tfrac{1}{2}(z_{12} + z_{23} - z_{13}), \quad z_3 = \tfrac{1}{2}(z_{13} + z_{23} - z_{12}).$$

This special case should not become the ontology for an arbitrary multiwinding transformer. For general n_x, the exact reference-coordinate matrix is $(n_x - 1) \times (n_x - 1)$ and is generally full. A diagonal n_x-arm star is therefore a restricted target model, not the automatic meaning of complete pairwise data. The full derivation and its round-trip test are in Multiwinding leakage reference compilation.

Even for three windings, a star arm can have negative reactance while the reference reactance matrix remains positive semidefinite. The composed witness uses positive pair-test reactances $(x_{12}, x_{13}, x_{23}) = (1, 1, 3)\ \Omega$. Its generated star coordinates are $(-0.5, 1.5, 1.5)\ \Omega$ in reactance, while the eigenvalues of the reference reactance matrix are $(0.5, 1.5)\ \Omega$. Reinterpreting the negative arm as a conventional line would therefore trigger an invalid componentwise passivity rejection even though the invariant matrix guard passes. The guard belongs to the compiled matrix relation, not to the visual intuition of three ordinary lines.

An evaluated four-winding companion

The three-winding construction is a useful diagram, but it is not the limit of the architecture. The companion artifact `experiments/generated/four-winding-lowering-witness.json` evaluates a four-winding factor with four ordered ports: three WYE ports and one DELTA port. Its reference-coordinate impedance has dimension 3×3 and is deliberately non-diagonal. The pairwise short-circuit data are generated from that full matrix and recovered after compilation, so the example does not silently replace a general reference matrix by independent star arms.

The evaluated target keeps the following objects separate:

Source meaning	Evaluated target	Why it remains explicit
winding and connection identity	ordered WYE/DELTA port maps	terminal dimensions and incidence differ
shunt on winding 2	WYE coil-coordinate shunt map	placement is not a generic line parameter
shunt on winding 3	DELTA coil-coordinate shunt map	the same numerical matrix has a different connection meaning
neutral grounding	two internal terminal grounding maps	grounding scope is not inferred from a bus label
tap and phase settings	two pointwise equation operators	a decision state is not frozen into one source object
winding current limits	recovery and constraint maps	limits remain attached to winding identity

The control states are intentionally finite and explicit: one state uses a lower tap on winding 2 and a leading phase shift on winding 4, while the other uses a higher tap and a lagging shift. Each state is compiled with fixed coefficients, and the witness checks that the resulting terminal operators differ, that both decision identities survive, and that terminal currents and complex power recover through the leakage, shunt, and grounding maps.

This is a pointwise exact compilation family, not a claim that a single static Y contains the whole decision problem. The source decision domain remains outside each fixed-state operator. Nor does the availability of a terminal operator imply an ordinary-edge transformer surrogate: the witness records the ordinary-edge branch as *not asserted* until an independent n-port realizability condition is supplied. The four-winding result is registered as $ARCH - LOWER - 002$.

Executable composition and evidence boundary

The generated artifact `experiments/generated/five-bus-transformer-lowering-witness.json` is registered as `ARCH-FIVEBUS-XFMR-001`. It hash-binds and composes three existing evidence objects:

- the five-bus cycle-space analysis;
- the ordered three-winding terminal lift; and

- the exact pairwise-leakage reference compilation.

Its checks verify the local and embedded cycle counts, one-factor/three-port identity, the declared three-winding special case, the numerical negative-arm guard, winding identity, and the continued presence of grounding and current-limit observations. This is direct evidence for the structural maps and loss ledger. It is not a new AC solve, a general n-port realizability theorem, or permission to compile every transformer into ordinary edges.

The detailed terminal connection and current-recovery equations remain in Multiwinding terminal leakage assembly. The broader formulation alternatives remain in Circuit formulations and the lowering boundary.

Chapter 12

Scope and thesis

Page status: reader-facing scope contract and methodological thesis.

Evidence boundary

Scope: the book's proposed representation and preservation vocabulary for steady-state and quasi-steady power-network models. **Evidence:** definitions, scoped derivations, and repository-level executable witnesses. **Numerical optimality:** not applicable to the thesis; local solver witnesses do not establish global decision equivalence. **Unresolved boundary:** a universal representation theorem and external validation of the proposed architecture.

The problem

Much power-system analysis starts from a bus–branch graph

$$G = (V, E),$$

Here V and E are semantic vertex and edge sets; they are not assumed to be the integer positions of a stored adjacency matrix. An implementation may choose enumerations κ_V and κ_E later. Where buses are vertices and lines or transformers are edges, this is useful, especially when the network and study satisfy the assumptions of a balanced transmission model. It is not a universal physical or decision model.

A simple graph cannot distinguish parallel circuits. A scalar-weighted edge cannot retain full conductor coupling. An ordinary edge cannot directly express a three-winding transformer, a coupled multi-circuit corridor, or a device with several electrical and control ports. A topology graph alone does not state which variables, limits, decisions, and constitutive relations are attached to its incidence structure.

Power-system shorthand

In this book, *the graph* is never a complete model by itself. The intended graph, active state, terminal quantities, and retained constraints must be named before a connectivity or reduction claim is interpreted.

The most consequential omissions are often constraints on quantities that were eliminated algebraically. In the running four-conductor Kron witness, the retained phase relation is exact to numerical precision, but the recovered neutral current is $43.0\ \mathrm{A}$ against a declared $42.6\ \mathrm{A}$ limit:

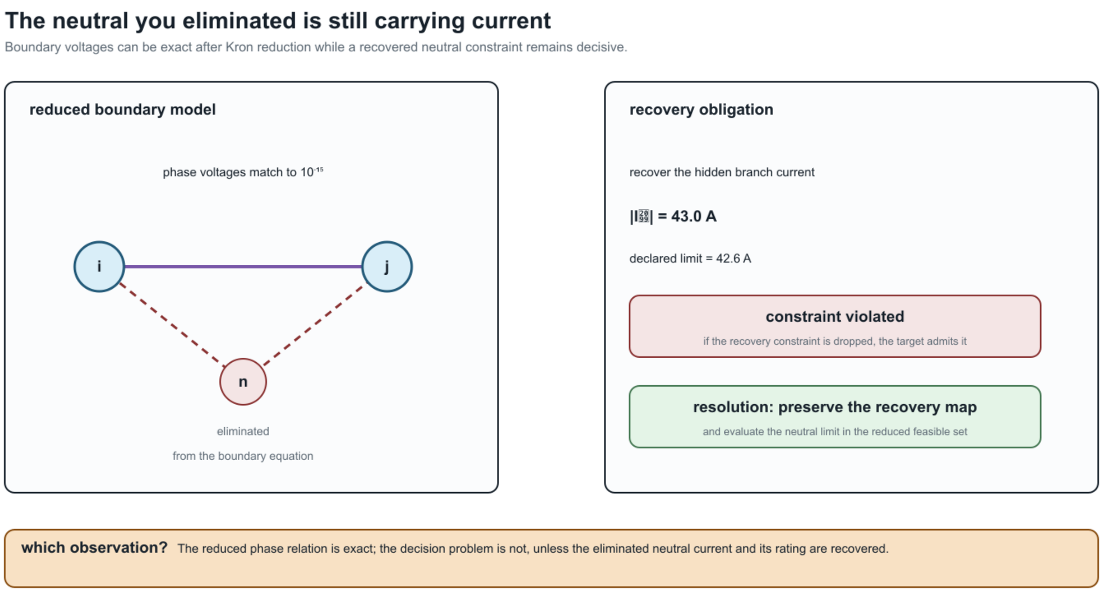

Figure 12.1: A Kron-reduced boundary model still needs the recovered neutral-current constraint.

The resolving phrase is **what must be recovered?** A reduced equation is not a decision certificate until every eliminated current, voltage, limit, and observation required by the study has a recovery or preservation map.

The limitations become consequential in decision problems. Suppose parallel branches ℓ have terminal relation

$$\mathbf{I}^{\mathrm{s}}_{\ell ij} = \mathbf{Y}_{\ell}\Delta\mathbf{U}$$

and individual feasible current sets $\mathcal{C}_{\ell}$. Their aggregate admittance

$$\mathbf{Y}_{\mathrm{eq}} = \sum_{\ell}\mathbf{Y}_{\ell}$$

preserves aggregate terminal current, but the original feasible voltage-difference set is

$$\{\Delta\mathbf{U} :\; \mathbf{Y}_{\ell}\Delta\mathbf{U} \in \mathcal{C}_{\ell} \quad \forall \ell\}\,.$$

There need not be one conventional edge rating that reproduces this set. Independent switching, contingency, maintenance, and investment variables make the loss still more apparent. Line-limit-preserving equivalents have been studied precisely because ordinary equivalents do not automatically retain these decision constraints [4].

The general baseline

The book treats the general steady-state network as multiconductor and multi-terminal. A source model may contain:

- buses with different ordered terminal sets;
- explicit phases, neutrals, voltage references, and grounding impedances;
- full series and shunt coupling matrices;
- conductor permutations and phase discontinuities;
- parallel assets with separate identity, state, and limits;
- multiwinding transformers and other arbitrary-port devices;
- continuous controls and discrete switch, tap, outage, or investment decisions;
- measurements, protection boundaries, hierarchy, and provenance.

This is not synonymous with a distribution feeder. It is a modelling baseline that does not assume away distinctions before the study question is known.

When transmission models are sufficient

Much of the complexity collapses under conditions common in transmission studies: compatible phase sets, approximate balance, transposition or sequence symmetry, negligible or externally resolved neutral behaviour, predominantly two-terminal equipment, and study questions insensitive to per-conductor or internal-device constraints.

Under a declared contract, a positive-sequence bus–branch model may then be exactly the right representation. The methodological error is not using such a model; it is treating its assumptions as universal power-network semantics. A central task of this book is to state the map from the general model to the simpler one and identify what makes the map admissible.

Central thesis

Proposal. A graph transformation for a power network is meaningful only relative to declared observations, constraints, and decisions. No representation is universally correct or universally minimal. Source data should retain typed physical and terminal structure, and simpler graphs should be generated as traceable, purpose-specific views.

The book investigates a linked reference architecture with three semantic layers:

1. **Identity:** what physical, logical, and generated objects exist?
2. **Interconnection:** which ordered terminals share variables or obey conservation relations?
3. **Behaviour and decisions:** which constitutive, limit, control, measurement, objective, and discrete-state relations connect the terminal variables?

A typed asset/property model records the first layer. A typed hierarchical port–factor incidence model is the principal candidate for the second and third. This architecture is a research proposal to be tested against actual representation families, software mappings, counterexamples, and decision problems—not an assumed canonical truth.

Not one hierarchy

There is no single total order from *most expressive* to *least expressive*. The asset and electrical views can be incomparable: an asset graph can retain ownership and construction history while omitting virtual electrical nodes; a compiled electrical graph can contain virtual transformer buses that have no physical asset identity.

The meaningful order is relative to a query or observation family Q. Write

$$M_1 \succeq_Q M_2$$

when every question in Q answerable from M_2 can also be answered from M_1 through a declared transformation. The order can change when Q changes from power flow to protection, asset management, fault location, optimal switching, or expansion planning.

This produces a **partial order of representations relative to preservation contracts**. The Representation taxonomy makes the independent comparison axes explicit.

Decision preservation

A transformation can preserve selected voltages while changing the feasible set or optimum. The book therefore evaluates, where relevant:

- equality and inequality feasibility;
- per-conductor, per-asset, and per-winding limits;
- continuous controls;
- discrete switching, tap, outage, contingency, and investment choices;
- objective values and active constraints;
- optimal or admissible decisions;
- recovery of eliminated source quantities;
- source-to-target provenance.

Claims such as *equivalent*, *limit preserving*, or *decision preserving* are incomplete unless the interface, operating domain, observation map, and recovery obligations are stated.

Boundaries of the first edition

The first edition concentrates on:

- steady-state and quasi-steady electrical networks;
- arbitrary multiconductor and explicit-neutral models;
- transmission and distribution topology processing;
- multi-terminal and multiwinding devices;
- projections used in power flow, OPF, state estimation, selected fault studies, and planning;

- exact and approximate reductions;
- preservation of operational constraints and decisions;
- typed normalization rules, provenance, and recoverability.

EMT, harmonics, thermal dynamics, communications, markets, protection logic, geographic asset systems, and graph learning initially appear as boundary cases. Later editions can develop them where the core language proves useful.

Study-family coverage boundary

The vocabulary is broader than the executable evidence. The current status is:

Study or exchange family	Current treatment	Explicitly not claimed yet
steady-state PF and OPF	executable multiconductor fixtures, decision cases, and compiled views	global optimality or universal solver performance
topology processing and switching	formal node–breaker/state-resolved quotient definitions	an executable mixed-integer switching study on the running fixture
state estimation	observation-map and preservation vocabulary; measurement fields in the source model	a solved estimator, bad-data detector, or covariance-preserving reduction
fault and grounding studies	grounding taxonomy, terminal-current relations, and selected reduction guards	a complete short-circuit/protection calculation across all fault classes
contingency and maintenance	semantic asset/state/provenance requirements and parallel-member counterexamples	a validated N-1/N-k engine or maintenance scheduler
protection	protection boundaries and relay/limit ownership are retained as dependencies	relay coordination, zone reach, or protection-operation equivalence
data exchange	CIM/CGMES, PowerModelsDistribution, OpenDSS, and MATPOWER crosswalk with adapter obligations	conformance to every profile, round-trip guarantee, or standards certification

This table is a scope contract for the first edition. A future executable claim in one of these families must add a versioned fixture or source-backed result rather than silently upgrading a conceptual crosswalk into an implementation claim.

Intended contribution

The proposed contribution is not another isolated reduction algorithm. It is a common language for stating:

- source and target model categories;
- whether a transformation is a projection, compilation, normalization, exact behavioural reduction, or approximation;
- what is preserved and what is forgotten;
- which assumptions make the transformation valid;

- how original quantities, limits, objectives, and decisions are recovered;
- which questions become unanswerable afterward.

That language should support both scientific results and an implementable transformation system. The Notation and modelling conventions and The running multiconductor network provide the common vocabulary and adversarial case on which the proposal will first be tested.

Part II

Representation obligations

Chapter 13

Representation frameworks

Page status: formal framework definitions with a checked structural architecture witness; evaluated-factor coverage remains open.

Purpose and status

This chapter gives a first mathematical specification of the principal representation frameworks used in the book. The definitions are adopted book conventions. The claim that the linked asset and hierarchical port–factor models are an adequate common source remains a proposal to test.

The frameworks are not one chain from detailed to simple. Three describe different electrical resolutions, one records orthogonal physical and organizational relations, and another family is compiled for computation.

The companion representation taxonomy is the reader-facing level map; this chapter is the normative mathematical specification for those levels and their linked source structures. The implementation status of the executable facade and fixture matrix is kept in the representation implementation record, rather than being a competing architecture chapter. In particular, equation and sparsity graphs are computational projections, not a fifth semantic level, and the asset/dependency model is linked to—but not nested inside—the electrical port–factor model.

Framework	Primary purpose	Status
simple topology graph	connectivity, islands, partitioning, and algorithms that cannot use parallel edges	derived quotient
oriented attributed multigraph	identified two-terminal equipment and conventional bus–branch models	derived engineering view
hierarchical port–factor incidence model	multiconductor, multi-terminal, coupled, ideal, constrained, and controlled equipment	proposed canonical electrical source model
asset/dependency relation model	structures, ownership, protection, failures, maintenance, and provenance	orthogonal companion source model
equation and sparsity graphs	algebraic coupling, ordering, decomposition, and solver structure	compiled computational views

Power-system shorthand

There is no context-free *network graph*. Each row above defines a different object with different admissible queries. The translation-traps chapter gives controlled replacements for this and other familiar shorthand.

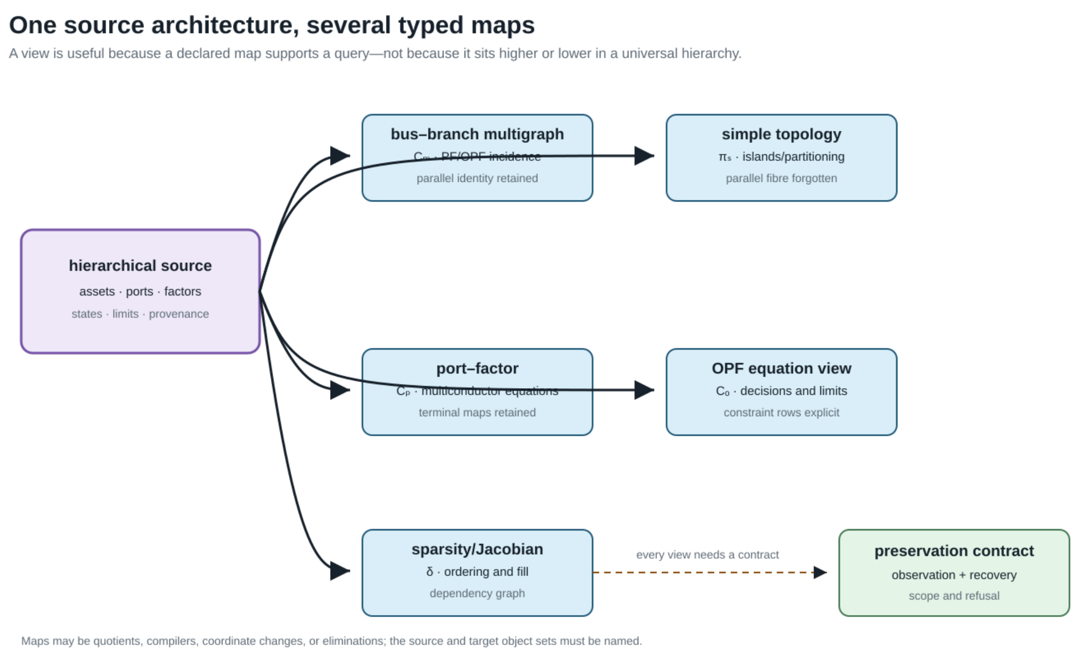

Figure 13.1: The source architecture feeding typed representation maps.

Vocabulary bridge

A node, edge, or message graph supplied to a learning architecture is a compiled computational view. It is not the source ontology unless its map from assets, ports, factors, states, limits, and provenance is declared. Heterogeneous node and edge types help express that map, but do not prove that parallel identity or n-port semantics survived compilation.

Here *canonical* means the selected source formalism for this book. It does not mean that adequacy or uniqueness has already been established.

These two plates belong here as well as in the detailed map chapter. This is the PDF-facing definition of the objects; the map chapter supplies the fuller map vocabulary and proofs. The arrows are query- and contract-indexed, not a single “more detailed” ordering.

The wider literature contains other valid graph and equation families. Their selection and relationship to these rows is surveyed in the literature map and the circuit formulations and lowering boundary. The definitions below are therefore the book's scoped source/view contract, not a claim that every power-network study should use the same formalism.

The normative finite-multigraph object, including flags, loops, degree and matrix conventions, is defined in Multigraphs for expert modelers. This chapter specializes that object to the loopless two-terminal engineering views used by the current executable cases; it does not introduce a competing multigraph convention.

Simple topology graph

Definition. A loopless undirected simple topology graph is a pair

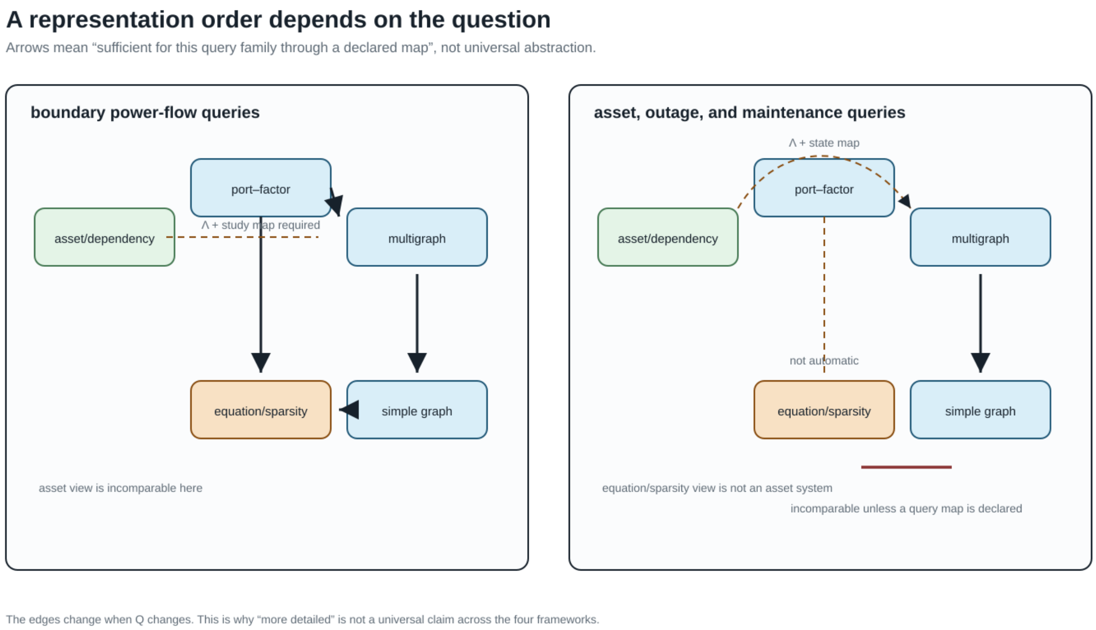

Figure 13.2: Query-relative partial orders: different query families induce different useful views.

$$G_{\mathrm{s}} = (\mathcal{B}, E_{\mathrm{s}}), \qquad E_{\mathrm{s}} \subseteq \big\{\{i,j\} : i,j \in \mathcal{B},\ i \neq j\big\}.$$

An optional weight map $w : E_{\mathrm{s}} \to \mathcal{W}$ does not restore the identities of several source elements mapped to the same edge. Its codomain and aggregation rule must be declared: a conductance, distance, capacity, and binary adjacency have different semantics.

Let a loopless identified multigraph, in the normative flag convention, have line set $\mathcal{L}$ and derived unordered endpoint map $\partial : \mathcal{L} \to \binom{\mathcal{B}}{2}$, where the codomain is the set of two-element subsets of $\mathcal{B}$. Its simple projection is

In this loopless specialization $\mathcal{L}^\circ = \mathcal{L}$; the superscript is retained on the map domain to make the non-loop restriction explicit.

$$\mathrm{simp} : \mathcal{L}^\circ \to E_{\mathrm{s}}, \qquad \mathrm{simp}(\ell) = \partial\ell, \qquad E_{\mathrm{s}} = \operatorname{im}\partial.$$

Thus $\ell_1 \sim_{\mathrm{simp}} \ell_2$ exactly when the two lines have the same unordered endpoints.

Proposition. If the loopless multigraph has c connected components, then its cycle rank and that of its simple projection satisfy

$$\begin{aligned}
\mu_{\mathrm{M}} &= |\mathcal{L}| - |\mathcal{B}| + c, \\
\mu_{\mathrm{s}} &= |E_{\mathrm{s}}| - |\mathcal{B}| + c, \\
\mu_{\mathrm{M}} - \mu_{\mathrm{s}} &= \sum_{e \in E_{\mathrm{s}}} \big(|\,\mathrm{simp}^{-1}(e)| - 1\big).
\end{aligned}$$

Proof. Collapsing parallel identity does not change the vertex set, adjacency relation, or connected components. Subtracting the two standard cycle-rank identities gives $|\mathcal{L}| - |E_{\mathrm{s}}|$. Partitioning $\mathcal{L}^{\circ}$ into the nonempty fibres of simp gives the final sum.

The projection therefore preserves connectivity and islands, but not the line-indexed cycle space, member states, or member constraints. A simple topology graph is also not automatically a nodal-admittance sparsity graph: after electrical stamping, cancellation or terminal-coordinate structure can make the matrix support different from bare adjacency.

Oriented attributed multigraph

Definition (loopless engineering specialization). An oriented attributed multigraph is the normative flag object

$$G_{\mathrm{M}} = (\mathcal{B}, \mathcal{L}, \mathcal{F}, s, \mathrm{ed}, o, a),$$

restricted here to edges whose two flags map to different buses. The maps $s : \mathcal{F} \to \mathcal{B}$ and $\mathrm{ed} : \mathcal{F} \to \mathcal{L}$ record incidence, each fibre $\mathrm{ed}^{-1}(\ell)$ contains two flags, o orders those flags as tail and head, and a is a family of typed attribute maps. The derived endpoint functions ∂^- and ∂^+ return the buses of the ordered tail and head flags. Parallel elements remain distinct members of $\mathcal{L}$ even when both derived endpoint functions agree.

The incidence matrix associated with the selected orientation is

$$A_{i\ell} = \begin{cases} -1, & i = \partial^-(\ell), \\ +1, & i = \partial^+(\ell), \\ 0, & \text{otherwise.} \end{cases}$$

Reorienting a line negates its incidence column. It does not change its physical incidence or assert a change in operating-point power transfer. The precise distinction between physical incidence, reference orientation, terminal signs, and power direction is developed in Orientation, terminal quantities, and power transfer.

Typical attributes include terminal maps, a symmetric element impedance $\mathbf{Z}_\ell$, end-specific shunts $\mathbf{Y}^{\mathrm{sh}}_{\ell ij}$, states, limits, and provenance. The multigraph becomes a PF or OPF model only after constitutive relations, injections, constraints, controls, and an objective or observation map are attached. A genuinely multi-terminal device belongs here only after an explicit two-terminal compilation with provenance.

Hierarchical port–factor incidence model

Definition. A hierarchical port–factor incidence model is a tuple

$$\mathfrak{P} = (\mathcal{Q}, \mathcal{J}, \Phi, j, f, \mathcal{H}, \{\mathcal{X}_q\}_{q\in\mathcal{Q}}, \{\mathcal{R}_\phi\}_{\phi\in\Phi}).$$

Here:

- $\mathcal{Q}$ is a finite set of typed, ordered ports;
- $\mathcal{J}$ is a finite set of junctions;
- Φ is a finite set of behavioural factors;

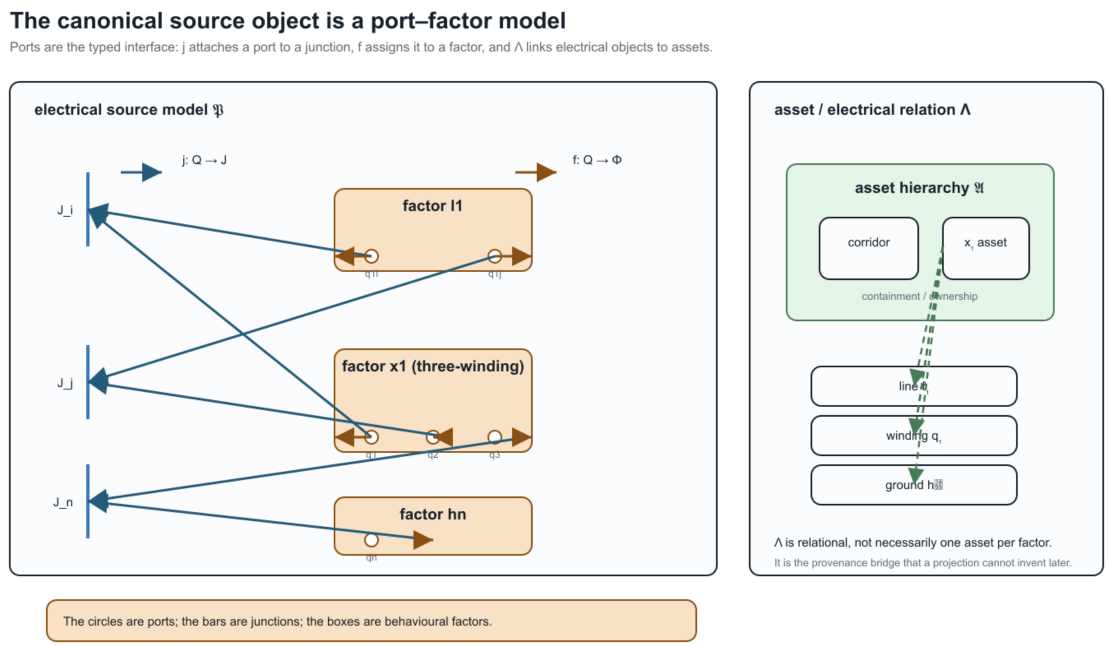

Figure 13.3: Canonical port–factor source object with explicit ports, junction attachments, factor ownership, and the asset relation "\Lambda".

- $j : \mathcal{Q} \to \mathcal{J}$ attaches each port to a junction;
- $f : \mathcal{Q} \to \Phi$ assigns each port to its owning factor;
- $\mathcal{H}$ is a rooted containment forest with declared subsystem boundaries;
- $\mathcal{X}_q$ is the variable space carried by port q;
- $\mathcal{R}_\phi$ is the factor relation on the ordered ports $\mathcal{Q}_\phi = f^{-1}(\phi)$.

A static factor relation may be written

$$\mathcal{R}_\phi \subseteq \prod_{q \in \mathcal{Q}_\phi} \mathcal{X}_q \times \mathcal{U}_\phi \times \Theta_\phi,$$

where $\mathcal{U}_\phi$ contains continuous or discrete decisions and Θ_ϕ contains fixed parameters. Equations, inequalities, measurements, and uncertainty sets are all relations rather than new graph edge types.

The circles in this figure are not decorative edge endpoints. They are the typed ports in $\mathcal{Q}$. The two incidence maps have different domains and meanings: j attaches a port to a junction, while f assigns that same port to its owning factor. The dashed Λ links are many-to-many provenance relations to the asset model. This is the glyph vocabulary reused by the later factor and lowering diagrams.

For junction k, let $\mathcal{Q}_k = j^{-1}(k)$. Its junction relation $\mathcal{R}_k^J$ enforces compatible effort variables after the declared terminal-coordinate maps and conservation of signed flow variables. For electrical phasor ports these are voltage compatibility and KCL.

Given boundary ports $\partial\mathcal{Q}$, the external behaviour is

$$\mathfrak{B}(\mathfrak{P}) = \operatorname{proj}_{\partial\mathcal{Q}} \left\{ z:\ z_{\mathcal{Q}_\phi} \in \mathcal{R}_\phi\ \forall\phi,\quad z_{\mathcal{Q}_k} \in \mathcal{R}^{\mathrm{J}}_k\ \forall k \right\}.$$

Here $\mathfrak{B}(\mathfrak{P})$ denotes external behaviour; the bus set $\mathcal{B}$ used by the simple and multigraph definitions is a different object. This definition makes an ordinary two-terminal line one factor of arity two, not the template for every device. A multiwinding transformer, coupled line group, converter, grounding relation, or shared control can retain its natural port arity. Hierarchy determines ownership of internal variables and the boundary across which behavioural reduction is defined.

Minimal executable witness

The first executable architecture slice is recorded in `experiments/generated/port-factor-architecture.json`. It instantiates $\mathfrak{P}$ for the two heterogeneous parallel lines, the three-port transformer x_1, and the neutral grounding factor h_n. The validator checks that every port has a declared junction and owning factor, that the three winding ports remain one factor of arity three, and that the relation

$$\Lambda \subseteq (\mathcal{V}_A \cup \mathcal{R}_A) \times (\mathcal{Q} \cup \mathcal{J} \cup \Phi)$$

contains both one-to-one realizations and the four relations from asset x_1 to its transformer factor and winding ports. This is a structural data witness, not yet a numerical factor evaluator: the relation signatures are declared strings and the electrical equations are tested by the existing transformation artifacts.

This witness is the evidence object for `ARCH-PORT-001`, whose claim is that the typed incidence, multiwinding factor, explicit grounding factor, and many-to-many Λ link can be represented together without collapsing their identities. The claim is deliberately limited: it validates the structure and its declared link signatures, not the adequacy of the proposed architecture for arbitrary asset models or a general evaluated-factor semantics. The generated artifact is indexed in the knowledge-base evidence register.

The same construction is now applied directly to the five-bus line-identity multigraph in `experiments/generated/five-bus-port-` It creates five bus junctions, fourteen endpoint ports, and seven distinct two-port scalar line factors. In particular, q and r remain separate factors even though their endpoints project to the same simple-graph edge. This is claim `ARCH-PORT-002`: a structural fixture lift that preserves identity and orientation, not a numerical factor evaluator or a new AC model.

The companion artifact `experiments/generated/five-bus-conductor-terminal-lift-witness.json` makes the scalar special case explicit: each of the seven identified lines has two scalar endpoint ports attached to one of five terminal junctions. The q/r parallel fibre remains visible in the terminal incidence and retains the extra line-identity cycle dimension. This is claim `ARCH-CONDUCTOR-002`; it is a structural lift, not a multiconductor, switch, or transformer calculation.

Asset and dependency relation model

Definition. An asset/dependency relation model is a typed attributed multi-relational structure

$$\mathfrak{A} = (\mathcal{V}_A, \mathcal{R}_A, \tau_V, \tau_R, \iota, \alpha).$$

The type maps τ_V and τ_R classify entities and relations, α stores typed properties, and

$$\iota : \mathcal{R}_A \to \bigcup_{n \geq 1} \mathcal{V}_A^n$$

gives each relation an ordered finite incidence. Binary relations recover ordinary source and target maps. The incidence defines relations such as `contains`, `mounted_on`, `protected_by`, `owned_by`, `located_at`, `shares_failure_mode_with`, and `derived_from`. Relations that are naturally many-way therefore remain hyperrelations rather than being forced into one untyped simple edge.

The link to the electrical model is generally a relation

$$\Lambda \subseteq (\mathcal{V}_A \cup \mathcal{R}_A) \times (\mathcal{Q} \cup \mathcal{J} \cup \Phi),$$

not a function. One asset may generate several electrical factors, one factor may depend on several assets, and generated factors may have no independent physical-asset identity. The asset model has no electrical behaviour merely because its relations are drawn as edges.

Equation, incidence, and sparsity graphs

For an equation system $F(x) = 0$ and inequalities $g(x) \leq 0$, a variable–relation incidence graph has one vertex class for variables, another for relations, and an edge whenever a relation depends on a variable. A matrix sparsity graph instead follows the nonzero pattern of a declared matrix such as the compound nodal operator $\mathbf{Y}^{\mathrm{N}}$, a Jacobian, or a KKT matrix.

The normative block- and scalar-support definitions are in Two topology levels and the nodal projection. These graphs need separate definitions because their vertices may be buses, terminals, scalar variables, vector blocks, equations, or constraints. Schur elimination can remove variables while adding fill edges. A nonzero-pattern edge means algebraic coupling and is not evidence of a physical line.

Typed maps rather than a ladder

The proposed source pair is $(\mathfrak{A}, \mathfrak{P}, \Lambda)$. From it, different guarded maps can produce

$$(\mathfrak{A}, \mathfrak{P}, \Lambda) \longrightarrow G_{\mathrm{M}} \longrightarrow G_{\mathrm{s}},$$

while a study compiler produces

$$\mathfrak{P} \longrightarrow \text{equations and constraints} \longrightarrow \text{sparsity graphs.}$$

Neither row is a universal abstraction order. The asset relation model remains linked sideways because its ownership, protection, and failure questions are incomparable with electrical boundary behaviour.

The relevant within-framework morphisms, orientation actions, cross-framework transformations, and query-relative notion of expressiveness are defined in Maps between representation frameworks.

Remaining formal work

This first definition pass does not yet settle:

- categorical composition of hierarchical open systems;
- realizability of a general reduced multiport in a restricted device library;
- a type system for units, bases, conductor coordinates, and state spaces;
- the exact boundary between a factor relation and a study constraint;
- machine-checkable correspondence between the mathematical objects and data schemas.

Those are explicit foundation tasks, not assumptions hidden behind the word *graph*.

Chapter 14

Representation taxonomy

Page status: foundational taxonomy; categories are analytical, not a standards claim.

This page is a classification card, not a second transformation specification. The authoritative registry of typed views, lowering maps, state-conditioned surgery, and diagnostics is From source graphs to views and graph surgery. The taxonomy below names the families and their typical losses; it does not redefine their object maps or preservation contracts.

The phrase *power-network graph* covers several mathematical and data structures that retain different facts. This taxonomy prevents a physical asset graph, an equation graph, and a solver sparsity graph from being treated as interchangeable merely because all have vertices and edges.

Five questions before naming a graph

For a representation M, ask:

1. **Identity:** which physical, logical, and generated objects remain distinguishable?
2. **Interconnection:** what counts as a terminal, junction, edge, hyperedge, or port?
3. **Behaviour:** where do constitutive equations, controls, limits, and uncertainty live?
4. **State:** which topology and equipment states are fixed, variable, or omitted?
5. **Purpose:** which observations, analyses, and decisions is the representation intended to support?

Two structures with identical unlabelled topology can answer these questions differently.

Representation families

Asset and property models

Vertices represent physical or organizational entities; typed relations record containment, ownership, construction, location, protection, measurement, and provenance. Such a model may retain two electrically equivalent parallel assets as distinct while omitting the internal variables of a compiled electrical equivalent.

Terminal-connectivity models

Objects expose ordered terminals or conductor bundles. Junctions express equality and conservation without yet requiring every device to be an ordinary edge. This family is the natural baseline for explicit phases, neutrals, grounding, phase discontinuities, and switchgear.

Node–breaker and bus–branch models

A node–breaker model retains switching equipment and detailed connectivity. A state-resolved bus model quotients nodes connected by closed ideal switches. An oriented bus–branch multigraph adds identified two-terminal branches, where orientation is normally a coordinate choice rather than a physical transfer direction. A simple graph additionally forgets parallel identity.

These are different representations, not interchangeable names for resolution levels. The node–breaker row is specialized in Node–breaker, bus–breaker, and topology processing; its state-resolved quotient is one concrete instance of the general view/surgery contract.

Four principal levels and the orthogonal companions

For the main argument, the families above are organized into four principal levels. The word *level* refers to the dominant electrical or identity query, not to a universal refinement order:

Principal level	What it answers	What it forgets or delegates
Simple graph	connectivity, islands, sparsity-free partitioning	parallel identity, terminal coordinates, device relations
Directed/oriented attributed multigraph	conventional bus–branch PF/OPF with each two-terminal asset retained	arbitrary-port behaviour and most internal conductor structure
Hierarchical port–factor incidence graph	multiconductor, multi-terminal, coupled, controlled, and ideal equipment	asset-lifecycle relations unless linked
Asset/dependency relation graph	ownership, protection, maintenance, failure, and shared-structure queries	electrical behaviour unless linked to factors and terminals

The first two rows are derived electrical views; the third is the canonical electrical source formalism adopted for this book; and the fourth is an orthogonal companion rather than a more detailed circuit graph. Terminal, node–breaker, bus–branch, equation, and sparsity views are named specializations or computational projections around these four levels. They should not be counted as additional rungs in a single ladder. The rigorous objects and maps are specified in Formal representation frameworks.

The diagram is the classification in one view. The computational projections hang off the electrical source model because their vertices and edges are chosen from a declared equation or matrix support; the asset model is drawn sideways because ownership and maintenance are not a refinement of electrical boundary behaviour.

Port, factor, and hypergraph models

A behavioural factor relates variables on an arbitrary ordered set of ports. Ordinary branches are arity-two special cases. Multiwinding transformers, converters, mutual couplings, shared controls, and measurement relations need not be decomposed into artificial pairwise edges. The formal relation between a two-uniform flag multigraph, an arbitrary-arity incidence structure, a bipartite incidence graph, and a typed port–factor relation is fixed in Multigraphs for expert modelers. This taxonomy names the representation family; it does not use *hypergraph* and *factor graph* as interchangeable data models.

Algebraic and equation models

Incidence, admittance, Laplacian, Jacobian, KKT, and constraint matrices induce graphs from their nonzero structure. Their vertices may represent variables, equations, blocks, or matrix rows rather than physical objects. They are indispensable for analysis and computation, but their edges do not automatically carry physical meaning.

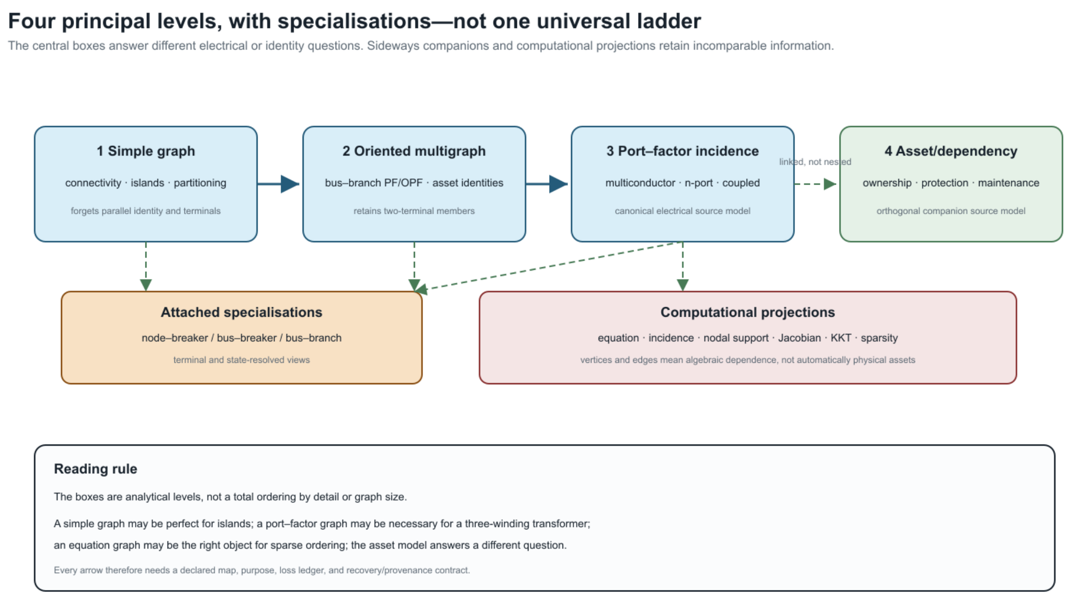

Figure 14.1: Four principal representation levels with terminal and node-breaker specialisations, computational projections, and the orthogonal asset model.

Equation-graph choice is coupled to formulation choice: nodal support, modified nodal, sparse tableau, branch-current, and port/factor targets can induce different variable and sparsity graphs from the same source model. The guarded formulation boundary is defined in Circuit formulations and the lowering boundary.

The block- and scalar-support graphs of the compound nodal operator are defined once in Two topology levels and the nodal projection. This taxonomy records their role without introducing a competing support definition. Likewise, a reduced/Kron row in this table names a view family; its elimination and recovery conditions belong to the compiled-view and reduction chapters.

Study-specific compiled models

A power-flow, OPF, state-estimation, fault, or decomposition model selects variables and relations for a task. Compilation may create virtual buses and branches, eliminate explicit currents, fix states, relax constraints, or introduce auxiliary variables. Graph size alone therefore says nothing reliable about physical or decision expressiveness.

Comparison by retained meaning

No row is globally maximal. An asset graph and a compiled equation graph may retain incomparable information.

A simple topology graph must not be identified with a sparsity graph merely because both are simple graphs. Their vertices and adjacency rules differ: one records a quotient of physical incidence, while the other records nonzero algebraic dependence in a declared matrix or equation system.

Family	Primary objects	Strongest use	Characteristic omission
Asset/property	equipment, owners, locations, records	identity and lifecycle	electrical state equations
Terminal connectivity	terminals, junctions, switches	conductor-aware interconnection	device behaviour unless linked
Bus–branch multigraph	buses and identified branches	conventional network algorithms	internal device and conductor structure
Simple topology or weighted graph	vertices and quotient edges	connectivity, visualization, and generic graph methods	parallel identity, controls, limits
Port–factor/hypergraph	ports and behavioural relations	multi-terminal composition	asset meaning unless linked
Equation/sparsity graph	variables, equations, nonzeros	numerical solution and decomposition	most physical identity

Required question	Representation obligation	Common failure
Which asset is out of service?	stable element identity and state	aggregated edge has no member state
Is every conductor within its rating?	terminal/conductor currents and limits	scalar branch rating replaces several limits
May a switch be operated?	switch identity, admissible states, control ownership	closed-switch quotient discards the choice
Is grounding represented correctly?	neutral, earth, reference, and impedance semantics	neutral and ground are conflated
Is an OPF decision equivalent?	feasible-set, objective, and decision maps	terminal voltages agree but active constraints change
Can source results be reconstructed?	recovery map and provenance	virtual or eliminated objects have no source correspondence

Comparison by decision support

Axes, not one ladder

Representations should be located along several independent axes:

- physical identity retained or forgotten;
- conductor and terminal resolution;
- factor arity and device vocabulary;
- hierarchy and subsystem boundaries;
- fixed versus variable topology state;
- equations only versus equations plus feasible sets and decisions;
- exact versus approximate behaviour;
- source provenance and recoverability;
- physical versus computational interpretation.

A representation can be simpler on one axis and richer on another. Compiling a transformer into a loss network may increase the number of vertices while reducing the device vocabulary. Eliminating internal buses may reduce the vertex count while creating dense coupling and more complicated constraints.

The general multiconductor baseline

The book begins with arbitrary ordered terminal sets and full conductor coupling. A transmission bus–branch model appears when additional conditions justify several collapses, for example:

- the retained buses have compatible phase sets and ordering;
- the system is sufficiently balanced for the declared observations;
- sequence domains decouple under the element models used;
- neutral and grounding behaviour is absent, externally fixed, or validly reduced;
- equipment can be represented as identified two-terminal branches;
- per-conductor and device-internal constraints do not affect the decisions of interest;
- parallel aggregation retains all relevant member states and all nonredundant limits, with any removed constraint certified as implied, or those distinctions are outside the contract.

This explains why many transmission studies can use compact graphs successfully without treating their assumptions as universal power-network semantics.

Proposed linked reference architecture

The working hypothesis of this book is that two linked structures provide a useful source from which the major representation families can be generated:

1. a typed asset/property model for stable physical and organizational identity;
2. a typed hierarchical port–factor model for electrical interconnection, behaviour, limits, and decisions.

This is a proposal to test. Evidence must include faithful mappings of multiconductor lines, multiwinding transformers, switchgear, grounding, parallel assets, controls, and study-specific models—not only an abstract claim of generality.

A representation card

Every representation chapter and knowledge-base entry should record:

Field	Question
Object types	What becomes a vertex, edge, port, factor, or attribute?
Identity	Which objects remain distinguishable?
Coordinates	How are terminals, conductors, orientation, units, and bases declared?
Behaviour	Which equations and inequalities are native?
State	Which continuous and discrete states are represented?
Purpose	Which analyses or decisions motivate the view?
Loss	What cannot be answered from this view alone?
Recovery	Can source quantities or objects be reconstructed?
Provenance	How are generated objects related to source objects?
Evidence	Is the mapping proved, tested, standardized, observed in practice, or proposed?

Chapter 15

Multigraphs for expert modelers

Page status: normative mathematical conventions and literature-backed reference; core matrix identities have an executable finite witness; electrical adequacy remains representation- and query-dependent.

Purpose

This chapter fixes the graph-theoretic object used when the identities of several two-terminal members must survive. It is written for readers who know ordinary graph theory but need to determine which generalizations remain valid for power-system models.

Three habits organize the chapter:

1. declare the graph object before asking a graph question;
2. qualify every overloaded quantity—especially *degree*, *loop*, *parallel*, and *Laplacian*—by its convention; and
3. describe simplification by a map and its fibres, not by saying that duplicate edges were removed.

The conventions are compatible with standard graph and matrix treatments, but not every text chooses the same definition for loops, adjacency diagonals, or paths. We therefore state the convention used here rather than appealing to a supposed universal one [5–7]. The representation-framework chapter locates this object among the book's other models. The cycles chapter develops the engineering consequences; it does not replace the definitions below.

The finite undirected multigraph

Edge-end definition

An **undirected multigraph with loops** is a tuple

$$G = (V, E, \mathcal{F}, s, \mathrm{ed}),$$

where V is a finite vertex set, E is a finite set of identified edges, $\mathcal{F}$ is a finite set of edge ends or **flags**, and

$$s : \mathcal{F} \to V, \qquad \mathrm{ed} : \mathcal{F} \to E, \qquad |\,\mathrm{ed}^{-1}(e)| = 2 \quad (e \in E).$$

Each edge therefore owns two distinct flags. If the two flags of e map under s to different vertices, e is a non-loop edge. If both map to the same vertex, e is a **graph loop**. Distinct edges may have the same two endpoint vertices; their identity in E makes them parallel members rather than one edge with an unexplained multiplicity.

The edge-end definition is slightly more elaborate than an endpoint pair, but it prevents two common ambiguities. A loop has two incident ends even though it has one endpoint vertex, and a later port model can map the two ends to distinct terminals owned by the same component. It also separates incidence from edge attributes.

For compact notation define the endpoint multiset

$$\partial(e) = \{\{s(h) : h \in \mathrm{ed}^{-1}(e)\}\} \in \mathrm{Sym}^2(V).$$

For a non-loop edge this is the unordered pair $\{u, v\}$; for a loop it is the repeated multiset $\{\{v, v\}\}$. The endpoint map is derived from the flag model. It is not a substitute when individual ends or terminals matter.

An **attributed multigraph** adds declared maps such as

$$\alpha_V : V \to \mathcal{A}_V, \qquad \alpha_E : E \to \mathcal{A}_E, \qquad \alpha_{\mathcal{F}} : \mathcal{F} \to \mathcal{A}_{\mathcal{F}}.$$

Names, impedances, ratings, phases, switch states, owners, provenance, and decision variables belong to these typed maps or to linked models. They are not encoded merely by drawing an edge.

This adopted source-object convention is registered as `GRAPH-MULTI-001`.

Orientation is additional structure

An orientation chooses, for each non-loop edge, one flag as tail and the other as head. A graph loop may also have its two flags ordered, but its tail and head vertex coincide. Orientation is a coordinate convention. It does not assert the sign of current, power, or causality in an operating solution.

Reorienting one edge negates the corresponding signed-incidence column and any edge-coordinate written in that orientation. It does not change connectivity, cycle rank, or the underlying physical member. In particular, for diagonal edge weights W, the assembled scalar operator BWB^{T} is unchanged when an edge orientation is reversed, provided the edge-coordinate convention is changed consistently.

The word degree is insufficient

For $v \in V$ define the **incidence degree**

$$d_{\mathrm{inc}}(v) = |s^{-1}(v)|.$$

A graph loop contributes two because it has two flags at v. This is the degree used by the handshake identity

$$\sum_{v \in V} d_{\mathrm{inc}}(v) = 2|E|.$$

Other useful counts are different:

- the **non-loop incidence count** counts flags at v belonging to edges with two different endpoint vertices;
- the **distinct-neighbour degree** counts vertices $u \neq v$ joined to v by at least one edge and ignores multiplicity;
- the **member count** counts identified incident non-loop equipment, so parallel members count separately;
- a **terminal count** belongs to a terminal or port model and need not equal any graph degree; and
- a matrix row nonzero count belongs to a compiled sparsity graph and can include diagonal terms or couplings that are not bus–branch edges.

Consequently, statements such as "remove degree-two buses" are incomplete. They must say which representation supplies the vertices, which degree is used, whether loops and open members count, and what electrical conditions make the proposed elimination valid.

Matrix conventions

Let $n = |V|$ and $m = |E|$. Fix an ordering of vertices and edges and an orientation of every non-loop edge.

Signed incidence matrix

The signed incidence matrix $B \in \{-1, 0, 1\}^{n \times m}$ has a -1 at the tail and $+1$ at the head of each non-loop edge. A graph-loop column is zero because its two signed incidences occur at the same vertex and cancel. Changing all signs in a column is an equivalent orientation convention.

Graph-theory trap

This zero column does **not** say that a physical self-connected or grounded device has no effect. It says only that an ordinary graph loop contributes nothing to the signed boundary operator.

Multiplicity adjacency and degree matrices

Let $\mu(u, v)$ be the number of edges with distinct endpoints u and v, and let $\lambda(v)$ be the number of graph loops at v. This book uses the multiplicity adjacency matrix

$$A_{uv} = \mu(u, v) \quad (u \neq v), \qquad A_{vv} = 2\lambda(v).$$

With $D = \operatorname{diag}(d_{\text{inc}}(v))$, the row sums of A equal the incidence degrees and

$$L = D - A = BB^{\mathsf{T}}.$$

The graph loops cancel between D and A. Some authors instead place $\lambda(v)$ or zero on the adjacency diagonal and modify associated degree or Laplacian definitions. Those are legitimate conventions, but identities must not be moved between them without adjustment [7, 8].

For diagonal nonnegative edge weights W, the weighted series Laplacian is

$$L_W = BWB^{\mathsf{T}}.$$

Ordinary graph-loop weights again vanish because the loop columns of B are zero. Complex branch stamps, ideal transformer ratios, mutual coupling, phase coordinates, and controlled factors generally require a richer assembly than this scalar expression. A power-system nodal admittance matrix is therefore not, without qualification, "the graph Laplacian."

Circuit-theory trap

A nodal-admittance diagonal may contain shunts, grounded loads, eliminated terminal factors, or other compiled contributions. Its diagonal support is not evidence that the source model contains graph self-loops, and its off-diagonal support is not a complete asset-level graph.

Graph loops are not shunts

Four objects that are often drawn similarly must remain distinct:

Object	Formal location	Incidence/Laplacian effect
graph loop	an edge whose two flags map to one graph vertex	zero signed-incidence column; cancels from $D - A$ under this convention
shunt to reference	a one-terminal constitutive relation, or an edge to an explicit reference vertex before grounding	contributes a grounded diagonal or primitive stamp
contraction-created loop	an ordinary edge whose endpoints become identified under a quotient	records information created by contraction; may later be deleted only by a declared map
diagonal dependency	a diagonal term in an operator or sparsity pattern	algebraic self-dependence; not automatically an asset edge

For a scalar grounded network one may write

$$Q = BWB^{\mathsf{T}} + Y_{\mathrm{sh}},$$

where Y_{sh} is a declared grounded diagonal contribution. Calling the second term a graph loop and then applying the zero-column convention would erase precisely the grounding effect being modeled.

The compatible incidence/adjacency/Laplacian identity and the graph-loop versus grounded-shunt distinction are registered as `GRAPH-MATRIX-001`.

Self-loops, circuit branches, and collapsed two-port factors

The word *loop* changes meaning at the boundary between graph theory and circuit theory. A graph self-loop is one identified edge whose two flags map to one vertex. An electrical loop or mesh is a closed branch path used by a particular circuit formulation. A graph self-loop may be a one-edge circuit in the graphic-matroid sense, but it is not thereby an ordinary circuit branch between two distinct retained nodes.

Many incidence-based network texts use loopless graphs as their circuit specialization: every ordinary branch has a distinct source and sink node, and the incidence column records the difference of the two node potentials [9]. This is a useful specialization, not a universal definition of every graph used in electrical modeling. Network-reduction literature also uses *loopy Laplacian* for diagonal grounded terms, including positive or negative differential conductances; that usage is a matrix-level abstraction and must not be silently identified with the graph-loop convention above [10, 11].

The book therefore uses the following boundary:

Situation	Retain in the source model	Possible compiled view
ordinary two-terminal branch with distinct retained terminals	identified factor and both terminal maps	loopless bus–branch edge or terminal relation
graph self-loop in the multigraph	edge identity, flags, state, and provenance	omitted from distinct-node connectivity, or retained as a declared degenerate factor
one-terminal shunt or grounded load	constitutive relation and attachment map	diagonal nodal stamp or explicit reference edge
two-terminal factor after terminal identification	original factor, its two ports, and the quotient map	a compiled one-terminal relation only after the factor equation is assembled

Thus a self-loop produced by node contraction is not automatically deleted and not automatically called a shunt. The safe order is:

1. retain the source factor and its terminal maps;
2. apply the topology-state quotient to those maps;
3. compile the factor in the declared equation target; and
4. simplify only after checking the requested observations, limits, controls, and recovery map.

This order matters because graph projection and constitutive compilation need not commute. A loopless connectivity view may omit the self-loop while the compiled factor still contributes a diagonal term or a constraint.

Exact collapse of a fixed linear π section

Let a fixed two-port π factor have series admittance Y_s and terminal shunt admittances Y_a and Y_b. Its terminal admittance relation is

$$\mathbf{i}_\pi = Y_\pi \mathbf{v}_\pi, \qquad Y_\pi = \begin{bmatrix} Y_a + Y_s & -Y_s \\ -Y_s & Y_b + Y_s \end{bmatrix}.$$

For retained node-voltage coordinates $\mathbf{u}$, let T_π map retained voltages to the two terminal voltages. The nodal contribution is the standard terminal-map assembly

$$Y_\pi^{\mathrm{N}} = T_\pi^{\mathsf{T}} Y_\pi T_\pi.$$

If topology processing identifies both terminals with the same retained coordinate and there is no transformer or coordinate conversion, then $T_\pi = [1\ 1]^{\mathsf{T}}$ and

$$Y_\pi^{\mathrm{N}} = \begin{bmatrix}1 & 1\end{bmatrix} Y_\pi \begin{bmatrix}1\\1\end{bmatrix} = Y_a + Y_b.$$

The series term cancels because its terminal voltage difference is zero. The two shunts remain and combine into one constant-admittance, equivalently constant-impedance, one-terminal nodal contribution. This is consistent with power-system π models, where the line is a two-port and shunt terms enter the nodal diagonal rather than the off-diagonal bus coupling [9, 12].

Circuit-theory trap

The two shunts in this collapse are assumed to use the same retained ground/reference and compatible terminal coordinates. If the sections use different references, or the retained coordinates identify more than the declared terminal pair, the one-terminal formula must be rederived from the full terminal relation.

This identity does **not** license a universal "self-loop deletion" rule. It does not preserve a branch-current observation, member rating, loss allocation, switching decision, or source provenance. It also does not apply without further analysis when the factor contains an ideal voltage source, zero-impedance constraint, nontrivial transformer ratio, mutual coupling, dependent source, control law, dynamic state, or incompatible terminal coordinates. Such factors belong in modified nodal, tableau, or port–factor formulations until an exact elimination and recovery map has been established [13, 14].

The loopless circuit specialization, the literature distinction between graph self-loops and loopy diagonal terms, and the guarded π-collapse identity are registered as `GRAPH-SELF-LOOP-001`, `GRAPH-LOOPY-001`, and `GRAPH-PI-COLLAPSE-001`.

Connectivity and cycles

Graph loops do not connect distinct vertices. Connected components are therefore determined by non-loop edges. If c is the number of components, then over any field of characteristic zero

$$\operatorname{rank} B = |V| - c, \qquad \dim\ker B = |E| - |V| + c.$$

Each graph loop contributes one zero column and hence one cycle-space dimension. Each additional edge in a parallel class contributes another dimension after the first member needed for connectivity. Thus multigraph cycles include:

- a one-edge circuit formed by a graph loop;
- a two-edge circuit formed by two parallel non-loop edges; and
- the familiar longer circuits.

Over $\mathbb{F}_2$, a cycle is often represented by an even-degree edge set and orientation signs disappear. Over $\mathbb{R}$ or $\mathbb{C}$, cycle coordinates retain signed coefficients. The dimension formula agrees, but the coefficient field and intended use must be stated.

The inclusion-minimal supports of nonzero vectors in $\ker B$ are the circuits of the graphic matroid. This definition handles loops and parallel members without adding exceptions [15, 16]. It also explains why a cycle basis is not unique: it is a vector-space basis, not a canonical set of physical loops.

The classical connection between incidence matrices, cut spaces, cycle spaces, and electrical network equations is developed in network-matrix texts such as [9]. Electrical interpretation still requires constitutive laws: a topological circuit does not prove that a circulating current exists at a particular operating point.

Simple projection is a quotient

Let

$$E^\circ = \{e \in E : \partial(e) = \{u, v\},\ u \neq v\}$$

be the non-loop edges. The loopless underlying simple graph is

$$\overline{G} = (V, \overline{E}), \qquad \overline{E} = \big\{\{u, v\} : e \in E^\circ,\ \partial(e) = \{u, v\}\big\}.$$

Its simple-projection map

$$\mathrm{simp} : E^\circ \to \overline{E}, \qquad \mathrm{simp}(e) = \partial(e)$$

has fibres $\mathrm{simp}^{-1}(\{u, v\})$ containing all identified members between the same endpoint vertices. Constructing $\overline{G}$ therefore performs two operations: graph loops are omitted, and every non-loop parallel class is replaced by one adjacency. Neither operation is invertible without retained fibre and loop records.

This is not a data-cleaning rule. Two records with the same endpoints may be two valid circuits, while two records with different bus identifiers may become parallel only after topology processing identifies their terminal nodes.

What the projection preserves

Preservation is query-dependent. For the unweighted loopless simple projection above:

The structural preservation boundary of this quotient is registered as `TR-GRAPH-SIMPLIFY-001`.

Weighted aggregation adds a second transformation whose rule depends on the query. Under suitable independence and coordinate assumptions, parallel scalar admittances add, capacities add for a maximum-flow query, shortest-path lengths take a minimum, and independent component reliabilities combine as $1 - \prod_k (1 - r_k)$. None of those operations is a universal edge merge, and none preserves member identity, individual limits, switching decisions, or coupling. The preservation-contract chapter provides the required transformation vocabulary.

Worked parallel fibre: four queries, four rules

Take two identified members e_1, e_2 between u and v. Give them scalar admittances $y = (10, 1)\,\mathrm{S}$, current limits $\bar{i} = (100, 100)\,\mathrm{A}$, route lengths $\ell = (10, 1)$, and independent availability probabilities $r = (0.9, 0.8)$. The simple projection records only the adjacency $\{u, v\}$. A useful weighted image depends on the question:

The capacity image is not an exact electrical-limit image. At a common voltage difference of $15\,\mathrm{V}$, the member currents are $(150, 15)\,\mathrm{A}$. Their sum $165\,\mathrm{A}$ satisfies the aggregate $200\,\mathrm{A}$ bound while e_1 violates its own $100\,\mathrm{A}$ bound. The same fibre therefore admits a sum, a minimum, a probability product, or no adequate scalar replacement depending on the exactness object.

This example is intentionally elementary. Matrix admittances require compatible terminal coordinates, capacity addition requires the declared flow model, and the reliability expression requires independent member availability. The numbers make the operation visible; they do not broaden its assumptions.

Query or object	Status under simple projection	Qualification
vertex set and connected components	preserved	loops and multiplicity do not join new vertex pairs
adjacency and distinct-neighbour sets	preserved by definition	edge identity is not preserved
unweighted vertex-to-vertex distance	preserved	edge-level routes and member choices are not
articulation vertices	preserved	use the conventional vertex-deletion query on the underlying loopless graph
vertex properties defined on the underlying simple graph, such as treewidth	preserved by definition	this says nothing about member-level constraints
edge count, incidence degree, and cycle-space dimension	not preserved	loops and excess parallel members change all three
bridges and edge connectivity	not preserved	a parallel pair contains no bridge, while its image may be a bridge
spanning-tree count and Eulerian parity	not preserved	multiplicity changes member choices and degrees
line graph and edge-disjoint routing	not preserved	both depend on identified edges
member outages, ratings, losses, controls, provenance, and decisions	not preserved	these live on source members or linked factors

Query	Image value	What remains outside the image
unconstrained linear terminal current	$y_{\mathrm{eq}} = 10 + 1 = 11\,\mathrm{S}$	member currents, ratings, losses, identity
endpoint maximum-flow capacity	$c_{\mathrm{eq}} = 100 + 100 = 200\,\mathrm{A}$	the electrical sharing law and member constraints
shortest route	$\ell_{\mathrm{eq}} = \min(10, 1) = 1$	the nonselected route and route identity
at-least-one-member availability	$r_{\mathrm{eq}} = 1 - (1 - 0.9)(1 - 0.8) = 0.98$	common-cause dependence, repair states, member provenance

Deletion, contraction, and matroid structure

For $e \in E$, deletion removes e and its two flags. Contraction of a non-loop edge identifies its two endpoint vertices and removes the contracted edge. Other edges incident to those vertices are carried through the quotient. That operation can create parallel edges and graph loops even when the input was simple.

The category of multigraphs is therefore the natural working space for repeated deletion and contraction. Insisting that every intermediate object remain simple silently inserts a further simplification after each contraction and can destroy information needed by later queries.

Worked deletion and contraction

Let e_1, e_2 be parallel edges between u and v and let e_3 join v to w while e_4 joins w to u. The connected multigraph has three vertices, four edges, and cycle rank two.

- Deleting e_1 leaves e_2 between u and v; connectivity is unchanged and the cycle rank falls to one.

- Contracting e_1 identifies u and v as x and removes e_1. The image of e_2 is a graph loop at x; the images of e_3 and e_4 are parallel between x and w. The contracted multigraph has two vertices, three edges, and cycle rank two.
- Simplifying that contracted graph deletes the loop and retains one of the two x–w members. The result has cycle rank zero.

The first two operations have standard edge-level meanings. The third is a separate quotient that loses both circuits. This is why algorithms based on minors naturally permit loops and parallel classes even when their input and final presentation are simple.

The graphic matroid $M(G)$ has ground set E and circuits given by the minimal cycle supports. In this language:

- a graph loop is a matroid loop;
- two parallel non-loop edges form a two-element circuit;
- a bridge is a coloop and belongs to every spanning forest basis; and
- matroid simplification deletes loops and retains one representative from each parallel class.

This language is especially useful when the query concerns independence, bases, cuts, or cycle supports rather than endpoint geometry. It does not carry terminal coordinates, impedances, controls, or power-flow equations by itself. Matroid equivalence is consequently not electrical equivalence.

Topology processing illustrates the distinction. Closing a zero-impedance switch may cause a node-identification contraction; opening a branch is closer to deletion; and bus–branch assembly may subsequently project parallel identified devices into one adjacency. Each step has a different map and a different recovery obligation. The topology-processing chapter develops the state semantics.

Power-system specialization

Same endpoints do not settle electrical parallelism

Two identified branches are graph-parallel when their endpoint vertices agree in the declared multigraph. They are electrically parallel for a particular formulation only if they also act on compatible retained terminal coordinates. Different phase sets, terminal permutations, transformer connections, mutual coupling, controls, or internal states can invalidate scalar parallel formulas even though the bus endpoints agree.

Conversely, elements can contribute to the same reduced nodal block only after terminal identification or coordinate lowering. Their shared matrix support is a fact about that compiled operator, not proof that the source assets were one edge.

The multiconductor parallel AC decision case is a concrete failure witness: a naive scalar aggregate serves more load than the source-member model while violating a member current limit, whereas the exact lifted and pruned formulations retain the source optimum. The example is why endpoint agreement is a necessary graph condition but not, by itself, an electrical aggregation certificate.

Loads and generators: source role versus compiled placement

Whether a load or generator “belongs to the graph” has no context-free answer. In this book, typed loads and generators belong to the canonical source model and attach through nodal or terminal maps. A selected graph view may represent them as factor vertices, terminal relations, one-terminal elements, edges to an explicit reference, or no graph edges at all.

A constant-admittance load can be compiled into a nodal-admittance matrix. A constant-power load generally remains a nonlinear current-injection or power constraint. A solution method may also place a linearized or constant-impedance component in the matrix and retain a compensation current outside it. OpenDSS, for example, uses primitive admittances and compensation currents in its current-injection solution architecture, with study-mode-specific equations [14, 17, 18].

These alternatives can produce different legitimate nodal matrices for the same source system, operating point, initialization, or study. Therefore:

1. matrix membership does not determine source ontology;
2. the nodal matrix must be labelled by formulation, study mode, state, and linearization point where applicable; and
3. moving a typed element into a matrix stamp must not discard its identity, model class, limits, provenance, or recovery map.

The load-model chapter gives the full engineering treatment. The circuit-formulation chapter distinguishes source factors from solver assembly.

Beyond multigraphs

A multigraph edge has exactly two flags. An n-port device, shared magnetic relation, protection dependency, or multiway constraint may not have a faithful two-terminal edge representation. Three common alternatives are:

- a bipartite incidence graph with explicit factor vertices;
- a hypergraph or incidence structure whose relation can own more than two flags; and
- a hierarchical port–factor model retaining typed coordinates and internal variables.

Replacing an n-port relation by a clique or star is a compilation choice. It can create graph cycles or auxiliary vertices that are absent from the source relation, so its preservation contract must identify which observations and constraints survive. The multigraph remains essential for identified two-terminal members, but it is not the universal source ontology.

The three-winding transformer is the useful mental model before the formal generalization: one physical relation owns three typed ports, while a star or terminal-clique graph is only a chosen lowering. The incidence construction below makes that distinction explicit and records why the resulting cycle structure belongs to the compiled view rather than automatically to the device inventory.

Incidence structures and typed n-port relations

The flag construction generalizes without forcing pairwise edges. Define a finite incidence structure

$$\mathcal{I} = (X, R, \mathcal{F}, s, \mathrm{rel}), \qquad s : \mathcal{F} \to X, \qquad \mathrm{rel} : \mathcal{F} \to R,$$

where X is a set of junction or object vertices and R is a set of identified relations. Unlike a multigraph, the fibre $\mathrm{rel}^{-1}(r)$ may have any declared finite cardinality. A two-uniform incidence structure, in which every relation owns exactly two flags, recovers the multigraph object. A relation with three flags is natively ternary rather than three unexplained pairwise edges.

For mathematical modeling, incidence alone is usually insufficient. Give each flag f a typed variable space $\mathcal{X}_f$, give each relation an ordering or role map ω_r on $\mathrm{rel}^{-1}(r)$, and attach a constitutive or constraint relation

$$\mathcal{R}_r \subseteq \prod_{f \in \mathrm{rel}^{-1}(r)} \mathcal{X}_f.$$

This is the flat core of the book's hierarchical port–factor object. The hierarchy additionally records ownership, internal variables, subsystem boundaries, and source provenance. A bare hypergraph records the multiway incidence set; a bipartite incidence graph represents each $r \in R$ as a factor vertex adjacent to its incident flags or junctions; the typed port–factor model retains the relation spaces and roles. These are related representations, not synonyms.

This arbitrary-arity incidence and typed-relation boundary is registered as `GRAPH-NPORT-001`.

For example, a three-winding transformer relation with ports h, m, l is one arity-three factor. A star lowering introduces an internal vertex and three two-terminal branches. Eliminating the star vertex can produce a three-edge terminal clique. The incidence graph is acyclic locally, the star is acyclic, and the clique has cycle rank one, yet all three can encode views of one device. The cycle-rank change belongs to the compilation, not to the physical transformer inventory.

Declaration checklist

Before using a graph-derived result, record:

1. **Vertices:** buses, connectivity nodes, terminals, conductors, factors, equations, variables, or another declared object?
2. **Edges:** identified assets, closed-state members, terminal incidences, nonzero matrix blocks, or dependencies?
3. **Multiplicity and loops:** allowed, omitted, or produced by a quotient?
4. **Orientation:** which endpoint order is stored, and which quantities change sign under reorientation?
5. **Degree:** incidence, member, distinct-neighbour, terminal, or sparsity degree?
6. **Matrix convention:** how are loop and diagonal terms represented?
7. **State and study:** which switch state, formulation, mode, and operating or linearization point apply?
8. **Map:** what source-to-view transformation was applied, and what are its fibres?
9. **Query:** connectivity, cycle support, flow, feasibility, reliability, optimization, or another objective?
10. **Recovery:** which source identities, constraints, and results remain recoverable?

If these entries are absent, a mathematically correct theorem about the chosen graph can still yield an engineering conclusion about the wrong object.

Evidence boundary

The definitions and elementary matrix identities above are adopted conventions; the repository checks them on a finite loop-and-parallel-edge witness in `experiments/test/multigraph_conventions.jl`. The cited books establish the general graph, matrix, and matroid background. The power-system sections are a representation synthesis linked to this book's executable cases and specialist chapters.

This chapter does not claim that every power-system device should be encoded as an edge, that every nodal matrix is a Laplacian, that endpoint-parallel members can always be aggregated, or that graph/matroid preservation implies electrical or decision preservation. Those are precisely the shortcuts the declared maps are intended to prevent.

Chapter 16

Maps between representation frameworks

Page status: formal vocabulary and scoped map definitions.

Why the arrows need types

The book uses several mathematical structures, so an arrow between two models cannot be called a *graph map* without further qualification. A renaming inside one framework, a quotient that forgets parallel identity, a compiler that introduces virtual objects, and a behavioural elimination have different domains, codomains, and proof obligations.

This chapter fixes a first map vocabulary. It distinguishes:

- **morphisms**, which preserve the declared structure inside one framework;
- **isomorphisms**, which are reversible morphisms and therefore express a change of names or coordinates rather than a loss of meaning;
- **cross-framework transformations**, which carry an explicit preservation contract and need not be morphisms in either endpoint category.

The definitions below are book conventions. They are intentionally strong for isomorphism and deliberately modest for general morphisms; later chapters may specialize them for particular data standards or equation languages.

The map-of-maps view is the practical reading rule for this chapter: a quotient, compiler, coordinate change, or elimination is justified by the query and contract attached to its arrow.

Simple-topology maps

Let $G = (\mathcal{B}, E)$ and $G' = (\mathcal{B}', E')$ be loopless undirected simple graphs.

Definition. A simple-graph homomorphism is a vertex map $h_{\mathcal{B}} : \mathcal{B} \to \mathcal{B}'$ such that

$$\{i, j\} \in E \quad \Longrightarrow \quad \{h_{\mathcal{B}}(i), h_{\mathcal{B}}(j)\} \in E'.$$

It is an isomorphism if $h_{\mathcal{B}}$ is bijective and adjacency is preserved in both directions. Because both graphs are loopless, a homomorphism may identify **nonadjacent** vertices, but it cannot identify the endpoints of an edge: their image would be a singleton rather than an edge of G'. An isomorphism only renames vertices.

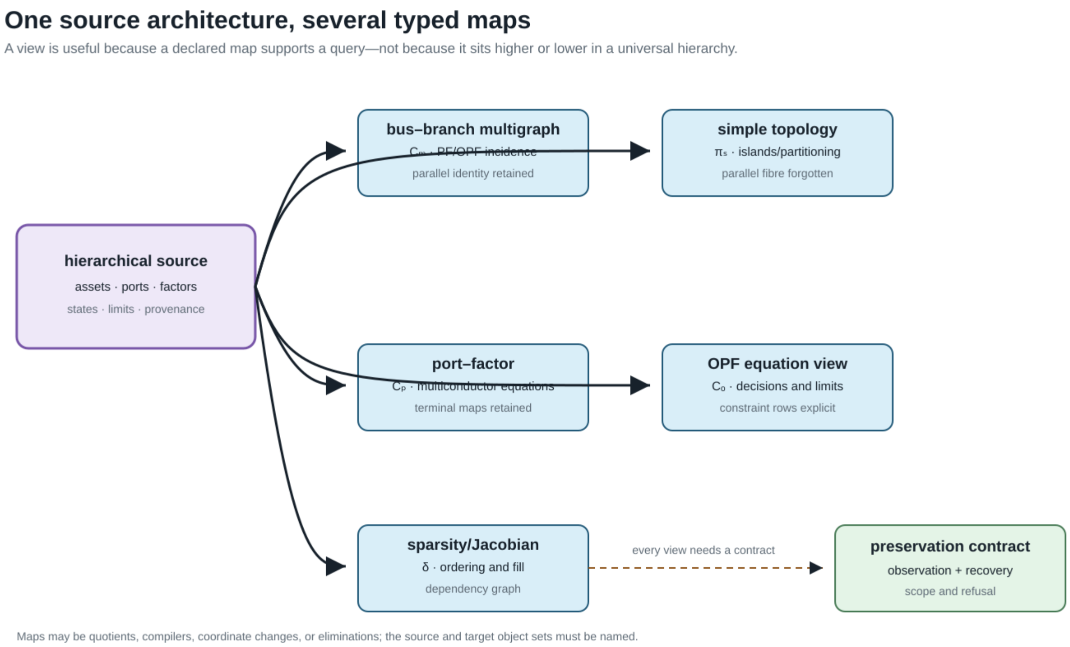

Figure 16.1: One source architecture feeding several typed maps.

This restriction matters in power-system topology processing. Let $F_\sigma \subseteq E$ be the closed ideal-switch edges in a resolved state, and let $\sim_\sigma$ be the equivalence relation generated by their endpoint pairs. The component map

$$\pi_\sigma : \mathcal{B} \longrightarrow \mathcal{B}/{\sim_\sigma}, \qquad i \longmapsto [i]_\sigma,$$

is an **edge-contraction quotient**, not a loopless simple-graph homomorphism of the preceding type. Each retained identified edge ℓij maps to an edge between $[i]_\sigma$ and $[j]_\sigma$ when those classes differ. An edge whose endpoints enter the same class becomes a loop under raw contraction; a loopless bus view deletes that loop while recording its source fibre. Parallel retained members remain parallel unless the separate quotient Q_{MS} is subsequently applied. In ordinary graph language, contracting F_σ and deleting the induced loops is a minor-type operation. In this book it is additionally typed by switch state, equipment class, and provenance. The complete power-system contract is given in Node–breaker, bus–breaker, and topology processing.

Oriented-multigraph maps

Consider oriented multigraphs G_{M} and G'_{M} with bus sets $\mathcal{B}, \mathcal{B}'$ and identified element sets $\mathcal{L}, \mathcal{L}'$.

Definition. An orientation-aware structural morphism consists of maps

$$h_{\mathcal{B}} : \mathcal{B} \to \mathcal{B}', \qquad h_{\mathcal{L}} : \mathcal{L} \to \mathcal{L}',$$

and a sign $\epsilon_\ell \in \{-1, +1\}$ for each $\ell \in \mathcal{L}$. For every stored triple ℓij, its image has endpoints

$$\begin{cases} h_{\mathcal{L}}(\ell)\, h_{\mathcal{B}}(i)\, h_{\mathcal{B}}(j), & \epsilon_\ell = +1, \\ h_{\mathcal{L}}(\ell)\, h_{\mathcal{B}}(j)\, h_{\mathcal{B}}(i), & \epsilon_\ell = -1. \end{cases}$$

Typed attributes must be carried by declared attribute maps. Reversal acts on end-specific quantities, for example by interchanging the ℓij and ℓji terminal data, but leaves element-intrinsic data such as $\mathbf{Z}_\ell$ attached to the same element identity.

The morphism is an isomorphism when the bus and element maps are bijective, the attribute maps are invertible, and the inverse obeys the same incidence and reversal rules. This definition makes an arbitrary reversal of reference orientation an isomorphism. It does **not** identify parallel elements: doing so is a quotient transformation, not a multigraph isomorphism.

Port–factor maps

Let $\mathfrak{P}$ and $\mathfrak{P}'$ be hierarchical port–factor models. A structural map contains functions

$$h_{\mathcal{Q}} : \mathcal{Q} \to \mathcal{Q}', \qquad h_{\mathcal{J}} : \mathcal{J} \to \mathcal{J}', \qquad h_\Phi : \Phi \to \Phi'$$

that commute with attachment and ownership:

$$h_{\mathcal{J}} \circ j = j' \circ h_{\mathcal{Q}}, \qquad h_\Phi \circ f = f' \circ h_{\mathcal{Q}}.$$

It must also preserve port types and the declared ancestor relation in the containment forest. A port-coordinate map $T_q : \mathcal{X}_q \to \mathcal{X}'_{h_{\mathcal{Q}}(q)}$ accompanies each port.

Definition. A port–factor isomorphism has bijective object maps, invertible coordinate maps, and exact transport of every factor and junction relation. If T_ϕ is the product of the coordinate maps over the ordered ports of factor ϕ, then

$$T_\phi(\mathcal{R}_\phi) = \mathcal{R}'_{h_\Phi(\phi)},$$

with the same condition for junction relations, decisions, and parameter types. Equality of relation images is important: merely mapping feasible source points into the target would establish refinement, not isomorphism.

Factor compilation, factor aggregation, and hidden-variable elimination are therefore not silently admitted as port–factor isomorphisms. They are typed transformations with their own behavioural claims.

Asset/dependency maps

For asset structures $\mathfrak{A}$ and $\mathfrak{A}'$, a typed relation morphism consists of entity and relation maps h_V, h_R that preserve types and ordered incidence:

$$\tau'_V \circ h_V = \tau_V, \qquad \tau'_R \circ h_R = \tau_R, \qquad \iota'(h_R(r)) = h_V^{\times n}(\iota(r))$$

for each n-ary relation r. Attribute translation must be declared field by field. An isomorphism requires bijective maps and invertible attribute translation.

The electrical link Λ is relational, so compatibility means

$$(a, e) \in \Lambda \quad \Longrightarrow \quad (h_A(a), h_E(e)) \in \Lambda'.$$

It does not require one asset per electrical factor. That requirement would exclude compilations, shared assets, and multi-asset factors by definition.

Equation-incidence and sparsity maps

An equation-incidence graph is bipartite. Its morphisms must preserve the variable and relation vertex classes as well as incidence. An isomorphism may rename or reorder variables and relations, but it cannot turn a variable into a constraint.

For a square matrix $\mathbf{M}$, simultaneous reordering by a permutation matrix $\mathbf{P}$ gives

$$\mathbf{M}' = \mathbf{P}^{\mathsf{T}}\mathbf{M}\mathbf{P}.$$

This is a sparsity-graph isomorphism. Schur elimination is not: it removes a block and can create fill. Algebraically equivalent equation systems can also have non-isomorphic incidence graphs after auxiliary variables are introduced.

The principal cross-framework transformations

The most important arrows used in the book are consequently named rather than folded into one generic graph homomorphism.

Map	Domain and codomain	Required declaration
Q_{MS}	identified two-terminal multigraph to simple topology	edge fibres and forgotten member data
π_σ	connectivity-node graph to the state-conditioned component quotient	resolved switch state, contracted edge set, loop policy, retained-member fibres, and isolated-node policy
C_{PM}	port–factor model to oriented bus–branch model	supported factor library, virtual objects, and provenance
C_{PE}	port–factor model to equations/constraints	study formulation, variable coordinates, and constraint ownership
S_{EM}	equation system to a matrix or sparsity graph	matrix choice, blocking, ordering, and numerical-zero policy
R_∂	open port–factor model to boundary behaviour	boundary, eliminated variables, internal-injection model, admissible inputs, and recovery
Λ	asset/dependency structure to electrical structure	a many-to-many relation, not a compulsory function

For the simple quotient,

$$Q_{\mathrm{MS}}(\ell) = \{i, j\} \quad \text{for every} \quad \ell ij,$$

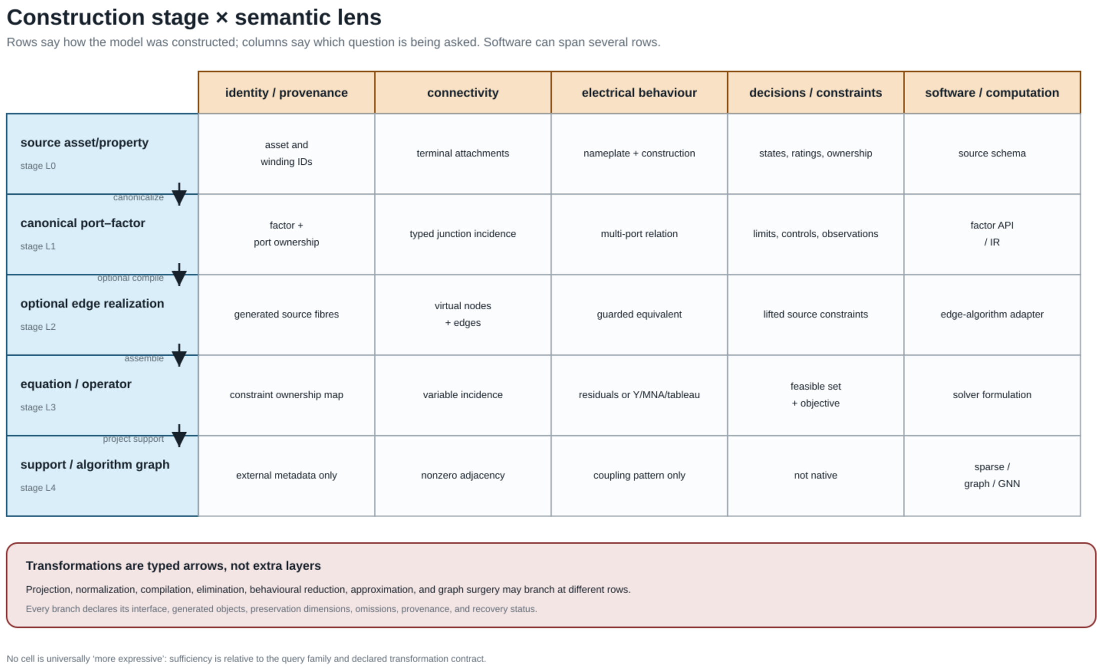

Figure 16.2: Construction stages form the rows and semantic questions form the columns; transformations branch between rows rather than becoming additional levels.

so the fibre $Q_{\mathrm{MS}}^{-1}(\{i,j\})$ is precisely the set of parallel source members. The fibre is part of the provenance record even though it is absent from the simple graph itself.

The compiler C_{PM} is partial. A two-terminal line maps directly, but a multiwinding transformer requires either a target that supports a multi-terminal object or a declared compilation into two-terminal factors. There is no canonical clique expansion.

The boundary map R_∂ is likewise relative to an internal-injection model. For example, a Schur complement with fixed i_I gives an exact affine boundary relation under that assumption; replacing the eliminated injection by a voltage-dependent or constant-power law defines a different source relation. Naming only the eliminated variables and admissible boundary inputs is therefore insufficient to establish exactness.

Construction stages crossed with semantic lenses

The preceding maps should not be rearranged into one universal ladder. It is more precise to separate two axes:

1. a **construction stage**, which records how the present object was derived;
2. a **semantic lens**, which records the question being asked of that object.

The five construction stages used by this book are:

$$L_0 = \text{source asset/property}, \quad L_1 = \text{canonical port–factor}, \quad L_2 = \text{optional edge realization}, \quad L_3 = \text{equation/operator}, \quad L_4 =$$

The columns ask about identity/provenance, connectivity, electrical behaviour, decisions/constraints, and software/computation. A software package may expose objects in several rows, so package names are annotations in this matrix rather than representation levels.

Projection, normalization, compilation, elimination, behavioural reduction, approximation, and state-conditioned surgery are typed arrows. They may branch from different rows and need not pass through L_2. In particular, direct factor stamping $L_1 \to L_3$ is the default route for an arbitrary-arity factor; $L_1 \to L_2 \to L_3$ is an optional compatibility path for an edge-only algorithm.

For each arrow $T : L_a \to L_b$, record an interface ledger

$$\mathcal{I}_T = (\mathcal{B}_T, \mathcal{X}_T, \mathcal{S}_T, \mathcal{O}_T, \mathcal{C}_T, \mathrm{prov}_T, R_T),$$

where the entries declare boundary quantities, coordinate spaces, state, observations, constraints, provenance, and recovery. This is an editorial interface schema, not a claim that every representation category has the same morphisms. Its purpose is to prevent a target that preserves terminal equations from silently inheriting asset, constraint, or decision semantics it does not contain.

The five-bus multi-port lowering applies the matrix to one three-winding transformer and shows why an acyclic star and a cyclic terminal support graph can both be valid derived structures.

Concrete interface smoke test

The generated `experiments/generated/layer-lens-api-witness.json` provides a small, executable crosswalk for claim `ARCH-LENS-001`. It keeps the construction stages and the semantic lenses separate:

Interface	Construction stages	What the smoke test demonstrates	What it does not infer
Versioned JSON3 data	L_0-L_1	asset, terminal, and source-fibre fields can be read from a versioned object	equation coefficients or solver variables
JuMP model	L_0 and L_3	a decision variable, bounds, an explicit constraint, and an objective remain visible in an optimization model	asset provenance unless a map supplies it
`SparseArrays.SparseMatrixCSC`	L_3-L_4	operator nonzeros expose a support relation that can be used by sparse algorithms	the source factor decomposition or ownership of an edge
Graph-learning tensor contract	L_4	`node_features`, `edge_index`, and `edge_attr` have declared shapes and labels	physical, circuit, or feasibility semantics without explicit features and maps

The test deliberately does not import a particular graph-learning package: the tensor names are an interface vocabulary, not a claim that every package uses the same data model. Likewise, a JuMP model is an optimization interface that can consume a compiled equation view; it is not itself a new graph layer. The preferred path for an arbitrary-arity factor remains direct $L_1 \to L_3$ stamping. The ordinary-edge row L_2 is an optional compatibility route and must retain a source fibre when it is used. A support graph can therefore be useful for algorithms while remaining unable to recover asset identity, terminal behaviour, or a feasible decision set on its own.

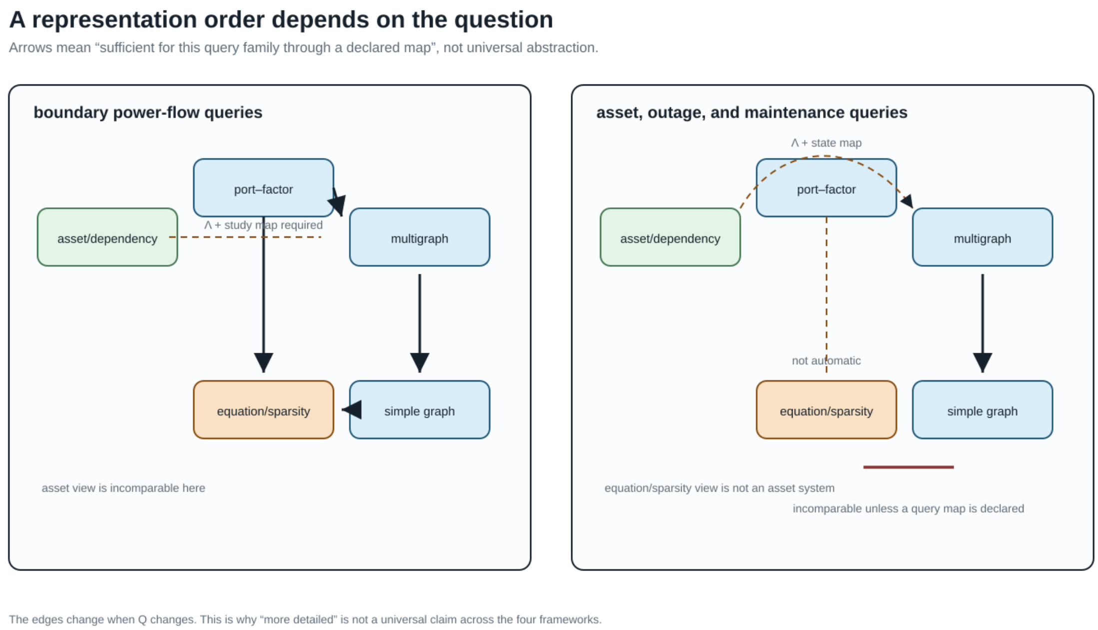

Figure 16.3: Partial orders under boundary power-flow queries and asset/outage queries.

Expressiveness relative to questions

There is no useful total ordering of these frameworks. Let $T : M \to N$ be a transformation and let $\mathcal{Q}$ be a family of queries on M.

Definition (exact query sufficiency). The target N is exactly sufficient for $\mathcal{Q}$ through T if every $q \in \mathcal{Q}$ factors through T: there is a query $\widehat{q}$ on N such that

$$q = \widehat{q} \circ T.$$

Connectivity queries factor through Q_{MS}. A query for the rating of ℓ_1 generally does not. Boundary-current queries can factor through exact Kron reduction, whereas an ownership query does not. This factorization is the exact case of the book's query-relative meaning of *fit for purpose*.

Approximate and one-sided contracts require different statements. On a declared domain $\mathcal{D}$ with an observation metric d_q, a scenario-approximate target may instead satisfy

$$\sup_{m \in \mathcal{D}} d_q(q(m), \widehat{q}(T(m))) \leq \varepsilon_q \qquad (q \in \mathcal{Q}).$$

For a feasible-set query, inner and outer status is expressed by containment, not by factorization. After applying the declared recovery or embedding map R, an inner approximation satisfies $R(\mathcal{F}_N) \subseteq \mathcal{F}_M$ and an outer relaxation satisfies $\mathcal{F}_M \subseteq R(\mathcal{F}_N)$ in the common comparison space. The domain, norm or set map, tolerance, and containment direction are therefore part of sufficiency. Preservation contracts defines the exact, inner, outer, and scenario-approximate classes used throughout the book.

The two panels make the non-ladder claim concrete. Under boundary power-flow queries, a typed port–factor view can compile to an oriented multigraph and a simple projection, while an equation/sparsity view answers a different compiled question. Under asset, outage, and maintenance queries, the asset/dependency view is primary and any electrical compilation requires an explicit Λ and state map. The arrows therefore change with the query family; they are not a universal “more detailed” order.

The running network through the maps

The fixture provides a concrete test of the distinctions.

Source object or relation	Port–factor view	Multigraph view	Simple quotient	Equation/sparsity view
$\ell_1 i_1 i_2$ and $\ell_2 i_1 i_2$	two factors with separate limits	two parallel identified edges	one adjacency with a two-member provenance fibre	separate device and limit blocks; possible shared voltage columns
$\ell_4 i_3 i_4$	one factor with unequal endpoint terminal maps	one oriented attributed edge	one adjacency, with conductor map forgotten	crossed terminal selection appears in coefficients
three-winding x_1	one factor with three winding-port bundles	not an ordinary edge without compilation	outside this quotient's domain until compilation	one high-arity block or several compiled blocks
grounding h_n	a shunt factor at the neutral junction	an attribute or explicit shunt, depending on target vocabulary	normally invisible	a nodal stamp and corresponding nonzeros

The generated provenance artifact now includes a seventh, non-visual `simple_topology` map in addition to the six illustrated views. Its edge $i1 - -i2$ maps back to both `line/l1` and `line/l2`; `x1` is explicitly outside that quotient until a transformer compilation is chosen.

What remains open

These definitions still leave substantive work:

- composition and equivalence of compilers that introduce different virtual objects;
- compositional laws for unresolved or decision-dependent families of state-conditioned contractions;
- refinement orders between nonlinear factor relations with decisions;
- machine-checked commutative diagrams tying fixture schemas to the mathematical maps;
- categorical boundary gluing for typed multiconductor open systems.

Those questions now have named objects and arrows on which later results can operate.

Chapter 17

Translation traps: graphs, circuits, and power-system language

Page status: reviewed explanatory synthesis and controlled vocabulary.

Why familiar words become dangerous

Graph theory, circuit theory, power-system practice, software and data modelling, mathematical optimization, and graph machine learning each have internally useful vocabularies. Trouble begins when a statement is moved from one vocabulary to another without also moving its assumptions. The short five-community vocabulary bridge introduces the communities and their characteristic false friends; this chapter develops the failures that matter for power-network models.

For example, *the line is directed from i to j* may be harmless data shorthand for the stored triple ℓij. It is false if it is read as a claim that active power must be nonnegative at that terminal. Similarly, *the feeder is radial* may describe the simple bus projection while the identified line multigraph contains a two-edge cycle formed by parallel circuits.

This chapter is an early warning map. Later chapters give the detailed definitions and proofs. Here the objective is to replace an underspecified phrase by a statement whose representation, variables, and study meaning are testable.

A disciplined translation pattern

When a familiar sentence carries mathematical weight, ask four questions:

1. **Which representation?** Name the simple graph, identified multigraph, port–factor incidence model, equation graph, asset relation, or another declared object.
2. **Which quantity?** Name the terminal current, terminal power, internal series current, voltage, state, constraint, or decision.
3. **Which state and domain?** State switch status, outage scenario, frequency, approximation, and admissible operating set.
4. **Which consequence?** State whether the claim concerns connectivity, equations, feasible sets, limits, objectives, or physical assets.

The controlled replacement for *power flows on edge ℓij*, for example, is: *at this operating point, $P_{\ell ij}$ is the active-power injection into terminal i of member ℓ, using the stored orientation ℓij.* That sentence remains valid if the operating transfer reverses.

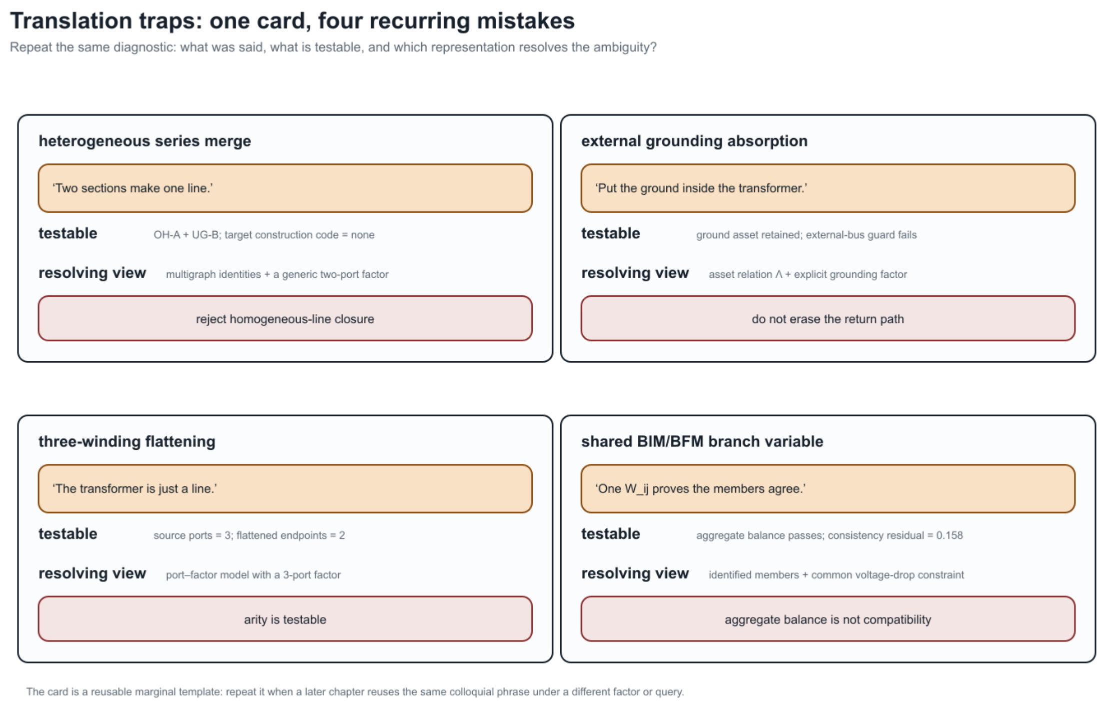

Figure 17.1: Reusable translation-trap card: what was said, what is testable, and which representation resolves the ambiguity.

The card is a deliberate template rather than a summary of four isolated mistakes. When a later chapter reuses a colloquial phrase, it should be able to repeat the same three fields in a margin or caption: quote the phrase, name a checkable quantity or guard, and identify the graph or factor model that makes the distinction visible.

Highest-priority translations

- *The network graph* becomes **the named graph derived for the stated query**.
- *Line ℓ is directed i to j* becomes ℓij **is its stored orientation**, unless direction is an intrinsic admissibility relation.
- *Power on line ℓ* becomes **the terminal-power pair $(\mathrm{S}_{\ell ij}, \mathrm{S}_{\ell ji})$ and its sign convention**.
- *Current is conserved on the edge* becomes **KCL holds at junctions; terminal-current antisymmetry depends on the device factorization**.
- *Power is conserved at the bus* becomes **terminal power balance follows from compatible voltages and KCL; device losses appear in sums over device terminals**.
- *The network has a cycle* becomes **the named representation has a specified cycle or cycle-space element**.
- *These lines are parallel* becomes **state whether parallelism is topological, terminal, electrical, operational, or homogeneous**.
- *This feeder is radial* becomes **the named active simple graph or identified multigraph is a forest**.

- *This bus is a leaf* becomes **its degree is one in the named graph; this alone does not authorize elimination**.
- *The reduced branch is equivalent* becomes **the reduced factor preserves a declared boundary observation and may not represent a physical asset**.

These replacements are deliberately a little longer. The cost is small compared with an invalid reduction, a missing terminal limit, or a topology claim made on the wrong graph.

Flows, signs, and conservation

An arrow is not an operating direction

An ordinary passive line has unordered physical incidence $\partial\ell = \{i, j\}$. Selecting ℓij fixes a reference orientation and terminal order. It does not imply

$$P_{\ell ij} \geq 0 \quad \text{or} \quad \ell \text{ permits transfer only from } i \text{ to } j.$$

The sign of $P_{\ell ij}$ is an operating-point result. A genuinely directed relation instead needs asymmetric physics or admissibility, such as a one-way control dependency. The complete distinction is developed in Orientation, terminal quantities, and power transfer.

Power-system shorthand

A branch arrow in a one-line diagram, data record, or optimization model often means *stored first end and second end*. Do not infer the sign of current or power from it.

A lossy branch has terminal powers, not one conserved flow

With currents defined into a two-terminal series element,

$$I^{\mathrm{s}}_{\ell ji} = -I^{\mathrm{s}}_{\ell ij}, \qquad S_{\ell ij} = U_i (I^{\mathrm{s}}_{\ell ij})^*, \qquad S_{\ell ji} = U_j (I^{\mathrm{s}}_{\ell ji})^*.$$

For $Z_\ell = R_\ell + \mathrm{j}X_\ell$, these statements imply

$$S_{\ell ij} + S_{\ell ji} = Z_\ell |I^{\mathrm{s}}_{\ell ij}|^2, \qquad P_{\ell ij} + P_{\ell ji} = R_\ell |I^{\mathrm{s}}_{\ell ij}|^2.$$

Thus opposite series currents do not produce opposite terminal powers unless the relevant loss is neglected. A nominal-π factor is more subtle: currents at the two composite terminals need not be negatives because the factor also contains paths through its shunts. The missing current has not vanished; it is accounted for inside the composite factor.

Circuit-theory trap

A branch carries one flow is a commodity-flow abstraction, not the semantics of a general AC multiport. Report both terminal observations and the device balance. A single antisymmetric active-power variable is a declared lossless approximation.

KCL is a junction statement; power balance needs voltage compatibility

Let currents be defined into all factors attached to a junction i. KCL is

$$\sum_{q \in \mathrm{ports}(i)} \mathbf{I}_q = \mathbf{0},$$

after every port current has been mapped into the junction's conductor coordinates. If those ports also share the compatible junction voltage $\mathbf{U}_i$, multiplication by that common voltage yields the corresponding complex-power balance,

$$\sum_{q \in \mathrm{ports}(i)} \mathbf{1}^{\mathsf{T}} \left(\mathbf{U}_i \circ \mathbf{I}_q^* \right) = 0.$$

This does not say that power is conserved *through each edge*. At a junction, power balance is derived from KCL plus voltage compatibility and consistent terminal maps. In a device, the sum of terminal powers records absorption, generation, or storage according to the constitutive model.

Branch ratings reinforce the terminal view. Sending-end current, receiving-end current, series-conductor current, thermal state, and apparent power are not interchangeable limits, especially for nominal-π and multi-conductor models.

Topology is not an operating story

A cycle is not a loop flow

A cycle is a property of a declared graph or incidence structure. It may support a cycle-space coordinate, but topology alone does not assert a nonzero circulating current or power transfer at an operating point. Parallel members form a two-edge line-identity cycle even though their simple projection has a single adjacency. Conversely, a cycle produced by the clique projection of one multi-terminal factor need not be an alternative physical path.

> **Graph-theory trap**
>
> Separate the existence of a cycle, a chosen cycle basis, a nonzero cycle coordinate, and a physical circulating flow. These are four different claims.

Radiality, leaves, and bridges depend on the graph and state

A feeder may be adjacency-radial in its simple projection and not member-radial in its identified multigraph. It may also be meshed in the asset inventory and radial in one active switching state. The terms *leaf*, *degree-two bus*, *bridge*, and *radial tail* likewise require a graph and an active state.

None of those predicates alone authorizes elimination:

- a leaf can own a load, grounding factor, measurement, control, or boundary observation;
- a degree-two junction can have a shunt, phase change, or terminal mismatch;
- a bridge can be essential to service, protection, reliability, or an investment decision;
- an open line can remain an asset with maintenance, restoration, and future state semantics even when it is absent from the active electrical graph.

Adjacency does not imply direct electrical coupling

A physical connection can be open in the active state, or its terminal maps can leave some conductors unconnected. Conversely, mutual impedance, a shared neutral, a multiwinding factor, or eliminated internal variables can couple nodal equations whose bus vertices are not adjacent in a selected topology graph. Connectivity, energization, constitutive coupling, and matrix nonzero patterns must therefore be tested separately.

In particular, *connected* does not mean *energized*. Energization requires a state-dependent path to an admissible source, together with the device and grounding semantics used by the study.

Equivalence is always relative to a question

The same nodal admittance is not the same decision problem

Two models can have the same unconstrained terminal admittance while differing in member ratings, outages, switches, controls, ownership, measurements, or investment choices. They then need not have the same feasible set or optimum. The parallel-line examples in this book make this failure executable.

Likewise, small voltage error on a scenario set is evidence about one observation metric. It is not, by itself, a bound on feasibility error, objective error, active-limit error, or discrete-decision error.

> **Decision-model consequence**
>
> Never promote equality of Y-bus matrices, a small state error, or a correct base-case power flow into decision equivalence without a constraint map and the relevant feasible-set or error certificate.

An equivalent branch need not be a line

Kron elimination produces a boundary relation. A Ward construction adds a realization of the eliminated external system for a specified study. Either can introduce dense couplings, shunts, sources, or general multiport factors. Calling each off-diagonal block a *line* can accidentally assign physical meaning, ratings, failures, or ownership that the reduced object does not possess.

A recovery map can still evaluate source quantities from retained variables. That is often more faithful than inventing limits on artificial reduced branches. Kron, Ward, and optimized network equivalents separates boundary reduction from target-library realization.

Equality of matrices is coordinate- and model-dependent

Matrix properties need precise names. Reciprocal multiconductor impedance or admittance matrices are commonly **complex symmetric**, $\mathbf{Y}^{\mathsf{T}} = \mathbf{Y}$. They are not generally Hermitian, $\mathbf{Y}^{\mathsf{H}} = \mathbf{Y}$. Positive-real or passivity conditions concern the Hermitian part $\mathrm{He}(\mathbf{Y})$ and should not be replaced by a vague symmetry claim.

Similarly, a per-unit conversion is a coordinate transformation only when the voltage, current, power, impedance, transformer, and terminal bases are moved consistently. A neutral reduction is an elimination or grounding operation, not the graph-theoretic deletion of a spare vertex. Both transformations need typed maps and stated domains.

Devices that resist the ordinary-edge picture

An ideal transformer, a phase-shifting transformer, and a multiwinding transformer are not adequately described as an ordinary lossy edge with one impedance. They can impose voltage and current coordinate actions, grounding relations, winding constraints, and controllable ratios. Compiling a multi-terminal factor into pairwise edges or a virtual star may create or remove apparent graph cycles while preserving a declared terminal relation.

The canonical electrical object in this book is therefore the hierarchical port–factor incidence model. The directed attributed bus–branch multigraph is an important engineering view, and the simple graph is a useful quotient, but neither is asked to carry facts that it cannot represent.

Recurring callouts used in the book

Later chapters use five controlled callout labels:

- **Graph-theory trap:** a correct graph concept has been applied to an unnamed or inappropriate representation.
- **Circuit-theory trap:** a conservation or device statement has been applied outside its factorization or constitutive assumptions.
- **Power-system shorthand:** a familiar phrase is useful in context but mathematically underspecified.
- **Decision-model consequence:** a representation or reduction changes what can be constrained, chosen, observed, or recovered.
- **Vocabulary bridge:** a familiar term crosses a community boundary and its object, qualifier, or unsafe inference must be made explicit.

These callouts do not replace definitions or proofs. Each should give a precise replacement statement and link to the chapter that establishes it.

Scope and controlled vocabulary

The current controlled vocabulary makes these ten translations explicit:

1. one physical system can have several legitimate graphs;
2. a branch arrow does not determine operating flow direction;
3. a lossy branch owns a pair of terminal powers;
4. KCL is not a claim that one power commodity is conserved on every edge;
5. a graph cycle is not an operating loop flow;
6. topological parallelism is weaker than electrical or operational aggregability;
7. radiality, leaves, and bridges require a representation and active state;
8. *bus* must distinguish physical, connectivity, topological, and reporting objects;
9. equal admittance or small voltage error does not establish decision equivalence;
10. a reduced equivalent factor is not automatically a physical asset.

Further scope questions remain for connectivity versus energization, neutral elimination, per-unit coordinates, ideal transformations, and cycles created by multi-terminal compilation; the enacted definitions and witnesses in the current sections delimit what is established so far.

First executable witnesses

The first three distinctions now have package-independent executable witnesses. They are intentionally small: each isolates one translation rather than pretending to validate a complete power-system solver.

Connectivity versus energization

The inventory contains the path $source--bus_a--bus_b--load$, but the bus_a--bus_b member is open in the active state. The load is connected in the asset graph and not energized in the active electrical graph. This is the minimum counterexample to treating inventory connectivity as an operating-state claim.

Complex symmetry versus Hermitian structure

The witness uses a reciprocal complex matrix $\mathbf{A}$ satisfying $\mathbf{A}^{\mathsf{T}} = \mathbf{A}$ but not $\mathbf{A}^{\mathsf{H}} = \mathbf{A}$. Its Hermitian part is positive semidefinite. This separates reciprocity, conjugate symmetry, and passivity tests without relying on a particular network package.

Terminal-specific ratings

For a scalar nominal-π factor, the series current is $I^{\mathrm{s}} = Y^{\mathrm{s}}(U_i - U_j)$ while the composite terminal currents also include their respective shunts. The witness chooses a voltage pair for which the two terminal current magnitudes differ, then places an illustrative rating between them. One terminal therefore violates the rating while the other does not. This is not a proposed rating rule; it is a guard against silently replacing terminal-specific limits with one edge scalar.

Executable anti-pattern witnesses

The same distinctions can be tested rather than left as warnings. The extended translation-trap witness records four negative cases in `experiments/generated/translation-trap-witnesses.json`:

Anti-pattern	Executable observation	Correct interpretation
heterogeneous series merge	pure-series elimination succeeds, but the target is marked outside the homogeneous physical-line class	keep the generic two-port composite and its source identities, or prove stronger line-class guards
external grounding absorption	the transformer compiler rejects a grounding object whose scope is `external_bus`	retain the grounding relation as a separate bus/grounding object
line–transformer flattening	a three-port factor is projected to two line endpoints	the two-terminal view loses winding incidence and cannot stand in for the transformer factor
BIM/BFM index loss	aggregate branch balance holds while the member consistency residual is nonzero	branch identities or the common-voltage-drop relation must remain in the formulation

These are deliberately *negative* witnesses: they do not show that every composition is impossible. They show that a tempting shorthand fails a named guard, or changes the model class, even when a smaller behavioural statement still looks plausible.

Run the witness and its tests from the repository root:

```
julia --project=experiments experiments/run_translation_traps.jl
julia --project=experiments experiments/test/runtests.jl
```

The generated result is `experiments/generated/translation-trap-witnesses.json`. The source module is `experiments/transformations/TranslationTraps.jl`; the anti-pattern extensions are implemented in `experiments/transformations/AntiPatternWitnesses.jl`.

Chapter 18

Two topology levels and the nodal projection

Page status: literature-backed definitions with executable structural and recovery witnesses; inverse recovery is reported by explicit identifiability status and remains non-canonical without additional structure.

The missing middle between a one-line diagram and $\mathbf{Y}^{\mathrm{N}}$

Power engineers routinely move between a bus–branch diagram, a multiconductor circuit, and a nodal admittance matrix. Those three objects are related, but they do not have the same vertices, edges, or cycles. Calling all three *the network graph* hides two distinct topology levels and a many-to-one algebraic projection:

1. the **equipment/terminal topology** records identified equipment and its high-level attachments;
2. the **conductor/port–factor topology** records the electrical terminals, junctions, conductor coordinates, and constitutive factors;
3. the **nodal-operator support graph** records which retained voltage coordinates are coupled by the assembled matrix.

The first two are retained source structure in this book. The third is a derived computational view. The asset/dependency model remains an orthogonal companion to both: ownership, protection, common-mode failure, and maintenance are not entries of a nodal admittance matrix.

The same boundary applies to formulation choice. A compound $\mathbf{Y}^{\mathrm{N}}$ is an exact target for a declared class of fixed linear factors, but it is not a universal representation of a power network. General power-network studies may need modified nodal or sparse-tableau variables, branch currents, multi-terminal factor relations, switch states, controls, and limits that are not recoverable from nodal support alone. See Circuit formulations and the lowering boundary for the guarded compilation alternatives.

The figure uses parallel lines because they expose the loss immediately. Both identified factors attach to the same electrical junction coordinates, and their matrix contributions occupy the same nodal block. The assembly preserves their combined linear boundary relation but forgets how that contribution was split between assets.

The middle panel is a port–factor incidence view, not a bare bipartite graph: the small open circles are ports, the junction bars are the codomain of j, and the factor boxes are the codomain of f. The labelled j and f families are intentionally distinct. A later compilation may replace this structure by ordinary edges, but that is a new target view with a provenance obligation rather than an identity of the source object.

Parallel lines make decomposition loss visible, while a multiwinding transformer makes graph-class change visible. The companion five-bus multi-port lowering shows one three-port factor as an acyclic incidence/star object and, after internal elimination, a cyclic terminal clique. That new support cycle is fill or terminal coupling, not an additional physical transformer loop.

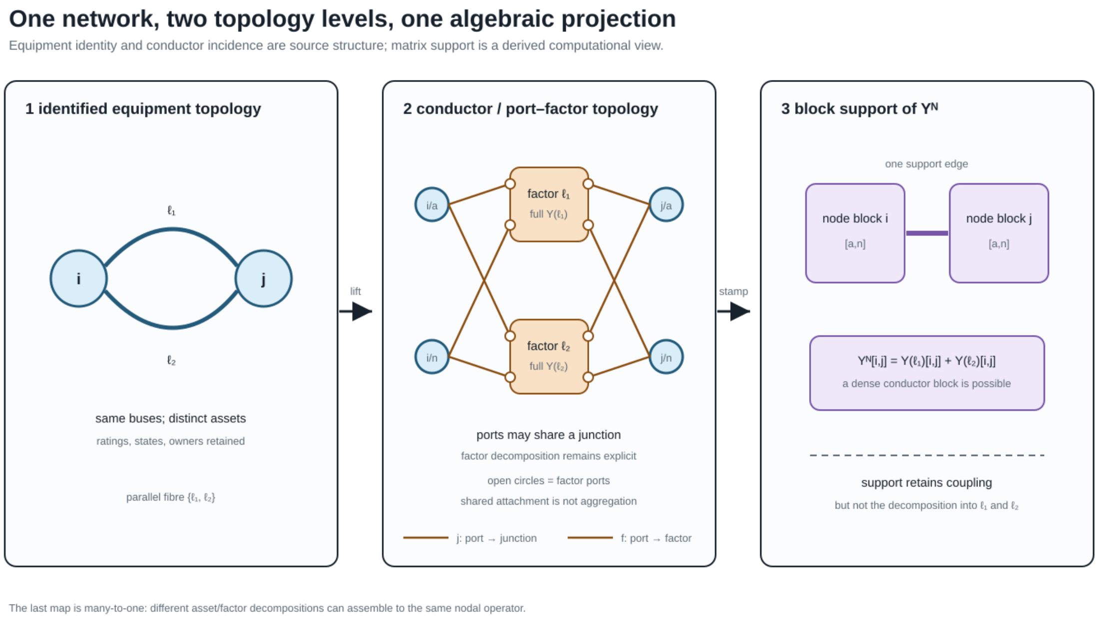

Figure 18.1: Two source topology levels and their many-to-one nodal-admittance projection.

Level 1: identified equipment and high-level attachments

For the two-terminal subset of a network, let

$$G_{\mathrm{M}} = (\mathcal{B}, \mathcal{L}, \partial), \qquad \partial(\ell) = \{i, j\}.$$

This is an identified multigraph: ℓ_1 and ℓ_2 remain distinct when $\partial(\ell_1) = \partial(\ell_2)$. A stored reference orientation is written ℓij and does not make the physical line a directed edge. The element-intrinsic impedance or primitive matrix keeps the symmetric element index $\mathbf{Z}_\ell$ or $\mathbf{Y}_\ell$; terminal observations use ℓij and ℓji.

The multigraph is only a high-level skeleton. A transformer x with winding set $\mathcal{K}_x$ is naturally multi-terminal, and a jointly coupled line group may own more than two port bundles. Such objects belong in an port–factor incidence structure, not in G_{M} unless an explicit two-terminal compilation has been selected. The orthogonal asset/dependency model links equipment records to this electrical structure through Λ; it is not this multigraph. Consequently, *radial at equipment level* must name both the selected object class and any multi-terminal compilation used to obtain an ordinary graph.

At this level, parallelism means repeated high-level attachment. It says nothing yet about terminal order, phase availability, grounding, mutual coupling, state, or whether currents may be added in a common coordinate space.

Level 2: conductor junctions and electrical factors

The canonical electrical model is the hierarchical port–factor object $\mathfrak{P}$ defined in Formal representation frameworks. For the present discussion, its important incidence maps are

$$j : \mathcal{Q} \to \mathcal{J}, \qquad f : \mathcal{Q} \to \Phi,$$

where j attaches a typed port to an electrical junction and f assigns the port to its owning factor. A junction may represent a scalar conductor coordinate such as i/a or a typed bundle such as $i/[a,b,c,n]$. The factor relation retains the full conductor coupling, terminal maps, internal variables, limits, state, and decisions.

Two distinct ports may attach to the same junction; indeed, the map j is deliberately many-to-one because that is how KCL composes devices. A different relation is needed for construction detail. Let

$$c : \mathcal{C}_{\text{phys}} \to \mathcal{Q}$$

map physical conductors or sub-conductors to their electrical ports. This map need not be injective: bundled conductors, paralleled cables landing on one terminal, and other construction-level multiplicities can share a port. The distinction is therefore precise: j composes electrical ports at a junction; c realizes a port from physical conductor objects.

Implementation note. A particular importer or line-constant tool may use a narrower source schema. The current BMOPFTools line-geometry compiler, for example, checks one geometry conductor per terminal label. That is a boundary of that adapter's present primitive, not a definition of c. A richer source must remain expanded or pass through a declared bundle-compilation rule before entering the adapter.

This level has its own notions of parallelism:

- factors can be parallel because all of their boundary port spaces and attachment maps coincide;
- physical conductors can be parallel inside one factor while sharing an electrical terminal at one or both ends;
- two factors can share high-level buses but fail to be terminal-parallel because their phase sets, neutral connections, or terminal maps differ;
- mutual coupling can place two nominal assets in one joint factor, so they are not independent edges even when the asset inventory lists them separately.

The high- and low-level decompositions are therefore related by an explicit lineage/refinement relation, not by an assumed one-edge-to-one-wire rule.

A coupling relation over line-section identities

Jointly coupled circuits expose why the identified multigraph and the factor model are both needed. Let $\mathcal{S}$ be a set of oriented electrical line sections, each with lineage to a stable line asset, and let

$$H_{\text{cpl}} = (\mathcal{S}, \mathcal{C})$$

be a typed relation graph whose edge $c \in \mathcal{C}$ identifies two spatially overlapping sections and carries their mutual blocks, coordinate maps, orientation, and overlap interval. The vertices of H_{cpl} are therefore section identities that appear as edges or refinements in the equipment view. This is a relation *over line sections*, not another bus graph.

Each connected component Γ of H_{cpl} compiles into one joint electrical factor ϕ_Γ. Its primitive is assembled before inversion or nodal stamping. This order preserves indirect coupling created by the inverse of a block matrix and avoids pretending that each line owns an independent constitutive factor. Ratings, states, and decisions remain attached to the source line or section quantities through the factor-to-asset lineage.

The coupled multi-voltage corridor case gives the smallest exact witness. Its two source line assets can belong to different voltage systems, while their joint four-port factor lowers to a generated terminal lattice with cross-voltage edges. Those edges belong to nodal support, not to galvanic connectivity or the asset inventory.

From factors to a compound nodal operator

Stack the retained junction-voltage coordinates into $\mathbf{U}$. Let Φ_{lin} be the declared subset of fixed linear unconstrained electrical factors. A load or generator does not enter this set merely because it attaches to the same network, but a declared constant-admittance part, Norton equivalent, fixed shunt, or iteration-specific linearization may enter it. The nodal operator is therefore a study- and state-specific part of the model from which some decision semantics may already have been omitted.

This distinction is important for one-terminal devices. Their attachment map belongs to the port–factor or terminal-connectivity model even when their linear contribution is stamped into $\mathbf{Y}^{\mathrm{N}}$. Conversely, a constant-power or controlled device may have no fixed admittance stamp while still contributing an injection function, control equation, limit, or optimization decision. Inclusion in Φ_{lin} is thus a formulation choice with a declared mode, state, and preservation contract; it does not decide whether the device belongs to the source graph.

For each $\phi \in \Phi_{\mathrm{lin}}$, let $\mathbf{A}_\phi$ be a real select/permute/sign matrix for its ordered terminal voltages, so

$$\mathbf{u}_\phi = \mathbf{A}_\phi \mathbf{U}, \qquad \mathbf{i}_\phi = \mathbf{Y}_\phi \mathbf{u}_\phi.$$

After mapping terminal currents back to the junction coordinates, linear assembly has the form

$$\mathbf{Y}^{\mathrm{N}} = \sum_{\phi \in \Phi_{\mathrm{lin}}} \mathbf{A}_\phi^{\mathsf{T}} \mathbf{Y}_\phi \mathbf{A}_\phi.$$

The factor primitive $\mathbf{Y}_\phi$ may already contain series, shunt, ideal-connection, transformer, or multiport structure. Incidence-matrix assembly of compound polyphase networks is developed by Kettner and Paolone [19]; nested primitive, winding, and connection maps for general multiphase transformers provide a particularly clear device-level example [20].

The transpose in this assembly is intentional. Because $\mathbf{A}_\phi$ is real, the power-dual current map is $\mathbf{A}_\phi^{\mathrm{H}} = \mathbf{A}_\phi^{\mathsf{T}}$. If a voltage map contains complex ratios or phase actions outside $\mathbf{Y}_\phi$, its current map uses the conjugate transpose instead. This is the same distinction used for transformer maps elsewhere in the book.

The equation is an **assembly identity**, not a unique factorization of $\mathbf{Y}^{\mathrm{N}}$. For two electrically aligned parallel factors $\ell_1 ij$ and $\ell_2 ij$, the same off-diagonal nodal block contains

$$\mathbf{Y}^{\mathrm{N}}_{ij} = \mathbf{Y}_{\ell_1,ij} + \mathbf{Y}_{\ell_2,ij} + \sum_{\phi \notin \{\ell_1,\ell_2\}} \mathbf{Y}_{\phi,ij}.$$

Even when the last sum is empty, $\mathbf{Y}^{\mathrm{N}}_{ij}$ does not identify its two summands. Let $\mathcal{D}$ be the linear difference space admitted by the declared aligned factor class. Every admissible Δ gives the same assembled block through

$$(\mathbf{Y}_1, \mathbf{Y}_2) \longmapsto (\mathbf{Y}_1 + \Delta, \mathbf{Y}_2 - \Delta).$$

Block structure and coordinate expansions

The phrase **vector-valued multigraph** is useful as a translation bridge for the two-terminal case: each high-level edge retains its identity and carries a vector of terminal quantities, while its constitutive data are matrices. It is not the canonical name of the general source model. In particular, an n-winding transformer is a typed port–factor relation, not an ordinary edge with a longer vector label.

Suppose each retained bus has c ordered conductor coordinates. The compound nodal operator is then naturally a block matrix,

$$\mathbf{Y}^{\mathrm{N}} = \begin{bmatrix} \mathbf{Y}_{11} & \cdots & \mathbf{Y}_{1N} \\ \vdots & \ddots & \vdots \\ \mathbf{Y}_{N1} & \cdots & \mathbf{Y}_{NN} \end{bmatrix}, \qquad \mathbf{Y}_{ij} \in \mathbb{C}^{c\times c}.$$

The block labels i, j are bus identities, not necessarily integer positions. For a finite bus set $\mathcal{B}$, an enumeration $\kappa_{\mathcal{B}}$ turns the labelled family $(\mathbf{Y}^{\mathrm{N}}_{ij})_{i,j\in\mathcal{B}}$ into the stored block array

$$\left[\mathbf{Y}^{\mathrm{N}}\right]_{\kappa_{\mathcal{B}}(i),\kappa_{\mathcal{B}}(j)} = \mathbf{Y}^{\mathrm{N}}_{ij}.$$

This is the matrix analogue of the labelled edge–cycle construction in the opening example. The enumeration can change without changing the block operator, its support, or its factor provenance.

For one two-terminal series factor with a full conductor admittance $\mathbf{Y}^{\mathrm{s}}_{\ell}$, the off-diagonal blocks are dense in general, while endpoint shunts add to the diagonal blocks. A nominal-π factor is therefore one matrix-valued factor whose local stamp may be written schematically as

$$\begin{bmatrix} \mathbf{Y}^{\mathrm{sh}}_{\ell i} + \mathbf{Y}^{\mathrm{s}}_{\ell} & -\mathbf{Y}^{\mathrm{s}}_{\ell} \\ -\mathbf{Y}^{\mathrm{s}}_{\ell} & \mathbf{Y}^{\mathrm{sh}}_{\ell j} + \mathbf{Y}^{\mathrm{s}}_{\ell} \end{bmatrix}.$$

This compact block view and the expanded scalar view answer different questions:

View	Vertices or indices	What a nonzero means
factor multigraph	identified assets/factors	a declared factor attachment
block support	buses or junctions	one or more conductor coordinates are coupled
scalar support	bus–conductor coordinates	a particular coordinate pair is coupled
numerical matrix	ordered rows and columns	a coefficient is numerically nonzero at this operating/model point

Two parallel matrix-valued factors can occupy the same block support and be summed by assembly. A dense $c \times c$ factor can produce a clique in the scalar support graph even when the bus-level equipment graph is a tree. The clique is an algebraic coupling pattern, not a claim that every pair of conductors is a physical line. Eliminating an internal bus or factor can add new nonzero blocks by Schur-complement fill; it does not create new source assets.

The complex phasor representation can also be realified for software that expects real arrays. For $\mathbf{Y} = \mathbf{G} + \mathrm{j}\mathbf{B}$, one common coordinate embedding is

$$\mathcal{R}(\mathbf{Y}) = \begin{bmatrix} \mathbf{G} & -\mathbf{B} \\ \mathbf{B} & \mathbf{G} \end{bmatrix}.$$

Realification doubles coordinates, not buses, conductors, or assets. The block support relation is normally the same under this embedding, while the scalar real coordinate graph has twice as many coordinate vertices and may show a different fine-grained pattern. Say **complex-labelled graph** or **realified coordinate graph**, not "the complex graph" or "the real graph," unless the coordinate convention is explicitly part of the sentence.

One important invariant does not survive unchanged. Even when $\mathbf{Y}$ is complex-transpose-symmetric, $\mathcal{R}(\mathbf{Y})$ is generally not an ordinary symmetric real matrix unless $\mathbf{B} = 0$. Realification preserves the coordinate relation and its support, but it changes which matrix symmetry statement is appropriate; do not silently transfer reciprocity claims from the complex representation to the realified one.

The four-view bridge in How to read power-network diagrams and equations is the reader-facing summary of this distinction. Its scoped correspondence is registered as `ARCH-BLOCK-001`: the witness checks the block assembly, support projection, and realification of one declared two-bus four-wire factor, while deliberately treating factor identity as separate provenance rather than inferring it from nonzero matrix entries.

These distinctions also explain why a current-injection solver and a backward–forward sweep can use different computational views of one declared four-wire model. The former may factor a block nodal operator; the latter may traverse element-wise impedance relations on a radial view. Neither algorithmic graph replaces the source factor and terminal semantics.

For reciprocal factors, $\mathcal{D}$ may require $\Delta^{\mathsf{T}} = \Delta$. Adding passivity constraints such as $\mathrm{He}(\mathbf{Y}_k) \succeq 0$ intersects the affine family with a convex feasible set, but does not generally make it bounded: reactive or other unconstrained parameter directions can remain unbounded. Bounded ambiguity needs catalog limits, sign restrictions, measurements, or another coercive guard. Ratings, outage states, investment variables, owners, and the two-edge line-identity cycle are absent unless retained separately.

This is the central many-to-one result registered as `ARCH-NODAL-001`. The companion executable witness constructs two distinct passive reciprocal parallel splits with an identical assembled operator.

When is assembly injective?

Let Θ be the admissible family of typed factor collections and define

$$\mathcal{S}(\{\mathbf{Y}_\phi\}) = \sum_\phi \mathbf{A}_\phi^{\mathsf{T}} \mathbf{Y}_\phi \mathbf{A}_\phi.$$

The exact criterion is

$$\mathcal{S}|_\Theta \text{ is injective} \quad \Longleftrightarrow \quad \ker \mathcal{S} \cap (\Theta - \Theta) = \{0\}.$$

This criterion matters more than a blanket claim that nodal data are or are not recoverable. A practical sufficient special case is available when:

1. topology, factor types, terminal coordinates, and active state are known;
2. at most one two-terminal factor occupies each unordered junction-block pair;
3. that factor's off-diagonal block uniquely determines its complete local stamp within the declared class; and
4. after subtracting those stamps, the remaining diagonal residual has a unique decomposition among the declared shunt and grounding classes.

Then the source primitives are recoverable from $\mathbf{Y}^{\mathrm{N}}$ within that model class. Familiar line-impedance extraction is an instance of this support-separated case, not a contradiction of `ARCH-NODAL-001`.

The principal failures are now named: **multiplicity**, when parallel or overlapping stamps share support; **elimination**, when internal coordinates have already been removed; and **over-parameterization**, when different factor parameters produce the same terminal stamp. Recoverability from a boundary response can nevertheless hold for restricted classes; circular planar critical resistor networks are a major positive theory [21]. The model class and injectivity proof are therefore part of any recovery claim.

A scoped recovery vocabulary

The inverse question should return a status, not just a matrix. For an observed operator $\mathbf{Y}^{\mathrm{N}}$ and declared model class Θ, classify the preimage of the restricted assembly map as follows:

The three statuses are:

- **identifiable** — one admissible source primitive and declared residual decomposition are recovered. A typical case has known support, one two-terminal factor per block pair, and a unique local stamp map.
- **set-identifiable** — the terminal primitive is fixed, but internal parameters or construction records remain an equivalence class. This is the expected result for an over-parameterized local construction with the same terminal stamp.
- **non-identifiable** — distinct source primitives or topologies remain observationally indistinguishable. Parallel multiplicity and eliminated internal coordinates are the basic examples.

This vocabulary is deliberately relative to Θ. Tightening the class with catalog bounds, switch state, measurements, construction metadata, or grounding declarations can turn a non-identifiable class into an identifiable one; silently assuming those facts does not. Conversely, a numerically unique optimizer solution is not evidence that the source map is injective.

The recovery contract is therefore:

1. validate the coordinate order, factor class, active state, and admissible parameter domain before attempting inversion;
2. return a unique recovered primitive only for the identifiable class;
3. return a representative together with an explicit equivalence or affine ambiguity for set-identifiable and non-identifiable classes; and
4. retain the source/provenance record as the authority for asset identity.

The generated `experiments/generated/nodal-source-recovery-witness.json` exercises all four statuses. Its support-separated scalar case recovers a series primitive and declared diagonal shunts. Its parallel case exhibits two distinct member splits with the same operator; its elimination case matches a hidden three-node chain to a direct two-node boundary factor; and its over-parameterized case fixes the terminal primitive while leaving two local parameter vectors indistinguishable. These are deliberately finite witnesses, not a claim that every practical line or transformer belongs to one class.

This scoped classification is registered as `ARCH-RECOVERY-001`. It turns the warning “do not invert $\mathbf{Y}^{\mathrm{N}}$ canonically” into a usable engineering interface: declare the model class, report the recovery status, and preserve the ambiguity whenever the data do not identify the source.

Which guards actually lift the ambiguity?

Extra information should be treated as an additional observation map, not as an informal reason to choose one reconstruction. If $\mathcal{M}$ records member currents, grounding metadata, state observations, or another declared measurement, define the augmented map

$$\mathcal{F}(\theta) = \big(\mathcal{S}(\theta), \mathcal{M}(\theta)\big).$$

The same restricted-kernel test applies:

$$\mathcal{F}|_\Theta \text{ is injective} \quad \Longleftrightarrow \quad \ker \mathcal{F} \cap (\Theta - \Theta) = \{0\}.$$

Four small cases make the distinction operational:

- **Catalog bounds** can make a parallel-split ambiguity compact without making it unique. A bounded interval is still a set of admissible source models.
- **Member-current observations** can lift parallel ambiguity when the voltage drop is nonzero and each member current is observed in the same coordinates. The total nodal operator and the member observations then jointly determine the scalar member admittances in that restricted class.
- **Grounding declarations** can resolve a diagonal residual only when the declaration identifies the grounding contribution and its terminal. A finite grounding impedance and an unspecified local shunt otherwise remain an attribution ambiguity.
- **Transformer or switch states** must be part of the declared state space. A known tap lets a state-conditioned primitive be recovered; an unknown tap can trade off against the primitive parameter and produce multiple state–parameter pairs with the same effective terminal relation.

The generated `experiments/generated/nodal-recovery-guards-witness.json` records these four outcomes. Its statuses are deliberately more informative than a binary "recoverable" flag: `bounded-non-identifiable`, `identifiable`, `identifiable-with-declaration`, and `identifiable-with-state`. The witness is scalar and finite; it establishes the guard logic, not a general theorem for all multiconductor transformers, nonlinear grounding, or state-dependent AC models. This extension is registered as `ARCH-RECOVERY-002`.

Matrix-valued observations need excitation rank

The scalar member-current example does not generalise automatically to a multiconductor factor. For a factor ℓ with an $m \times m$ primitive, voltage snapshots $V \in \mathbb{C}^{m\times r}$, and member-current snapshots I_ℓ, the observation equation is

$$I_\ell = Y_\ell V.$$

Full primitive recovery from this relation requires $\operatorname{rank}(V) = m$ and observation of every row of I_ℓ. In the square full-rank case, $Y_\ell = I_\ell V^{-1}$; more generally the declared experiment must span the retained conductor space. If $r < m$, any nonzero admissible Δ with $\Delta V = 0$ gives the same member currents. If only selected current rows are observed, a nonzero Δ can instead satisfy $P\Delta V = 0$ for the row-selection map P. Reciprocal symmetry does not remove these directions by itself.

The generated `experiments/generated/multiconductor-recovery-witness.json` makes all three cases explicit: two independent voltage snapshots with complete current vectors recover two reciprocal two-conductor factors; one complete snapshot leaves a nonzero reciprocal ambiguity; and full-rank snapshots with only one measured current phase still leave the unmeasured phase ambiguous. This is a linear identifiability witness, not a claim about nonlinear measurement design, noise, or estimator consistency. It is registered as `ARCH-RECOVERY-003`.

Noise changes recovery into a certified uncertainty set

With noisy current snapshots

$$I_\ell = Y_\ell V + E, \qquad \|E\|_{\mathrm{F}} \le \varepsilon,$$

the full-rank pseudoinverse estimate $\widehat{Y}_\ell = I_\ell V^\dagger$ is an estimator, not an exact source recovery. For square invertible V (and, with the corresponding qualification, for a full-row-rank snapshot matrix),

$$\|\widehat{Y}_\ell - Y_\ell\|_{\mathrm{F}} \le \varepsilon \, \|V^\dagger\|_2.$$

Thus experiment conditioning is part of the recovery certificate. The same noise radius can produce a tight uncertainty set for well-conditioned voltage snapshots and a much larger set for nearly dependent snapshots. Physical symmetry or passivity constraints may intersect that set, but they should be reported as additional guards rather than used to hide the measurement uncertainty.

The generated `experiments/generated/noisy-multiconductor-recovery-witness.json` compares well-conditioned and nearly collinear two-conductor voltage snapshots under the same Frobenius noise radius. Both estimates satisfy the deterministic bound; the ill-conditioned experiment amplifies the bound by more than two orders of magnitude. This is a finite error certificate, not a statistical consistency or estimator-optimality result. It is registered as `ARCH-RECOVERY-004`.

Rank, reference, and grounding

The nodal operator is not automatically invertible. Under the connectivity, common polyphase-coordinate, and passive-component hypotheses of Kettner and Paolone, a connected shunt-free compound network has the common-mode nullspace and rank

$$\operatorname{rank}(\mathbf{Y}^{\mathrm{N}}) = (|\mathcal{B}| - 1)m.$$

Under their corresponding grounded/shunted hypotheses, an effective nonzero shunt removes that gauge freedom and the compound nodal operator is full rank [19]. For several galvanic components, absent references can contribute separate nullspaces. Ideal devices, singular terminal maps, missing phases, and more general factors require their own rank analysis.

This is why *choose a voltage reference* and *model physical grounding* must remain distinct instructions. Deleting a coordinate to fix a numerical gauge does not add a grounding asset; conversely, a finite grounding impedance is a physical factor. The running-network numerical export illustrates the conditional case: its passive 20×20 operator has reported numerical rank 18 at the declared tolerance, not because every nodal operator must be singular but because reference and grounding structure remain in that model.

Is nodal admittance a simple-graph concept?

Not in the physical sense. A nodal admittance matrix is a linear operator on a chosen ordered voltage space. From it one can derive simple support graphs at several granularities. The graph-loop, shunt, incidence, adjacency, and Laplacian conventions used to distinguish those objects are fixed in Multigraphs for expert modelers.

Definition (block support). Given bus blocks $\mathbf{Y}^{\mathrm{N}}_{ij}$, define

$$G_Y^{\mathrm{blk}} = (\mathcal{B}, E_Y^{\mathrm{blk}}), \qquad \{i,j\} \in E_Y^{\mathrm{blk}} \iff \mathbf{Y}^{\mathrm{N}}_{ij} \neq \mathbf{0}.$$

Definition (scalar support). Given retained coordinates $\mathcal{C} = \{(i,p)\}$, define

$$G_Y^{\mathrm{sc}} = (\mathcal{C}, E_Y^{\mathrm{sc}}), \qquad \{(i,p),(j,q)\} \in E_Y^{\mathrm{sc}} \iff Y^{\mathrm{N}}_{(i,p),(j,q)} \neq 0.$$

These support graphs are simple by construction: a matrix position is either zero or nonzero. But that does not make the source network a simple graph. Several factor stamps sum into one position, and their decomposition is not encoded in matrix support. If an algorithm needs the decomposition, it can use a **stamp multigraph** whose identified members are the separate $\mathbf{A}_\phi^{\mathsf{T}} \mathbf{Y}_\phi \mathbf{A}_\phi$ contributions. That multigraph is extra data; it cannot generally be recovered from the sum. This separation is registered as `ARCH-SUPPORT-001`.

The support relation also requires qualifications:

- a dense off-diagonal block can be produced by mutual conductor coupling inside one physical line;
- several contributions can occupy one block, including parallel factors and multi-terminal compilations;
- exact cancellation can make a matrix entry zero even though source factors touch both coordinates;
- a diagonal block combines incident series terms, local shunts, grounding, and possibly several compiled factors;
- changing coordinates can change scalar support without changing the underlying external relation.

Thus G_Y^{blk} is often an excellent sparsity and decomposition view, while G_Y^{sc} exposes within-block coupling. Neither is an asset register.

Why a radial network can acquire cycles

Gan and Low show that a multiphase radial network can be represented as an equivalent scalar network that is radial at the macro level but has a clique associated with each line [22]. Their companion multiphase BIM/BFM work similarly treats each bus–phase pair as a coordinate of an equivalent scalar circuit [23]. The observation is valuable here because it makes the level distinction impossible to ignore.

For an m-conductor two-terminal factor with a dense $2m \times 2m$ terminal stamp, its scalar support can contain a clique on the $2m$ endpoint coordinates. A triangle or larger cycle inside that clique is an algebraic coupling cycle. It is not evidence that operators can open an alternative physical route, that power is circulating, or that the bus-level feeder is meshed. If some primitive entries are structurally zero, the clique loses the corresponding support edges; the matrix pattern, not the word *multiconductor*, decides the scalar support.

There is also a constructive counterpart. Let the simple bus-level graph be a tree, give every bus the same m retained coordinates, and suppose each tree edge has one structurally dense two-terminal stamp whose support is the clique on its two endpoint blocks. Assume the assembled nonzeros do not cancel.

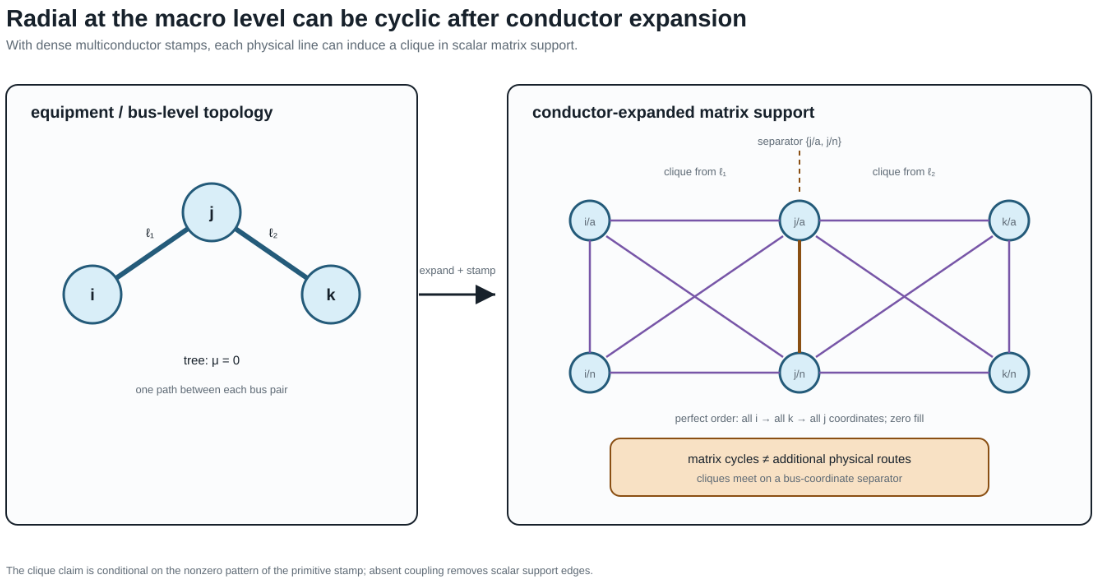

Figure 18.2: A bus-level tree and the cyclic scalar support induced by dense multiconductor line stamps.

Proposition (tree of dense stamps). The resulting scalar support graph is chordal. Eliminating all coordinates of a leaf-bus block and proceeding inward is a perfect elimination ordering and creates zero structural fill.

Proof. At a leaf bus i with parent j, every later neighbour of a coordinate (i,p) lies in $(\{i\}\times\mathcal{P})\cup(\{j\}\times\mathcal{P})$. The dense edge stamp makes that set a clique, so (i,p) is simplicial. Eliminating the entire leaf block removes one leaf from the bus tree and leaves the same construction on a smaller tree. Induction gives a perfect elimination ordering. A perfect elimination ordering adds no fill.

The line-stamp cliques meet on bus-coordinate separators. Their clique tree is inherited from the bus tree, but is not literally isomorphic to it in general: a tree with n buses has $n-1$ line cliques before any maximal-clique coalescence. The two-line figure, for example, has two cliques joined through the separator $\{j/a, j/n\}$.

This result, registered as `ARCH-CHORDAL-001`, explains why the cycles are benign for the declared sparse computation and why Gan and Low can exploit chordal structure in multiphase radial OPF. It is conditional: multi-terminal factors, missing coupling entries, cancellation, or a meshed bus graph can change the support and its elimination properties.

This produces several useful apparent paradoxes:

These are not contradictions. Each sentence changes the graph without saying so.

Three cycle questions, not one

The cycles and radiality chapter defines the corresponding graph objects in detail. The practical crosswalk is:

A cycle basis computed in one row is not automatically a basis for another. In particular, a parallel pair gives a two-member cycle in the identified multigraph but one edge in block support, while a dense line stamp can give many scalar-support cycles without any asset-level cycle.

Statement	Resolution
a radial feeder has cycles	equipment topology can be a tree while conductor-expanded matrix support contains cliques
two parallel lines become one edge	their stamps add in one block-support edge; asset identity has been projected away
one line becomes many edges	one dense multiconductor factor produces many scalar nonzeros
a transformer creates a triangle	a clique compilation of one multi-terminal factor creates a support cycle, not three transformer assets
a new edge appears after reduction	Kron fill-in is an equivalent boundary coefficient, not a discovered line
a physical relation has no matrix edge	cancellation, coordinate choice, or eliminated variables can hide the relation

Cycle question	Graph or incidence object	What it can support
Is there an alternative route through identified two-terminal members?	equipment/bus multigraph	switching, outages, member radiality, line-identity cycle bases
Is there repeated incidence through conductor junctions and factors?	conductor/port–factor graph or a declared compilation	terminal connectivity, factor decomposition, conductor-resolved equations
Does the assembled or reduced operator have cyclic sparsity?	block or scalar matrix-support graph	chordal decomposition, ordering, fill, sparse numerical algorithms

Executable projection and elimination witness

The generated `experiments/generated/topology-projection-witness.json` checks the two central mechanisms without relying on a power-flow solver.

- Two distinct reciprocal passive two-conductor splits assemble to a byte-identical 4×4 nodal operator. Both have zero normalized Frobenius assembly error, so the consistency certificate cannot identify the source attribution. The small negative minimum passivity eigenvalues (approximately -1.4×10^{-16} at worst) are recorded as floating- point roundoff against a 10^{-12} tolerance.
- A three-bus two-conductor bus-level tree has macro cycle rank zero and scalar structural-support cycle rank six. The declared leaf-block perfect order produces zero fill, while eliminating the separator block first produces four fill edges.

The source is `experiments/transformations/TopologyProjectionWitness.jl`; the focused test is `experiments/test/topology_projection_witness.jl`. These finite witnesses test `ARCH-NODAL-001`, `ARCH-SUPPORT-001`, and `ARCH-CHORDAL-001`; they do not claim that every factor library is passive, identifiable, or chordal.

Kron reduction adds a fourth source of apparent adjacency

Partition retained boundary coordinates B and eliminated internal coordinates I. When $\mathbf{Y}_{II}$ is invertible, Kron reduction gives

$$\widehat{\mathbf{Y}}_{BB} = \mathbf{Y}_{BB} - \mathbf{Y}_{BI}\mathbf{Y}_{II}^{-1}\mathbf{Y}_{IB}.$$

The Schur-complement term can introduce a nonzero retained block between two boundary nodes that shared no source factor. This fill edge belongs to the reduced operator support. It does not belong retrospectively to the source asset or conductor topology. Dörfler and Bullo characterize this graph effect for electrical-network Kron reduction [10], while the compound polyphase setting requires the relevant block-rank conditions [19].

Any eliminated current, voltage, or limit that still matters to a decision problem must be evaluated through a recovery map. This includes neutral- conductor current limits: the disappearance of the neutral coordinate from the retained operator does not remove the conductor's thermal constraint.

Maintain the decomposition; do not promise inversion

The canonical record should retain at least:

- stable asset and factor identities;
- ordered ports, junction attachments, and conductor/terminal maps;
- factor class and full primitive relation or a reproducible construction record;
- active-state, rating, grounding, control, and decision ownership;
- each assembly, compilation, coordinate, and reduction map;
- provenance from every matrix block or generated object back to its source factors;
- recovery maps for eliminated quantities that remain observable or constrained.

For a supplied nodal operator and claimed source decomposition, define each assembled stamp $\mathbf{S}_\phi = \mathbf{A}_\phi^\mathsf{T}\mathbf{Y}_\phi\mathbf{A}_\phi$ and the Frobenius-norm assembly backward error

$$\eta_{\mathrm{asm}} = \frac{\left\|\mathbf{Y}^{\mathrm{N}} - \sum_\phi \mathbf{S}_\phi\right\|_{\mathrm{F}}}{\|\mathbf{Y}^{\mathrm{N}}\|_{\mathrm{F}} + \sum_\phi \|\mathbf{S}_\phi\|_{\mathrm{F}}} \le \tau_{\mathrm{asm}}.$$

The denominator makes the test dimensionless and remains informative when large stamps nearly cancel; the all-zero case is handled separately. The certificate must also record the coordinate order, units, states, factor types, norm, and threshold.

This test verifies **assembly consistency**, not **source attribution**. It is invariant under every admissible regrouping in $\ker\mathcal{S}$ and is therefore structurally blind to the parallel-split ambiguity established above. Two decompositions can both achieve zero backward error while assigning different primitives, ratings, or identities to the members. Attribution requires provenance or independent identifying information.

Recovery from $\mathbf{Y}^{\mathrm{N}}$ alone is an inverse problem. It is non-identifiable whenever the restricted assembly criterion above fails. Additional catalog constraints, construction priors, measurements, switch states, or asset records can narrow the candidate set, but an estimator must report the remaining affine or constrained ambiguity rather than inventing line identity. The safe engineering objective is therefore:

> preserve the two-level source structure through compilation, and validate every derived nodal operator against it; attempt recovery only as a separately scoped inference problem.

That direction supports both rigorous proofs and practical data standardisation. Power engineers can work with familiar bus blocks and line triples, while the retained maps make clear which topology, constraints, and physical meanings survive each transformation.

Chapter 19

Cycles, parallelism, and radial structure

Page status: representation-scoped graph definitions with executable invariant witnesses, including conductor-terminal and state-conditioned lifts; broader multi-terminal compilation families remain open.

Why these words need a representation

Power-system discussions often say that a network has a cycle, a parallel line, or a radial end as if these were properties of the physical system alone. They are first properties of a declared mathematical representation. A simple topology graph, an identified line multigraph, a junction–factor incidence graph, and a compiled equation graph can give different answers while all being legitimate views of the same network.

This chapter applies the normative object and matrix conventions in Multigraphs for expert modelers to cycle, parallelism, bridge, leaf, and radiality questions. The electrical and decision meaning of a topological statement is a second question: constitutive equations, limits, states, and observations must still be attached.

The preceding two-level topology chapter explains why asset, conductor/port–factor, and nodal-operator support graphs can have different edges and cycles. Here those distinctions become explicit cycle, parallelism, bridge, leaf, and radiality definitions.

Simple cycles and line-identity cycles

Let $G_{\mathrm{s}} = (\mathcal{B}, E_{\mathrm{s}})$ be a loopless undirected simple graph. A simple cycle is a closed walk with at least three distinct vertices and no repeated edge or internal vertex. Its cycle-space dimension is

$$\mu_{\mathrm{s}} = |E_{\mathrm{s}}| - |\mathcal{B}| + c,$$

where c is the number of connected components. The equation counts a vector-space dimension; it does not select a unique cycle basis.

For the loopless specialization of the normative flag multigraph, write

$$G_{\mathrm{M}} = (\mathcal{B}, \mathcal{L}, \partial), \qquad \partial : \mathcal{L} \to \binom{\mathcal{B}}{2}.$$

Choose an arbitrary stored orientation for each line and let A be its oriented incidence matrix. We use the following representation-aware definition.

Definition. A **line-identity cycle** is a nonzero vector $z \in \ker A$ whose support is inclusion-minimal among nonzero vectors in $\ker A$. A cycle basis is any basis of $\ker A$. Reorienting a line negates one incidence column and changes the coordinates of z but not the underlying cycle or its dimension.

This is the circuit definition of the graphic matroid written in incidence coordinates. It includes the two-edge cycle made by parallel lines. If $\ell_1 ij$ and $\ell_2 ij$ have the same unordered endpoints, their two incidence columns are equal up to sign, so a vector supported on $\{\ell_1, \ell_2\}$ lies in $\ker A$. The cycle does not require three distinct buses.

For a loopless multigraph with c connected components,

$$\mu_{\mathrm{M}} = |\mathcal{L}| - |\mathcal{B}| + c.$$

The simple projection $\mathrm{simp} : \mathcal{L}^{\circ} \to E_{\mathrm{s}}$ forgets line identity and retains the unordered endpoint pair. In this loopless specialization $\mathcal{L}^{\circ} = \mathcal{L}$; the superscript makes the non-loop domain explicit. The cycle-rank difference is then

$$\mu_{\mathrm{M}} - \mu_{\mathrm{s}} = \sum_{e \in E_{\mathrm{s}}} \big(|\operatorname{simp}^{-1}(e)| - 1 \big).$$

Thus each additional identified member in a parallel fibre contributes one line-identity cycle dimension, even though the simple graph sees only one adjacency. The five-bus example makes this loss explicit: its source rank is three and its simple projection rank is two because the fibre over the endpoint pair $\{q, r\}$ contains two lines.

The cycle vector is a topological object. A nonzero cycle coordinate in a branch-current parameterization is not automatically a circulating current, and a topological cycle does not imply that power is flowing around it at a particular operating point.

Graph-theory trap

Do not use *cycle*, *cycle-basis coordinate*, and *loop flow* as synonyms. The first two are properties or coordinates of a declared graph; the last is an operating statement requiring electrical variables and equations.

The three panels are deliberately side by side: the triangle is a simple cycle, the parallel fibre contributes a cycle only when line identity is retained, and the tail is a maximal bridge path ending at a leaf.

Conductor-terminal and state-conditioned lifts

The running-network conductor-terminal witness carries the same distinction one level closer to the electrical factors. Each ordered conductor map becomes a member edge between terminal vertices such as $i1/a$ and $i2/a$; the line identity is retained, so parallel phase or neutral members remain visible. The artifact `experiments/generated/conductor-terminal-lift-witness.json` reports member and simple-projection cycle ranks for open and closed switch states. An unknown switch is represented by both admissible realizations rather than being silently treated as open, so radiality is a state-conditioned observation.

This is still a terminal-incidence witness, not a universal rule for compiling every multi-terminal factor. A three-winding transformer can be represented as a factor node, a star compilation, or a clique compilation; those choices are separate views whose cycle spaces must not be conflated with the physical terminal graph.

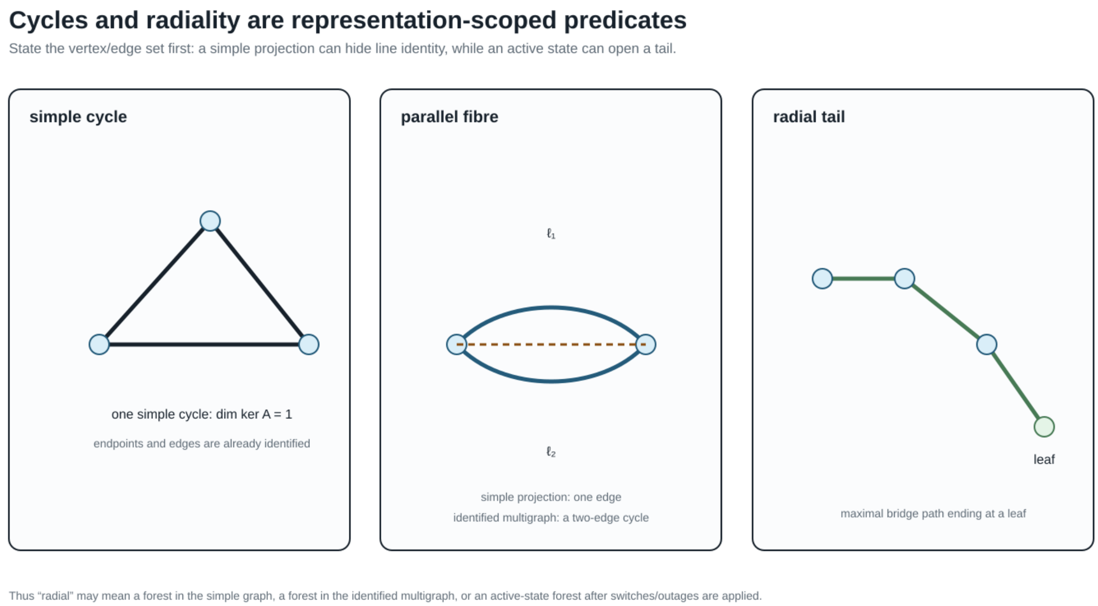

Figure 19.1: Simple cycles, line-identity cycles, and radial tails.

Parallelism has levels

The phrase **parallel lines** is underspecified. In particular, spatial co-location is not one rung in a chain from weak to strong electrical parallelism. Two circuits can share a corridor without sharing endpoints, and two graph-parallel lines can be electromagnetically independent. The book therefore distinguishes the following facets:

Facet	Condition	Where it can be decided
physical parallelism	line sections overlap spatially on common towers, poles, trench, or right of way	asset, geometry, and section-lineage model
topological parallelism	same unordered bus pair $\partial(\ell_1) = \partial(\ell_2)$	identified multigraph
terminal parallelism	endpoint terminal spaces and terminal maps align, up to declared coordinate actions	terminal-connectivity or port model
electrical parallelism	both factors see the same boundary voltage variables and their currents can be summed	port–factor or equation model
operational parallelism	both members are active in the declared switch, outage, investment, and state scenario	asset/state and decision model
homogeneous parallelism	construction, grounding, parameter, and constraint guards support a physical merge	asset plus electrical model

Here *physically parallel* follows line-modeling usage and means spatially co-located; it does not mean a transfer corridor. Mutual coupling adds another orthogonal relation. Physically parallel sections can belong to one joint electromagnetic factor even when their endpoint pairs, conductor sets, and nominal voltages differ. The coupled multi-voltage corridor case develops this situation and its optional ordinary-edge lowering.

Only topological parallelism is visible in a bare multigraph. In a simple graph it is not an internal relation at all:

it survives only as the fibre

$$\mathrm{simp}^{-1}(\{i,j\}) \subseteq \mathcal{L}^{\circ}$$

of the multigraph-to-simple-graph quotient. In a port–factor model, the two lines remain separate factors attached to the same junction variables. That representation can test whether a phase permutation, neutral connection, grounding scope, control, and limit really align.

Electrical parallelism permits a terminal relation such as

$$\mathbf{I}_{ij}^{\mathrm{total}} = \sum_{\ell \in \mathrm{simp}^{-1}(\{i,j\})} \mathbf{I}_{\ell ij},$$

but it does not permit replacing member constraints by a summed constraint. That replacement needs an explicit implication or lifting certificate. A parallel pair can be electrically aggregable for an unconstrained nodal admittance and still be operationally non-aggregable because its members have different ratings, outages, owners, or investment decisions.

Degree, leaves, bridges, and radial ends

For a bus i, define the distinct-neighbour degree in the simple projection and the incidence degree in the loopless identified multigraph by

$$d_{\mathrm{nbr}}(i) = |\{j : \{i,j\} \in E_{\mathrm{s}}\}|, \qquad d_{\mathrm{inc}}(i) = |s^{-1}(i)| = \sum_{e \in E_{\mathrm{s}} \colon i \in e} |\,\mathrm{simp}^{-1}(e)|.$$

These are different measurements. Because this chapter's identified multigraph is loopless, d_{inc} also equals its incident-member count; the equality would require adjustment if graph loops were admitted. A bus joined to one neighbour by two parallel lines has $d_{\mathrm{nbr}}(i) = 1$ but $d_{\mathrm{inc}}(i) = 2$. Calling it a leaf or calling it degree one without naming the convention is therefore ambiguous.

We use these precise terms:

- a **simple leaf bus** has $d_{\mathrm{nbr}}(i) = 1$ in the declared simple projection;
- a **multigraph leaf bus** has $d_{\mathrm{inc}}(i) = 1$ in the declared loopless identified multigraph;
- a **pendant line** is a bridge incident to a leaf in the declared graph;
- a **radial tail** is a maximal path of bridges ending at a leaf, with any internal buses of the path having degree two in that graph;
- a **pendant subnetwork** is a region attached through one articulation bus; it need not itself be radial.

An edge is a **bridge** if deleting that identified edge increases the number of connected components. A simple edge e represents a parallel fibre $\mathrm{simp}^{-1}(e)$ in the multigraph.

Proposition. For a line ℓ in a loopless identified multigraph, ℓ is a bridge if and only if $\mathrm{simp}(\ell)$ is a bridge in the simple projection and $|\,\mathrm{simp}^{-1}(\mathrm{simp}(\ell))| = 1$.

Proof. If another line shares the same endpoint pair, deleting ℓ leaves that pair connected, so ℓ cannot be a bridge. Conversely, suppose the fibre of $e = \mathrm{simp}(\ell)$ is a singleton but ℓ is not a bridge. Then there is a multigraph path between the two sides after deleting ℓ. Projecting that path and removing repeated vertices gives a walk, hence a path, in the simple graph with e deleted. This contradicts that e is a simple-graph bridge. Therefore a singleton fibre over a simple bridge is a multigraph bridge, and the two conditions are equivalent.

Corollary. The identified multigraph is a forest if and only if its simple projection is a forest and every nonempty edge fibre is a singleton.

This gives two useful but different radiality predicates:

$$\begin{aligned} \text{adjacency-radial} &:\Longleftrightarrow G_{\mathrm{s}} \text{ is a forest}, \\ \text{member-radial} &:\Longleftrightarrow G_{\mathrm{M}} \text{ is a forest}. \end{aligned}$$

Adjacency-radial does not imply member-radial. A feeder with two parallel circuits can have a tree-shaped simple projection and still contain a line-identity two-cycle. Conversely, a simple graph can be meshed even when a particular operating state opens enough members to make the active multigraph radial.

Power-system shorthand

Radial feeder is incomplete unless it names both the graph and the active state. In particular, a tree-shaped simple projection can hide parallel member cycles.

“Radial end” should therefore be replaced in technical writing by the graph, state, and object being tested: for example, “the m end is a pendant bridge in the active identified multigraph” or “the simple bus projection has a leaf at m.”

Upstream and downstream are state-conditioned hierarchy labels

For a resolved radial state σ, choose a source root r and orient the active tree away from r. The resulting rooted forest is a useful derived view:

$$\mathcal{R}(\sigma, r) = \big(G_M^{\sigma}, r, \mathrm{par}_{\sigma,r}, \mathrm{depth}_{\sigma,r}\big).$$

It supports parent/child, ancestor/descendant, feeder-head and downstream subtree queries. It is not a fifth canonical electrical graph: it is a state-specific algorithmic view over the identified graph.

The labels become non-unique or undefined when:

- a component has more than one source or no designated root;
- the active graph contains a cycle;
- a switch state is unknown;
- a tie closes or a parent branch opens;
- a topology decision changes the energized component.

For a meshed state, a selected spanning forest may still provide a convenient parent map, but the omitted chords remain cycle edges. Their directions are bookkeeping choices and their equations must remain in the model. Calling all mesh branches upstream or downstream silently replaces the source graph by an unstated tree projection.

Switching therefore requires recomputing the hierarchy for every resolved state. A transformation or recursion that uses parent/child structure must carry σ, the root choice, the tree or forest identity, and a recovery map to the stable ℓij reference orientation. It is valid for that declared state domain, not automatically for the inventory or for every future switching action.

> **Power-system shorthand**
>
> In a technical statement, write "downstream in the active identified tree rooted at r under σ," or use "parent edge," "source-distance," and "operating transfer direction." Unqualified upstream/downstream labels are not stable semantics on a meshed or reconfigurable network.

Multi-terminal factors change the question

A multiwinding transformer represented as one factor is not an ordinary graph edge. The arbitrary-arity flag/incidence definition and its typed relation spaces are given in Multigraphs for expert modelers. Its natural incidence structure can be displayed as a bipartite graph:

```
bus i  --  factor x  --  bus j
                    \--  bus k
```

This junction–factor incidence graph is a tree. A clique projection onto bus vertices creates the triangle ij, jk, ki; a star compilation through a virtual junction remains a tree. The apparent cycle can therefore be created by the representation rather than by an alternative physical transfer path.

The same warning applies to hyperedges, grounding factors, shared controls, and equation graphs. A cycle in a factor-incidence graph means repeated incidence through factors and junctions. A cycle in a nodal-admittance sparsity graph means algebraic coupling. Neither should be silently called a physical power-transfer loop.

Consequences for decisions and reductions

The graph predicates above do not authorize a transformation by themselves:

- a bridge may carry a load, generator, measurement, grounding relation, protection boundary, or investment decision;
- a leaf may be the boundary of a nontrivial factor or a retained observation;
- a radial tail may be reducible for one boundary relation and indispensable for another;
- a parallel pair may provide outage redundancy even when its simple projection contains only one bridge;
- opening a chord is a topology decision, not a coordinate change;
- summing parallel admittances can preserve Y-bus behavior while changing member-level feasible sets.

For a declared graph transformation, the preservation contract should record at least:

1. the graph whose cycles, degrees, bridges, or forests are being tested;

2. whether members are identified, aggregated, or merely projected;
3. the active state and admissible future states;
4. the boundary quantities and source constraints that must be recovered;
5. the provenance fibre for every quotient edge or compiled factor.

The scope rule is simple:

> A cycle, parallel relation, bridge, leaf, or radiality claim must name its representation. None of these graph properties, alone, authorizes an electrical reduction.

Executable active-state certificate

The package-independent witness in `experiments/generated/active-radiality-witness.json` compares a four-member inventory with an active state in which one of two parallel members is open. The inventory's simple projection is radial, but its identified multigraph is not: the parallel fibre contributes a line-identity cycle. After the outage, the active identified multigraph is a tree, so both active radiality predicates agree. This is the small counterexample needed before using radiality as a guard for a conductor-coordinate or phase-to-phase reduction.

The running-network witness extends this counterexample to declared switch and line-outage states. Each state row records the active line, switch, and transformer-winding inventories before reporting cycle ranks. Opening the switch therefore changes the active inventory without silently deleting the switch asset, while an $l2$ outage removes exactly that line member. The transformer remains represented by explicit winding members with factor provenance. Radiality is consequently a predicate of the state-conditioned member inventory, not a permanent property of the asset inventory.

The five-bus companion `experiments/generated/five-bus-active-radiality-witness.json` makes the same distinction on the chapter's line-identity example. Its inventory has member cycle rank 3 and simple-projection cycle rank 2; the declared spanning tree $\{r, s, w, u\}$ is radial at both levels. This is claim `TR-GRAPH-ACTIVE-001`: the active state must be named before calling the network radial.

Run it with:

```
julia --project=experiments experiments/run_active_radiality.jl
julia --project=experiments experiments/run_five_bus_active_radiality.jl
```

Chapter 20

Orientation, terminal quantities, and power transfer

Page status: foundational definitions and terminology.

Why an arrow is ambiguous

Power-system diagrams, data formats, and optimization models use arrows for several different purposes. A line drawn from i to j can mean an arbitrary stored orientation, a terminal-current sign convention, a positive reference direction for active power, an observed operating-point transfer, or a genuinely one-way admissible action. These meanings must not be inferred from one another.

Direction concept	Mathematical data	Dependence
physical incidence	unordered endpoints $\partial\ell = \{i, j\}$	equipment model
reference orientation	one selected arc $o(\ell) = \ell ij$	arbitrary coordinate choice
terminal-current sign	current defined into or out of each terminal	modelling convention
operating-point transfer	sign of $P_{\ell ij}$ at a solution	voltage, state, and controls
causal or permitted direction	asymmetric control or feasible relation	physical or study semantics

A conventional passive AC line is therefore an undirected physical connection with an orientation, not intrinsically a directed physical edge.

Power-system shorthand

An arrow on a passive branch normally fixes storage order, terminal names, or a positive reference. It does not predict the sign of current or active power at a solution.

The terminal-arc double cover

For every two-terminal element ℓ with $\partial\ell = \{i, j\}$, introduce the two terminal arcs

$$\overrightarrow{\mathcal{L}} = \{\ell ij, \ell ji : \ell \in \mathcal{L}\}.$$

The reversal map

$$\rho : \overrightarrow{\mathcal{L}} \to \overrightarrow{\mathcal{L}}, \qquad \rho(\ell ij) = \ell ji,$$

is an involution with no fixed points. A reference orientation chooses one arc from each pair; the bidirected terminal-arc view retains both. The line remains one member of $\mathcal{L}$.

With the book's incidence convention, the selected arc ℓij gives -1 at i and $+1$ at j. Choosing ℓji instead negates the column. Incidence rank, the undirected cycle space, and physical solutions are invariant under that coordinate change, while signed cycle coordinates change accordingly.

A stored orientation is not an operating direction

The stored triple ℓij determines which endpoint is written first. It does not imply any of the statements

$$P_{\ell ij} \geq 0, \qquad |P_{\ell ij}| \geq |P_{\ell ji}|, \qquad \text{or} \qquad \ell \text{ can transfer only from } i \text{ to } j.$$

Those are operating or device claims requiring separate equations. Active power can reverse between operating points, and in an unbalanced multiconductor model its sign can differ by conductor.

Reversing the stored orientation swaps end-specific records:

$$(\mathbf{N}_{\ell i}, \mathbf{N}_{\ell j}, \mathbf{Y}^{\mathrm{sh}}_{\ell ij}, \mathbf{Y}^{\mathrm{sh}}_{\ell ji}, \mathbf{I}_{\ell ij}, \mathbf{I}_{\ell ji}) \longleftrightarrow (\mathbf{N}_{\ell j}, \mathbf{N}_{\ell i}, \mathbf{Y}^{\mathrm{sh}}_{\ell ji}, \mathbf{Y}^{\mathrm{sh}}_{\ell ij}, \mathbf{I}_{\ell ji}, \mathbf{I}_{\ell ij}).$$

It does not blindly negate both terminal currents. Antisymmetry belongs to a particular internal series-current coordinate, not to every terminal quantity.

Rooted-tree orientation is a derived view

When an active network is radial, practitioners often orient every branch from the feeder source toward the leaves and call the resulting arcs *upstream* and *downstream*. This is useful, but it is not the stored orientation ℓij and it is not an intrinsic direction of a passive line.

For an active identified graph G^{σ}_{M} and a selected source root r in each component, a rooted-tree view adds a parent map

$$\mathrm{par}_{\sigma,r} : V \setminus \{r\} \longrightarrow V$$

defined by the unique root-to-node path. The resulting parent-to-child arcs, depths, ancestors and descendants are a **state- and root-dependent algorithmic view**. They are appropriate for feeder recursions, backward/forward sweeps and radial branch-flow notation, but they are not new asset attributes.

> **Graph-theory trap**
>
> A parent-to-child arc is not an operating power-flow direction. Closing a tie can create a chord, reverse power can change the sign of $P_{\ell ij}$, and a different source or spanning tree can change the parent map without changing any line asset.

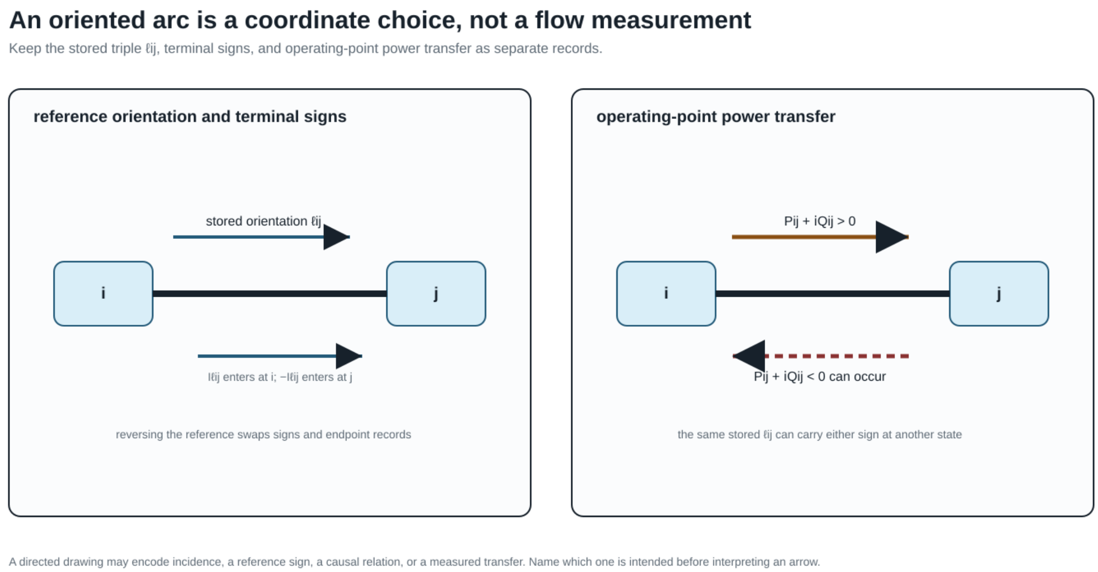

Figure 20.1: Stored orientation and operating-point power transfer are separate records.

If G_M^σ is meshed, choose a spanning forest only if an algorithm needs one. Tree edges then receive parent-child roles, while chords retain their cycle equations and must not be called upstream or downstream without a separate convention. A spanning-tree orientation is therefore a coordinate or algorithmic choice, not a claim that the physical network is radial.

The figure makes the sign discipline explicit: reversing ℓ_{ij} changes the coordinate convention, while changing the operating point can reverse $P_{\ell ij}$ without changing the stored asset orientation.

Series current and terminal power

For a reciprocal scalar series element, use current into the element at each terminal:

$$I^{\mathrm{s}}_{\ell ij} = Y_\ell(U_i - U_j), \qquad I^{\mathrm{s}}_{\ell ji} = -I^{\mathrm{s}}_{\ell ij}.$$

The corresponding terminal complex-power injections are

$$S_{\ell ij} = U_i(I^{\mathrm{s}}_{\ell ij})^*, \qquad S_{\ell ji} = U_j(I^{\mathrm{s}}_{\ell ji})^*.$$

Although series current is antisymmetric, terminal power is not conserved:

$$S_{\ell ij} + S_{\ell ji} = (U_i - U_j)(I^{\mathrm{s}}_{\ell ij})^* = Z_\ell |I^{\mathrm{s}}_{\ell ij}|^2.$$

For $Z_\ell = R_\ell + \mathrm{j}X_\ell$ with $R_\ell \geq 0$, the active-power absorption is

$$P_{\ell ij} + P_{\ell ji} = R_\ell |I^{\mathrm{s}}_{\ell ij}|^2 \geq 0.$$

A lossy branch therefore owns two terminal powers and a loss relation, not one conserved scalar flow. Calling $S_{\ell ij}$ the *power flow from* i is a useful shorthand only after its terminal sign convention is fixed.

The familiar single-flow picture is recovered in a declared lossless approximation, where $P_{\ell ij} = -P_{\ell ji}$. It should be presented as a special collapse rather than the semantics of a general edge.

Circuit-theory trap

Opposite series currents do not imply opposite terminal powers. Voltage differs across the impedance, so the two terminal powers sum to the element's complex absorption. A nominal-π factor can additionally have terminal currents that are not negatives because it contains shunts.

Nominal-pi elements

For a scalar nominal-π element, define

$$\begin{aligned} I_{\ell ij} &= I^{\mathrm{s}}_{\ell ij} + Y^{\mathrm{sh}}_{\ell ij} U_i, \\ I_{\ell ji} &= -I^{\mathrm{s}}_{\ell ij} + Y^{\mathrm{sh}}_{\ell ji} U_j. \end{aligned}$$

Then

$$I_{\ell ij} + I_{\ell ji} = Y^{\mathrm{sh}}_{\ell ij} U_i + Y^{\mathrm{sh}}_{\ell ji} U_j,$$

which is generally nonzero. Current has not been destroyed. It has been diverted through shunt paths retained inside the composite line factor. A conductive shunt absorbs active power, while inductive or capacitive shunts exchange reactive power.

The terminal-power balance is

$$\begin{aligned} S_{\ell ij} + S_{\ell ji} = Z_\ell |I^{\mathrm{s}}_{\ell ij}|^2 \\ + |U_i|^2 (Y^{\mathrm{sh}}_{\ell ij})^* + |U_j|^2 (Y^{\mathrm{sh}}_{\ell ji})^*. \end{aligned}$$

For a multiconductor member the same distinction is expressed by the complete two-end primitive

$$\begin{bmatrix} \mathbf{I}_{\ell ij} \\ \mathbf{I}_{\ell ji} \end{bmatrix} = \begin{bmatrix} \mathbf{Y}_\ell + \mathbf{Y}^{\mathrm{sh}}_{\ell ij} & -\mathbf{Y}_\ell \\ -\mathbf{Y}_\ell & \mathbf{Y}_\ell + \mathbf{Y}^{\mathrm{sh}}_{\ell ji} \end{bmatrix} \begin{bmatrix} \mathbf{U}_i[\mathbf{N}_{\ell i}] \\ \mathbf{U}_j[\mathbf{N}_{\ell j}] \end{bmatrix}.$$

Full coupling makes a story about independent power commodities travelling on individual conductor edges still less reliable. The invariant object is the declared multiport relation and its terminal power balance.

Internal versus explicit shunts

The same fixed nominal-π behaviour can be factorized in two ways:

1. one composite two-port line factor containing series and shunt terms;
2. one series factor plus two explicit shunt factors attached to the endpoint junctions.

In the second factorization the series currents are negatives, and the shunt factors separately account for current to ground or neutral. In the first, the composite line's terminal currents are not negatives. Moving between the two is an exact compilation only when terminal coordinates, grounding scope, parameters, limits, ownership, and provenance are mapped explicitly.

This is why statements such as *current is conserved on every edge* are representation dependent. KCL is conserved at the complete network level; which internal path is called an edge depends on the factorization.

When a directed graph is physical

A directed edge is appropriate when order is intrinsic to the represented relation, for example:

- a one-way communication or control dependency;
- a protection logic dependency;
- a device with a genuinely asymmetric admissible transfer set;
- a study graph of causal or optimization dependencies.

Even then, a multiport factor may be more faithful than a directed ordinary edge. A controllable converter can have oriented information flow, terminal power variables at several ports, losses, and bidirectional feasible operating regions at the same time.

Direction contract used here

An arrow in this book is interpreted through the five direction concepts above. Unqualified phrases such as *directed line*, *power on the edge*, and *reverse current* are therefore replaced by the physical incidence, stored orientation, terminal quantity, or operating-point statement actually meant.

Chapter 21

Circuit formulations and the lowering boundary

Page status: formulation guide and equivalence contract; the definitions and guards below are book conventions supported by circuit and power-system precedents. General equivalence theorems beyond the declared linear scopes remain future work.

The same power network can be compiled into several equation systems. A nodal admittance matrix is one particularly useful target, but it is not a universal representation of general *power networks*. A numerical $\mathbf{Y}$ may be available while asset identity, switch states, branch currents, grounding, multi-terminal behaviour, controls, limits, or decisions have already been discarded. Conversely, a faithful source model may require a tableau, modified nodal system, branch-current variables, or an unevaluated port–factor relation.

This chapter supplies the formulation boundary for From source graphs to views and graph surgery. The source object and its view registry remain authoritative there; this chapter owns equation targets, assembly identities, solvability guards, and formulation equivalence. Two topology levels and the nodal projection owns topology, support, projection, and nodal-operator non-identifiability. The representation maps chapter owns the cross-framework arrow vocabulary.

Formulation families

Let $\mathcal{M}$ be a declared power-network model with voltage, current, power, control, state, and parameter variables. A formulation is a choice of unknowns x, equations $F(x; u, \theta) = 0$, inequalities $g(x; u, \theta) \leq 0$, and an observation map h. The formulation is therefore more than a matrix: the variable ownership, limits, state domain, and recovery map are part of its contract.

The circuit literature distinguishes nodal analysis from modified and tableau extensions precisely because some elements do not admit a convenient admittance-only branch relation [13, 24]. Power-system work makes the same boundary operational: sparse tableau formulations retain multi-port element variables and node–breaker actions that would otherwise require changing Y_{bus} between states [25].

These formulations may be mathematically equivalent after elimination on a restricted domain. They are not interchangeable source models: eliminating a current variable can remove the direct place to attach a thermal limit, a switch state, or a protection relation.

Numerical footprint of the formulation choice

The choice of equation target also changes the symbolic problem seen by a solver. Schur elimination can add fill edges, while a Jacobian dependency graph records equation–variable relations rather than physical incidence. These are formulation-level effects, so they belong here beside the lowering boundary rather than only in the later executed numerical witnesses.

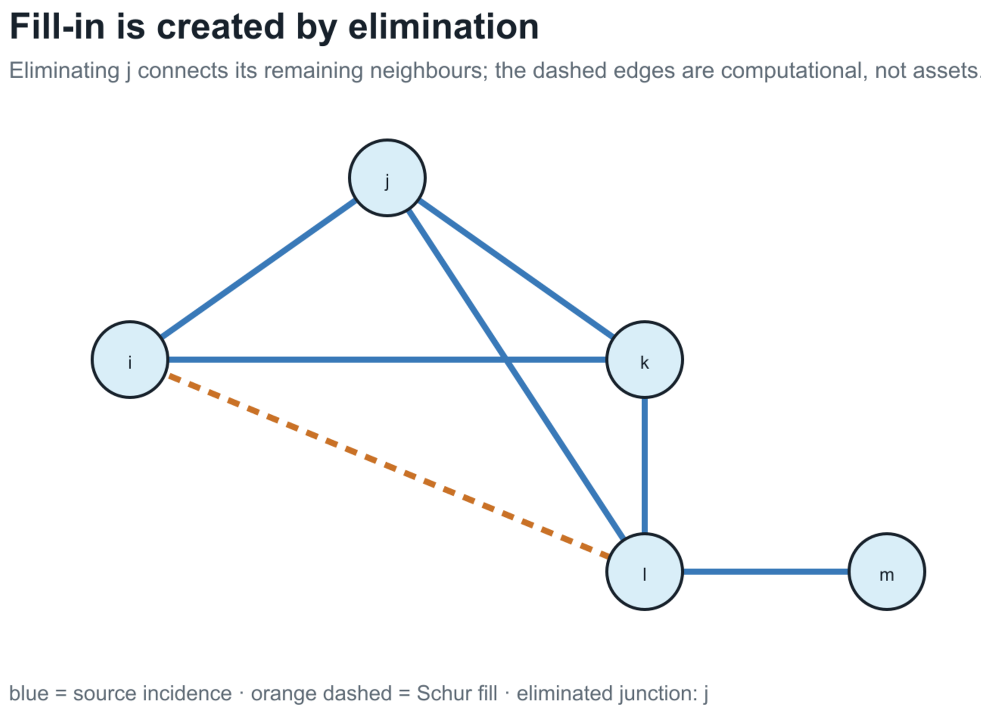

Figure 21.1: Schur elimination creates structural fill-in.

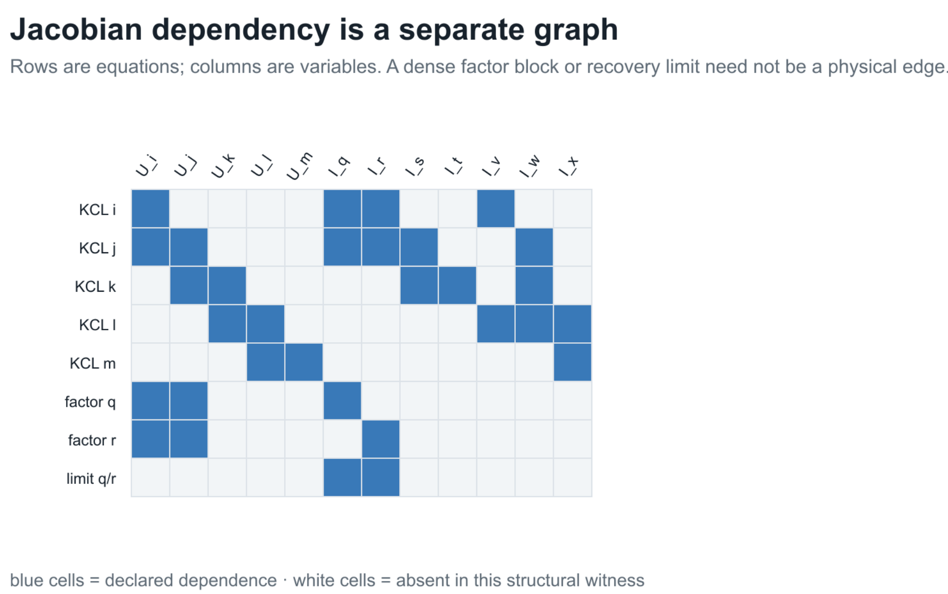

Figure 21.2: Jacobian dependency is a separate graph from physical incidence.

Formulation	Primary unknowns	Strong use	Typical loss or guard
nodal admittance	retained node voltages	fixed linear network stamping and PF linear subproblems	requires admittance-form branch relations; hides branch variables and asset identity
modified nodal analysis	node voltages plus selected branch currents	ideal voltage sources, current-controlled elements, switches, and mixed devices	larger saddle-point system; source provenance still has to be attached
sparse tableau	node, branch, port, and device variables	OPF limits, node–breaker models, multiport factors, and fixed sparsity	more variables and equations; elimination is a separate transformation
branch-current or branch-flow	voltages, currents, and often powers on identified members	explicit ratings, radial identities, and device-level constraints	equations depend on chosen branch model and orientation conventions
hybrid/port parameters	selected boundary efforts and flows	multiport composition and boundary equivalents	a representation is local to declared ports and may be singular at some frequencies or states
general port–factor relation	ordered typed ports and behavioural relations	nonlinear, controlled, dynamic, and multi-terminal equipment	not itself a single sparse matrix or ordinary graph

The figures are conceptual structural views. The pinned numerical export and nonlinear KKT witnesses remain in Numerical consequences of representation and reduction, where their fixture-specific counts, residuals, and solver limitations are recorded.

When an exact nodal admittance target exists

For a declared set Φ_{lin} of fixed linear factors, choose a retained voltage vector $\mathbf{U}$. If each factor has an admittance-form terminal relation

$$\mathbf{i}_\phi = \mathbf{Y}_\phi \mathbf{A}_\phi \mathbf{U}$$

and its current injection maps back through $\mathbf{A}_\phi^\intercal$, then assembly gives

$$\mathbf{Y}^{\text{N}} = \sum_{\phi \in \Phi_{\text{lin}}} \mathbf{A}_\phi^\intercal \mathbf{Y}_\phi \mathbf{A}_\phi, \qquad \mathbf{i}^{\text{inj}} = \mathbf{Y}^{\text{N}} \mathbf{U}.$$

Coupled branches and the equivalent lattice

Mutually coupled two-terminal sections are an instructive exact lowering. The source relation is not a sum of independent branch factors: the complete coupling group first supplies a joint impedance $\mathbf{Z}_\Gamma$. When it is invertible, $\mathbf{Y}_\Gamma = \mathbf{Z}_\Gamma^{-1}$ and one incidence map for all participating section drops gives

$$\mathbf{Y}_\Gamma^{\text{N}} = \mathbf{A}_\Gamma^\intercal \mathbf{Y}_\Gamma \mathbf{A}_\Gamma.$$

For two scalar sections on four distinct terminals, this stamp admits an exact six-edge lattice realization. Four of those edges cross between the two source endpoint pairs and two can carry the negative of the mutual admittance under the stored orientation. The lattice is therefore an ordinary weighted graph, but not a physical bus–branch inventory: its cross-voltage edges are generated couplings and its source line currents require the recovery map $\mathbf{i}_\Gamma = \mathbf{Y}_\Gamma \mathbf{A}_\Gamma \mathbf{U}$.

The coupled multi-voltage corridor case derives the full sign pattern, per-unit scaling, partial-overlap sections, and state guards. If the joint primitive is singular or the target edge library cannot carry the generated weights, the compiler must retain a direct factor or tableau rather than forcing this lattice.

Here Φ_{lin} is a formulation subset, not the entire source inventory. It contains factors whose parameters and states are fixed for the declared solve. A factor carrying an unfixed tap, switching state, control law, or other decision remains in the equation/constraint operator unless the target is explicitly rebuilt pointwise over that decision domain.

This is an assembly identity, not a claim that $\mathbf{Y}^{\mathrm{N}}$ is a canonical factorization or a complete power-network model. A useful sufficient guard for exact nodal stamping is:

The important degenerate case is covered in Multigraphs for expert modelers: identifying the two terminals of a fixed linear π factor can leave an exact one-terminal shunt stamp, but only after the factor has been assembled through its terminal map. Deleting a graph self-loop before that compilation is not an equivalent operation in general.

1. the retained variables are node-voltage coordinates sufficient for the declared observation and decision queries;
2. every included factor has a well-defined linear relation in those voltage coordinates, or has already been exactly eliminated with a declared Schur complement and recovery map;
3. voltage-source, current-controlled, dynamic, and algebraic-constraint variables that cannot be eliminated are retained elsewhere in the target;
4. the topology and device state used for assembly are fixed, or the target is rebuilt for each declared state;
5. grounding, reference, terminal maps, and coordinate order are declared; and
6. every limit, control, and decision that the study queries has a surviving variable or an explicit recovery/constraint map.

Source factors versus study-specific nodal splits

The source model and the matrix used by an iterative solver are related but are not the same object. Let μ denote a study mode, σ an active equipment state, and k an operating point or iteration. A current-injection solver may use the split

$$\mathbf{Y}_{\mathrm{lin}}^{(\mu,\sigma,k)}\mathbf{V}^{k+1} = \mathbf{I}_{\mathrm{src}}^{(\mu,\sigma)} + \mathbf{I}_{\mathrm{comp}}^{(\mu,\sigma)}(\mathbf{V}^{k}).$$

The linear operator may contain passive delivery elements, fixed shunts, constant-admittance load or generator parts, or a declared Norton equivalent. The compensation term carries the remaining nonlinear or operating-point dependent current. A constant-impedance load can therefore appear in the diagonal of $\mathbf{Y}_{\mathrm{lin}}$ while a constant-power load at the same bus appears through $\mathbf{I}_{\mathrm{comp}}$. A generator may be a PV/PQ constraint, a Norton source, a dynamic equivalent, or a current-limited injection depending on the study.

This is not automatically a Newton–Raphson step. Newton's method forms a residual Jacobian for an increment Δx,

$$\mathbf{J}(\mathbf{x}^k)\Delta\mathbf{x} = -\mathbf{F}(\mathbf{x}^k),$$

whereas a fixed-point current-injection method can keep $\mathbf{Y}_{\text{lin}}$ fixed and update only the compensation current. Setting the compensation current to zero for a direct or initial solve is a linear starting model, not proof that the nonlinear source model is itself constant impedance.

OpenDSS documents this separation operationally: power-conversion elements use a constant primitive-admittance part plus compensation currents, and its direct or admittance solution can include load and generator equivalents in the system matrix [14, 17]. Its fault-study equations construct a mode-specific nodal model with source and generator equivalents and the portion of load current not already included in the matrix [18]. These are legitimate formulation targets, not competing definitions of the physical network graph.

The practical rule is to qualify every nodal matrix with its source class, study mode, active state, coordinate order, and linearization or reduction point. Prefer names such as $\mathbf{Y}_{\text{passive}}$, $\mathbf{Y}_{\text{direct}}$, $\mathbf{Y}_{\text{fault}}$, or $\mathbf{Y}_{\text{lin}}^{(k)}$ to an unqualified claim that the system has one uniquely meaningful Y_{bus}.

Rank and sign contract

Let $\mathbf{Y}^{\text{N}}(\sigma, \gamma)$ denote the assembled operator after the declared active state σ and grounding/reference map γ have been applied, and let n_V be the retained voltage dimension. The nodal target passes its basic regularity guard only when

$$\operatorname{rank}\big(\mathbf{Y}^{\text{N}}(\sigma, \gamma)\big) = n_V.$$

A reference label, a nominal ground, or a removed datum does not establish this rank by itself. An isolated component, a disconnected grounding element, a singular shunt, or a state-dependent topology can leave a nontrivial nullspace. The rank must be checked on the assembled compound operator for the declared state and coordinates. This is the scoped diagnostic supported by [19], not a universal assertion that every grounded power-system model is nonsingular. The executable witness records both a disconnected network with a declared reference and a regular grounded case.

Failure of these guards does not make the network invalid. It means that a bare $\mathbf{Y}$ target is unavailable, incomplete, or semantically lossy for the declared study. A reduced $\mathbf{Y}$ can still be useful for a narrower boundary-voltage query, provided that the omitted variables and limits are recorded.

A nodal matrix is not a complete power-network graph

The support of $\mathbf{Y}^{\text{N}}$ records nonzero coupling among the retained voltage coordinates. It does not, by itself, record:

- which parallel assets contributed to one block;
- whether a coupling came from a line, transformer, grounding factor, or eliminated internal node;
- which switch or breaker state produced the assembled topology;
- branch currents and member-specific ratings;
- multi-terminal factor identity or internal winding variables; or
- the feasible-set and objective semantics of an OPF decision problem.

This is why Two topology levels and the nodal projection treats nodal support as a derived computational view. A support graph can be useful and exact for one query while being insufficient as the source of another.

Modified nodal and sparse tableau targets

Modified nodal analysis augments node voltages with currents through elements whose constitutive relation is not naturally an admittance injection. A common linear schematic form is

$$\begin{bmatrix} \mathbf{Y} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix} \begin{bmatrix} \mathbf{U} \\ \mathbf{I}_{\mathrm{e}} \end{bmatrix} = \begin{bmatrix} \mathbf{J} \\ \mathbf{e} \end{bmatrix}.$$

The blocks encode KCL, selected element currents, voltage-source or control relations, and any declared algebraic coupling. The matrix can be sparse even when eliminating $\mathbf{I}_{\mathrm{e}}$ would create dense fill-in.

The sign convention is part of the formulation contract. With $\mathbf{B}$ the signed incidence of selected element currents into the KCL rows, use

$$\mathbf{Y}\mathbf{U} + \mathbf{B}\mathbf{I}_{\mathrm{e}} = \mathbf{J}, \qquad \mathbf{C}\mathbf{U} + \mathbf{D}\mathbf{I}_{\mathrm{e}} = \mathbf{e}.$$

Thus a positive component of $\mathbf{I}_{\mathrm{e}}$ follows the stored element/terminal orientation encoded by $\mathbf{B}$; $\mathbf{J}$ is the positive KCL injection on the right-hand side, and $\mathbf{e}$ is the right-hand side of the selected voltage or device relation. Reversing an element orientation changes the corresponding incidence signs and current coordinate, not the physical factor. The witness uses this convention for its ideal voltage source.

A sparse tableau keeps this idea at the factor boundary. Let $\mathbf{z}$ collect node, terminal, branch-current, power, and device variables. A tableau target records

$$\mathbf{F}_{\mathrm{KCL}}(\mathbf{z}) = 0, \qquad \mathbf{F}_{\mathrm{KVL}}(\mathbf{z}) = 0, \qquad \mathbf{F}_{\mathrm{device}}(\mathbf{z};\theta) = 0, \qquad \mathbf{g}(\mathbf{z};\theta) \le 0.$$

This is especially useful when the study needs branch currents, multi-port device variables, switch constraints, or member-level limits. The sparse tableau node–breaker precedent keeps breaker actions in component constraints instead of silently replacing each action with a new fixed Y_{bus} [25].

The tableau is not automatically “more physical” or “more canonical.” It is a better target only for the declared questions. If the study asks only for a fixed linear boundary voltage relation, eliminating tableau variables may be appropriate; if it asks for switching, protection, or member ratings, the uneliminated variables are part of the preservation contract.

Branch-current, branch-flow, and chain formulations

A branch-current formulation retains an oriented current for each identified member and writes KCL together with the member constitutive equations. A branch-flow or BFM formulation instead promotes selected complex powers, voltage magnitudes, and current/power products to variables; its balance equations and relaxations depend on the declared branch model and on whether parallel member identities are retained. An aggregate branch-flow equation is therefore not automatically equivalent to a member-current formulation.

For a two-port partition, a chain or ABCD representation may write a transfer relation such as

$$\begin{bmatrix} u_i \\ i_{ij} \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} u_j \\ i_{ji} \end{bmatrix}.$$

This can be useful for cascading declared two-port sections, but it is a coordinate choice with its own invertibility and termination guards. It is not a universal substitute for a multi-conductor or multi-terminal factor. Hybrid effort/flow variables and scattering variables can avoid a singular chosen partition in some applications; they change the target coordinates and must carry their own recovery and observation contracts.

Structural solvability of ideal-source blocks

Ideal voltage-source loops and ideal current-source cutsets deserve a small diagnostic before numerical solution. After declaring the branch ordering and sign convention, write the affected KVL or KCL rows as an affine block $Aq = b$. The scoped check used here compares $\operatorname{rank}(A)$ with $\operatorname{rank}([A\ b])$:

rank result	interpretation
$\operatorname{rank}([A\ b]) > \operatorname{rank}(A)$	contradictory source constraints; no solution for that state
equal ranks, but $\operatorname{rank}(A) <$ number of rows	consistent redundant constraints; retain them or remove them with provenance
equal full row rank	consistent independent constraints

The distinction matters for MNA/tableau assembly. A redundant ideal-source row is not automatically an error, while a contradictory loop or cutset makes the declared state infeasible. The generated witness $experiments/generated/circuit-formulation-witness.json$ records both cases for a voltage-source loop and a current-source cutset. This is a rank- consistency diagnostic only; it is not a general statement about DAE index, dynamic regularity, or every possible dependent-source formulation.

Lowering from a source model

The formulation-aware compilation boundary is:

$$\mathcal{M} \xrightarrow{C} \mathcal{M}_{\mathrm{port}} \xrightarrow{A} \mathcal{E} = (\mathbf{F}, \mathbf{g}, \mathrm{obs}, \mathrm{prov}),$$

where $\mathcal{E}$ is a declared equation/constraint operator. Optional targets are then guarded lowerings:

$$\mathcal{E} \xrightarrow{L_{\mathrm{MNA}}} \mathcal{E}_{\mathrm{MNA}}, \qquad \mathcal{E} \xrightarrow{L_{\mathrm{Y}}} (\mathbf{Y}^{\mathrm{N}}, \mathrm{recover}) \quad \text{when the nodal guards hold.}$$

Direct factor stamping into $\mathcal{E}$ is the default. L_Y is not an automatic final pass after every factor compiler. It may fail because a factor needs extra unknowns, because an elimination block is singular, because a state-dependent switch changes the target structure, or because the resulting matrix would omit a queried constraint. The compiler must return a diagnostic and retain the source-to-target map rather than inventing a virtual admittance.

Every lowering record should state:

1. the source ports, factors, states, and coordinate order;
2. the target variables and equations;
3. the eliminated variables and the condition for elimination;
4. the preserved observations, constraints, and decisions;
5. the omitted semantics and unresolved guards; and
6. the recovery and provenance maps.

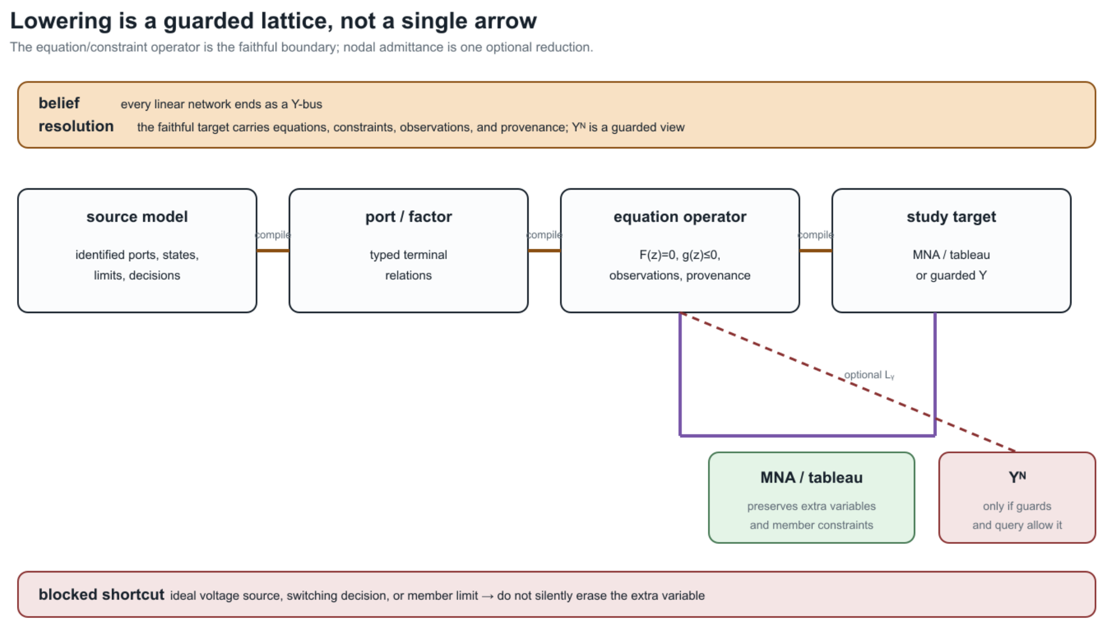

Figure 21.3: The formulation-lowering lattice: the equation/constraint operator is the faithful boundary, while MNA/tableau and nodal admittance are guarded study targets.

The diagram is intentionally asymmetric. The source model first lowers to an equation/constraint operator that still has somewhere to attach observations, limits, and decisions. MNA or a sparse tableau can retain those variables directly. A nodal $\mathbf{Y}$ target is a separate diagonal branch: it is available only when the declared formulation guards and query contract permit the extra variables and constraints to be eliminated. The crossed shortcut is the common but unsafe mental model in which every linear factor is silently turned into an admittance edge.

This is the formulation-specific instance of the general compiled-view contract. It also explains why a compiler pipeline can legitimately stop at a tableau or factor operator even when a smaller numerical matrix could be constructed for a narrower query.

Formulation equivalence is query-relative

Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be two equation/constraint operators with solution sets $\mathcal{S}_1$ and $\mathcal{S}_2$. A lowering map $T : \mathcal{S}_1 \to \mathcal{S}_2$ and a recovery map R are not sufficient to call the formulations equivalent without saying what is observed. For an observation family H (voltages, member currents, powers, states, or decisions), we call the pair *equivalent for H* when the declared domains are mapped consistently and

$$H_1(x) = H_2(Tx), \qquad \text{or, after recovery,} \qquad H_1(x) = H_2(RTx).$$

For a decision problem, this contract must also map feasible sets, objective values, and state domains. An algebraically eliminable variable may therefore be harmless for a boundary-voltage query and load-bearing for a member-current limit or switching decision. Approximate maps use the same vocabulary, but must state the error measure, domain, and which observations are only bounded or one-sided preserved.

The executable witness makes the distinction concrete. MNA and a plain nodal description agree for H_{voltage} in the ideal-source example, while they are not equivalent for $H_{\text{voltage+source-current}}$ because the source current is an additional retained unknown. Likewise, summing aligned parallel admittances preserves the terminal voltage relation but not the member-current-limit observation. The rank and sign guards above are part of the domain of these equivalence statements; they are not optional annotations. The rank claim is registered as $FORMULATION - NODAL - 003$.

Scope collapses and literature alternatives

Several familiar power-flow models are valid collapses when their guards are declared:

Starting target	Collapse	What must be checked
fixed linear port factors	$\mathbf{Y}^{\mathrm{N}}$	admittance-form relations, regular elimination, fixed grounding and state
MNA/tableau	nodal admittance	all retained extra variables are eliminable and unqueried
multiconductor model	scalar or positive-sequence Y	phase symmetry, balance, grounding, limits, and observations
node–breaker tableau	bus–branch Y_{bus}	state fixed and switch/protection decisions outside the query
port–factor source	ordinary-edge graph	declared n-port realizability and retained provenance

The alternatives are therefore not a contest for one universally best graph. They are formulation choices indexed by the study's observations, constraints, and decisions. The representation taxonomy classifies their retained meanings; the transformation semantics register records the guards for moving among them.

Power-system shorthand

“The network has a Y-bus” usually means that one study formulation has assembled a nodal operator for one declared state and variable set. It does not mean that the full power-network source model is a simple graph, a two-terminal multigraph, or a lossless collection of admittance edges.

Evidence boundary

The circuit and power-system precedents establish that nodal, modified-nodal, and sparse-tableau formulations are established alternatives. The book's stronger claim—that a formulation-aware lowering contract can preserve the declared power-network decisions and provenance—is a proposed architecture. The generated artifact `experiments/generated/circuit-formulation-witness.json` supplies three minimal failure cases:

1. an ideal voltage source whose source current requires an MNA variable;
2. a floating two-node network whose nodal operator is singular without a reference or shunt; and
3. aligned parallel members whose aggregate admittance is exact for the terminal relation but cannot carry their distinct current limits.

The first case solves two right-hand sides in MNA form, showing identical source voltage but different source currents. The other cases separate numerical singularity from semantic loss. These are scoped formulation witnesses, not a universal impossibility result for every circuit or power-network model.

Chapter 22

From source graphs to views and graph surgery

Page status: proposed architecture with a package-independent compiled-view, degeneracy, phase-selective switching, and state-conditioned zone-surgery witness; broader transformation algebra remains open.

This chapter makes one boundary explicit: a power-network model is not one picture that is repeatedly relabelled. It is a typed source object together with views and state-conditioned operations. The same asset may therefore appear in a single-line diagram, a multi-line diagram, a port–factor graph, a bus–branch quotient, and a nodal-support graph without those drawings having the same semantics.

This chapter is the bridge between formal representation frameworks, node–breaker topology processing, cycles and radial structure, and the transformation semantics register.

Identity-bearing source graphs

The canonical electrical source is the hierarchical port–factor model $\mathfrak{P}$ defined in Formal representation frameworks. For this chapter only, use the following view-level shorthand: $\mathcal{E}$ names identified factors, $P_e = f^{-1}(e)$ their ordered port sets, ι is the canonical attachment map j, and $\mathcal{R}$ collects the canonical containment, coordinate, state, and factor relations. Thus the shorthand

$$\mathcal{M}_{\text{view}} = (\mathcal{J}, \mathcal{E}, \{P_e\}_{e\in\mathcal{E}}, \iota, \sigma, \mathcal{R})$$

is a notation convenience, not a second canonical tuple. A two-terminal line has $|P_e| = 2$. A three-winding transformer has $|P_e| = 3$ (or more when conductor ports are explicit). Its identity is not recovered by looking at three pairwise edges after the fact.

An **identity fibre** is a set of source objects that a later view represents by one object. For a view v write

$$q_v : \mathcal{M} \longrightarrow \mathcal{V}_v, \qquad \mathrm{fib}_v(x) = q_v^{-1}(x).$$

The fibre may be a singleton, a parallel family, or a mixed family of ports and factors. The quotient is useful only when the query is invariant over the fibre, or when the omitted distinctions are carried as guards and provenance.

This is why the book's ℓij notation retains line identity: ℓ labels the physical member while i and j describe its endpoints. A quotient may forget ℓ; it must not pretend that the member never existed.

A visualisation registry

The six initial view classes are recorded in the artifact `experiments/generated/compiled-views-surgery-witness.json`.

The same artifact carries four explicit source-to-view map records. These are small contracts, not inferred drawing relationships: each names its source and target object IDs, map kind, preserved and forgotten semantics, and reverse-map status.

This registry is the book's scoped application of typed algebraic graph transformation: maps have declared matching conditions, omitted structure, and negative conditions such as "do not contract an unknown switch." The broader double-pushout theory, rewrite composition, and confluence vocabulary are available in [26]; this chapter does not claim to implement that entire calculus.

Two views are equal only up to the equivalence declared for their framework: an isomorphism may rename objects and coordinates, while a quotient or lowering is equal only when its preservation contract and source fibres agree. This uses the morphism/isomorphism distinction in Maps between representation frameworks, rather than treating two drawings that look alike as the same view.

View	Typical purpose	What it can forget
single-line	equipment-level communication and planning	conductor coordinates and internal factor equations
multi-line	identified conductors and phase/neutral paths	internal n-port equations
port–factor	canonical coupled electrical model	asset ownership if it is not attached
node–breaker	switching and connectivity decisions	compiled bus equations
nodal support	matrix coupling and ordering	factor identity, multiplicity, and switch decisions
reduced/Kron	retained-port equivalent	eliminated coordinates and unmapped member limits

Standards boundary and the house visual contract

The visual language has several useful standards precedents, but no single standard specifies the full path from a power-system asset to a multiconductor factor graph or a compiled equation system. IEC 60617 supplies a maintained library of electrotechnical graphical symbols [27], while IEC 61082 supplies rules for preparing and presenting electrotechnical documents and diagrams [28]. These standards constrain how symbols and documents should be presented; they do not decide whether a three-winding transformer is represented as one factor, a star, a complete leakage graph, or an ordinary-edge realization.

The exchange standards answer a different question. IEC 61970-453 links diagram-layout definitions to CIM objects and supports management of schematic revisions [29]. IEC 61850-6 exchanges substation and IED configuration structures [30]. Transformer terminal, neutral, polarity, and connection conventions are covered by IEEE C57.12.70 [31]. We therefore use these standards as interoperability anchors, not as evidence that a drawing alone preserves the source model.

Every maintained diagram in this book should declare the following contract:

This makes a symbol a typed visual interface rather than an unqualified icon. The same x_1 identifier should survive the asset, multi-line, factor, and compiled views even when the latter introduces virtual nodes or edges.

Equipment visual-language matrix

The first maintained equipment plate uses the following matrix. It is a design guide for future figures, not a claim that every cell is a separate graph class.

Contract field	Question the caption or registry must answer
object level	Is this an asset, terminal, conductor, factor, equation, or operational overlay?
identity fibre	Which source objects are represented by the drawn object?
preserved semantics	Which connectivity, terminal, constitutive, limit, state, and provenance facts remain?
omitted semantics	What has been hidden: neutral, shunt, winding identity, control, or internal state?
map type	Is the relationship a refinement, quotient, decomposition, compilation, or approximation?
reverse map	Is recovery total, partial, state-conditioned, or unavailable?
state scope	Is a state asset-wide, ganged, per-winding, per-conductor, or per-pole?
edge provenance	Is a drawn edge a physical asset, terminal relation, factor, support edge, or control relation?

Equipment	Single-line identity view	Multi-line terminal view	Factor/equation view	Operational overlay
line ℓij	one identified branch with model badge	ordered conductors, neutral, endpoint shunts	$\mathbf{Z}_\ell$, $\mathbf{Y}_\ell$ and coupling blocks	outage, rating, orientation convention
transformer x	one asset with winding count and connection group	winding bundles, WYE/DELTA/zig-zag, neutral and earth ports	transfer maps, leakage, excitation, grounding	tap, phase shift, control mode
regulator	transformer-like asset plus control marker	per-phase or mechanically coupled taps	$\mathbf{T}_x(t)$ and control equations	tap state, deadband, decision status
switch/breaker	one connectivity device	pole-wise contacts and gang relation	ideal constraint or state-conditioned quotient	open, closed, unknown, phase-selective
shunt/ground	attached auxiliary symbol or explicit badge	separate branch to neutral/earth/reference	admittance or grounding factor	switched state, protection, ownership

This plate is generated by `experiments/render_visual_language_equipment_plate.py`. It is deliberately a cross-view legend rather than a standards-compliance drawing: the persistent IDs identify the source objects, while the dashed maps show where a view or lowering introduces derived factors. The warning panel is part of the figure's semantics—single-line compression does not certify a scalar model, and a factor edge does not become a physical asset.

The visual grammar uses line style and labels in addition to colour: solid connectors denote declared electrical attachment, double strokes denote a conductor bundle, dashed arrows denote a view or lowering map, and a separate control arrow denotes a decision or operating relation. An arrow on an ℓij branch is an orientation convention, not a claim about the sign of operating current or complex power.

The three-winding transformer expansion is deliberately progressive:

1. one asset glyph x_1;
2. explicit winding and conductor ports;
3. ideal-transfer, leakage, excitation, grounding, and control factors;
4. a target-specific block or ordinary-edge compilation.

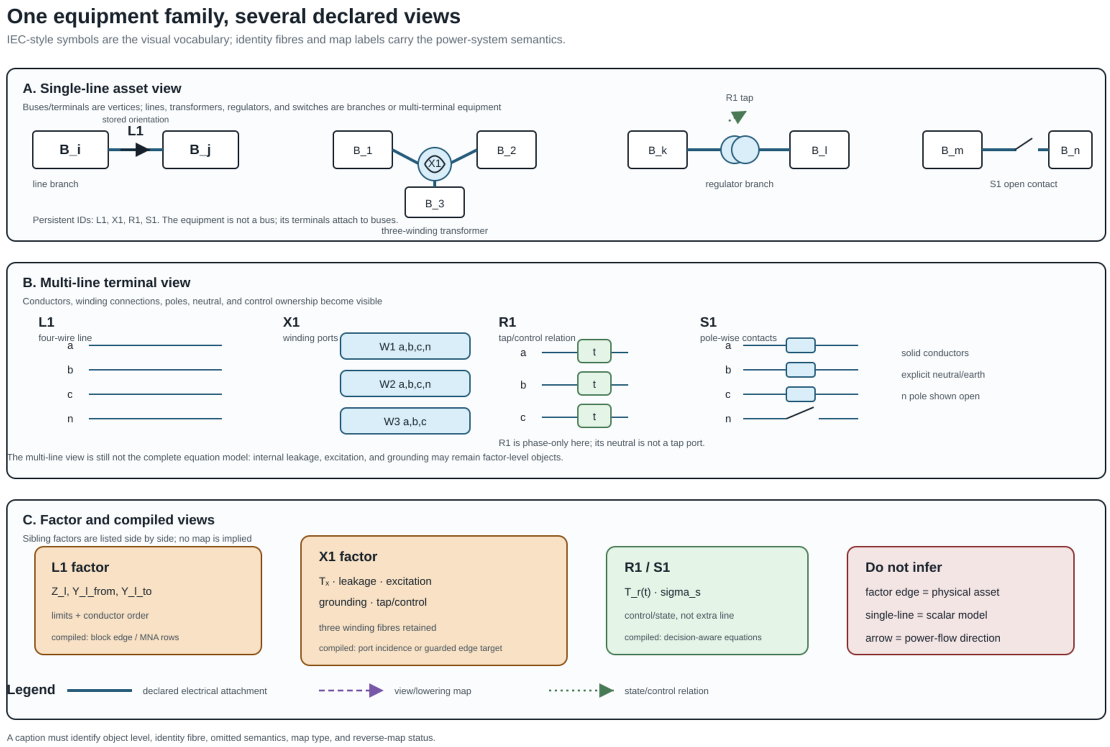

Figure 22.1: A line, three-winding transformer, regulator, and switch across single-line, multi-line, factor, and compiled views.

If a leakage decomposition uses a complete pairwise graph between winding states, its edges must be labelled *leakage-factor edges*. They are not physical conductors and must not be fed to an asset outage or ownership algorithm as if they were independent lines. This is the critical visual lesson borrowed from the transformer decomposition literature and made explicit in the book.

Special semantic overlays

The companion plate below makes four omissions especially difficult to miss. Neutral grounding is an explicit factor with its own current and recovery obligations. A nominal-π line has two terminal shunts, so absorbing them into a line or transformer changes the terminal-power and ownership semantics. A phase-selective switch has a vector state σ_S rather than one scalar open/closed bit. Finally, an n-winding leakage graph may contain pairwise computational factors whose edges have no asset identity. These are different kinds of edge, even when a target graph library draws them with the same stroke.

For review and reuse, the shared legend is:

The rule is operational: before a lowering or graph-surgery algorithm runs, record state scope and edge provenance in the map certificate. If either is unknown, retain the richer factor representation or refuse the transformation.

The registry is not merely a list of drawing conventions. Each view declares its object level, preserved semantics, forgotten semantics, and reverse-map status. A caption should therefore say “quotient view” or “lowered view” when that is what the reader is seeing. A visually plausible single-line diagram is not evidence that a neutral, grounding factor, switch state, or multiport identity survived.

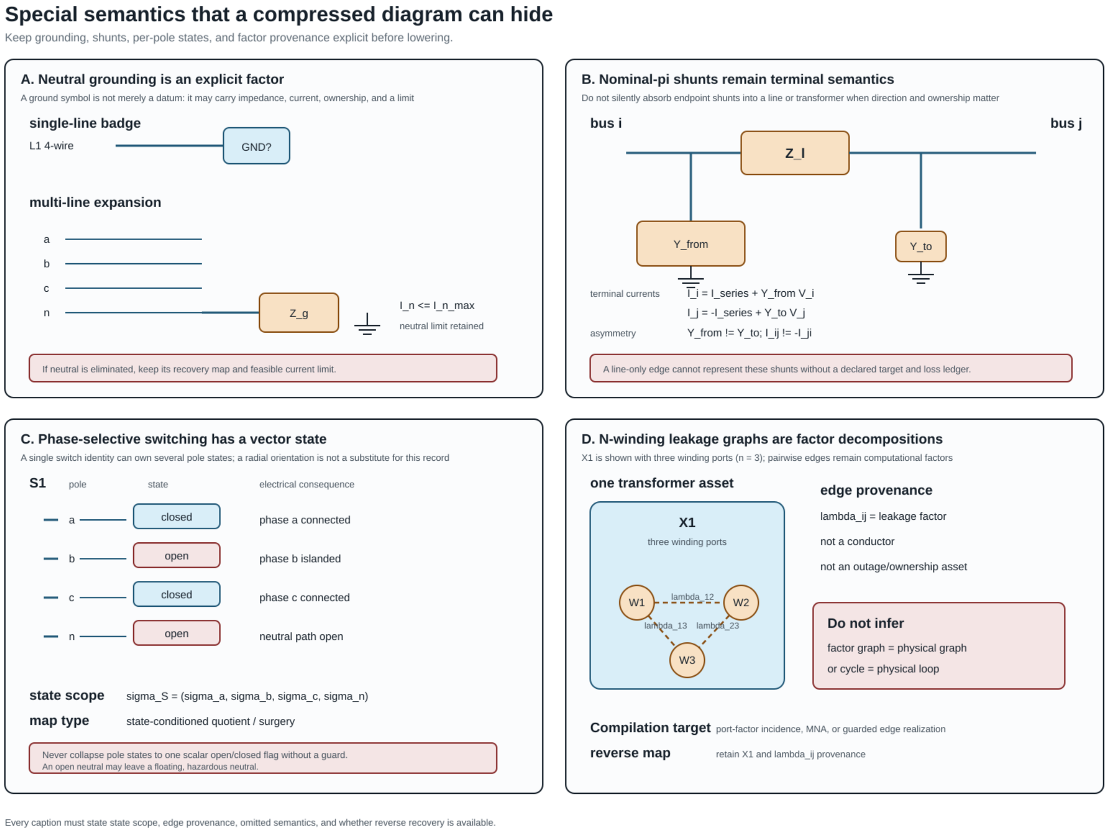

Figure 22.2: Neutral grounding, nominal-pi shunts, phase-selective switching, and n-winding leakage-factor provenance.

Mark	Meaning in the plates	Do not infer
bus rectangle	retained connectivity node or terminal boundary	an equipment asset
equipment glyph	identified physical asset	one scalar constitutive equation
orange factor box	constitutive or computational factor	an independently switchable asset
dashed λ_{ij} edge	pairwise leakage/coupling factor	a conductor, outage edge, or physical cycle
open/closed state chip	declared per-pole state	one state for every pole or winding
earth branch	explicit reference, grounding, or neutral path	a removable datum with no current consequence

The map is often one-way. A nodal support graph can show that two terminal coordinates couple, but it cannot identify which parallel members produced the nonzero block. A reduced circuit can preserve a retained-port relation while leaving no source-level current limit for an eliminated neutral. This is a semantic limitation, not a rendering defect.

The industrial node–breaker, bus–breaker, and bus–branch progression in CIM and PowSyBl is a useful precedent for this registry [32, 33]. The present registry extends that idea to multiconductor ports, n-terminal

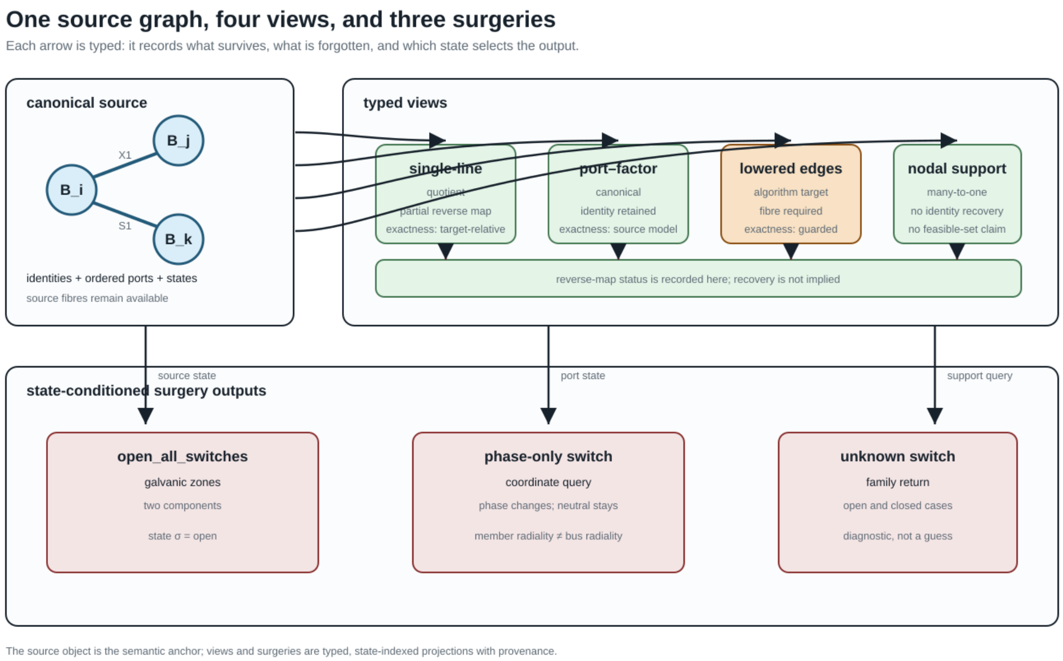

Figure 22.3: One source graph, four representative typed views, and three state-conditioned surgeries.

factors, grounding, and reduction provenance.

The upper row is a representative subset of the six-entry registry: the arrows may be quotient, refinement, lowering, or many-to-one assembly maps. The node–breaker and reduced/Kron entries are omitted from this compact figure. The lower row is a surgery family: the same source can produce different active graphs, and an unknown state produces a family rather than a silently selected result.

Lowering as a typed compilation boundary

For a declared algorithm, use the following typed pipeline. Direct factor stamping into a declared equation/constraint operator is the default; a nodal admittance target and ordinary-edge lowering are optional guarded branches. The formulation choices and exact-$\mathbf{Y}$ guards are developed in Circuit formulations and the lowering boundary.

$$\mathcal{M} \xrightarrow{C} \mathcal{M}_{\text{port}} \xrightarrow{A} \mathcal{E} = (\mathbf{F}, \mathbf{g}, \text{obs}, \text{prov}), \qquad \mathcal{M}_{\text{port}} \xrightarrow{L} \mathcal{G}_{\text{edge}} \xrightarrow{A_{\text{edge}}} \text{target algorithm.}$$

When the nodal guards hold, $\mathcal{E}$ may additionally lower to a nodal operator $(\mathbf{Y}^{\mathrm{N}}, \mathbf{J})$. Otherwise the faithful target may be MNA/tableau or a direct factor relation; the compiler must not invent a $\mathbf{Y}$ merely because a downstream library expects one.

C completes the canonical port–factor representation. L lowers a factor to the ordinary-edge or incidence objects expected by a graph algorithm. A assembles equations or sparsity. Every arrow must record:

1. the source object and coordinate order;
2. the target object IDs and source fibres;

3. the relations and limits that are preserved;
4. the semantics intentionally omitted; and
5. whether a reverse map is total, partial, or unavailable.

For a three-port transformer, a lowerer may introduce three ordinary edges in an incidence graph, but this is exact only under a declared realizability condition. For a reciprocal terminal admittance $\mathbf{Y}_\phi$ with terminal-space all-ones vector $\mathbf{1}$, an edge-only realization requires the floating/no-ground-path condition $\mathbf{Y}_\phi\mathbf{1} = 0$ (together with the target library's reciprocity and coordinate assumptions). Otherwise the exact complete-graph construction needs the pairwise series blocks and residual shunts

$$\mathbf{Y}^{\mathrm{s}}_{pq} = -\mathbf{Y}^{\mathrm{K}}_{pq}, \qquad \mathbf{Y}^{\mathrm{sh}}_{p} = \mathbf{Y}^{\mathrm{K}}_{pp} - \sum_{q\neq p} \mathbf{Y}^{\mathrm{s}}_{pq}.$$

This is the realizability proposition in Kron, Ward, and optimized network equivalents, not an automatic transformer-to-three- lines rule. Magnetizing, grounding, or other retained shunt current must not be silently dropped. Even when the expansion is exact, the ordinary-edge graph is not the canonical equipment graph: removing its provenance fibre would make later operations unable to distinguish a transformer from three unrelated lines.

The analogy with compiler lowering is useful because it sets the right expectation: lowering changes the representation so an algorithm can run. A nanopass compiler similarly uses many small, typed intermediate passes rather than one opaque rewrite [34]. The analogy does not grant permission to infer source semantics from the lowered code.

State-conditioned graph surgery

Graph surgery is a family of queries and transformations indexed by a declared state σ. For a surgery operation S write

$$S(\mathcal{M}, \sigma) = (\mathcal{V}_\sigma, \mathcal{D}_\sigma, \mathrm{prov}_\sigma),$$

where $\mathcal{V}_\sigma$ may be one graph or a finite family of graphs, $\mathcal{D}_\sigma$ contains diagnostics, and prov_σ maps each output node, edge, port, and zone to its source objects.

Useful operations include:

- open*all*switches, which removes switch conductance while retaining the switch assets as possible future actions;
- galvanic_zones, which computes connected components using only the declared zero-impedance or closed-switch relation;
- energized_subgraphs, which additionally uses sources and a declared energization rule. The term is intentionally inherited from the book's terminology and translation-traps chapters; in a multiconductor model the query is port-indexed, so one phase may be energized while the neutral remains floating;
- active_radiality, which reports member, endpoint, conductor, and compiled bus predicates separately; and

- eliminate_switches, which is a lowering operation only after its state and zero-impedance assumptions are fixed.

An unknown switch state must return a family or an explicit unknown result. It must not be silently treated as open. Two-terminal and n-terminal surgeries also differ: a phase-only switch acts on phase ports without necessarily opening the neutral or earth path. Consequently, a bus-level tree can coexist with a disconnected phase-terminal graph, or with a neutral path that remains connected.

For n binary unknown switches, explicit completion enumeration can contain up to 2^n active graphs. Enumeration is appropriate for small certificates, but it is not the default representation for a large substation. The default summary is three-valued for each queried pair or port set:

- **certainly connected** in every admissible completion;
- **certainly separated** in every admissible completion; or
- **undetermined**, connected in some completions and separated in others.

This summary is the intersection/union view of the state-conditioned quotient π_σ from node–breaker topology processing. A caller can request explicit completions when the summary is insufficient.

For an n-terminal factor, the surgery result should name the active and isolated port sets. Opening the lv port of a three-winding transformer is not the same operation as deleting one of three pairwise lines: the factor identity, the remaining hv/mv relation, and the isolated-port status remain explicit. If the port state is unknown, the result is a family of port sets.

Degenerate and under-determined models

Some modelling problems cannot be resolved from the graph alone. In the generated witness, two identified ideal switches have identical terminals and the same state domain. Electrically, their parallel closed quotient is perfectly well-defined. The ambiguity is in the orthogonal asset/dependency model: the quotient cannot say which device owns protection, maintenance, or failure semantics. The correct result is therefore an **asset-attribution diagnostic** with both source identities retained, not a claim that the electrical model is ill-posed.

The same rule applies to duplicate factor/terminal sets, missing grounding or reference declarations, and singular active-state maps. These are not ordinary graph errors: they are model-quality findings that require a source declaration, an additional observation, or a deliberately restricted query.

In particular, a four-wire coordinate list without a grounding or reference declaration must not acquire one by convention. Likewise, a rank-deficient active-state map must not be inverted merely because a downstream algorithm expects an inverse. A restricted endpoint-voltage query, an explicit pseudoinverse policy, or a source-level grounding declaration may make a well-scoped operation possible; the default result is a diagnostic.

Executable scope

The package-independent witness `experiments/generated/compiled-views-surgery-witness.json` checks six small cases:

1. a three-port transformer lowered to ordinary edges with a retained source fibre;
2. two parallel ideal switches reported as under-determined rather than collapsed;
3. a four-wire phase-only switch whose phase connectivity changes while the neutral path remains connected; and

4. an open/closed/unknown switch surgery that returns one-zone, two-zone, or state-family results;
5. a port-selective n-terminal surgery that retains the isolated port; and
6. missing-reference and singular-active-map diagnostics.

These are architecture witnesses, not claims that every utility data model uses the same view classes. The next extensions are richer n-terminal factors, energization and protection semantics, and independently reviewed adapters to external standards.

Graph-theory trap

When a diagram appears to contradict a familiar statement such as “the feeder is radial,” first ask: radial in which view, at which state, and with which conductor or member identities retained?

Part III

Canonical model and valid collapses

Chapter 23

From source data to a canonical network model

Page status: semantic-projection and validation contract for entering a network into the book's representation architecture.

The graph seen by a study is rarely copied directly from a source file. A utility export, an OpenDSS deck, a CIM profile, or a solver dictionary carries different mixtures of identifiers, terminals, implicit references, defaults, device models, and study state. Before asking whether a graph transformation preserves anything, we must know what network the source data actually means.

This chapter adapts the most useful BMOPFTools lesson to the book's broader setting: ingestion is a **semantic projection**, not a promise of byte-level round-trip identity. The output should be a canonical, typed network model with explicit losses, inferences, and provenance.

The projection contract

Let D be a source document and C a canonical network model. An adapter is a partial map

$$P : D \longrightarrow (C, F, \Pi),$$

where F is a finding ledger and Π is a provenance manifest. The map is partial because an input may be malformed, incomplete, or outside the supported factor library. A successful parse means only that P returned a candidate model; it does not mean that every study question supported by D is preserved in C.

The canonical model should make at least these objects explicit:

The book's preferred identifiers retain the BMOPFTools-style ownership rule: ℓ identifies an element, while ℓij identifies an oriented terminal attachment. A converter may reorder arrays or create virtual objects, but it must publish the map that explains the change.

Validation gates before graph construction

The following gates are deliberately ordered. A later graph cannot repair a failure that should have been caught at an earlier semantic layer.

These checks should return stable finding codes and machine-readable details, not only prose warnings. A graph quotient may be mathematically valid while the input is semantically unfit for the intended decision problem.

Source concern	Canonical object	Why it matters for graphs
stable equipment identity	asset/property record	keeps parallel and replacement assets distinct
endpoint and conductor names	ordered terminal maps	prevents phase/neutral permutations from hiding in matrices
connectivity and switching	state-resolved incidence	separates inventory topology from active topology
constitutive relation	line, shunt, transformer, load, or factor relation	prevents an edge from standing in for unknown physics
ratings and owners	typed constraint observations	keeps limits attached to the object they constrain
defaults and inferred meanings	finding with confidence/provenance	makes assumptions auditable

Gate	Question	Typical failure
schema	are fields and shapes structurally recognised?	malformed matrix or unknown component block
complete-ness	are required fields present for this factor subtype?	transformer winding lacks a terminal map
domain	are values physically and numerically plausible?	negative rating, impossible tap, invalid angle unit
integrity	do references and dimensions agree?	line points to a missing bus or mismatched conductor count
confor-mance	does the object satisfy study-specific rules?	WYE winding without a declared neutral or multiple active sources
readiness	is the declared decision problem well posed?	no voltage bounds, slack-only objective, or missing limit owner

The ordering is semantic, not cosmetic: a later graph view cannot repair a missing terminal map, an invalid domain value, or an ownerless limit that should have been rejected earlier.

Inference is not identity

Practical formats often encode meaning positionally or through defaults. An adapter may infer that terminal 4 is a neutral, that two named objects are parallel members, or that a regulator is a transformer with a control loop. Such inferences can be useful, but they are not source facts until recorded as provenance-bearing findings. The canonical record should distinguish:

1. **declared** values copied from the source;
2. **derived** values computed from declared data;
3. **inferred** values introduced by an adapter rule; and
4. **unsupported** values that could not be represented.

This distinction is essential when a later transformation claims preservation. An inferred terminal permutation may be harmless for a scalar connectivity query but fatal for a conductor-current limit.

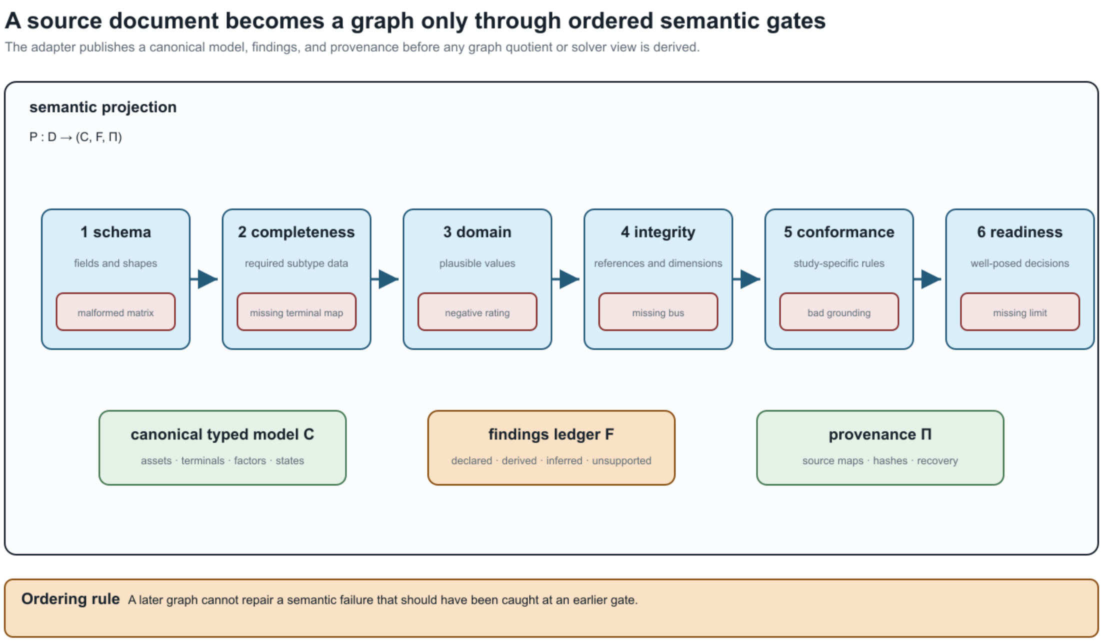

Figure 23.1: The ordered source-to-canonical validation pipeline.

What a safe adapter publishes

For every source-to-canonical map, record:

1. source format, profile, and version;
2. stable asset and terminal identifiers;
3. units, bases, phase, neutral, earth, and grounding conventions;
4. state, scenario, and control treatment;
5. factor, rating, and objective mappings;
6. generated objects and their source parents;
7. unsupported or lossy fields;
8. validation findings and their severity; and
9. round-trip or recovery checks for the declared observations.

The result is a transformation certificate at the data boundary. It is the same preservation-contract language used later for Kron, aggregation, and positive-sequence maps, but the source and target are data models rather than electrical equations.

Impedance data need one additional discipline: the canonical record should retain the full derivation context even when an adapter exports only a solver- friendly matrix. Geometry or linecode provenance, frequency, earth model, ordered conductors, series/shunt blocks, and matrix diagnostics belong to the source record; Kron, phase-to-neutral, sequence, and positive-sequence matrices are derived views with their own guards. The four-wire impedance-model ladder is the first executable example of this source-to-view contract.

Consequences for the graph views

The canonical model is not itself a single graph. It is the source from which the book derives:

- an asset/dependency graph for ownership, maintenance, and failures;
- a terminal-connectivity graph for state and grounding questions;
- a directed/oriented multigraph for bus–branch equations;
- a port–factor graph for coupled and multi-terminal devices; and
- equation and sparsity graphs for a chosen formulation.

If the adapter collapses two assets, loses a neutral, or silently grounds a terminal, every downstream view inherits that loss. Conversely, a solver may create virtual buses or auxiliary factors that are useful computationally but must map back to the canonical source objects before a decision or limit is interpreted.

Running-network application

The running fixture is intentionally small enough to audit. Its canonical record declares four lines, a three-winding transformer, a switch, ordered conductor sets, explicit neutral semantics, ratings, and state ownership. The generated adapter crosswalk checks those obligations against the pinned CIM/CGMES, PowerModelsDistribution, OpenDSS, and MATPOWER descriptions. That crosswalk is not an import claim: it is a checklist of what an actual adapter would have to preserve or mark unsupported.

The practical rule is therefore:

> Do not draw the graph first and infer the model later. Declare the canonical objects, validate them, and derive each graph as a named view with provenance.

This chapter supplies the missing front door for the representation maps. The next chapters show how constitutive load and impedance choices can change the feasible decision problem even when that front-door graph is unchanged.

Chapter 24

Load models and decision dependence

Page status: constitutive-model reference chapter with scoped numerical load-model, connection-map, and continuation witnesses plus independent reproduction; broader nonlinear and network-level certificates remain future work.

Two networks can have identical buses, edges, terminals, and ratings while defining different decision problems because their loads obey different voltage laws. A graph representation does not identify a load model. The constitutive relation belongs to the factor or asset layer and must be carried through any preservation claim.

Vocabulary bridge

Electrical state below means the continuous voltages and currents at an operating point. Equipment status, operating scenario, state-estimator metadata, load-model parameters, and a learned hidden state are separate objects even when software stores them in one state container.

Five common voltage laws

Let $v = |\Delta U|$ be the magnitude seen by one load connection and let v^{nom} be its declared anchor. For active power, the standard families can be written as

$$
\begin{aligned}
P_{\mathrm{CP}}(v) &= P^{\mathrm{nom}}, \\
P_{\mathrm{CI}}(v) &= P^{\mathrm{nom}} \frac{v}{v^{\mathrm{nom}}}, \\
P_{\mathrm{CZ}}(v) &= P^{\mathrm{nom}} \left(\frac{v}{v^{\mathrm{nom}}}\right)^2, \\
P_{\mathrm{ZIP}}(v) &= P^{\mathrm{nom}} \left(\alpha_Z \left(\frac{v}{v^{\mathrm{nom}}}\right)^2 + \alpha_I \frac{v}{v^{\mathrm{nom}}} + \alpha_P\right), \\
P_{\exp}(v) &= P^{\mathrm{nom}} \left(\frac{v}{v^{\mathrm{nom}}}\right)^{\gamma_P}.
\end{aligned}
$$

Reactive power may use its own coefficients or exponent. Constant power does not need v^{nom}; the other laws do. ZIP is a convex combination only when the declared coefficients are nonnegative and sum to one. Integer exponents $0, 1, 2$ make the exponential family coincide with a ZIP special case, but arbitrary exponents do not.

At $v = v^{\mathrm{nom}}$ the families agree by construction. Away from that point they do not. The same topological graph can therefore produce different currents, losses, voltage margins, active constraints, and optimal decisions.

Why this belongs in a graph-model book

Consider a fixed two-bus graph with series impedance Z and a load at the receiving bus. The power-flow relation is not just the graph incidence; it is

$$U_s - U_r = ZI, \qquad S_r = U_r I^*,$$

together with a law for S_r as a function of U_r. Replacing constant power by constant impedance changes the equation graph and the feasible set without changing the bus–branch multigraph. A transformation that preserves only Y or connectivity has not preserved the decision problem unless it also preserves the load relation and the observations that depend on it.

This is a useful counterweight to a common simplification: “the network graph is unchanged, so the OPF is unchanged.” The graph is one layer of the model; the load factor is another.

When a load or generator enters the nodal operator

The factor layer and the nodal operator should not be confused. A constant- impedance load is a one-terminal shunt factor, so a declared formulation may stamp its admittance into a diagonal nodal block. A ZIP or constant-power load may instead be represented by a fixed linear part plus a voltage-dependent compensation current. The same distinction applies to generators: a fixed Norton or dynamic equivalent may contribute a matrix block, while a PV/PQ control, current limit, or inverter control law remains an injection or constraint relation.

OpenDSS provides a useful engineering example. Its normal solution uses a system nodal matrix together with compensation currents from nonlinear power- conversion elements; its direct/admittance option solves with load and generator equivalents included in the matrix. The documentation also notes that loads may be switched to an admittance representation for fault studies and that the fault-study matrix has its own source, generator, and load-current composition [14, 17, 18, 35]. This is a solver and study choice, not a claim that the underlying load has become a line edge or that the source graph has changed.

The same warning applies to a two-terminal factor whose terminals become the same resolved node. A fixed linear π section can reduce exactly to a constant- admittance shunt after terminal-map assembly, but that compiled diagonal term is not a self-loop edge and does not erase the source factor, its controls, limits, or provenance. See Multigraphs for expert modelers for the derivation and refusal boundary.

The supplied application-directed distribution equivalent illustrates the same boundary at feeder scale. Its reduction first constructs a nodal equivalent and then aggregates PVs and loads at an equivalent node, with phase-shifting and shunt elements added to preserve the selected study responses. The resulting model is validated for declared power-flow and EMT observations; it is not presented as the unique graph of the original feeder [36].

For this reason, a nodal matrix should be annotated with at least:

- the source factor inventory and terminal/attachment maps;
- the study mode and active device state;
- which load, generator, shunt, or source components were stamped into the matrix;
- which nonlinear or controlled parts remain in injection and constraint maps;
- the operating point or iteration used for any linearization; and
- the observations, limits, decisions, and recovery maps that remain valid.

Decision consequences

Load-model choice becomes especially visible in three regimes:

1. **voltage regulation and CVR:** constant-power loads do not reduce demand when voltage is lowered, while impedance loads do;
2. **hosting capacity and voltage limits:** a voltage-dependent load can partially relieve a binding voltage constraint, changing the reported capacity; and
3. **weak feeders and collapse:** constant-power demand can create a nose point, so a local high-voltage solution may disappear as the load scale increases.

The relevant preservation contract must name the load family, its anchor, phase or connection map, and whether the decision varies load scale, voltage, or model parameters. A numerical result that changes after a load model swap is not necessarily a solver error; it may be the intended change in the physical problem.

Per-conductor and connection semantics

For a multiconductor load, v is not automatically a bus-voltage scalar. The factor must declare whether it uses phase-to-neutral, phase-to-phase, sequence, or another terminal combination. A delta load and a wye load can share the same bus terminals while seeing different voltage coordinates. The load model therefore belongs beside the terminal map, not as an unqualified attribute of a graph vertex.

Likewise, a three-phase load may be balanced in name but still have phase-specific P_p and Q_p values. A balanced positive-sequence view preserves it only when the load relation, grounding, limits, controls, and observations close under the same phase symmetry.

Executable connection-map probe (LOAD-CONNECTION-001)

The generated `connection_maps` witness in `experiments/generated/load-grounding-witnesses.json` uses ordered terminals (a, b, c, n) and applies two explicit linear maps to the same balanced bus:

$$T_Y \mathbf{v} = (V_a - V_n, V_b - V_n, V_c - V_n), \qquad T_\Delta \mathbf{v} = (V_a - V_b, V_b - V_c, V_c - V_a).$$

The wye observations have unit magnitude, while the delta observations have magnitude $\sqrt{3}$ for the recorded positive-sequence voltage. The bus and graph are unchanged; only the factor's terminal map changes. This is a small structural witness, not a complete unbalanced load-flow or rating model.

Modelling checklist

Before comparing two graph views or deleting a factor, record:

- the load connection and ordered terminal map;
- the active and reactive voltage law;
- the nominal voltage anchor and units;
- phase-specific coefficients or exponents;
- whether the law changes with state, control, or time;

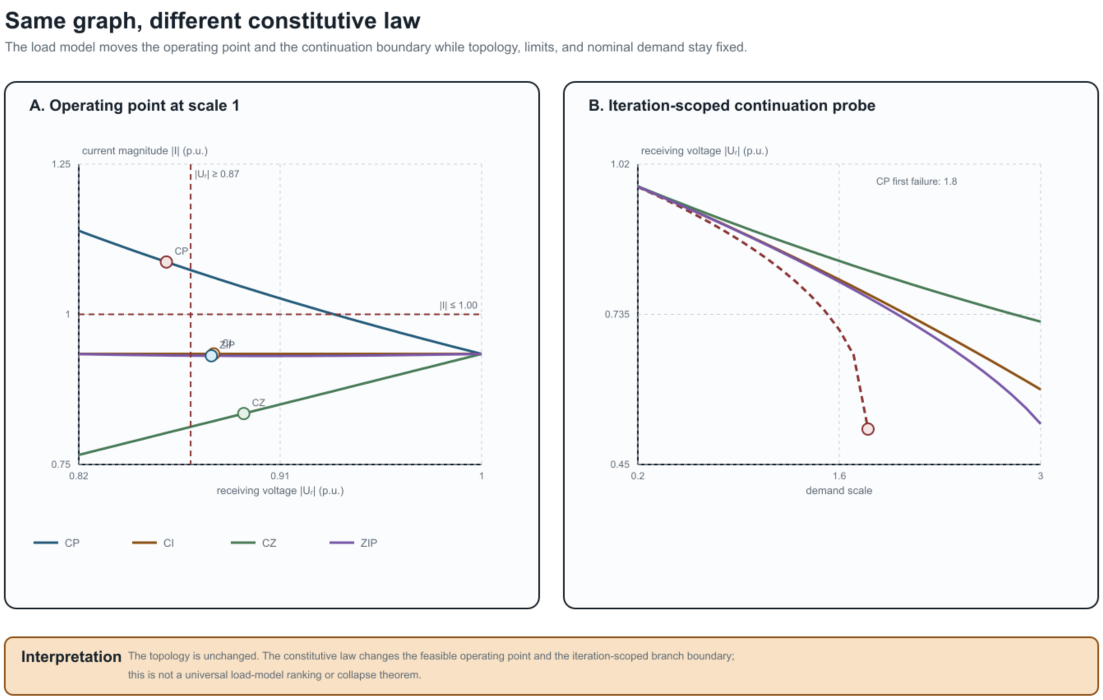

Figure 24.1: Operating-point divergence and continuation boundaries for four load laws.

- the voltage and current quantities used in limits; and
- whether the study seeks feasibility, load delivery, hosting capacity, voltage regulation, or a different decision.

The book's existing parallel-line and transformer cases preserve their load equations explicitly. This chapter makes the reason general: constitutive models are part of the representation contract, even when they are invisible in a simple graph drawing.

Scoped numerical witness

The generated artifact `experiments/generated/load-grounding-witnesses.json` solves one fixed two-bus network under three load laws. With the same source, series impedance, nominal demand, voltage limit $|U_r| \geq 0.87$ and current limit $|I| \leq 1.00$, the recorded high-voltage solutions are:

Load law	$\lvert U_r \rvert$	$\lvert I \rvert$	Voltage limit	Current limit
CP	0.8592	1.0872	fail	fail
CI	0.8803	0.9341	pass	pass
CZ	0.8937	0.8348	pass	pass
ZIP (active $(0.4, 0.3, 0.3)$, reactive $(0.2, 0.3, 0.5)$)	0.8792	0.9310	pass	pass

This is a decision witness, not a universal ranking of load models. It shows that a graph-preserving change of constitutive law can change both feasibility and the active constraint set.

The left panel places the recorded operating points against the same voltage and current limits. The right panel is the finite continuation probe: CP is the first family to fail the declared iteration test, while the other three families remain traceable through scale 3.0. Neither panel is a universal voltage-collapse or load-model-ranking theorem.

The artifact `experiments/generated/load-model-independent-reproduction.json` repeats the same damped fixed-point calculation with a separate standard-library Python implementation. It reproduces all four recorded rows and their voltage/current limit decisions. This supports reproducibility of the declared scalar fixture; it does not establish global solvability, a universal ranking of load laws, or the adequacy of CP/CI/CZ/ZIP for a particular utility study. The ZIP row uses nonnegative active and reactive coefficients that each sum to one; the two coefficient triples are deliberately distinct to expose that reactive demand need not follow the active-power law.

Continuation probe (LOAD-CONTINUATION-001)

The same scalar fixture also includes a demand-scale continuation in `load_continuation`. It tracks the previous high-voltage iterate over scales $0.2, 0.3, \ldots, 3.0$ with damping 0.5. The recorded CP branch converges through scale 1.7 and fails the declared iteration/residual test at scale 1.8; CI, CZ, and the ZIP branch remain converged through scale 3.0. This is useful as a warning that a load-model change can move a numerical branch boundary, but the boundary is solver- and continuation-specific. It is not a global voltage- collapse theorem, and the artifact does not claim that the first failed iterate is the exact saddle-node point.

Chapter 25

From conductor geometry to impedance fidelity

Page status: physical-fidelity reference chapter; a geometry-derived running-network certificate remains future work.

The matrix attached to a line is not an arbitrary edge label. It is often the result of a modelling ladder that starts with conductor geometry, earth-return assumptions, frequency, and wire data. The ladder explains why a graph can look unchanged while its electrical relation—and therefore its decisions—has changed.

The physical-to-matrix ladder

For a declared ordered conductor set or coupled corridor group, a typical steady-state construction is

$$\text{wire data + geometry + earth model + frequency} \longrightarrow (\mathbf{R}_\ell, \mathbf{X}_\ell, \mathbf{Y}_\ell^{\mathrm{sh}}) \longrightarrow \text{line or joint terminal factor} \longrightarrow \text{network equations ar}$$

The series matrix is

$$\mathbf{Z}_\ell = \mathbf{R}_\ell + \mathrm{j}\mathbf{X}_\ell.$$

Diagonal entries describe self effects; off-diagonal entries describe mutual coupling. Earth-return and grounding assumptions can affect both. A linecode that contains only a scalar positive-sequence impedance has already forgotten which conductor geometry and return path produced it.

The matrix is still not the complete factor. Terminal maps, from/to shunts, connection factors, ratings, and state ownership determine how it enters the network. For an independent line the book indexes intrinsic impedance by ℓ and terminal quantities by ℓij. For mutually coupled line sections, the joint primitive is instead indexed by a coupling group Γ and linked back to every participating line asset.

When the primitive spans several line assets

For two coupled sections with possibly different conductor counts, the series primitive has block form

$$\mathbf{Z}_\Gamma = \begin{bmatrix} \mathbf{Z}_{11} & \mathbf{Z}_{12} \\ \mathbf{Z}_{21} & \mathbf{Z}_{22} \end{bmatrix}.$$

The off-diagonal blocks describe cross-circuit coupling and can be rectangular. They should not be copied into two nominally independent line records and the self blocks should not be inverted separately. A source adapter should retain or construct oriented electrical sections, pairwise coupling records, and a coupling-group identifier; the group primitive is assembled first and then compiled into a direct factor, tableau, or guarded nodal stamp.

Different-voltage circuits also require cross-coordinate base data. With a common power base and compatible phase-voltage conventions, the mutual impedance base between voltage systems a and b is

$$Z_{\mathrm{b},ab} = \frac{U_{\mathrm{b},a}U_{\mathrm{b},b}}{S_{\mathrm{b}}}.$$

Using either circuit's self-impedance base for the cross block can break the declared reciprocity and scaling relation. The coupled multi-voltage corridor case develops the source ownership, per-unit map, partial-overlap semantics, and equivalent lattice.

A canonical impedance-data contract

Standardisation should make the source model richer than any one solver input. For each oriented branch ℓij, and for every coupling group in which one of its electrical sections participates, the canonical record should retain:

- the asset identity, ordered conductor set, and endpoint terminal maps;
- geometry, conductor material, length, temperature, frequency, and earth model;
- the derivation method (for example Carson, Pollaczek, finite element, or a fitted matrix) and its source version;
- the series matrix $\mathbf{Z}_\ell$ and endpoint shunt blocks $\mathbf{Y}^{\mathrm{sh}}_{\ell ij}$ and $\mathbf{Y}^{\mathrm{sh}}_{\ell ji}$ separately;
- units, base values, matrix ordering, coordinate convention, and reference ground semantics; and
- current, voltage, power, neutral, protection, and control observations that use the resulting coordinates.

A coupling-group record additionally retains participating section identities, lineage to stable assets, overlap intervals, orientation/dot conventions, cross-block coordinate maps, the joint matrix ordering, and whether the relation is phase-domain, sequence-domain, series-only, or full series-and- shunt data.

The canonical record is therefore a source of derived views, not merely a serialization of the values most convenient for one engine. A positive- sequence scalar or a phase-only matrix can be exported, but it must retain a pointer to the full record and the transformation path that produced it.

The first executable version of this contract is the generated `experiments/generated/four-wire-impedance-model-ladder.jso` fixture. It is a small deterministic matrix example rather than a geometry-identification claim. Its purpose is to make ordering, units, ground assumptions, shunts, recovery maps, and risk tags testable before importing larger authored case studies.

The source-backed follow-on is the Australian Carson reproduction. It lifts the construction fields that are actually present in the `ImpedanceModels.jl` history, regenerates the primitive and OpenDSS solve, and keeps the published Australian matrices in a separate comparison channel until their original construction mappings are recovered.

Symmetry and sequence coordinates

For three phase conductors, a perfectly transposed idealisation often has a circulant phase block: equal self terms and equal mutual terms under cyclic permutation. The Fortescue transform then diagonalizes that block into sequence coordinates. This is a legitimate coordinate specialization when the factor, grounding, boundary data, controls, limits, and observations all preserve the same symmetry.

An untransposed geometry generally breaks circulance. The transform still exists as a change of coordinates, but the transformed matrix is not diagonal; sequence channels mix. The distinction is important:

- **transform available:** any compatible phase vector can be expressed in sequence coordinates;
- **sequence decoupling valid:** the constitutive matrices and the rest of the decision model preserve the sequence subspaces.

The existing positive-sequence collapse chapter gives the exact restriction theorem. This chapter supplies the physical provenance that theorem needs.

Fidelity levels

Level	Retains	Typical questions it can answer	Typical loss
geometry-derived conductor matrix	wire positions, mutual terms, earth model, frequency	unbalance, neutral shift, conductor currents	detailed geometry may be unavailable or uncertain
fitted full matrix	coupled $\mathbf{Z}$ and shunt blocks	multiconductor PF/OPF and terminal limits	source geometry and identification uncertainty
diagonal phase matrix	separate phase self terms	decoupled approximations with explicit phase identity	mutual coupling and return-path effects
sequence matrix	zero/positive/negative sequence blocks	balanced or fault studies under closure assumptions	phase-specific geometry and many terminal decisions
scalar positive-sequence edge	one complex relation per bus pair	transmission-style balanced PF/OPF	neutral, phase, winding, and asset distinctions

Moving down the ladder is not automatically wrong. It is a projection whose admissibility depends on the study query. A scalar edge may be adequate for a balanced transmission objective and inadequate for a neutral-current limit or phase-specific protection observation.

Conditioning is part of fidelity

Geometry-derived matrices can be ill-conditioned because conductors are close, because a neutral is weakly grounded, or because the chosen coordinate basis contains nearly redundant modes. Per-unit scaling and coordinate changes may improve numerical conditioning without changing an invertible solution set; they do not restore information that a projection discarded. A reduction that removes a weakly observable neutral mode must report both its algebraic guard and its decision-observation consequences.

What the adapter must record

For every impedance matrix used in a transformation, record:

- conductor order and terminal maps;

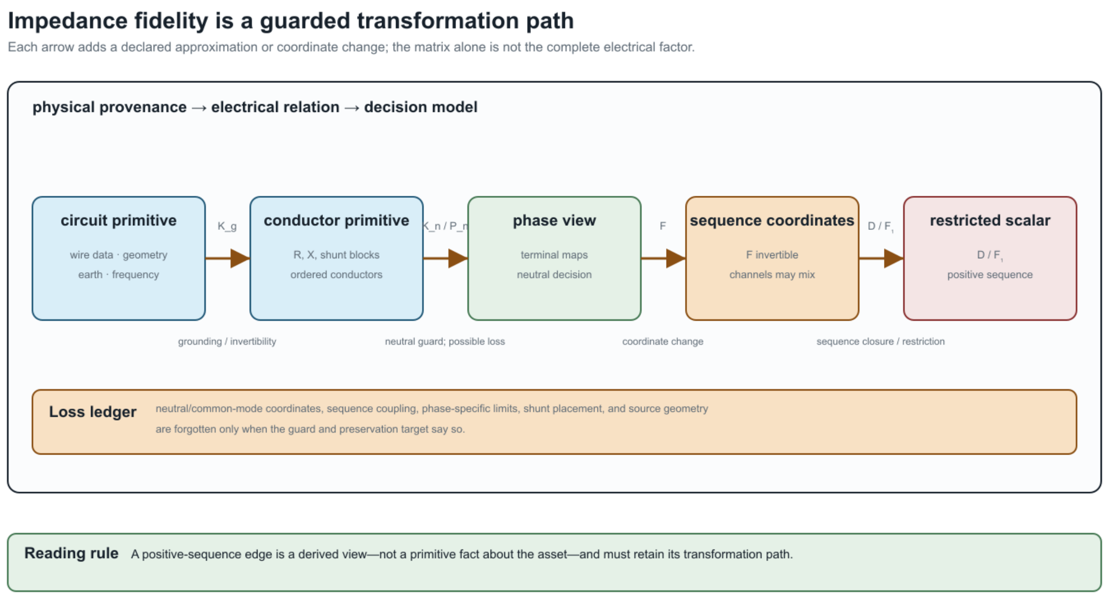

Figure 25.1: The impedance-fidelity ladder with guarded transformations and an explicit loss ledger.

- units, base values, frequency, and length convention;
- geometry or linecode provenance;
- earth-return and grounding assumptions;
- symmetry, reciprocity, passivity, and conditioning diagnostics;
- whether shunts are explicit or folded into a terminal primitive; and
- which limits and decisions use the resulting current coordinates.

This ledger connects the physical model to the graph model. It prevents a sequence or scalar edge from being mistaken for a primitive fact about the asset, and it gives a principled place to report uncertainty before an OPF comparison is made.

The model ladder is a transformation path

The common path is not a list of interchangeable names:

$$\text{circuit primitive} \xrightarrow{K_g} \text{conductor primitive} \xrightarrow{K_n \text{ or } P_n} \text{phase view} \xrightarrow{\mathbf{F}} \text{sequence coordinates} \xrightarrow{D \text{ or } \mathbf{F}_1} \text{restricted scalar view.}$$

K_g and K_n require declared grounding and invertibility assumptions; P_n can recover neutral current under its zero-ground-current guard but loses common-mode voltage; $\mathbf{F}$ is an invertible coordinate change; D deletes sequence coupling; and $\mathbf{F}_1$ retains only a positive-sequence subspace. The last two are therefore not harmless formatting operations. Their admissibility depends on the factor, boundary data, limits, controls, and observations.

The companion transformation register records these distinctions as typed edges. The four-wire ladder fixture checks the exact current-recovery relation, exposes nonzero sequence mixing for a non-circulant matrix, and keeps shunt deletion and positive-sequence use visibly guarded.

Relation to the running network

The running network deliberately retains full matrices, heterogeneous line terminal maps, and a three-winding factor. Its positive-sequence specialization is a derived view, not a replacement for the source. A future geometry-derived fixture can populate this ladder numerically; until then, the book treats the matrix provenance and symmetry assumptions as explicit contract fields rather than claiming a geometry-to-OPF validation result.

Chapter 26

A coupled multi-voltage corridor

Page status: literature-backed representation and exact fixed-linear lowering case with an executable scalar certificate; a geometry-derived multi-voltage certificate and decision-preservation tests remain follow-on work.

Two circuits can share towers or a right of way without sharing buses, terminals, or nominal voltage. Following the line-modeling literature, they are then *physically parallel sections*: spatially co-located, but not necessarily parallel edges in a bus multigraph. Here *physical* qualifies the shared route; it does not mean a transfer corridor or electrical interchangeability. Electromagnetic coupling nevertheless makes their longitudinal voltage drops and currents part of one constitutive relation.

This case separates three objects that a one-line diagram can make easy to conflate:

1. two identified line assets with their own ratings and states;
2. one joint electromagnetic factor over the coupled electrical sections; and
3. an optional ordinary-edge lattice that realizes the same fixed-linear terminal equation.

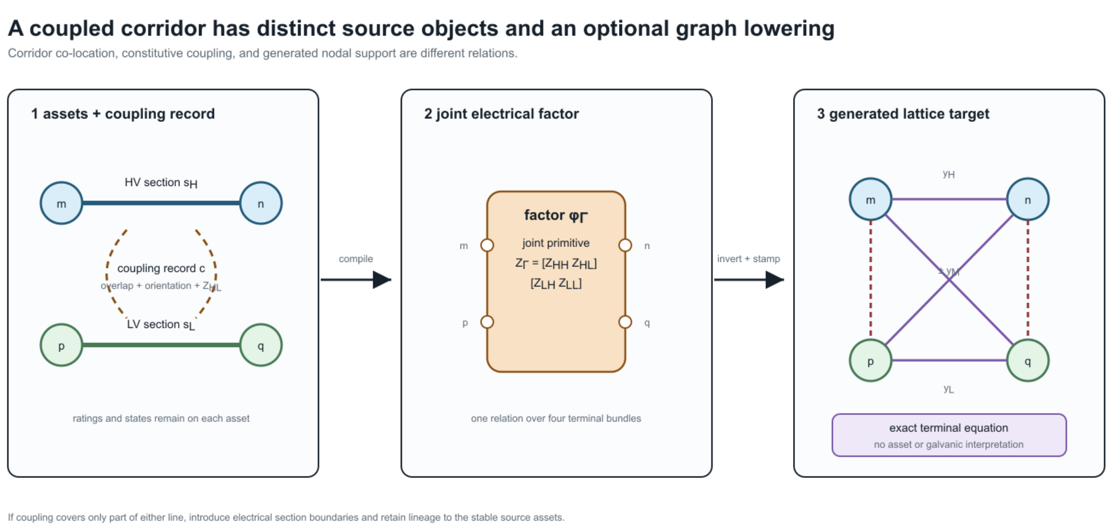

Figure 26.1: Two line assets compile through a joint coupling factor into a generated terminal lattice.

The right-hand lattice is an equation target. Its cross-voltage edges are not conductors, transformer windings, galvanic connections, or additional assets. Their weights can be signed or general complex quantities even when the joint primitive is reciprocal and passive as a whole.

Physical parallelism is not visible in the bus multigraph

Let the high-voltage section s_{H} join buses m and n, and let the lower-voltage section s_{L} join buses p and q. The endpoint pairs need not agree:

$$\partial(s_{\mathrm{H}}) = \{m, n\}, \qquad \partial(s_{\mathrm{L}}) = \{p, q\}.$$

The two sections may even belong to otherwise galvanically disconnected voltage systems. Their spatial co-location is therefore an asset/construction relation, while their mutual impedance is a constitutive relation. Neither is implied by the high-level bus–branch incidence map.

This is a limitation of that projection, not of graph representation in general. Refining each line asset into oriented conductor or wire sections makes the relevant granularity available. A separate **section-coupling graph** can then use those refined section identities as vertices and mutual-coupling records as edges. It captures spatial and constitutive relations that are absent from the bus multigraph, but it is not the original line-asset graph and its coupling edges are not galvanic connections. Nor is it the standard line graph of the bus multigraph: line-graph adjacency means that source edges share an endpoint, whereas section-coupling adjacency means that refined sections participate in a declared electromagnetic coupling relation.

A later weighted-lattice lowering is different again. Its vertices are terminal-voltage coordinates and its edges realize the assembled equation. That graph captures the electrical effect of coupling, not the physical parallelism itself; its generated edges are neither source line objects nor physical wires. The representation progression is therefore

```
line assets and bus attachments
    -> refined conductor sections plus a coupling relation
    -> optional generated equation edges.
```

For data and prose, this book uses the following terms:

- **physically parallel sections** overlap spatially on common towers, poles, trench, or right of way for a declared interval;
- a **mutual-coupling record** relates two oriented sections and carries their cross blocks, coordinate maps, overlap, and provenance;
- a **coupling group** Γ is a connected component of the resulting section-coupling relation; and
- the **joint primitive** $\mathbf{Z}_{\Gamma}$ is assembled for the whole group before conversion to an admittance form.

This ownership rule matters. Storing the same mutual block redundantly in two line records creates update and orientation ambiguity. Inverting each self block separately is also not equivalent to inverting the assembled joint matrix.

The joint series primitive

Let the two circuits have ordered conductor spaces of dimensions r_{H} and r_{L}; the dimensions need not agree. In physical voltage and current coordinates, choose orientations $m \to n$ and $p \to q$ and write

$$\begin{bmatrix} \Delta\mathbf{v}_{\mathrm{H}} \\ \Delta\mathbf{v}_{\mathrm{L}} \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{Z}_{\mathrm{HH}} & \mathbf{Z}_{\mathrm{HL}} \\ \mathbf{Z}_{\mathrm{LH}} & \mathbf{Z}_{\mathrm{LL}} \end{bmatrix}}_{\mathbf{Z}_\Gamma} \begin{bmatrix} \mathbf{i}_{\mathrm{H}} \\ \mathbf{i}_{\mathrm{L}} \end{bmatrix}.$$

Here $\mathbf{Z}_{\mathrm{HL}}$ can be rectangular. For a reciprocal model in compatible physical coordinates, $\mathbf{Z}_{\mathrm{LH}} = \mathbf{Z}_{\mathrm{HL}}^{\mathsf{T}}$. Ground-return, shield, neutral, bundle, transposition, and reduction assumptions belong to the provenance of these blocks, not to the bus graph.

This form covers the series part of a lumped steady-state model. A nominal- π or general fixed-linear terminal factor can additionally have full shunt blocks, including capacitive coupling between circuits. Distributed and frequency-dependent multiconductor models require a frequency-indexed or dynamic terminal relation; they should not be relabelled as this fixed series primitive without a separate approximation contract.

Exact nodal stamping

First take one scalar conductor on each circuit. Stack the terminal voltages as

$$\mathbf{u} = \begin{bmatrix} V_m & V_n & V_p & V_q \end{bmatrix}^{\mathsf{T}}, \qquad \mathbf{A}_\Gamma = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

Then $\Delta\mathbf{v} = \mathbf{A}_\Gamma \mathbf{u}$. If $\mathbf{Z}_\Gamma$ is nonsingular, define

$$\mathbf{Y}_\Gamma = \mathbf{Z}_\Gamma^{-1} = \begin{bmatrix} y_{\mathrm{H}} & y_{\mathrm{M}} \\ y_{\mathrm{M}} & y_{\mathrm{L}} \end{bmatrix}.$$

The terminal-current stamp is

$$\mathbf{i}^{\mathrm{inj}} = \mathbf{A}_\Gamma^{\mathsf{T}} \mathbf{Y}_\Gamma \mathbf{A}_\Gamma \mathbf{u},$$

with nodal block

$$\mathbf{Y}_\Gamma^{\mathrm{N}} = \begin{bmatrix} y_{\mathrm{H}} & -y_{\mathrm{H}} & y_{\mathrm{M}} & -y_{\mathrm{M}} \\ -y_{\mathrm{H}} & y_{\mathrm{H}} & -y_{\mathrm{M}} & y_{\mathrm{M}} \\ y_{\mathrm{M}} & -y_{\mathrm{M}} & y_{\mathrm{L}} & -y_{\mathrm{L}} \\ -y_{\mathrm{M}} & y_{\mathrm{M}} & -y_{\mathrm{L}} & y_{\mathrm{L}} \end{bmatrix}.$$

This is the classical building-block assembly for mutually coupled branches [37, 38]. The same construction extends blockwise to multiconductor circuits.

The generated lattice

For reciprocal $\mathbf{Z}_\Gamma$, the inverse $\mathbf{Y}_\Gamma$ is complex symmetric, so an undirected weighted graph is sufficient. A nonreciprocal primitive requires a directed or more general factor target. Under the book's weighted-Laplacian convention, the scalar nodal stamp is realized by six ordinary edges:

Circuit-theory trap

The following table is a generated scalar equation realization, not a line inventory. If copied out of context, every row must retain its `generated_from` provenance and `asset_interpretation = false`; the cross-voltage rows are not conductors or galvanic connections.

generated endpoint pair	weight
$m--n$	y_H
$p--q$	y_L
$m--p$	$-y_\mathrm{M}$
$m--q$	y_M
$n--p$	y_M
$n--q$	$-y_\mathrm{M}$

The count *six* is scalar-specific. In a coordinate-expanded block realization, the two cross pairs become up to $r_\mathrm{H} r_\mathrm{L}$ cross-coordinate pairs before sparsity is considered, and the self blocks can require further within-circuit coordinate edges.

The sign pattern depends on the two stored orientations. Reversing one section changes the sign of its mutual entries and permutes the terminal labels; it does not change the physical relation when the coordinate maps are transformed consistently.

This proves a scoped statement: an ordinary weighted graph can exactly realize the fixed-linear terminal equation when the joint admittance exists and the target edge library admits the required weights. It does **not** prove that the lattice is the source asset graph. The source branch currents are recovered through

$$\begin{bmatrix} i_\mathrm{H} \\ i_\mathrm{L} \end{bmatrix} = \mathbf{Y}_\Gamma \mathbf{A}_\Gamma \mathbf{u},$$

and member limits, outages, measurements, and decisions must remain attached to those recovered source quantities.

Decision-model consequence

The certificate establishes no OPF, protection, limit, or discrete-state equivalence. Signed generated weights can also violate positivity or convexity assumptions made by graph-based solvers. A decision study must keep the source constraints and decisions, use the current recovery map, and prove its own preservation contract; that test has not yet been run.

If $\mathbf{Z}_\Gamma$ is singular, or if a queried current/state cannot be eliminated, a tableau or direct joint-factor formulation is the faithful target. Singularity is a refusal condition for this admittance/lattice branch, not evidence that the coupled section is invalid. Consequently the lattice cannot be the canonical source representation: a canonical model must still represent a physically admissible coupled section when this optional lowering target is undefined.

Different voltage bases

The joint primitive is simplest in physical units. If per-unit coordinates are required, use a power-dual scaling over the complete coupling group rather than converting the two self blocks independently. With one common power base S_b and compatible phase-voltage conventions, the mutual impedance base between circuits a and b is

$$Z_{\mathrm{b},ab} = \frac{U_{\mathrm{b},a}U_{\mathrm{b},b}}{S_\mathrm{b}}.$$

The corresponding mutual-admittance base is its reciprocal. This cross-base rule preserves reciprocity under a common power base; arbitrary independent power bases need an explicit left/right coordinate transformation. Multi- circuit, multi-voltage line studies use this product-voltage base rather than either circuit's self-impedance base [39].

Partial overlap and state

Coupling is often present over only part of each line. The electrical model must then introduce section boundaries where the coupling starts or ends. The generated section nodes are not new line assets: a lineage map relates every section back to its stable source line. Smearing one mutual value over the uncoupled lengths can change fault currents and relay conclusions [40].

Opening a circuit terminal also does not automatically delete the coupling record. An open or grounded out-of-service conductor can still have induced voltage or current, depending on its terminal constraints and the retained series/shunt model. The coupling relation becomes inactive only when the declared physical or study state removes that interaction; ordinary breaker opening changes a boundary constraint.

Information-model precedent

Engineering exchange and analysis models commonly retain coupling as a separate relation keyed to both lines. CIM's $MutualCoupling$ class carries zero-sequence mutual parameters and four distances locating the coupled regions [41]. PowSyBl exposes a network-level line- coupling extension with two line identities and section intervals [42]. PowerWorld's sequence format similarly records two branch identities, dot convention, mutual impedance, and start/end fractions [43].

These precedents are narrower than the book's source contract: they are often zero-sequence and short-circuit oriented rather than full phase-coordinate series-and-shunt models. Their important architectural lesson is nevertheless general: coupling is a relation between identified sections, not an intrinsic scalar label owned independently by either line.

Pedagogical and evidence boundary

Yan and Saha study the concrete Australian case of an 11 kV three-wire circuit and a 415 V four-wire circuit sharing poles. Their phase-coordinate current-injection model shows that cross-voltage coupling can materially affect low-voltage magnitude and unbalance under plausible geometry and loading [44]. Kersting gives the complementary same-voltage derivation for physically parallel distribution circuits, including series and shunt coupling [45].

The generated `experiments/generated/coupled-corridor-lattice-witness.json` certificate checks joint-matrix inversion, the six-edge lattice stamp, orientation covariance of the mutual term and rebuilt edge table, source-current recovery, cross-voltage per-unit round trips, signed generated weights, and singular refusal. It does not yet reproduce either paper's numerical results or test a limit-, protection-, state-, or decision-preservation contract. The source-relation convention is registered as `COUPLED-CORRIDOR-001`, and the exact fixed-linear lattice lowering as `COUPLED-CORRIDOR-002`. The next executable tranche should add:

1. a deterministic three-wire/four-wire fixture with physical-unit and per-unit round trips; and
2. a geometry-derived 11 kV/415 V case with partial-overlap and open/grounded state probes.

Reproduction

```
julia --project=experiments experiments/run_coupled_corridor_lattice.jl
julia --project=experiments experiments/test/coupled_corridor_lattice.jl
```

Chapter 27

The running multiconductor network

Page status: semantic specification; numerical realization is versioned separately.

The book uses one synthetic reference network to compare representations and transformations. The network is deliberately small enough to draw and solve repeatedly, but it contains the features that disappear in a conventional balanced bus–branch diagram.

This page is the **semantic specification**. Its first numerical realization is the executable running network, version 0.1.0. The semantic contract remains authoritative where that fixture explicitly lists an unimplemented feature.

The layout below is the reference drawing for the rest of the book. Later figures should be read as views or overlays of this same source object set, not as unrelated replacement networks.

Design requirements

The running network must contain:

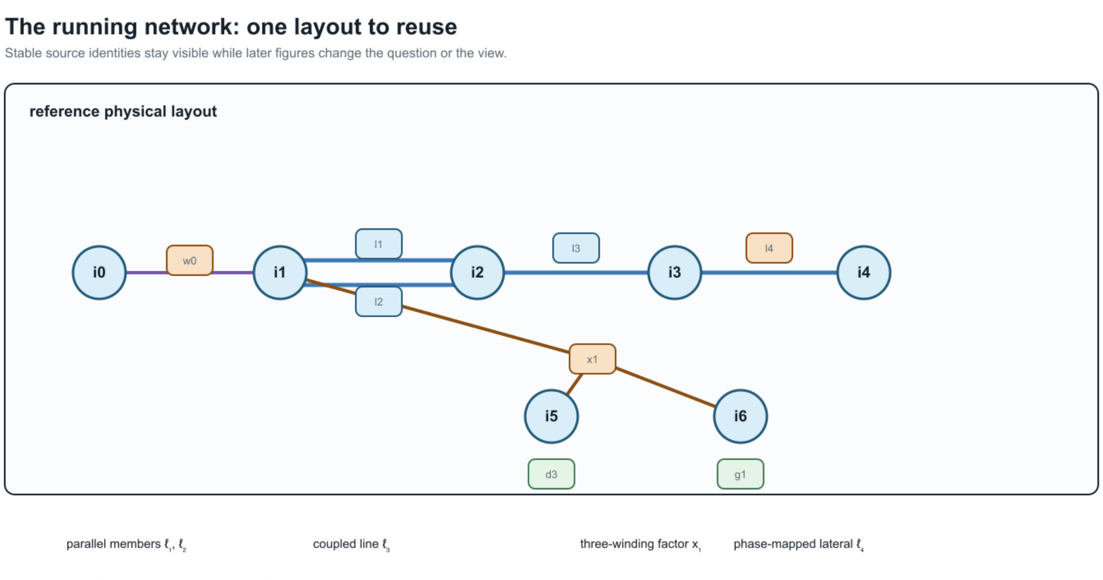

Figure 27.1: Reusable physical layout of the running multiconductor network.

- ordered, nonuniform bus terminal sets;
- a four-conductor line with full mutual coupling;
- an explicit neutral and impedance grounding;
- a conductor permutation or phase discontinuity;
- heterogeneous parallel branches with distinct limits and states;
- an ideal switch represented as an object rather than merged buses;
- a genuinely multiwinding transformer;
- at least one continuous control and one discrete decision;
- measurements and operational limits;
- stable source identities and provenance for every generated object.

Source objects

The provisional bus set is

$$\mathcal{B}^{\mathrm{R}} = \{i_0, i_1, i_2, i_3, i_4, i_5, i_6\}.$$

The superscript R denotes the running case, not an electrical quantity. Bus terminal vectors are:

Bus	Ordered terminals	Role
i_0	$[a, b, c, n]$	source-side substation bus
i_1	$[a, b, c, n]$	switched feeder head and transformer primary
i_2	$[a, b, c, n]$	grounded junction and parallel-corridor receiving bus
i_3	$[a, b, c, n]$	downstream four-wire bus
i_4	$[a, c, n]$	phase-discontinuous lateral bus
i_5	$[a, b, c, n]$	secondary winding bus
i_6	$[a, b, c]$	delta tertiary winding bus

These labels are stable source identifiers. A compiled view may create virtual buses but must not reuse or renumber these identities without a reversible map.

Two-terminal topology

The declared forward topology includes

$$w_0 i_0 i_1 \in \mathcal{T}^{W\rightarrow},$$

where w_0 is an ideal switch, and

$$\ell_1 i_1 i_2,\ \ell_2 i_1 i_2,\ \ell_3 i_2 i_3,\ \ell_4 i_3 i_4 \in \mathcal{T}^{L\rightarrow}.$$

Lines ℓ_1 and ℓ_2 are a heterogeneous parallel pair. They have distinct full impedance matrices, conductor ratings, construction records, outage states, and candidate investment decisions. An aggregated admittance may be a valid terminal-behaviour view while being invalid for the decision problem.

The generated line-identity witness makes the graph consequence explicit. In the scalar line projection of this multiconductor fixture, the four lines have one cycle-space dimension: the chord is ℓ_2 against the declared tree $\{\ell_1, \ell_3, \ell_4\}$, so the cycle is the two-member parallel fibre. Collapsing unordered bus pairs to a simple graph gives cycle rank zero. The switch and the three-winding transformer are deliberately excluded from this line-only calculation; their terminal and factor incidences belong to the port–factor view rather than being silently treated as scalar edges. The executable record is `experiments/generated/running-network-cycle-space-witness.json`.

Line ℓ_3 is a four-conductor section with nonzero mutual impedance and shunt admittance. Line ℓ_4 is a three-conductor lateral. Its terminal maps make the phase selection explicit:

$$\mathbf{N}_{\ell_4 i_3} = [a, c, n], \qquad \mathbf{N}_{\ell_4 i_4} = [c, a, n].$$

The deliberate permutation tests whether an implementation compares conductor coordinates before composing matrices or merging sections.

Grounding

The neutral terminal at i_2 is connected to earth through grounding element h_n with finite admittance y_{h_n}. It is neither a perfect voltage reference nor an annotation on the bus:

$$I_{h_n} = y_{h_n} U_{i_2,n}.$$

The source establishes the network voltage reference through a separately declared grounding condition at i_0. The fixture also contains a small reference shunt h_{ref} on tertiary terminal a at i_6; it is retained as an explicit object rather than folded into the transformer. These distinctions make the neutral-grounding counterexample observable: eliminating i_2 as a zero-injection degree-two junction is invalid if the grounding relation is forgotten.

Multiwinding transformer

Transformer x_1 has three winding ports

$$\mathcal{K}_{x_1} = \{1, 2, 3\}, \qquad \beta_{x_1}(1) = i_1, \quad \beta_{x_1}(2) = i_5, \quad \beta_{x_1}(3) = i_6.$$

The provisional connections are grounded wye, grounded wye, and delta. Each winding has its own ordered terminal map, reference voltage, current limit, and winding identity. Leakage data are specified as pairwise short-circuit impedances referred to a declared winding base. No derivation may silently replace the transformer by three independent two-winding devices.

A compiled loss network may introduce an internal virtual bus and two-terminal branches. Its certificate must map every generated object to x_1 and the relevant winding, preserve the terminal relation, state how winding limits are enforced, and provide a round-trip test. The first exact compilation of the fixture's pairwise leakage data is developed in Multiwinding leakage reference compilation. Its connection to all eleven external winding terminals is developed in Multiwinding terminal leakage assembly. The earlier five-bus multi-port lowering reuses this transformer's checked interface as a structural teaching extension; it does not replace or relocate the numerical running fixture. An explicitly marked illustrative extension then exercises excitation, transformer-internal grounding, and fixed transfer serialization in Fixed-linear transformer factor completion without changing the canonical BMOPFTools fixture. The associated parameterized transformer tap decision case then adds an explicitly illustrative discrete winding control and tests decision recovery without altering the fixture. The transformer tap AC decision case then embeds that factor in explicit network voltage, neutral-KCL, power-balance, and recovered-current constraints.

Nodal elements and controls

The case includes:

- a voltage source s_0 at i_0;
- an unbalanced controllable generator g_1 at i_3;
- an unbalanced four-wire demand d_1 at i_3;
- phase-discontinuous demands d_{2a} and d_{2c} at i_4;
- a secondary demand d_3 at i_5;
- a delta demand d_4 at i_6;
- voltage and current measurements at selected terminals;
- conductor-current limits on every line and winding.

Element connection configurations and terminal maps are explicit. A nodal attachment does not imply balanced three-phase or phase-to-neutral connection.

Decision problems

Continuous base problem

The first executable problem is a fixed-topology multiconductor AC OPF:

$$\min_{z \in \mathcal{F}_{M\mathrm{R}}} c_0(P_{s_0}) + c_1(\mathbf{P}_{g_1}) + c_{\mathrm{loss}} P_{\mathrm{loss}} + c_{\mathrm{shed}} P_{\mathrm{shed}}.$$

The feasible set includes terminal KCL, full conductor-coupled branch relations, transformer winding relations, grounding, voltage limits, per-conductor current limits, device capabilities, and any fixed equipment state.

Discrete extension

A later extension makes selected states decisions:

- open or close w_0;
- remove one member of the parallel pair as a contingency;
- choose a reinforcement or switching action;
- select a discrete transformer or regulator state where the device model supports it.

The continuous problem should be established first so that failures caused by representation are not confused with failures of a mixed-integer solver.

Required views

The case will be published in at least these forms:

View	Mandatory retained facts
Asset/property	stable element and winding identity, provenance, construction, state ownership
Terminal connectivity	ordered terminals, grounding, phase discontinuity, switch state
Port–factor	device relations, arbitrary winding arity, limits, controls
Bus–branch multigraph	parallel identity and compiled virtual-object provenance
Simple graph	explicitly documented forgotten distinctions
OPF model	variables, equations, feasible set, objective, source map
Sparsity graph	variable/constraint blocks with no unsupported physical interpretation

Counterexample variants

Small variants isolate one failure at a time:

1. **R-PAR:** retain only ℓ_1 and ℓ_2 and their endpoint buses; compare aggregate admittance with member current limits and outage decisions.
2. **R-GND:** retain the chain through i_2 and its grounding element; test degree-two elimination with and without the shunt relation.
3. **R-PERM:** retain ℓ_4 and its endpoint terminal maps; test conductor-coordinate normalization.
4. **R-XFMR:** retain x_1 and boundary injections; test multiwinding compilation, winding-limit recovery, and round trips.
5. **R-SW:** retain w_0 and neighbouring connectivity nodes; compare a fixed-state quotient with a model that preserves switching as a decision.

Each variant must be small enough for an independent analytic or exhaustive check.

Preservation questions

Every transformation applied to the running case must answer:

- Are terminal voltages and currents exact over the declared operating domain?
- Are per-conductor and per-winding limits preserved or lifted?
- Are switch, outage, tap, and investment choices retained?
- Is the target feasible set exact, inner, outer, or scenario approximate?
- Is the objective identical after lifting?
- Do optimal target decisions map to admissible source decisions?
- Can eliminated currents, voltages, losses, and active constraints be recovered?
- Are every generated object and parameter traceable to source identities?

BMOPFTools realization

BMOPFTools is the preferred first implementation platform where its component and decision models fit the case. The book-level source specification remains implementation independent. Any BMOPFTools fixture must record:

- the exact repository commit;
- input data and augmentation or conversion steps;
- terminal ordering and units;
- solver and environment;
- the source-to-implementation object map;
- known differences between the semantic specification and implemented model.

Chapter 28

Executable running network

Page status: executable fixture, local solver evidence, and a derived state-conditioned radiality witness; claims are versioned and verification-scoped.

Status and purpose

Empirical artifact, version 0.1.0. The semantic specification now has a numerical realization in `data/running-network/v0.1.0.json`. The fixture is synthetic. Its values are chosen to make representation choices observable, not to recommend equipment ratings or a particular feeder design.

BMOPFTools is the first implementation platform because its model retains ordered conductor terminals, full impedance matrices, explicit grounding, parallel branches, switches, and an independent `n_winding` transformer model. The book-level specification remains implementation independent.

What is realized

Object class	Source identities	Numerical feature under test
buses	$i_0, \dots, i_6$	nonuniform ordered terminal sets
switch	w_0	fixed closed state and per-conductor current limits
parallel lines	ℓ_1, ℓ_2	distinct full impedances and distinct limits
coupled line	ℓ_3	full four-conductor series matrix
mapped line	ℓ_4	$[a, c, n] \mapsto [c, a, n]$ conductor permutation
grounding	h_n	finite neutral-to-earth admittance at i_2
transformer	x_1	WYE–WYE–DELTA three-winding model with winding limits
loads	d_1, d_2, d_3, d_4	unbalanced wye, single-phase, and delta demand
generator	g_1	bounded continuous active and reactive decisions

BMOPFTools requires each single-phase load to be a two-terminal component. The source load d_2 is therefore compiled into `d2a` and `d2c`. That realization is recorded explicitly rather than silently redefining a partial-wye object.

Six views of the same source

The panels are not levels in a single hierarchy. The provenance artifact also contains a non-visual simple-topology quotient used by the cycle and radiality definitions:

1. the asset/property view retains stable physical identity;

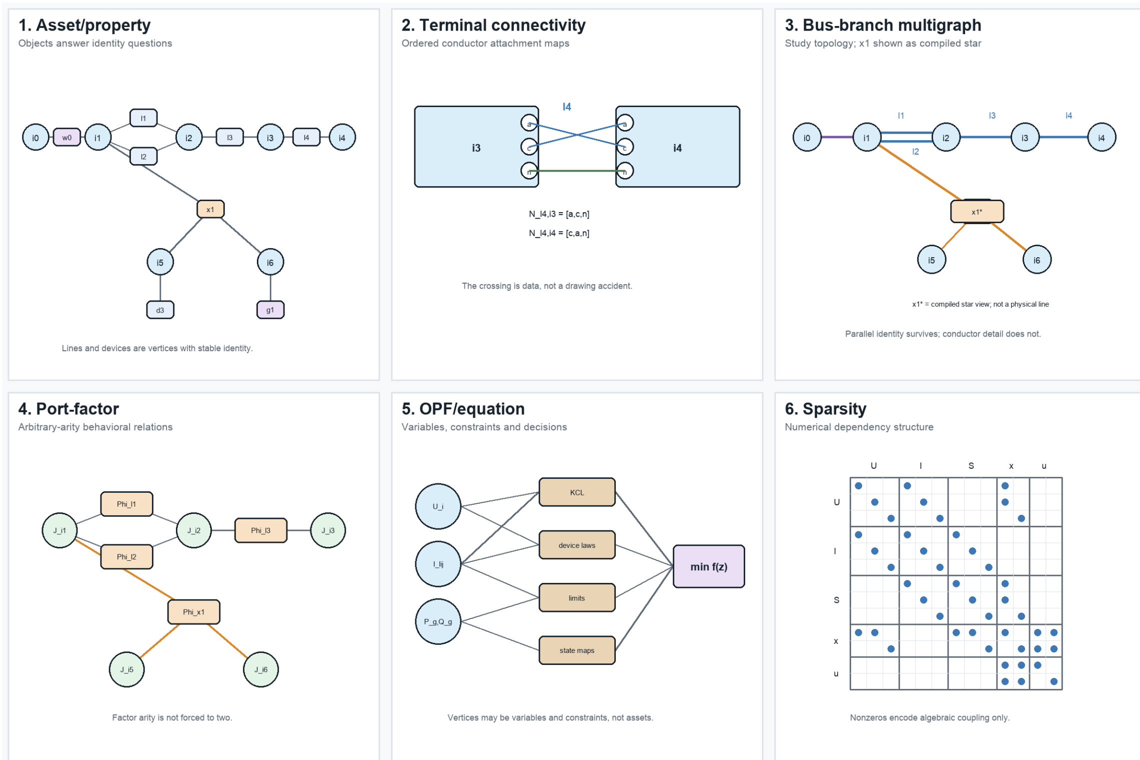

Figure 28.1: Six non-isomorphic views of the running network.

2. the terminal view exposes ordered conductor mappings and grounding;

3. the bus–branch multigraph supports conventional network algorithms while retaining ℓ_1 and ℓ_2 separately. In the figure, x_1^* is an explicitly labelled compiled star used to draw the multi-terminal device; it is not a claim that the transformer is physically three two-terminal lines;

4. the port–factor view represents x_1 as one factor of arity three;

5. the OPF graph makes variables, constraints, limits, and decisions explicit;

6. the sparsity graph records numerical coupling but does not claim that a nonzero is a physical branch.

Every generated view must retain a source map. A label such as `Phi_x1` is a generated factor identity with `source = x1`; it is not a replacement asset ID. The complete maps for the six illustrated views, together with the derived simple-topology quotient, are generated in `experiments/generated/view-source-maps.json`. Automated checks require every generated object to identify an existing fixture source and bind the map to the exact fixture and figure hashes.

The lineage is the operational companion to the view figure: virtual compiled objects are allowed, but source identity, map identity, and recovery remain available for limits, outages, maintenance, and decisions.

Executed checks

The generation script parses the exported BMOPF JSON, runs JSON-schema and model-conformance checks, solves a determined power flow, and solves an OPF with g_1 dispatchable. The current artifact reports:

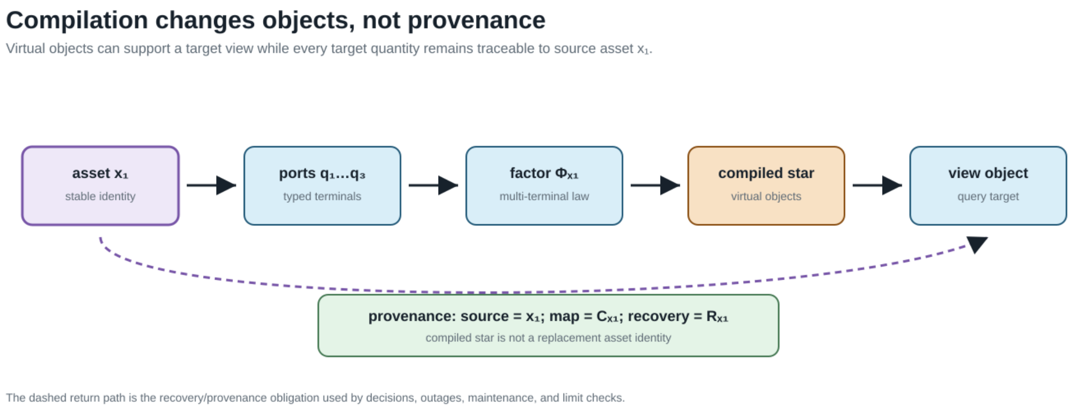

Figure 28.2: Compilation changes objects, not provenance.

Check	Result
schema valid	yes
validation errors	0
validation warnings	0
expected map diagnostic	`I.TMAP.CROSS_PHASE_LINE`
expected map diagnostic	`I.TMAP.PERMUTED_ORDER`
power-flow termination	`LOCALLY_SOLVED`
power-flow active loss	554.317 W
OPF termination	`LOCALLY_SOLVED`
OPF objective	-13.2000

The negative generator cost is a test device that drives g_1 to its active power bounds, making the continuous decision observable. The objective has no economic interpretation. Both the cost and this interpretation must change before the case is used for an economic study.

Reproducibility boundary

Run the complete slice from the repository root:

```
julia --project=experiments experiments/run_vertical_slice.jl
julia --project=experiments experiments/test/runtests.jl
```

The generated provenance record includes Julia and package versions, the local BMOPFTools commit, a tracked-diff hash, and hashes of untracked files. The normal development run records local modifications accurately.

An isolated reproduction now clones the committed BMOPFTools revision into a temporary directory, runs the fixture and all tests there, and verifies that the exported fixture is byte-identical to the canonical artifact:

```
bash scripts/reproduce_clean_fixture.sh
```

Fixture version 0.1.0 passes at clean BMOPFTools commit `b7aa9a1bb48bcc8b790d3bcf5417d6a32036352a`. The clean provenance, validation, PF, and OPF outputs are retained under `experiments/generated/clean-reproduction`.

This establishes a pinned local reproduction at that commit; it is not a tagged BMOPFTools release, a guarantee of bit-identical nonlinear solver iterates, or an independent-solver result.

Version 0.1.0 realization boundary

The versioned fixture is the current numerical contract, not a silent completion of every object in the semantic specification. It includes the declared d_4 delta demand and h_{ref} tertiary reference shunt, and it retains w_0 as a fixed closed switch. It does not add an undeclared g_2 generator or any other convenience object merely to make a continuous solve look more complete. Future fixture versions may promote semantic-only controls, but must change the version, source hashes, and provenance together.

Derived running-network topology states

The generated `experiments/generated/running-network-radiality-witness.json` lifts the topology predicates to this same fixture without changing version 0.1.0's continuous PF/OPF contract. It retains the four line identities, the w_0 switch, the two compiled transformer winding connections, and every ordered terminal map. Four derived states are compared: base switch closed, switch open, l_2 outage, and the combined switch-open/l_2-outage state.

The base and switch-open states remain member-nonradial because l_1 and l_2 are parallel members, even though their simple adjacency projection is radial. Removing l_2 removes that member cycle. The witness therefore reports both predicates and preserves transformer-factor provenance rather than treating the bus quotient as the source topology.

What remains semantic-only

The fixture currently treats w_0 as fixed closed. Discrete switching, contingency selection, investment choices, measurements, and protection zones remain in the semantic specification but are not claimed as executable in version 0.1.0. This boundary prevents a successful continuous OPF from being mistaken for validation of the full decision model.

Chapter 29

Earth, neutral, and reference model classes

Page status: scoped model-class taxonomy with a numerical E_2 explicit-earth/protection witness; asset-aware protection studies remain future work.

Evidence boundary

Scope: reduced-earth, neutral/grounding-factor, and explicit-earth model classes in the declared steady-state fixtures. **Evidence:** definitions, impedance constructions, and scoped numerical witnesses. **Numerical optimality:** no protection or network optimization result is claimed by the taxonomy or fixtures. **Unresolved boundary:** source-faithful earth-return data, asset-aware protection, uncertainty, and independent engineering review.

"Ground" is used for several different objects in power-system models. This book distinguishes a mathematical voltage reference, a neutral conductor, an earth-return path, and a physical grounding asset. A representation must state which of these it contains before a reduction or fault claim is interpreted.

Four distinct meanings

- **Reference:** a gauge choice such as $U_0 = 0$. It fixes coordinates but does not carry current or represent soil impedance.
- **Neutral:** an explicit conductor or terminal with its own voltage, current, impedance, and limits. It may be grounded at one or more locations.
- **Earth return:** a conductive path through soil, ground wire, shield wire, or a reduced earth-impedance model. It can carry current and couple phases.
- **Grounding asset:** an electrode, grid, bond, transformer grounding point, or protection object with identity, state, and maintenance semantics.

Conflating these meanings can make a phase-to-neutral reduction appear exact while deleting a ground-fault path or a neutral-to-earth constraint.

Model classes

The classes are not a strict accuracy ladder. E_1 may be the right model for a study whose observations are terminal voltages, while E_3 is required for a grounding-asset outage decision even if both use the same external admittance.

The ladder is generated by `experiments/render_earth_return_ladder.py`. Its arrows indicate added modelling commitments, not a universal accuracy ranking.

Class	Electrical representation	Typical use	What it cannot answer by itself
E_0 ideal reference	algebraic reference or gauge; no earth current variable	balanced transmission PF/OPF	earth-return current, grounding impedance, fault path
E_1 reduced earth return	earth effects embedded in sequence, Carson, or fitted impedance/shunt blocks	feeder PF, planning, approximate fault studies	identity and state of the physical earth path unless linked
E_2 explicit earth conductor	conductor or port with voltage/current and mutual coupling	four-wire, shield-wire, grounding, protection studies	soil detail beyond the selected conductor model
E_3 asset-aware grounding	E_1 or E_2 plus electrodes, bonds, grids, protection and maintenance relations	switching, outage, protection, asset decisions	none beyond the declared physical and study model

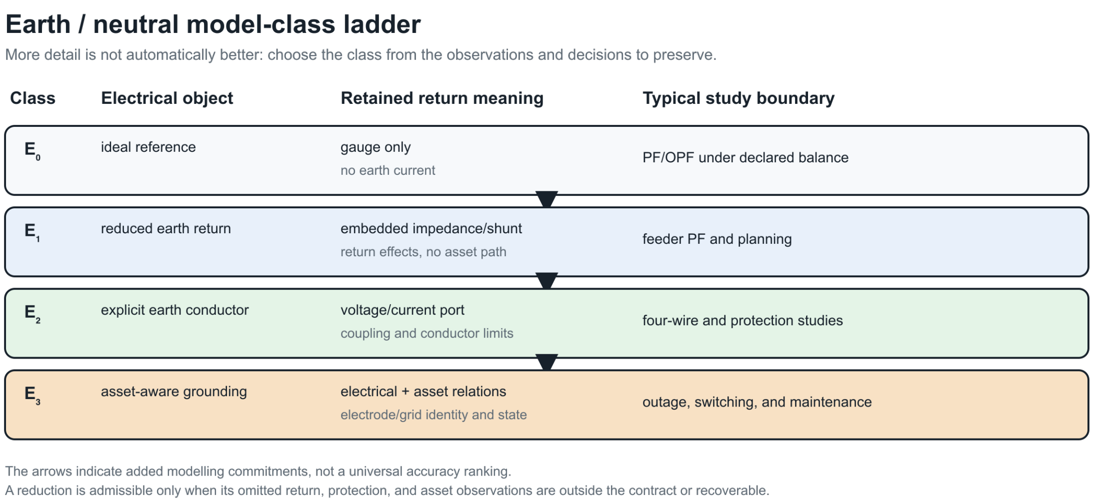

Figure 29.1: Earth and neutral model-class ladder, from an ideal reference to asset-aware grounding.

Scope contract for reductions

A transformation involving neutral or ground must record:

1. the reference convention and gauge;
2. whether neutral and earth are separate ports;
3. the earth-return class E_0–E_3;
4. the number and location of ideal or finite grounding points;
5. whether line shunts, mutual coupling, and ground-fault currents are retained;
6. which voltage, current, protection, and asset observations are preserved;
7. a recovery map for eliminated neutral or earth quantities.

For example, the phase-to-neutral map in the circuit-coordinate chapter can be exact for phase-to-neutral terminal behaviour under sparse compatible grounding, while being insufficient for E_2 neutral-to-earth voltage limits. The Geth–Heidari–Koirala reduction is useful precisely because it states grounding and radiality conditions rather than treating a three-wire picture as an automatic physical deletion [46].

Circuit-theory trap

Setting the reference potential to zero is not the same as setting a conductor voltage to zero, and neither operation removes an earth-return current. A reduced model must say which variable was fixed, eliminated, or merely not represented.

Consequences for the running network

The running fixture uses a reduced-earth E_1 model with an explicit neutral and grounding factor: h_n is a finite neutral-to-earth shunt, the four-wire line retains a neutral terminal, and the source reference is declared separately. It does not contain an explicit earth conductor or earth port, so it is not an E_2 model. The fixture does not claim a detailed soil or electrode model, so grounding-asset decisions remain outside its current numerical scope. Future explicit-earth cases should add a physical earth conductor or grounding-grid factor and test recovery of ground currents, neutral voltages, and protection observations. The separate E_2 witness below provides that minimal explicit-earth check without changing the classification of the running fixture.

A useful comparative case

The BMOPFTools grounding tutorial suggests a compact experiment that fits the book's preservation story. Hold the line matrix, load, and bus graph fixed and vary only the customer-end grounding relation:

State	Added relation	Observations that should change
floating neutral	no local neutral-to-reference path	neutral-to-earth voltage and return-current allocation
impedance grounded	finite electrode shunt	electrode current, neutral displacement, and phase-to-neutral voltage
perfectly grounded	ideal voltage constraint	neutral voltage by construction and a different current path
explicit earth return	earth conductor or factor with its own impedance	earth current, neutral current, and grounding-asset limits

This is not four different graphs in the simple-topology sense. It is one asset/connectivity view with four electrical and decision models. A study that only asks whether buses are connected sees no change; a study that asks for neutral limits, touch voltage, fault current, or electrode maintenance does. The comparison is now instantiated in the scoped artifact `experiments/generated/load-grounding-witnesses.json`. It keeps the same two-conductor bus–branch graph and varies only the customer-end relation. The recorded neutral-voltage magnitudes are approximately 0.14184 (floating), 0.10026 (finite impedance), and 0.00000 (ideal grounding); the corresponding ground-current magnitudes are 0, 0.27807, and 0.88999 per unit. These values make the decision point concrete without claiming that the three-state scalar fixture represents an explicit-earth-conductor or protection study.

The comparison is therefore evidence for the E_0-E_3 scope contract, not a reason to promote “grounded” to a single graph attribute.

Explicit earth conductor and a protection observation

The same generated artifact contains a three-node linear fixture with phase, neutral, and earth conductor voltages. A finite neutral-to-earth bond is retained as a separate relation. Three resolved states are compared:

State	Earth voltage magnitude	Earth-conductor current	Fault current	Protection observation
earth in service	0.04975	0.18540	0	no trip
earth conductor maintenance outage	0.14184	0	0	no trip
phase-to-earth fault	0.52173	1.94437	2.81114	trip
neutral-to-earth fault	0.07851	0.29260	0.25112	no trip

The outage and fault rows show why an explicit earth port cannot be replaced by an ideal reference: the earth conductor has its own current, availability, protection observation, and asset identity. The witness also records earth-node voltage as a touch-voltage observation: the declared 0.10 pu limit passes in service (0.04975 pu) and fails for both the maintenance outage (0.14184 pu) and the faults (0.52173 pu and 0.07851 pu). The earth-conductor maintenance decision is retained with its asset state and cost. Protection uses a CT ratio of 10 and a 0.20 pu secondary pickup: the phase fault measures 0.28111 pu, trips after 0.2466 s under the declared inverse-time curve, while the neutral fault measures 0.02511 pu and does not trip. A separate saturation probe caps the phase-fault secondary current at 0.18 pu and changes that trip decision. These are declared sensitivity models, not relay or CT standards. This is a deliberately small E_2/E_3 boundary witness and does not model relay curves beyond the illustrative curve, CT electromagnetic behaviour, or a complete fault-class enumeration.

Chapter 30

Node–breaker, bus–breaker, and topology processing

Page status: scoped topology-processing definitions with a generated node–breaker switch-state and radiality witness; larger running-network topology lifts remain future work.

This chapter specializes the general view and surgery contract in From source graphs to views and graph surgery. It is authoritative for the node–breaker state processor and its concrete κ/π_σ quotient, while the general view registry, lowering boundary, unknown-state family semantics, and degeneracy diagnostics are defined there. The examples below therefore explain this specialization rather than introducing a competing transformation vocabulary.

The general taxonomy in Maps between representation frameworks types π_σ as an edge-contraction quotient. In particular it is not an ordinary homomorphism between loopless simple graphs, because it deliberately identifies endpoints of closed switch edges. The resulting parallel members and contraction-created loops use the normative conventions in Multigraphs for expert modelers.

Topology processing is a state-conditioned compilation. It is not the same as deleting switches from a graph or replacing every closed switch by a zero impedance line without recording the state that justified the replacement.

Connectivity and topological nodes

Let $\mathcal{T}$ be the set of conducting-equipment terminals and let $\mathcal{C}$ be connectivity nodes. A connectivity model records a partial incidence map

$$\kappa : \mathcal{T} \rightharpoonup \mathcal{C}$$

for zero-impedance terminal connections. Switches, breakers, disconnectors, busbar sections, and jumpers are separate objects with a state σ_e. For a fixed active state σ, form the graph

$$G_{\mathrm{closed}}(\sigma) = (\mathcal{C}, \{e : \sigma_e = \mathrm{closed}\}).$$

The **topological nodes** are the connected components of $G_{\mathrm{closed}}(\sigma)$. Write

$$\pi_\sigma : \mathcal{C} \longrightarrow \mathcal{N}_\sigma$$

for the component quotient. A node–breaker bus–branch view is then compiled by attaching each terminal to $\pi_\sigma(\kappa(t))$ and retaining the conducting equipment whose terminals are connected to distinct or equal topological nodes. In the general registry this is a state-conditioned quotient with provenance; here the notation makes the switch-connectivity relation explicit.

This gives the state-resolved distinction:

Object	Retained meaning
connectivity node	a zero-impedance connection point in the detailed model
switch/breaker	an identified asset with state, control, and protection semantics
topological node	a component of closed connectivity for one declared state
bus–branch bus	a compiled algorithmic node, often a topological node but not necessarily a physical bus

An open switch remains an asset and a possible future action even though it is not an edge of $G_{\text{closed}}(\sigma)$. A closed switch may disappear from the compiled electrical equations for that state, but its provenance and decision identity must remain recoverable.

State-resolved compilation

Let M be a terminal or port–factor model with switch states. A topology compiler is a partial map

$$C_{\text{top}}(M,\sigma) = (G_{\text{bus}}(\sigma), \text{prov}_\sigma, \text{recover}_\sigma).$$

The compiler must declare:

1. the admissible state domain and whether unknown or failed states are allowed;
2. the zero-impedance relation used for contraction;
3. the component algorithm and treatment of isolated terminals;
4. the equipment classes that become bus–branch arcs;
5. the handling of loops, parallel members, multi-terminal factors, and grounding factors;
6. the map from each generated bus or arc to its source terminals and assets;
7. whether the compiled state is fixed, a scenario, or a decision variable.

For a fixed state, contracting closed ideal switches can preserve the declared electrical connectivity relation. It does not preserve a switching decision, protection boundary, maintenance identity, or an open-state contingency unless those are carried by prov_σ and the surrounding model. The general surgery result may be a graph or a family with three-valued connectivity summaries; the present compiler assumes a declared resolved state when it constructs $G_{\text{bus}}(\sigma)$.

Worked contraction state

Consider connectivity nodes a, b, c, d, a switch s_{ab}, and identified non-switch members

$$e_{ac}, \quad e_{bc}, \quad e_{ab}, \quad e_{cd}.$$

When s_{ab} is open, π_{open} leaves all four nodes distinct. When it is closed, the quotient identifies a and b as the topological node $u = [a, b]$. The member images are then

$$e_{ac} \mapsto \{u, c\}, \qquad e_{bc} \mapsto \{u, c\}, \qquad e_{ab} \mapsto \{\{u, u\}\}, \qquad e_{cd} \mapsto \{c, d\}.$$

Thus one state change simultaneously creates a parallel class $\{e_{ac}, e_{bc}\}$ and a graph loop e_{ab}. A loopless simple target would collapse the first pair and omit the loop, but those are two additional information-losing maps after the connectivity quotient. A provenance-complete compiler retains the three source-member identities and records whether the loop image is electrically redundant, represented as another factor, or rejected by the target formulation. It does not call the loop a shunt: both of its source terminals belong to the same quotient node, whereas a shunt has a declared reference or grounding relation.

If one of these source members is a two-terminal π factor, the contraction does not authorize deleting it as a graph loop. Compile its two terminal maps first; under the fixed linear assumptions in Multigraphs for expert modelers, the series contribution may cancel while its endpoint shunts combine into a one-terminal nodal stamp. Other factors may require modified nodal or tableau variables and must remain explicit.

Rooted feeder view after topology processing

After compiling a resolved state, a radiality check may construct a rooted feeder hierarchy. This is a derived map, not a replacement for the node–breaker model:

$$H_\sigma = \left(\mathcal{N}_\sigma, E_\sigma, r_\sigma, \operatorname{par}_\sigma\right).$$

The root r_σ is a declared source topological node and $\operatorname{par}_\sigma$ is defined only when each active component is a tree with one root. If a switching candidate closes a tie, opens a parent branch, creates an island, or introduces multiple sources, the hierarchy must be recomputed or reported as undefined. A frozen parent map must not be used to interpret the new state.

In a meshed candidate, retain a spanning forest and mark the remaining active members as chords. The chords preserve cycle constraints and outage choices; they are not downstream branches merely because an algorithm has assigned them an orientation.

Node–breaker and bus–branch are not competing truths

The node–breaker view is the natural source for topology decisions because it retains switching equipment and detailed connectivity. The bus–branch view is often the natural target for a solved PF or OPF instance because it has fewer nodes and a conventional incidence matrix. The transformation between them is state- and purpose-relative:

The simple graph of topological nodes is a further quotient. It may be useful for islands and partitioning, but it forgets member identity and should not be used as the source of switching or protection decisions.

Question	Node–breaker view	State-resolved bus–branch view
Which breaker is open?	direct	only through provenance/state metadata
Are two terminals connected now?	component query	same query after π_σ
Can a switch be operated?	direct decision object	lost if the quotient is frozen
What is the branch admittance?	attached factor or equipment	compiled arc relation
Are parallel assets distinct?	yes	yes only in a multigraph target
Is a multiwinding transformer native?	terminal/factor object	requires a declared compilation

Generated state witness

The artifact `experiments/generated/node-breaker-state-witness.json` uses four connectivity nodes, three fixed line members, and two switch assets. It enumerates four declared states: both switches open, a closed parallel switch, a closed chord, and an unknown switch. For each resolved state it reports member-radiality, simple adjacency-radiality, compiled-bus radiality, the closed-switch contraction components, and the surviving line members. For the unknown state it enumerates both admissible realizations rather than silently choosing open or closed.

The witness exposes a useful separation:

State	Member-radial	Adjacency-radial	Interpretation
both switches open	yes	yes	resolved tree
parallel switch closed	no	yes	member cycle hidden by simple projection
chord switch closed	no	no	visible adjacency cycle
switch unknown	unknown	unknown	report both admissible realizations

These are state-conditioned predicates, not properties of the equipment inventory alone. The compiled bus quotient can be radial even when the identified-member graph is not, because closed ideal switches contract connectivity components and remove self-loops from the bus view.

The active-state panel makes the qualification explicit: report both the simple-projection predicate and the identified-member predicate, together with the state σ that selects open and closed members.

The same physical drawing is therefore not one graph with four labels: each overlay retains a different object set and answers different queries.

Relation to CIM/CGMES and software topology views

CIM distinguishes equipment terminals and connectivity/topological nodes; the topological-node layer is derived from the currently connected state rather than being a replacement for equipment identity [32, 47]. PowSyBl similarly describes topology views as a state-dependent processing step [33]. These are useful external precedents for κ and π_σ; they do not by themselves specify multiconductor factor equations, member-level ratings, or decision-preserving reductions.

Decision-model consequence

A bus count that falls after switch contraction is not evidence that a switching problem has been preserved. The state and the contraction map are part of the model contract.

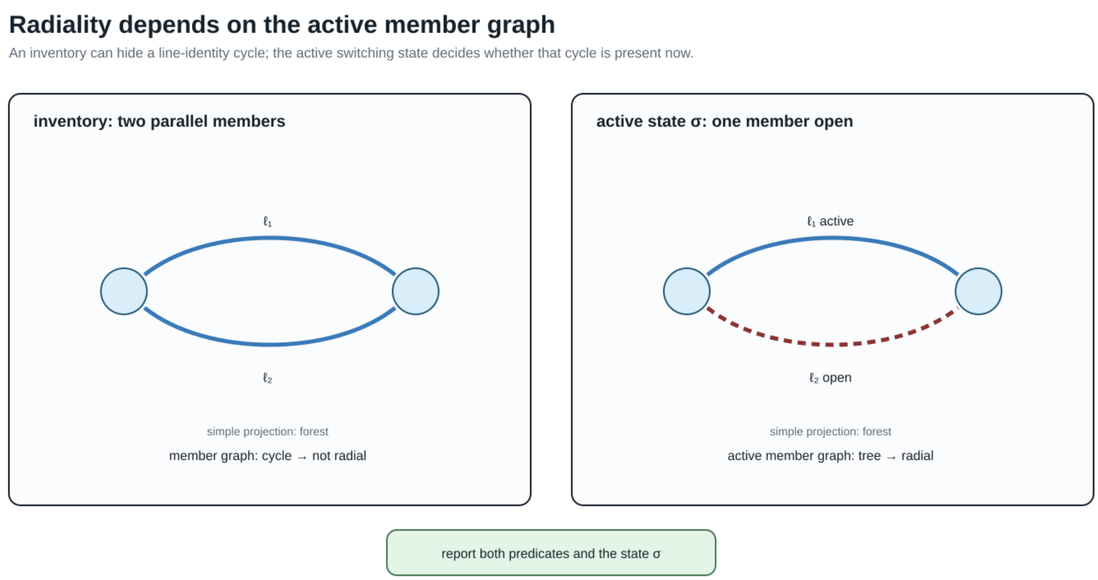

Figure 30.1: Inventory and active-state radiality differ.

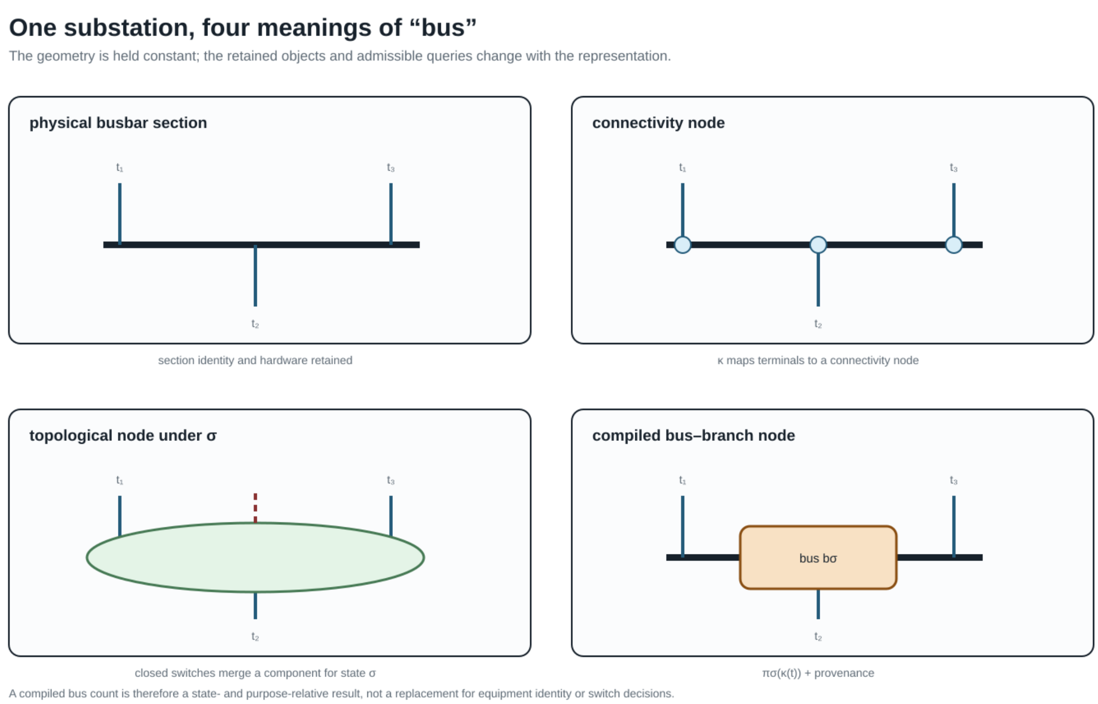

Figure 30.2: One substation shown as four bus representations.

Minimal running-case interpretation

In the running fixture, w_0 is a switch between i_0 and i_1. The fixed closed PF/OPF instance may compile it into a terminal connection, while the discrete extension must retain w_0 as an asset and state variable. The parallel lines ℓ_1 and ℓ_2 remain separate identified factors after topology processing; their common endpoint pair does not authorize aggregation.

Chapter 31

When the general model collapses

Page status: controlled positive-sequence specialization with factor- and network-level executable witnesses; global decision equivalence and external review remain open.

Evidence boundary

Scope: three-phase linear factors and networks with cyclic symmetry, balanced boundary data, and positive-sequence observations. **Evidence:** sequence-coordinate derivation, balanced transmission witness, and independent numerical reimplementation of that fixture. **Numerical optimality:** the witness is not a global nonlinear OPF result. **Unresolved boundary:** exactness for untransposed or unbalanced networks, phase-specific decisions, and external mathematical review.

The general multiconductor, multi-terminal model is a baseline for preserving meaning. It is not a claim that every study needs all of that structure. This chapter gives a positive derivation of the familiar balanced positive-sequence bus–branch model and lists the assumptions that make the collapse admissible.

The balanced invariant subspace

Let a three-phase two-terminal factor use the phase order (a, b, c) and the Fortescue matrix $\mathbf{F}$

This is the classical symmetrical-coordinate construction introduced by Fortescue [48]. The transform itself is always invertible; sequence decoupling is not. Decoupling requires the phase-domain operator, including connected series/shunt factors and grounding, to leave the sequence subspaces invariant.

$$\mathbf{v}_{abc} = \mathbf{F}\mathbf{v}_{012}, \qquad \mathbf{F} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}, \qquad a = e^{\mathrm{j}2\pi/3}.$$

The positive-sequence subspace is

$$\mathcal{V}_+ = \{\mathbf{F}[0, V_1, 0]^\mathsf{T} : V_1 \in \mathbb{C}\}.$$

An exactly balanced operating point has phase voltages and currents in this subspace, with the corresponding phase shifts. Balance is a restriction on the admissible state and injections; it is not created by drawing one edge per bus pair.

Factor-level assumptions

Consider a linear series or nominal-π factor with phase-domain relation $\mathbf{i} = \mathbf{Y}\mathbf{v}$. A sufficient exact-collapse contract is:

1. **compatible phase sets:** every retained terminal has the same ordered three-phase set, with no unresolved neutral or earth port;
2. **cyclic symmetry:** each series and shunt matrix commutes with the cyclic phase permutation, equivalently it is circulant in the declared phase coordinates;
3. **balanced boundary data:** sources, injections, and measurements are restricted to $\mathcal{V}_+$;
4. **sequence-compatible grounding:** zero- and negative-sequence return paths are either absent from the study or represented by fixed factors whose constraints are not queried;
5. **two-terminal closure:** each device is already a two-terminal factor, or a multi-terminal device has an explicitly verified positive-sequence compilation;
6. **decision symmetry:** controls are common to all phases, and limits, costs, and admissible states are invariant under the same phase symmetry;
7. **observation restriction:** the study does not ask for phase-specific, neutral-to-ground, negative-sequence, zero-sequence, or internal winding quantities.

Under these assumptions, $\mathbf{F}^{-1}\mathbf{Y}\mathbf{F}$ is diagonal in sequence coordinates, and the positive-sequence block is invariant:

$$\mathbf{Y}_{012} = \mathbf{F}^{-1}\mathbf{Y}_{abc}\mathbf{F} = \mathrm{diag}(Y_0, Y_1, Y_2), \qquad I_1 = Y_1 V_1.$$

Transposition alone is therefore not an exact positive-sequence reduction. It can support a balanced or approximately cyclic model, but exact restricted closure still requires the operator, grounding, devices, limits, controls, and observations to respect the declared sequence symmetry.

For a nominal-π factor, the same statement applies to the series and shunt blocks separately. The positive-sequence network is therefore a derived two-terminal scalar complex network with one voltage and current per bus/arc, provided the factor library and decision constraints close under the restriction.

The generated witness `experiments/generated/positive-sequence-collapse-witness.json` diagonalizes one circulant impedance matrix to numerical precision and records a non-circulant perturbation that mixes sequences. The companion `experiments/generated/balanced-transmission-witness.json` assembles a three-bus, two-arc nominal-π network with the same cyclic series and shunt factors. It solves the reduced phase-domain equations and the scalar positive-sequence equations independently, embeds the scalar voltages and branch currents, and checks their residuals. The result is a network-level positive example for the declared balanced observation family, not a global decision-equivalence theorem.

The generated comparison `experiments/generated/balanced-transmission-independent-reproduction.json` repeats the phase-domain and scalar solves with a separate standard-library complex Gaussian-elimination implementation. It matches the Julia voltages, residuals, and branch-current checks row-for-row. This is independent numerical reproduction of the declared fixture, not independent review of the modelling assumptions or a standards-aligned transmission validation.

The geometry is a projection aid: the circulant factor leaves the positive-sequence axis invariant, while the perturbed factor produces an off-axis residual. The residual ρ_+ is a coordinate diagnostic until propagated through constraints and decisions.

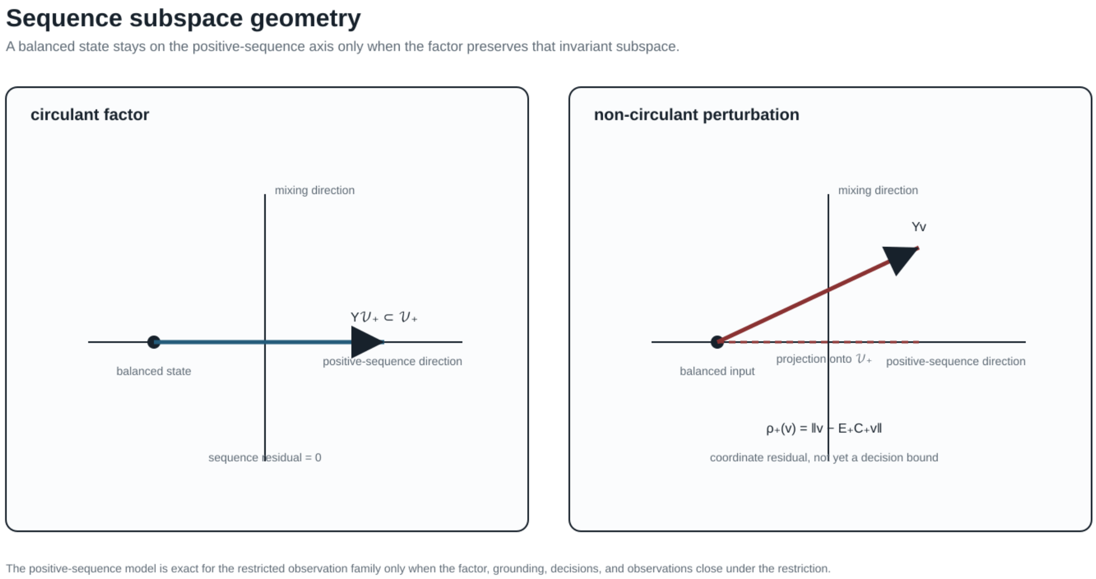

Figure 31.1: Positive-sequence invariant subspace and sequence mixing.

Network-level derivation

Let C_+ restrict a general model to the positive-sequence subspace and let E_+ embed a positive-sequence state into phase coordinates. Under the factor, boundary, and decision assumptions above,

$$\mathcal{F}_+ = C_+(\mathcal{F}_{abc}), \qquad E_+(\mathcal{F}_+) \subseteq \mathcal{F}_{abc},$$

and the declared observations agree on the embedded balanced states as

$$h_{abc} \circ E_+ = \widehat{h}_+.$$

This is a **restricted** factorization on $E_+(\mathcal{F}_+)$. It is not the stronger global identity $h_{abc} = \widehat{h}_+ \circ C_+$ on arbitrary phase-domain states.

Thus the positive-sequence model is exact for the restricted observation family H_+. It is not exact for the unrestricted phase-domain feasible set unless every feasible phase-domain point is balanced, which is a much stronger statement.

For an approximately balanced model, define the residual of a phase-domain state v by

$$\rho_+(\mathbf{v}) = \|\mathbf{v} - E_+ C_+ \mathbf{v}\| \,.$$

This is a coordinate residual, not a decision-error bound. A voltage residual must be propagated through the factor equations and constraints before it can support an engineering approximation claim.

General object	Positive-sequence image	Required qualification
three phase voltages/currents	one complex sequence variable	only balanced states are represented
circulant series and shunt matrices	scalar Y_1 blocks	non-circulant coupling produces sequence mixing
identical phase limits and common controls	one sequence limit/control	phase-specific constraints are outside the image
two-terminal factor	oriented bus–branch arc	multi-terminal devices need a verified compiler
neutral/earth factor	omitted or externally resolved	grounding and zero-sequence questions are forgotten
phase-specific measurements	aggregate sequence observation	not preserved by the quotient

What collapses, and what does not

The running fixture intentionally fails several guards: it has four-wire lines, an explicit grounded neutral, nonuniform terminal sets, a phase permutation, unbalanced loads, full coupled matrices, and a three-winding transformer. It is therefore a test of the general model, not a balanced positive-sequence witness. The transmission specialization must be a separately declared fixture or a parameterized subcase with its own residual and checks.

Power-system shorthand

"Use a positive-sequence model" is a modelling decision with assumptions, not a graph-theoretic simplification. State the balance, transposition, grounding, equipment, limit, and observation assumptions before treating the resulting bus–branch graph as exact.

Decision consequence

The positive-sequence collapse is exact for a restricted decision problem when the feasible-set inclusion, recovery embedding, and observation factorization above hold. If a contingency opens one phase, a relay observes zero sequence, a transformer has phase-dependent taps, or a conductor limit becomes active, the decision domain has left $\mathcal{F}_+$. The general port–factor model or an explicitly guarded intermediate model is then required.

Part IV

Preservation contracts

Chapter 32

Preservation contracts

Page status: foundational definitions; certificate requirements are normative proposals.

Evidence boundary

Scope: observation-indexed feasible-set equivalence and preservation contracts for declared model classes. **Evidence:** definitions, linear recovery derivations, and scoped certificate examples. **Numerical optimality:** certificate examples are not global optimality proofs. **Unresolved boundary:** general nonlinear, mixed-integer, uncertainty-aware, or externally reviewed preservation theorems.

Observation precedes equivalence

Two models are not simply "equivalent." They are equivalent with respect to a chosen observation map and admissible input set.

Let model M define feasible internal and boundary variables $(x, z) \in \mathcal{F}_M$ and let H be a declared family of observation maps. A reduction from M to $\widehat{M}$ is exact for H if, for every admissible input or decision $u \in \mathcal{U}$ and every observation $h \in H$, the observable feasible sets agree:

$$\{h(x, z, u) : (x, z, u) \in \mathcal{F}_M\} = \left\{\widehat{h}(\widehat{x}, u) : (\widehat{x}, u) \in \mathcal{F}_{\widehat{M}}\right\}, \qquad h \in H,\ u \in \mathcal{U}.$$

This definition deliberately includes feasible sets, not only an unconstrained terminal map. It can therefore distinguish electrical equivalence from optimization equivalence.

Decision-model consequence

Equality of a nodal admittance, a boundary voltage relation, or a sampled state trajectory does not establish equality of constrained feasible sets. Limits, controls, discrete states, objectives, and recovery are separate observation families.

The card is generated by `experiments/render_preservation_contract_card.py`. It is a compact reading aid for the certificate fields below; it is not a replacement for the quantified definition or the machine-readable schema.

The book now separates four layers that are often collapsed under the word “preservation”: typed structure, constitutive behaviour, decision semantics, and provenance/ownership. See Transformation semantics and register for the definitions, closure rules and anti-pattern register. A certificate should be explicit when it preserves only one of these layers.

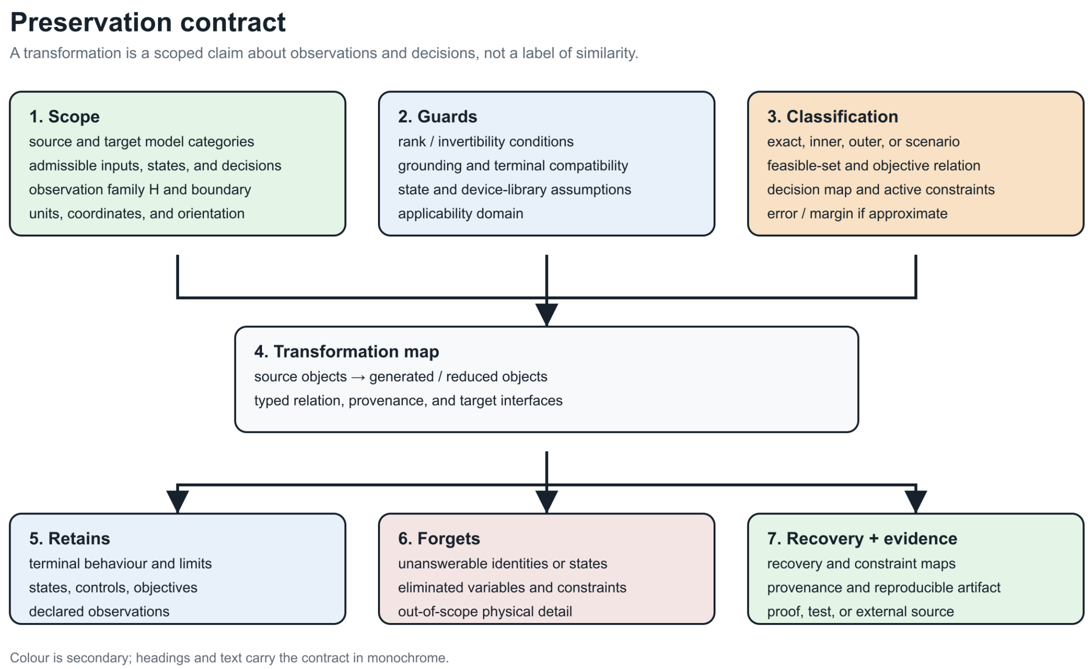

Figure 32.1: Preservation contract card: scope, guards, classification, retained and forgotten meaning, recovery, and evidence.

Preservation dimensions

A transformation certificate should identify which dimensions it preserves:

$$\begin{aligned}\Sigma = \{ &\text{connectivity},\ \text{terminal behavior},\ \ \text{phase/neutral behavior},\\ &\text{asset identity},\ \text{limits},\ \ \text{switching states},\ \text{measurements},\\ &\text{protection},\ \text{spatial provenance},\ \ \text{dynamics}\}.\end{aligned}$$

This is not a binary checklist. Each item needs a precise scope. For example, "terminal behavior" might mean:

- linear phasor behavior at one frequency;
- nonlinear steady-state behavior;
- an impedance function over a frequency interval;
- time-domain behavior for a specified initialization class;
- a scenario-bounded approximation to voltage magnitudes only.

Proposed certificate schema

The field table in this section is a **non-normative v1.2 proposal**. The machine-checked repository schema remains version 1.1.0 and deliberately does not yet require an $error_bound$ field or a normalized numerical-evidence object. Implementations must validate against the versioned JSON schema; the proposal below is a compatibility target for a future schema revision, not an implicit requirement on current certificates.

Every transformation record should contain:

Field	Meaning
`source_type`, `target_type`	model categories and schema versions
`rule_id`	stable identifier for the rewrite or compiler rule
`scope`	source objects affected
`preconditions`	structural, physical and state assumptions
`preserves`	formal or testable preservation claims
`forgets`	questions no longer answerable
`provenance`	source-to-target and target-to-source object maps
`recovery_map`	reconstruction of eliminated variables where possible
`constraint_map`	exact, conservative or approximate lifting of limits
`error_bound` **(v1.2 proposal)**	norm, domain and bound for approximate transformations
`evidence`	theorem, derivation, test suite, or external reference

Exact, conservative, and approximate maps

Operational constraints deserve a classification independent of the equations:

- **Exact:** reduced and original feasible observable sets coincide.
- **Inner/conservative:** every reduced feasible point lifts to an original feasible point, possibly excluding valid original points.
- **Outer/relaxed:** every original feasible observation is retained, but the reduced model may admit non-physical points.
- **Scenario approximate:** accuracy is measured over a specified sample or uncertainty distribution.

A claim that a reduction "preserves limits" is incomplete without identifying which of these meanings applies.

This is the visual version of the classification. The sets are observed sets, not raw internal state spaces: the observation map h is part of the contract. The $15V$ point is the scalar parallel-line witness from the first failure case; it lies in the outer target's extra region and therefore cannot be called a preserved member limit.

Recovery maps are first-class

If internal variables are eliminated by a Schur complement, their recovery is often available algebraically. For a partitioned linear nodal model,

$$\begin{bmatrix} i_B \\ i_I \end{bmatrix} = \begin{bmatrix} Y_{BB} & Y_{BI} \\ Y_{IB} & Y_{II} \end{bmatrix} \begin{bmatrix} v_B \\ v_I \end{bmatrix},$$

and $i_I = 0$, the internal voltage is

$$v_I = -Y_{II}^{-1} Y_{IB} v_B.$$

The left panel is the positive mechanism behind an exact lifted decision: solve or reduce in the target variables, reconstruct $z = R(\widehat{x})$, and evaluate the source constraints on the recovered quantities. The right panel is the failure mode. A target boundary relation can still be correct while the source member limits are uncheckable, so the target feasible set may be an outer relaxation.

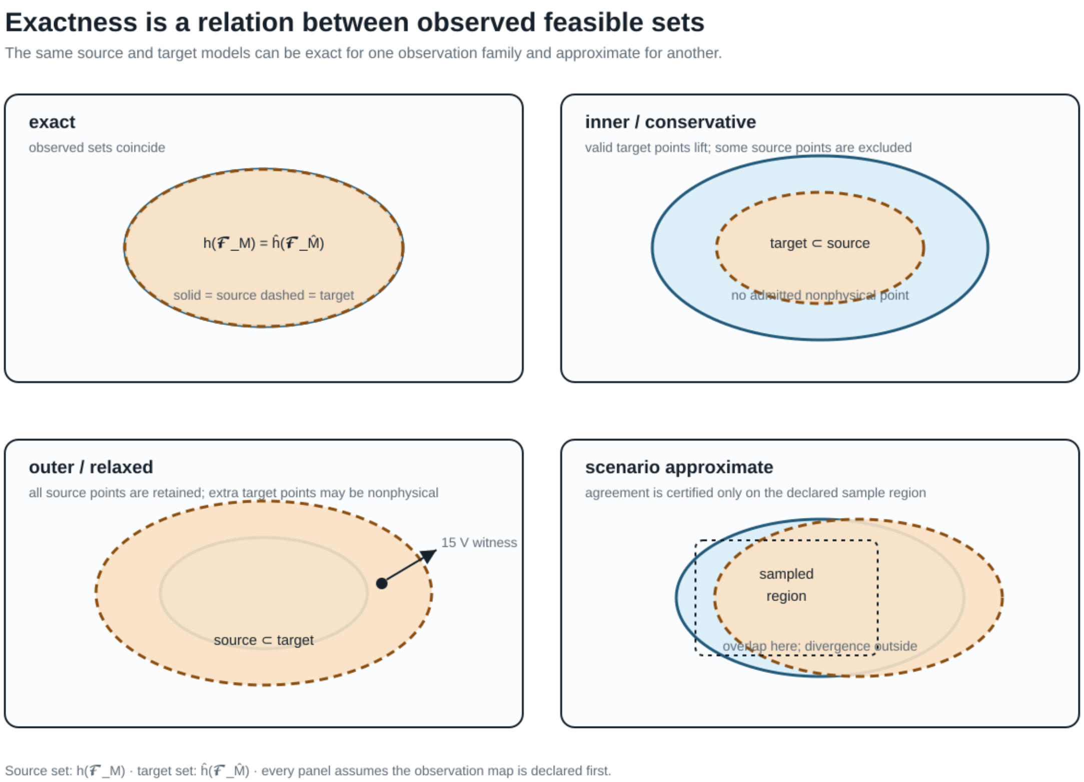

Figure 32.2: Exact, inner, outer, and scenario-approximate observed-set relations.

The reduced admittance is

$$Y_{\text{red}} = Y_{BB} - Y_{BI} Y_{II}^{-1} Y_{IB}.$$

Kron reduction preserves the selected boundary relation, and is well characterized for loopy Laplacians [10]. It does not by itself preserve equipment identity, sparsity, protection zones, or individual branch limits. Storing the recovery operator permits evaluation of some original quantities without pretending the reduced branches are physical assets.

Compositionality requirement

A strong preservation framework should be compositional: replacing a subsystem by a certified equivalent should remain valid when that subsystem is connected to an admissible environment through its declared ports. The black-box functor for passive linear networks formalizes this principle for terminal current–potential relations [49].

For power systems, the open problem is to extend this idea to typed conductors, nonlinear devices, discrete controls, limits, uncertainty, and decision variables without losing useful computational structure.

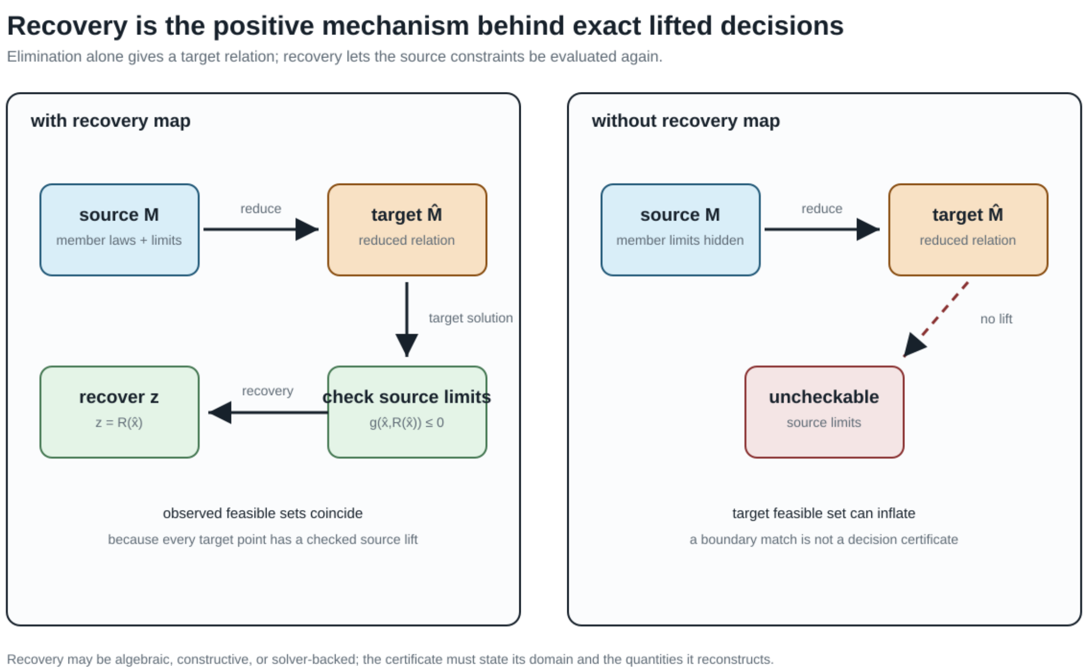

Figure 32.3: Recovery-map loop with and without source-constraint evaluation.

Chapter 33

Transformation semantics and register

Page status: normative vocabulary and first reader-facing transformation register.

The word *equivalent* is too small for the transformations in this book. A rewrite can preserve a terminal equation while deleting an asset, changing a relaxation, or losing a current limit. This chapter gives the vocabulary used to say exactly what survived.

Vocabulary bridge

Graph coarsening and learned pooling belong in this register when their outputs are used to answer a physical or decision query. Permutation equivariance can preserve the effect of renaming source objects; it does not by itself preserve member identity, equipment states, limits, feasible decisions, or recovery after a many-to-one pooling map.

Four preservation layers

Write a model as

$$M = (\mathsf{S}, \mathsf{R}, \mathsf{C}, \mathsf{P}),$$

where S is typed structure (objects, ports and incidence), R is the constitutive and control relation, C is the constraint and decision contract, and P is provenance and ownership.

A transformation $T : M \to \widehat{M}$ should state which of these layers it preserves:

Layer	The question a certificate must answer
Structure	Are object sorts, terminal identities, incidence, orientation and hierarchy preserved up to a declared map?
Behaviour	Do the declared terminal, current, voltage, power or dynamic relations agree?
Decision	Do feasible decisions, objectives, active limits and recoverable source quantities agree?
Provenance	Can the target still answer which source asset, winding, terminal or owner a quantity belongs to?

These layers are independent. A Kron reduction can preserve a boundary relation while changing structure and provenance. A coordinate permutation can preserve all four layers. A summed parallel limit can preserve an unconstrained terminal relation while changing the decision layer.

Structure-preserving versus structure-changing

Call T **structure-preserving** on a declared structure signature when there are maps on each object sort and port set that are bijective on the retained objects and commute with typed incidence, terminal maps and hierarchy. In other words, the diagram of objects and relations commutes after a change of names or coordinates. A permutation of conductors, for example, is a structure-preserving normalization when the matrices, terminal maps, limits and inverse permutation are transformed together.

An identification, elimination, aggregation or asset composition is instead **structure-changing**. It may still be exact for a smaller observation family, but it must name what was forgotten and what target factor class is allowed. The following distinctions are therefore deliberate:

Classification	Meaning	Typical example
Exact normalization	Same typed object, different coordinates or names	conductor permutation
Exact compilation	Different vocabulary, same declared relation and mapped constraints	a fixed multiwinding factor compiled into terminal factors
Exact behavioural reduction	Hidden variables or objects removed, with a recovery/observation contract	guarded pure-series elimination
Inner restriction	Target lifts to the source, but some valid source points are removed	fixing a tap or retaining only a conservative limit
Outer relaxation	Every source observation remains, but target may admit non-liftable points	summing member thermal ratings
Scenario approximation	Agreement is bounded only on a declared scenario or operating domain	operating-point Ward equivalent
Invalid composition	No declared factor class or a precondition is false	absorbing an external ground into a line or transformer without a contract

The last row is important: an algebraic matrix product is not automatically a well-typed network transformation.

Closure is part of the theorem

Suppose $\mathcal{L}$ is the library of physical line factors. A series relation may be exact in a larger library $\mathcal{F}$ of generic passive two-ports while failing to lie in $\mathcal{L}$. The correct statement is

$$T : \mathcal{L} \times \mathcal{L} \longrightarrow \mathcal{F}, \qquad T(\ell_1, \ell_2) \notin \mathcal{L} \text{ in general.}$$

For a nominal-π section, cascading two sections generally produces a general two-port rather than the naive nominal-π parameter sum. For a transformer, the target must retain turns ratio, winding incidence, grounding, excitation, taps and ratings. For an external grounding impedance, the target must retain the ground asset and its current observation unless the study explicitly declares those facts out of scope.

The first transformation register

This table is the reader-facing view of the certificate schema. It is kept small and explicit; generated certificates remain the evidence source.

Every row should eventually be backed by a certificate with `preconditions`, `preserves`, `forgets`, `constraint_map`, `recovery_map`, `target_type` and `evidence`. The register is therefore a navigation layer over the existing certificate system, not a competing schema.

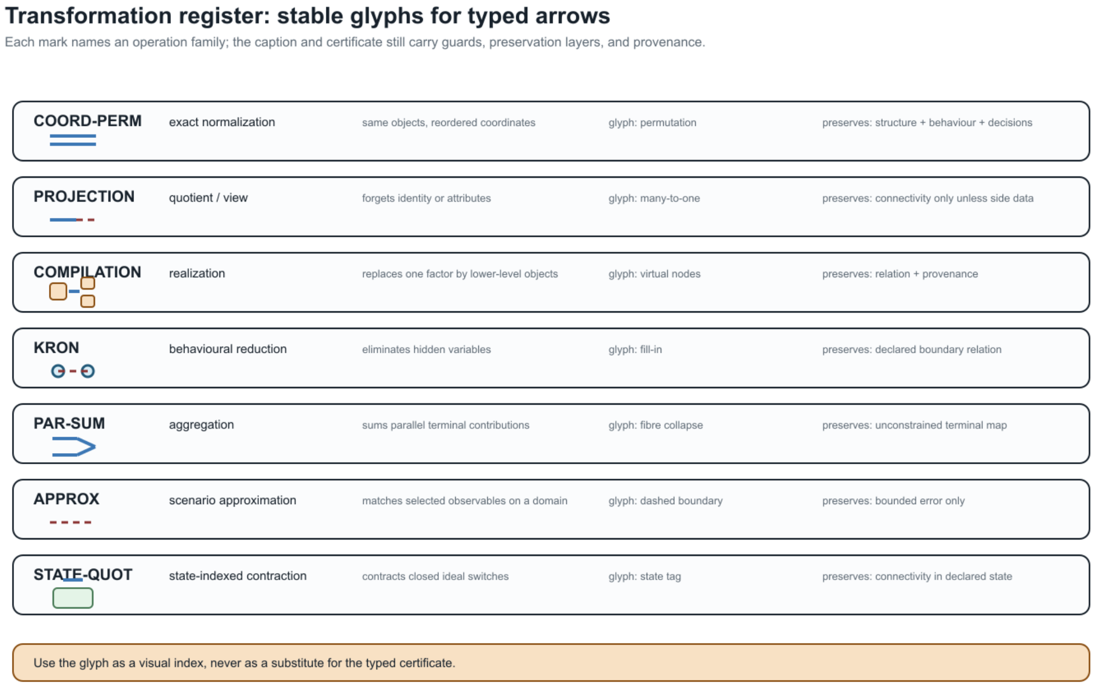

Figure 33.1: Typed transformation-register glyphs used as a visual index for operation families.

The glyphs are deliberately low-bandwidth. A solid or dashed arrow, a split factor, a fill edge, or a state tag helps the reader find the relevant row, but it never carries the guard or preservation theorem by itself. Those remain in the register and in the executable certificate.

Narrow circuit transformations: star–delta and shunt placement

The register also covers transformations that power engineers routinely use inside a line, load, or feeder model without calling them *graph transformations*. They still change the factorization and therefore need the same preservation language.

Scalar floating star–delta

For three scalar impedances z_a, z_b, z_c connected from terminals a, b, c to a **floating** internal star point, the terminal behaviour can be represented by a delta with

$$z_{ab} = z_a + z_b + \frac{z_a z_b}{z_c}, \qquad z_{bc} = z_b + z_c + \frac{z_b z_c}{z_a}, \qquad z_{ca} = z_c + z_a + \frac{z_c z_a}{z_b}.$$

The inverse uses the corresponding nonzero delta-sum denominators, for example

$$z_a = \frac{z_{ab} z_{ca}}{z_{ab} + z_{bc} + z_{ca}}.$$

These are exact terminal-equation transforms for a linear scalar network when the required impedances and sums are nonsingular. They are not automatically exact for a constant-power load, a switched capacitor bank, a grounded star point, branch-specific current limits, or a protection/measurement boundary. Those quantities either need recovery maps or remain explicit factors.

The familiar formulas also do not lift by replacing each scalar with an arbitrary coupled matrix: matrix products need not commute, and a matrix star point may have a different terminal coordinate space at each arm. Use a block Schur complement or a typed port–factor relation instead. A commuting or scalar-block specialisation can be declared separately.

Star and delta have different graph shapes: a star is a tree through an internal point, while a delta contains a three-edge cycle. That cycle can be a compilation artefact, not evidence of an additional physical corridor cycle. Conversely, eliminating a floating star point can hide a source identity or a grounding location. `STAR-DELTA` therefore preserves a declared terminal relation, not the source topology or asset semantics by default.

Endpoint shunts and tool asymmetry

For a two-terminal series factor with distinct endpoint shunts, write

$$
\begin{aligned}
I_{ij} &= Y_s(U_i - U_j) + Y_i^{\mathrm{sh}} U_i, \\
I_{ji} &= -Y_s(U_i - U_j) + Y_j^{\mathrm{sh}} U_j.
\end{aligned}
$$

The endpoint shunts may be unequal. A reciprocal primitive can therefore have different diagonal endpoint blocks while its off-diagonal blocks remain transpose paired. Endpoint asymmetry is not the same thing as non-reciprocity. Replacing Y_i^{sh} and Y_j^{sh} by their average because a software line object has one shared shunt field is an approximation (or an invalid encoding), not a coordinate change.

Absorbing a capacitor bank, a neutral grounding point, a magnetizing branch, or another explicit shunt into a line or transformer can preserve a fixed linear terminal equation. It can nevertheless delete the shunt's switching state, rating, owner, protection boundary, frequency dependence, or neutral/ earth-current observation. A safe lowering target is either an endpoint- augmented generic two-port with both shunts retained in provenance, or separate explicit shunt factors. If the target tool cannot represent unequal from/to shunts, retain the richer source and report the adapter limitation; do not silently symmetrize.

Power-system shorthand

"Put the capacitor/grounding into the line" can mean an exact fixed-state assembly of a nodal operator, or it can mean changing the equipment model. State which one is intended, and keep the shunt asset and recovery map when its state, limit, protection, or neutral-current meaning is in scope.

The same warning applies when a delta load is replaced by a wye load: the linear floating terminal relation may be preserved, while grounded-neutral current, phase-specific limits, unbalance, and switching semantics are not.

The executable companion at experiments/generated/narrow-circuit-transformations-witness.json checks the floating scalar identity, rejects a grounded-star use of that rule, and quantifies the residual introduced by an adapter that replaces unequal endpoint shunts with one shared field.

Risk-aware impedance paths

The four-wire impedance ladder provides a compact example of how the register should be used. Starting from a coupled conductor primitive, a path may contain the following edges:

$$
\hat{\mathbf{Z}}_\ell^{\mathrm{circ}} \xrightarrow{K_g} \bar{\mathbf{Z}}_\ell^{\mathrm{cond}} \xrightarrow{K_n \text{ or } P_n} \mathbf{Z}_\ell^{\mathrm{phase}} \xrightarrow{F} \mathbf{Z}_\ell^{012} \xrightarrow{D \text{ or } F_1} \mathbf{Z}_\ell^{\mathrm{restricted}}.
$$

The path should carry a *risk vector*, not a single score. For each of the four preservation layers, record one of `exact`, `guarded`, `bounded`, `not-preserved`, or `unknown`. For example:

This makes transformation sequences auditable. A later rule may not silently upgrade an earlier `not-preserved` layer to `exact`; composition must carry the weakest status and the union of unresolved guards. The generated `experiments/generated/four-wire-impedance-model-ladder.json` artifact records this path and checks that every edge has explicit risk tags. It also composes the main $K_g \to K_n \to F \to D \to F_1$ route and the phase-to-neutral branch. The composition record carries the weakest exactness label, the union of guards that have not been discharged, and the union of forgotten observations and risk tags. A deliberately mismatched source/target pair is rejected by the same executable helper, so a visually plausible chain cannot silently become a proof.

Anti-patterns worth showing explicitly

Adding different lines in series

Different impedances are not by themselves a counterexample. With a zero-injection, shunt-free junction and compatible conductor coordinates,

$$\mathbf{Z}_{\mathrm{eq}} = \mathbf{Z}_{\ell_1} + \mathbf{P}^{\mathsf{T}}\mathbf{Z}_{\ell_2}\mathbf{P}$$

is an exact behavioural composite. Calling that composite a longer instance of one physical line class is a separate claim. A different construction code, frequency basis, terminal map, rating owner, splice or protection boundary blocks that stronger claim.

Folding a line into a transformer

A fixed line–transformer cascade can be compiled into a generic multiport, but the result is not a line merely because it has two external buses. The voltage ratio, vector group, galvanic boundary, winding limits and control ownership are not decoration. If they matter to the study, flattening them into a line is a semantic loss.

Folding external grounding into a transformer

An internal transformer neutral branch belongs to the transformer model. An external grounding reactor or shunt belongs to the bus/grounding model. Combining them may be algebraically convenient, but it hides where neutral current flows and who owns the limit. The canonical transformation should refuse the merge or emit a composite factor with an explicit ground-port and recovery ledger.

Treating a formulation change as a graph proof

The BIM/BFM parallel-line example is a formulation-level warning. Replacing a shared W_{ij} by a branch-wise $W_{\ell ij}$ restores an index but weakens the BIM relaxation; omitting the BFM consistency relation admits flows that do not satisfy Kirchhoff's voltage law across the parallel members. Neither change is a graph isomorphism.

The executable negative cases are collected with the book's translation-trap witnesses. They are useful regression tests for the register: a proposed rule must either satisfy the named guard or be classified as a structure-changing, outer, inner, or scenario transformation.

Review checklist

Before calling a transformation safe, ask:

1. What are the source and target object sorts?

2. Which ℓij identities and terminal maps remain addressable?
3. Which constitutive relation is preserved, and on what domain?
4. Where do every current, power, voltage, thermal and protection limit go?
5. Can eliminated quantities be recovered for every feasible target decision?
6. Does the result remain in the claimed physical factor class?
7. Which asset, ownership, measurement and grounding questions are no longer answerable?
8. Does the proof concern the physical model, a relaxation, an objective value, or a particular optimizer?

If any answer is missing, the result is a candidate reduction, not a certified equivalence.

Rule	Source → target	Preserves	Forgets or changes	Target closure / status
COORD-PERM	conductor-labelled factor → permuted factor	structure, behaviour, decisions, provenance	only the original ordering as a presentation choice	same factor class; exact normalization
SERIES-PURE	$\ell_1 ib + \ell_2 bj$ → composite two-port	pure-series terminal relation; recovered currents; intersected current limits	junction and separate asset identities	generic series factor; exact only under guards
SERIES-LINE	two line assets → one homogeneous line	only when construction, model, ratings and ownership are closed under concatenation	otherwise deletes physical boundaries	conditional physical merge; normally reject
PAR-SUM	parallel members → summed terminal admittance	unconstrained terminal relation	member currents, member limits, outages, ownership	generic aggregate; outer if member constraints are summed
KRON	retained/internal port factor → Schur complement	selected boundary relation; recoverable internal states	internal assets, sparsity, unlifted limits and protection	general factor; exact for declared boundary observations
XFMR-COMPOSE	transformer plus connected factor → completed multiport	declared terminal relation and mapped winding limits	compact transformer identity if flattened	completed transformer factor; exact only with provenance
GROUND-ABSORB	external grounding factor + device → device-only factor	perhaps a fixed linear terminal relation	ground asset, neutral current, ownership, protection and topology dependence	not canonical by default; reject unless explicitly scoped
STAR-DELTA / DELTA-STAR	scalar floating three-terminal star ↔ delta	linear terminal relation under non-singular scalar guards	internal node, branch identities, grounding and branch limits unless mapped	exact behavioural circuit transform; not a general multiconductor matrix rule
SHUNT-ABSORB-ENDPOINT	explicit bus shunt + factor → endpoint-augmented factor	fixed linear terminal relation at the declared state	shunt asset, switching decision, ownership, protection and placement	exact only as a generic factor; reject for a narrower line/transformer class unless encoded
SHUNT-SYMMETRIZE	unequal from/to shunts → one shared shunt parameter	only if equality or a proved observation-specific approximation holds	endpoint asymmetry and from/to current semantics	approximate or invalid by default; never infer from tool field names
BIM/BFM-INDEX	formulation with branch index → formulation with shared/branch variables	only the declared projection or relaxation relation	branch identity or relaxation-tightness information	formulation change, not a graph isomorphism
ROOTED-TREE	active radial graph → rooted feeder hierarchy	parent/child and ancestor queries for the declared state and root	mesh chords, alternate paths, future switch states and root-independent meaning	derived algorithmic view; recompute after topology changes

Edge	Typical structural status	Typical decision risk
K_g	guarded compilation	earth-return and ground-potential observations may be lost
K_n	guarded reduction	neutral voltage and neutral-current limits need recovery constraints
P_n	guarded coordinate/recovery map	common-mode voltage and shunt-to-ground effects are out of scope
F	exact coordinate change	phase-specific constraints need a mapped coordinate contract
D	structure-changing approximation	sequence coupling and cross-channel limits are dropped
F_1	restricted approximation	unbalance, neutral and phase-specific decisions are not represented

Chapter 34

Projection, compilation, and reduction

Page status: foundational transformation definitions and scope boundaries.

Transformations that all produce a different graph can have fundamentally different meanings. The following vocabulary is adopted throughout this book.

Projection

Definition. A projection forgets attributes, types, distinctions, or internal structure without solving the governing equations.

Examples include:

- multigraph to simple graph by forgetting edge identity;
- conductor graph to phase-aggregated graph;
- detailed switchgear to a state-resolved bus view;
- weighted to unweighted topology.

A projection can be reversible only if the forgotten information is retained as side data or is uniquely inferable. Most are many-to-one.

Compilation or realization

Definition. Compilation replaces a high-level component with a network of lower-level components supported by a target mathematical formulation.

Compilation may introduce virtual nodes and branches. It is therefore not ordered by graph size. Multiwinding transformer compilation in PowerModelsDistribution is a practical example [50].

Compilation should ideally have:

- a semantics theorem or component-level equivalence test;
- stable provenance from virtual objects to the source device;
- a partial inverse that reconstructs the high-level object when the compiled subgraph still matches the compiler image;
- explicit handling of controls and limits attached to the source object.

Normalization

Definition. Normalization is a semantics-preserving rewrite into a chosen canonical form within an explicitly declared model class.

Examples might include:

- canonical conductor ordering and orientation;
- conversion of a grounding impedance annotation to an explicit shunt factor;
- merging two adjacent subdivisions of one homogeneous physical line;
- normalization of equivalent transformer parameter conventions.

Normalization is stronger than terminal equivalence because the target should remain a valid instance of the intended physical vocabulary. A general two-port equivalent is not necessarily a normalized line.

Exact behavioral reduction

Definition. Behavioral reduction eliminates hidden variables while preserving a declared external relation. It need not preserve internal physical structure.

Kron reduction is the principal linear example [10]. Schur elimination can introduce clique edges among the neighbors of an eliminated node. The reduced graph may therefore be denser and may contain branches with no physical counterpart.

Caliskan and Tabuada identify classes of homogeneous generalized electrical networks for which time-domain Kron reduction remains within compatible element classes [51]. Their result is especially important here: **closure under reduction is a physical/model-class condition, not a generic consequence of variable elimination.**

Approximate reduction

Definition. Approximate reduction preserves selected observables only up to a stated error over a stated operating domain.

Distribution-feeder reduction methods have addressed unbalanced phase models, mutual coupling, spatial variation of load and generation, and critical-bus voltage error [52]. Recent Opti-KRON work adds radiality and phase-connectivity objectives [53], while a separate extension targets radiality recovery [54]. These are valuable, but their use of "structure preserving" should not be confused with preservation of construction codes, asset identity, neutral grounding, protection, or individual decision constraints.

State-dependent topology quotient

Closed-switch contraction occupies a special category. It is exact for electrical connectivity under the assumption of ideal closed switches, but it is indexed by network state. Retaining only the quotient loses possible future switching states and switch-level operations. The source connectivity model and the quotient should therefore coexist [32, 33].

Why these transformations do not form one chain

1. Compilation can increase graph size while reducing device vocabulary.
2. Elimination can decrease vertex count while increasing edge density and constitutive complexity.

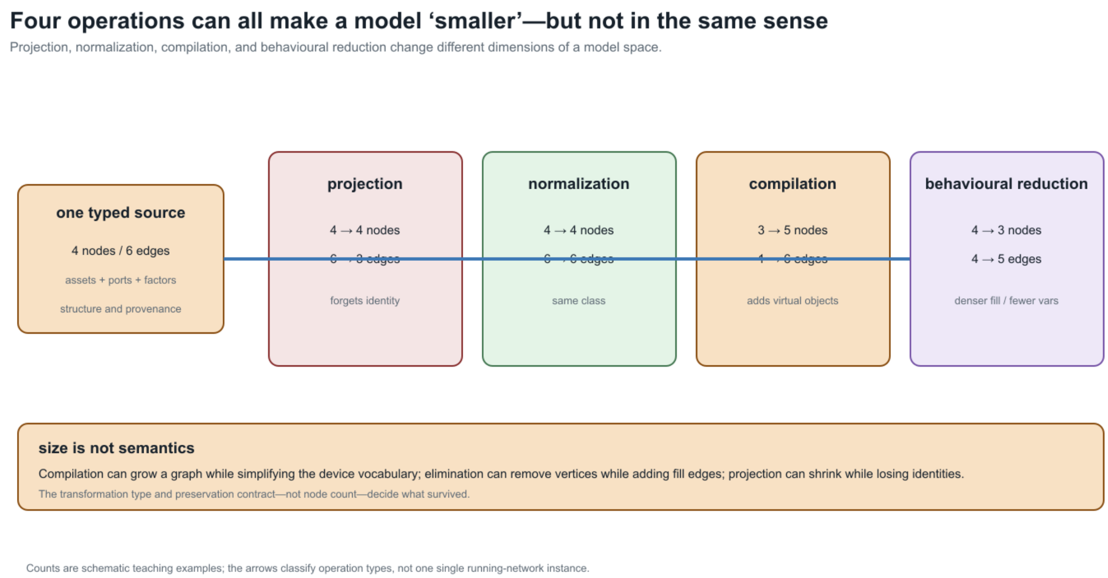

Figure 34.1: Projection, normalization, compilation, and behavioural reduction change different notions of size.

3. An asset projection and a behavioral equivalent can preserve incomparable information.
4. A state-dependent quotient is not a time-independent abstraction.
5. An approximate reduced model can outperform an exact terminal equivalent for a chosen application metric while being less generally valid.

The counts in this teaching plate are schematic. Its point is the inversion: compilation can add virtual nodes and edges, while behavioural reduction can remove variables and create denser fill. A smaller drawing is therefore not evidence of a stronger reduction, and a larger drawing is not evidence of a less faithful one. The preservation contract and target factor class decide what the arrow means.

The resulting mathematical object is better viewed as a graph of model spaces and typed transformations than as a linear ladder.

Chapter 35

Numerical consequences of representation and reduction

Page status: generated structural, linearized, symbolic KKT, and package-level solver-diagnostics crosswalk witnesses; solver-internal nonlinear exports remain future work.

A graph choice is also a numerical choice

Two representations can describe the same declared electrical behaviour while producing very different numerical problems. A quotient may reduce the number of variables but worsen scaling; a Kron reduction may preserve a boundary relation but introduce dense couplings; a factor model may expose physical structure while a bus–branch compiler produces a smaller, more familiar Jacobian. These are computational consequences, not afterthoughts.

The relevant comparison is therefore not only

same terminal map?

but also

same feasible observations, with what conditioning, sparsity, and recovery cost?

The statements below concern a declared operating point, coordinate system, and solver tolerance. They do not claim that one representation is uniformly better.

Vocabulary bridge

Electrical loss is power absorbed by a device model. Information loss is a failure of recovery under a representation map. An optimization or ML loss is an objective function. The shared word does not make these quantities comparable, and reducing one need not reduce either of the others.

Scaling and conditioning

Let a linearized or linear subproblem be

$$\mathbf{A}x = b.$$

Changing units or terminal coordinates is represented by nonsingular maps $x = \mathbf{D}_x\widetilde{x}$ and $b = \mathbf{D}_b\widetilde{b}$. The coefficient matrix becomes

$$\widetilde{\mathbf{A}} = \mathbf{D}_b^{-1}\mathbf{A}\mathbf{D}_x.$$

The solution set is unchanged, but the numerical condition number can change:

$$\kappa(\widetilde{\mathbf{A}}) = \|\mathbf{D}_b^{-1}\mathbf{A}\mathbf{D}_x\|\|\mathbf{D}_x^{-1}\mathbf{A}^{-1}\mathbf{D}_b\|.$$

This is why per-unit coordinates are a numerical convention as well as a reporting convention. A declared base scales voltage, current, power, and impedance together; it does not erase poor physical conditioning, singular topology, or an omitted conductor. A coordinate transformation is safe to call an exact rewrite only when its inverse and its power-dual action are retained, as in the coordinate-transformation chapter.

Circuit-theory trap

A dimensionless or per-unit matrix is not automatically well conditioned. The conditioning claim must name the norm, the bases, and the operating domain. A small residual in badly scaled coordinates can coexist with a large error in a physical voltage, current, or decision.

For a computed solution $\widehat{x}$ define the residual $r = b - \mathbf{A}\widehat{x}$. A backward-error measure is, for example,

$$\eta(\widehat{x}) = \frac{\|r\|}{\|\mathbf{A}\|\,\|\widehat{x}\| + \|b\|}.$$

Forward error bounds require additional assumptions. In a nonsingular linear problem, a standard normwise estimate has the form

$$\frac{\|\widehat{x} - x\|}{\|x\|} \lesssim \kappa(\mathbf{A})\,\eta(\widehat{x}),$$

up to higher-order terms and the chosen norm. Certificates should therefore store both a residual/backward-error quantity and a conditioning estimate, not just a solver termination flag.

Jacobians: physical topology is not matrix sparsity

For a nonlinear decision model $F(y, u) = 0$, Newton or interior-point methods use a Jacobian such as

$$\mathbf{J} = \frac{\partial F}{\partial y}.$$

The physical graph predicts some nonzeros, but the Jacobian graph is a graph of variable–equation dependence. A single multi-terminal factor can create a dense block between all of its ports. A nominal-π element can couple endpoint voltages through series terms and endpoint currents through shunts. Constraint rows for limits, controls, and recovery variables add edges that have no corresponding physical line. Conversely, a physical asset may be absent from a particular Jacobian block when its state is fixed or its equation has been eliminated.

Write G_phys for a chosen physical incidence graph and G_J for the bipartite Jacobian graph. In general there is no equality $G_phys = G_J$. The useful statement is a declared dependency map

$$\delta : \ (\text{factor}, \text{terminal}, \text{constraint}) \longmapsto (\text{equation rows}, \text{variables}),$$

from which G_J is generated. This map is the numerical analogue of source provenance: it explains why a fill-in or a dense block exists.

The five-bus structural witness makes the distinction concrete. Its source has seven line identities but only six edges after simple projection because q and r are parallel. Eliminating j then adds the i-l fill edge to the projected pattern. The fill argument and the dependency argument are shown separately so that a reader does not confuse a Schur-complement edge with a numerical derivative. The conceptual pair now appears in the formulation and lowering chapter; this chapter retains the executed witnesses and their fixture-specific counts.

The data and checks are recorded in `experiments/generated/numerical-structure-witness.json`; the renderer is `experiments/generate_numerical_structure_views.py`. This separation is intentional: both structural views can be checked without pretending that either is a solver-exported Jacobian.

Fill is not inevitable. The two-level topology chapter proves a positive structural case: dense two-terminal conductor stamps over a bus-level tree form a chordal scalar-support graph, and eliminating leaf-bus coordinate blocks inward is a perfect elimination order with zero symbolic fill. The warning here concerns arbitrary orderings, meshes, reductions, and factor libraries; it should not hide structure that a declared model makes exploitable.

The numerical witness now carries a crosswalk to `experiments/generated/five-bus-typed-kron-witness.json`. That crosswalk uses the non-pendant ℓ elimination, whose fill edges are $j - m$ and $k - m$, and records the small boundary residual together with the recovered u-branch limit observation. This ties the Schur-complement example to the numerical-structure discussion without conflating fill edges with physical assets or solver-private factorization output.

A pinned numerical export

The running fixture provides a second, numerical witness through BMOPFTools' passive and constant-Z linearized Y-bus exporters. In the exporter's node order, the passive matrix has 20 rows and 166 nonzeros. The constant-Z linearized matrix agrees with that passive matrix for this fixture. Realifying the complex current relation gives a 40-by-40 real matrix with 664 nonzeros. This is a coordinate embedding, not a promise that every complex invariant is preserved: the complex Ybus is transpose-symmetric to numerical precision, whereas the realified current-Jacobian embedding has a large transpose residual. That residual is expected for a nonzero imaginary part: the block matrix $\mathcal{R}(Y)$ with diagonal blocks $\Re Y$ and off-diagonal blocks $-\Im Y$ and $\Im Y$ is generally not symmetric. This is not a round-off failure. Realification preserves the coordinate support and the current relation, but symmetry claims must be stated in the representation in which they are checked (the witness residual is approximately 448.79):

$$\begin{bmatrix} \Re I \\ \Im I \end{bmatrix} = \begin{bmatrix} \Re Y & -\Im Y \\ \Im Y & \Re Y \end{bmatrix} \begin{bmatrix} \Re V \\ \Im V \end{bmatrix}.$$

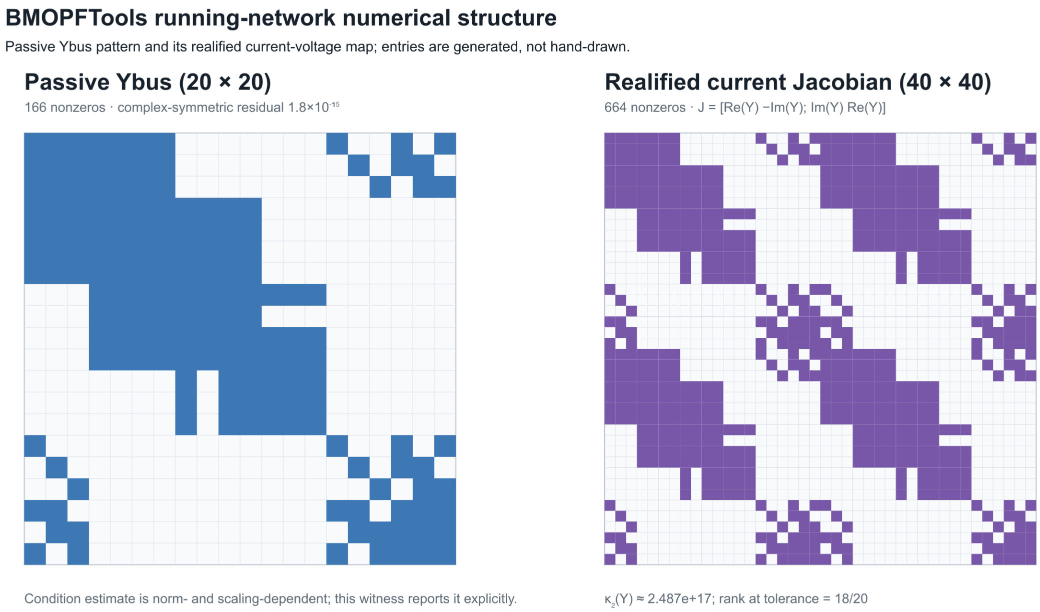

Figure 35.1: Generated passive Ybus and realified current-Jacobian patterns.

The ordinary 2-norm condition estimates are approximately 2.49×10^{17} (unscaled) and 2.36×10^{16} (after simple row/column equilibration). Because both matrices are numerically rank deficient (rank 18 of 20 at the witness tolerance), those finite values should not be read as meaningful condition numbers: they are dominated by singular values below the modelling tolerance. The witness therefore also reports the rank-aware effective estimates $\kappa_{\text{eff}} = \sigma_1/\sigma_{18}$ and its equilibrated counterpart. These are diagnostic, not universal constants: they depend on units, node ordering, norm, rank tolerance, and the chosen fixture. For this export the effective estimates are approximately 6.50×10^8 and 1.13×10^7 after equilibration. A solver must account for reference and grounding structure rather than treating any of these numbers as a graph invariant.

The complete export, including node order, nonzero entries, checks, and the linearization convention, is recorded in `experiments/generated/ybus-jacobian-witness.json`. The scripts `experiments/run_ybus_jacobian_witness.j` and `experiments/render_ybus_jacobian_view.py` regenerate it. This is a pinned linear current/Jacobian witness; it is not yet a nonlinear OPF KKT Jacobian or an independent-solver comparison.

Nonlinear decision Jacobian and KKT fill

The next witness adds the nonlinear term that makes the parallel-line warning decision relevant. For a fixed complex endpoint voltage U and member currents I_1, I_2, the residual includes

$$S(U, I_1, I_2) - \alpha S_0 = U(I_1 + I_2)^* - \alpha S_0.$$

The source formulation retains two explicit current laws and two current vectors. The aggregate formulation replaces them by one summed current law. At the recorded operating point both residuals are exactly zero, but their finite-difference Jacobians and KKT patterns differ:

For avoidance of a common counting error, the source witness has two complex member-current coordinates, represented as four real variables, in addition to the three retained real coordinates. It therefore has seven

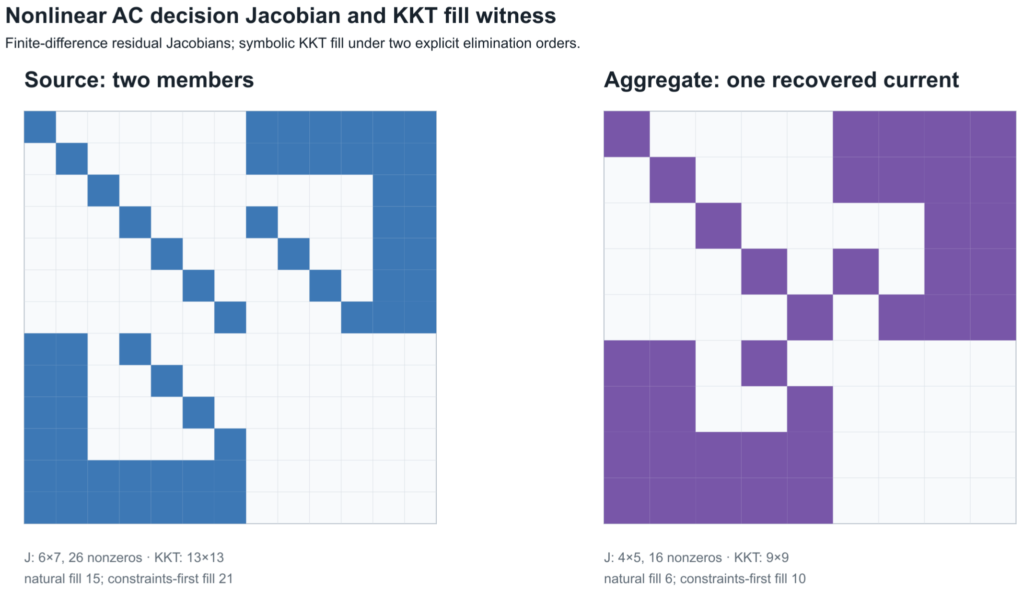

Figure 35.2: Nonlinear source and aggregate KKT sparsity and fill witness.

real source variables; the aggregate witness has five real variables. The reduction removes two real variables (one complex current), not eight. The source KKT system has 13 rows and columns because it adds six residual/constraint multipliers; that KKT dimension is a different count from the number of primal variables.

formulation	residual Jacobian	KKT dimension	natural fill	constraints-first fill
two explicit members	6×7 (26 nonzeros)	13	15	21
summed member current	4×5 (16 nonzeros)	9	6	10

The fill counts are symbolic graph counts, not floating-point factorization times. They show why an ordering is part of a numerical certificate: the same KKT graph has different fill under different elimination orders. The source and aggregate models also have different graphs, so a smaller KKT system does not by itself establish decision equivalence.

The complete finite-difference and symbolic-fill artifact is `experiments/generated/nonlinear-kkt-witness.json`, regenerated by `experiments/run_nonlinear_kkt_witness.jl` and `experiments/render_nonlinear_kkt_view.py`. It is deliberately labelled a nonlinear decision witness rather than a solver-internal Ipopt KKT export. A package-level crosswalk now binds it to the BMOPFTools Ybus witness, while actual solver-internal factorization diagnostics remain a separate boundary.

Package-level diagnostics crosswalk

The generated `solver-diagnostics-crosswalk.json` composes the two witnesses without pretending that either is a production solver export. It retains the physical node/terminal order, the BMOPFTools passive and constant-Z Ybus, the realified current Jacobian, the finite-difference nonlinear residual Jacobian, and the symbolic KKT graph under two declared orders. In the source formulation, for example, the natural and constraints-first orders produce different fill counts, so the crosswalk records ordering as part of the diagnostic identity.

The crosswalk also exercises the public `BMOPFTools.opf_checked_kkt_factorization` callback on a staged OPF context. The probe accepts a regular matrix and rejects a near-singular one. This is a checked callback boundary. A minimal parameterized OPF now passes that callback through DiffOpt: the forward sensitivity agrees with a central finite difference to the recorded tolerance, and the KKT diagnostic is accepted. The adapter also captures the matrix passed to the callback, recording its dimension and nonzero count, alongside the JuMP variable and constraint order used to account for its rows and columns. Four additional callback rows remain outside that declared JuMP block, so they are retained as an explicit DiffOpt/solver-internal boundary rather than silently assigned a physical label. The crosswalk still records `solver_internal_kkt_export = false` because solver-provided row labels, scaling, linear-solver choice, pivot or inertia data, and factorization statistics are not yet exported as a source/target comparison.

The same witness records BMOPFTools' differentiability report: inequality counts, active, near-active, weakly-active, and violated labels, minimum inactive slack, and qualifications. In this deliberately unconstrained fixture the inequality count is zero. The report's `ready` flag is state provenance for the selected local solve, not a proof of LICQ, strict complementarity, second-order sufficiency, global optimality, or branch stability.

The crosswalk now also builds a regular JuMP mirror and records its native affine/quadratic variable-support count alongside a separate `JuMP.NLPEvaluator` view. This gives an executable constraint-row/nonzero summary for the model-level derivative graph (23 supported variable entries in the current 19-row fixture), while leaving `native_nlp_export_is_solver_internal = false`: it is not Ipopt's private KKT ordering, pivoting, inertia, or factorization export.

For a deliberately narrow parallel-line fixture, the crosswalk also compares a two-member source against a single 0.25 Ω equivalent member. The tested voltage sensitivity is preserved, while the captured KKT matrix grows for the explicit source. The native JuMP support count changes as well, so the distinction is visible before any solver-private factorization. This is evidence that an exact electrical scalar equivalent can still change solver structure; it is not a general AC or multiconductor reduction certificate.

Numerical evidence boundary

The three numerical artifacts answer different questions and should not be collapsed into one solver claim:

Artifact	What is measured	What it does not establish
structural witness	incidence, dependency, and symbolic fill edges	numerical derivative values or solver performance
Ybus/Jacobian witness	pinned passive and constant-Z matrix patterns, rank, and conditioning diagnostics	nonlinear OPF sensitivities, active-set stability, or factorization timings
nonlinear KKT witness	finite-difference residual structure and symbolic fill under two declared orders	Ipopt's internal derivative graph, linear-solver pivoting, or global optimality
solver-diagnostics crosswalk	shared node/order provenance, a checked-KKT callback probe, a DiffOpt sensitivity cross-check, and JuMP ordering metadata	an exported source/target solver KKT comparison, solver-native row labels, or ordering/factorization metadata

An actual solver diagnostic would need to bind the exported derivative rows and columns to the source/target variable order, record scaling and tolerances, identify the linear solver and ordering, and retain the factorization or inertia report. Until that export exists, the chapter's KKT language is a symbolic comparison, not an implementation claim about a production solver.

Elimination and fill-in

Partition a sparse matrix by retained variables B and eliminated variables I. The Schur complement is

$$\mathbf{S} = \mathbf{A}_{BB} - \mathbf{A}_{BI}\mathbf{A}_{II}^{-1}\mathbf{A}_{IB}.$$

Even when $\mathbf{A}$ is sparse, $\mathbf{S}$ can be dense on the neighbours of I. In the symbolic graph, eliminating a vertex connects its remaining neighbours into a clique. The extra nonzeros are **fill-in**. The same mechanism appears in Kron reduction: preserving a boundary relation does not preserve sparsity, asset identity, or the cost of a subsequent solve.

The order of elimination matters. For a fixed matrix, two orders can have the same exact Schur complement but different intermediate fill, memory use, and factorization time. A certificate that reports a reduction should therefore record at least:

numerical field	required meaning
`ordering`	symbolic elimination or factorization order
`nnz_input`, `nnz_factor`, `nnz_fill`	structural nonzero counts under that order
`rank_guard`	rank/invertibility test for eliminated blocks
`conditioning`	estimate for the retained and eliminated solves
`recovery_cost`	work/storage needed to reconstruct omitted variables

These fields are properties of a compiled numerical problem, not of a simple graph in isolation.

Solver behaviour and decision margins

For a constrained decision problem, solver behaviour is part of the evidence but not the preservation theorem. A reduced model may converge faster because it has fewer variables, or slower because it has denser and more ill-conditioned blocks. A feasible point can also be numerically ambiguous when a constraint margin is comparable with the estimated forward error.

For a scalar inequality $g_k(y, u) \leq 0$, define the signed margin

$$m_k = -g_k(\widehat{y}, \widehat{u}).$$

If e_k is a declared bound or estimate for the error in that constraint, classify the result as:

certified feasible	$m_k > e_k,$
numerically ambiguous	$\lvert m_k \rvert \leq e_k,$
certified violated	$m_k < -e_k.$

This is deliberately stronger than reporting $g_k \leq 0$ at printed precision. The same classification applies to decision comparisons: if two objectives or tap choices differ by less than the propagated numerical error, the certificate should not claim a unique optimum.

Decision-model consequence

A reduction that preserves a boundary voltage to tolerance may still change the active member limit, the selected switch state, or the winning discrete decision. Record the decision margin and the recovery error for every claimed decision-preservation result.

What a transformation certificate should expose

The numerical fields extend the general preservation contract:

$$\mathcal{N} = (\text{coordinates}, \text{scaling}, \kappa, \eta, \text{dependency graph}, \text{ordering}, \text{fill}, \text{recovery cost}, \text{decision margins}).$$

An exact electrical map may therefore be computationally unattractive, while an approximate map may be useful if its error and decision domain are explicit. The running network's six generated views should eventually carry this record alongside their source maps; the current fixture records the source maps and solver outcomes but does not yet claim a complete cross-solver conditioning study.

Scope and open work

This chapter gives definitions and reporting requirements, not a universal solver benchmark. The next executable tranche should compare the running multiconductor network and its five-bus quotient under a pinned ordering and coordinate convention, report Y/Jacobian sparsity and fill-in, and test whether recovered current and rating margins remain decisive after reduction.

Chapter 36

Rating and limit semantics

Page status: normative rating and limit vocabulary; utility-field mappings remain future implementation work.

A rating is not a scalar decoration on an edge. It is a constraint attached to an object, a quantity, an operating domain, and often an ambient or protection assumption. A transformation can preserve the circuit equations while changing which rating is enforceable.

A typed limit

Represent a limit by a record

$$\lambda = (a,\, q,\, \mathcal{D},\, b,\, \chi,\, \omega),$$

where a is the constrained asset or terminal, q is the measured quantity, $\mathcal{D}$ is its domain, b is the bound, χ is the scenario/ambient validity condition, and ω is the provenance or protection owner. A feasible point x satisfies

$$q_a(x, u; \theta) \in \mathcal{D}, \qquad q_a(x, u; \theta) \le b_a(\theta, \chi)$$

for every active limit record. The notation permits scalar, vector, complex, thermal, apparent-power, and relay regions without pretending they are the same constraint.

Common distinctions

The running parallel cases use centered component-current discs as deliberately simple limit regions. They are not nameplate ratings and should not be read as continuous/emergency thermal limits. The certificate proves implication for the declared discs and matrices only.

Preservation obligations

When an object is projected, compiled, merged, or eliminated, the certificate must classify each source limit as one of:

- **retained:** the same quantity and domain remain explicit;

Distinction	Examples	Why it matters for a reduction
duration	continuous/normal, emergency, short-time	the admissible state and contingency set changes
environment	ambient-adjusted ampacity, seasonal, weather-dependent	the bound depends on χ rather than only the asset
location	conductor, terminal equipment, transformer winding, busbar	a recovered branch current may not be the protected quantity
quantity	current magnitude, complex power, apparent power, temperature, CT/relay pickup	equal admittance does not imply equal feasible regions
ownership	line owner, protection zone, operator, investment project	a shared or aggregated limit can lose the decision owner
uncertainty	nameplate tolerance, forecast interval, scenario bound	nominal exactness is not robust preservation

- **mapped:** a recovery or coordinate map gives an exact target constraint;
- **conservative:** every target-feasible point satisfies the source limit, but valid source points may be excluded;
- **relaxed:** all source-feasible observations remain, but target points may violate a source limit;
- **scenario/uncertain:** implication is certified only over declared scenarios or parameter sets;
- **forgotten:** the target cannot answer the rating query.

In particular, a neutral-current limit after Kron elimination is normally **mapped**, not forgotten: recover the eliminated neutral voltage or current from the retained variables and impose the original magnitude or multi-conductor norm bound. A boundary Y matrix without that recovered constraint answers terminal behaviour but not the neutral-rating decision.

For a parallel-line transformation, summing admittances preserves an aggregate terminal relation but does not map two member limits to one scalar limit unless the member feasible regions and their state/ownership semantics support that map. The exact-pruning certificates in the guarded transformation cases retain the member factors and remove only limits proved redundant under their declared nominal domains.

Power-system shorthand

"The line is rated at 200 A" is incomplete. Identify the member, terminal, conductor set, duration, ambient condition, measured quantity, and owner of the limit before comparing it with a transformed constraint.

Vocabulary bridge to preservation contracts

The limit dispositions above describe what happens to an individual constraint; the preservation-contract vocabulary describes the resulting feasible-set relation. They are related, but not interchangeable:

This table is a translation aid, not a theorem: a single transformation can retain some limits, map others, and forget the rest, so the contract must be checked record by record.

limit disposition	usual contract interpretation
retained	exact, with the same query still explicit
mapped	exact, provided the recovery map is part of the certificate
conservative	inner approximation
relaxed	outer approximation
scenario/uncertain	scenario-approximate
forgotten	no decision-level preservation claim

Chapter 37

Data-model crosswalk

Page status: versioned practice crosswalk with a checked running-fixture contract; external package imports and full round-trip adapters remain open.

The book's objects are semantic categories, not a claim that every software or standard uses the same schema. This crosswalk records where common ecosystems provide a direct correspondence, where an adapter is needed, and what meaning must not be inferred from a successful import.

Crosswalk

Interpretation notes

CIM/CGMES is primarily an information and exchange model. Its distinction between terminals, connectivity nodes, and topological nodes is especially useful for the book's node–breaker compiler, but profile selection and state processing remain part of the exchange contract [32, 47].

PowerModelsDistribution exposes an engineering data model and a conversion to a mathematical model; the conversion may create several mathematical objects, apply phase projection, Kron reduction, per-unit conversion, or multinetwork expansion [50, 55]. Those are precisely the kinds of maps that this book requires to carry guards, source identities, and recovery data.

OpenDSS is a practical circuit/equipment language with reduction operations. Its reduction documentation is evidence of useful engineering procedures, not a universal proof that limits, states, grounding, or provenance are preserved [56].

Mutual coupling demonstrates why a crosswalk also needs relations whose endpoints are equipment sections rather than buses. CIM retains a distinct $MutualCoupling$ object with two terminal associations and four distances for the coupled portions [41]; PowSyBl's line-coupling extension and PowerWorld's sequence records similarly identify both lines and their overlap intervals [42, 43]. These implementations are useful precedent, but their scalar zero-sequence or short-circuit scope must not be generalized silently to a full phase-coordinate series-and-shunt primitive.

MATPOWER's case format is intentionally compact: version-2 cases package $baseMVA$, bus, gen, $branch$, and optional $gencost$ matrices, and the conventional branch model combines lines, transformers, and phase shifters in a common two-terminal representation [57]. The modern data-model documentation makes the port/node connection idea more explicit [58], but arbitrary multiconductor factors, asset lifecycle relations, and many-to-many provenance still require an adapter.

Adapter contract

An adapter should publish:

1. source format and version/profile;
2. identifier and terminal maps;
3. unit, base, phase, neutral, and grounding conventions;
4. state and scenario treatment;
5. factor and rating mappings;
6. generated objects and their provenance;
7. unsupported or lossy fields;
8. round-trip or recovery tests for the declared observations.

The crosswalk is therefore a starting vocabulary, not a certification that a file imported successfully is semantically equivalent to the source model.

The same boundary applies to study semantics: an exchange adapter can preserve identifiers and terminal maps while still omitting estimator covariances, protection-zone logic, contingency state, or profile-specific validity rules. Those omissions must be reported as unsupported fields in the adapter record; successful parsing is not evidence that a PF, OPF, state-estimation, fault, or protection study has been preserved.

Version-pinned contract witness

The generated `experiments/generated/data-model-crosswalk-witness.json` checks the adapter obligations on the running fixture (`data/running-network/v0.1.0.json`) for four pinned documentation profiles: CIM/CGMES, PowerModelsDistribution, OpenDSS, and MATPOWER. It verifies unique asset identifiers, terminal provenance, rating owners and units, state-field retention, and an explicit MATPOWER two-terminal compilation boundary.

This is deliberately a contract test rather than an import test. The repository does not claim that any external package is installed, that a particular publisher profile is the latest, or that a successful parse preserves a study result. The remaining adapter work is to run these same checks against selected version-pinned external files and record unsupported fields and round-trip recovery results.

Book object	CIM/CGMES	PowerModelsDistribution	OpenDSS	MATPOWER	Mapping status
asset/entity $\mathfrak{A}$	$IdentifiedObject$ and typed equipment/asset classes	engineering dictionaries and source identifiers	named circuit objects and classes	row identity plus case metadata	partial; stable identity and lifecycle provenance need an adapter
terminal/port q	$Terminal$ and phase attributes	engineering terminal maps and conductor sets	bus names with node/terminal suffixes	bus/branch endpoints, usually scalar bus indices	direct for topology; multiconductor semantics vary
connectivity node c	$ConnectivityNode$	engineering bus/connectivity data	bus connectivity	bus row in $mpc.bus$	direct only after phase/ground conventions are declared
topological node n_σ	$TopologicalNode$ and state-derived topology	transformed mathematical bus/node set	active circuit connectivity	compiled bus set	state-resolved derived view
two-terminal factor	$ConductingEquipment$ with terminals	line, switch, transformer, load, or generator object	line, transformer, reactor, load, generator	branch/gen/load rows	direct only within each tool's native factor library
multi-terminal factor	equipment plus multiple terminals	multiwinding transformer and coupled engineering objects	multiwinding/terminal object, often compiled for analysis	no native arbitrary-port object	requires a declared compiler and provenance
line-section coupling relation	$MutualCoupling$ links two line terminals and coupled-distance intervals, principally for zero-sequence short circuit data	no general source mapping currently established	can represent a joint conductor primitive when circuits are compiled as one multi-conductor element; separate-line provenance needs an adapter	no standard branch-pair mutual-coupling record	relation, section orientation, group assembly, and study scope require explicit adapter support
factor relation $\mathcal{R}_\phi$	electrical parameters plus profile-specific attributes	engineering-to-mathematical conversion and JuMP formulation	primitive stamping and solver model	standard π branch/gen/load equations	equations are not fully represented by topology data
rating/limit	equipment, operational limit, and	thermal/current/voltage	line/transformer/relay properties	branch/gen limits	quantity, duration

Part V

Transformations and recovery

Chapter 38

Circuit coordinate transformations

Page status: guarded transformation definitions with executable witnesses.

Why these are transformations, not graph deletions

The attached four-wire and three-wire manuscripts make an important distinction explicit: reducing the number of voltage variables can be a change of electrical coordinates, an elimination of an unobservable mode, or an approximation that discards a physical return path. These are not the same operation as deleting a neutral vertex from a graph.

The source model in this book retains ordered conductor terminals and factors. The transformations below are maps on terminal variables and currents, with a separate statement of exactness, recovery, and what is no longer observable.

Circuit-theory trap

A three-variable target is not automatically a three-wire physical model. It may be a phase-to-neutral coordinate realization of a four-wire factor, a phase-to-phase quotient of a three-wire factor, or a Kron-reduced boundary relation. The target type and discarded mode must be named.

Two reductions that are often conflated

For a phase/neutral partition of a nodal impedance or admittance relation, neutral elimination and a phase-to-neutral coordinate map are different operations. Writing the impedance in blocks,

$$\mathbf{Z}_{abc,n} = \begin{bmatrix} \mathbf{Z}_{pp} & \mathbf{Z}_{pn} \\ \mathbf{Z}_{np} & \mathbf{Z}_{nn} \end{bmatrix},$$

the Schur-complement reduction is

$$\boxed{\mathbf{Z}_{abc}^{\mathrm{Kron}} = \mathbf{Z}_{pp} - \mathbf{Z}_{pn}\mathbf{Z}_{nn}^{-1}\mathbf{Z}_{np}}.$$

It eliminates a declared neutral variable and requires $\mathbf{Z}_{nn}$ to be invertible. By contrast, the phase-to-neutral map changes the voltage coordinates and lifts currents on a declared zero-sum subspace:

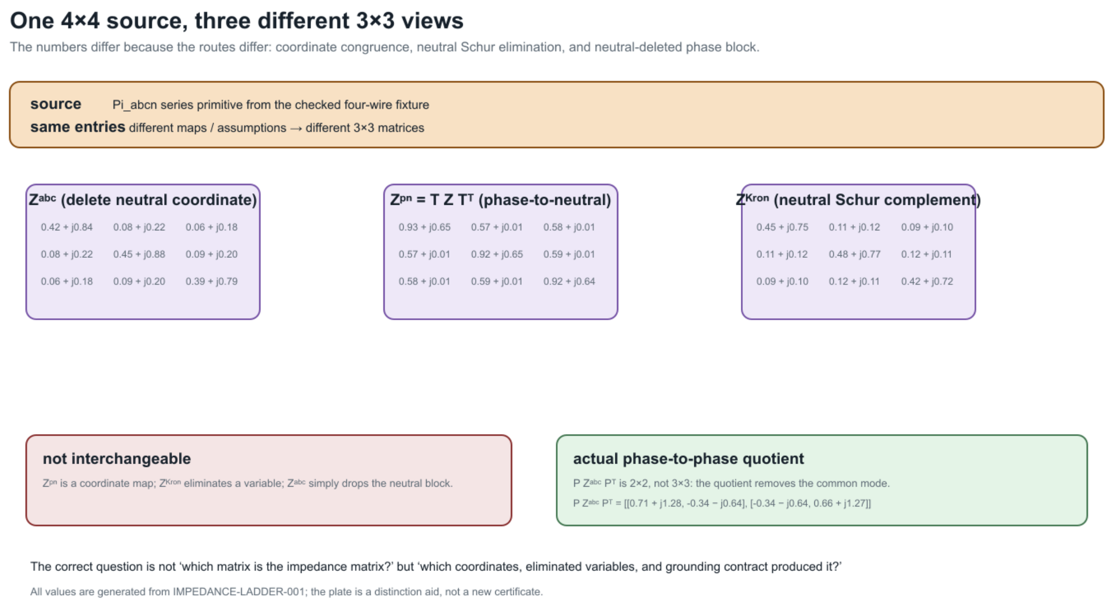

Figure 38.1: One checked four-wire primitive produces three different 3×3 views under three different routes.

$$\boxed{\mathbf{Z}^{\text{pn}} = \mathbf{T}\mathbf{Z}_{abc,n}\mathbf{T}^{\mathsf{T}}}, \qquad \mathbf{T} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

The latter does not assert that the neutral voltage is zero or that a physical neutral conductor has been removed. The two expressions agree only under additional grounding, shunt, and current-subspace assumptions, together with a declared recovery map. A numerical example should therefore report which formula was evaluated, the grounding convention, the invertibility condition, and the residual of the discarded mode.

The plate is deliberately not a catalogue of interchangeable impedance matrices. Z^{pn} is a phase-to-neutral coordinate congruence, Z^{Kron} eliminates the neutral block, and Z^{abc} is merely the neutral-deleted phase block. The actual phase-to-phase quotient is shown as a 2×2 inset because the common-mode quotient removes one dimension. This distinction is the practical reason to record the route, grounding contract, and discarded mode alongside every exported matrix.

Four-wire phase-to-neutral transformation

Let the phase-to-ground voltage vector at bus i be

$$\mathbf{U}_i = \begin{bmatrix} U_{i,a} & U_{i,b} & U_{i,c} & U_{i,n} \end{bmatrix}^{\mathsf{T}}.$$

Define the phase-to-neutral map

$$\mathbf{U}_i^{\mathrm{pn}} = \mathbf{T}\mathbf{U}_i, \qquad \mathbf{T} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

For a four-wire line with total current $\mathbf{I}_{\ell ij}$, series current $\mathbf{I}^{\mathrm{s}}_{\ell ij}$, and shunt current $\mathbf{I}^{\mathrm{sh}}_{\ell ij}$,

$$\mathbf{I}_{\ell ij} = \mathbf{I}^{\mathrm{s}}_{\ell ij} + \mathbf{I}^{\mathrm{sh}}_{\ell ij}.$$

When the shunt contribution can be neglected for the declared study, the neutral current is determined by the phase currents under the zero-ground injection condition:

$$I_{\ell ij,n} = -(I_{\ell ij,a} + I_{\ell ij,b} + I_{\ell ij,c}).$$

The current map is therefore

$$\mathbf{I}_{\ell ij} = \mathbf{T}^{\mathsf{T}}\mathbf{I}^{\mathrm{pn}}_{\ell ij}, \qquad \mathbf{I}^{\mathrm{pn}}_{\ell ij} = \begin{bmatrix} I_{\ell ij,a} & I_{\ell ij,b} & I_{\ell ij,c} \end{bmatrix}^{\mathsf{T}}.$$

For a full four-by-four series impedance $\mathbf{Z}^{\mathrm{s}}_{\ell}$, left-multiplication by $\mathbf{T}$ gives the transformed series relation

$$\mathbf{U}_j^{\mathrm{pn}} = \mathbf{U}_i^{\mathrm{pn}} - \underbrace{\mathbf{T}\mathbf{Z}^{\mathrm{s}}_{\ell}\mathbf{T}^{\mathsf{T}}}_{\mathbf{Z}^{\mathrm{pn}}_{\ell}}\mathbf{I}^{\mathrm{pn}}_{\ell ij}.$$

This is a congruence transformation on the declared zero-sum current subspace, not a claim that the neutral conductor has zero voltage everywhere. A neutral voltage recovery map can be retained separately when the topology and grounding conditions make it unique.

Exactness contract

The phase-to-neutral target is exact for the full four-wire equations under the following sufficient conditions, matching the attached computational study [46]:

1. line shunt admittances to ground are zero or explicitly retained in a representable transformed factor;
2. connected devices inject negligible current into ground except at declared grounding factors;
3. each continuous neutral section has at most one ideal earth reference; and
4. the recovery traversal encounters no disconnected or multiply grounded neutral component.

Under these guards, phase-to-neutral voltages, phase currents, terminal powers of zero-sequence-free devices, and series losses can be recovered exactly. The neutral-to-ground voltage and neutral-to-earth limits are not represented by the three voltage variables alone; they require the recovery map and retained grounding data.

With sparse grounding, line charging, or multiple grounding points, the same map can still be a useful approximation, but the discarded earth-coupled current must be named and bounded. A small voltage error in a sample is not a general decision-preservation theorem.

Decision-model consequence

Eliminating the neutral can remove voltage-to-ground limits, grounding decisions, fault paths, and protection observations even when the phase-to-neutral power flow looks accurate. Keep the source factor and a recovery map if any of those questions remain in scope.

Three-wire phase-to-phase reduction

For an electrically isolated three-wire section, let

$$\mathbf{U}_i = \begin{bmatrix} U_{i,a} & U_{i,b} & U_{i,c} \end{bmatrix}^\top, \qquad \mathbf{U}_i^{\mathrm{pp}} = \mathbf{P}\mathbf{U}_i, \qquad \mathbf{P} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}.$$

The kernel of $\mathbf{P}$ is $\mathrm{span}(\mathbf{1})$. The phase-to-phase target therefore removes the common-mode voltage, while its transpose maps reduced currents into the zero-sum subspace:

$$\mathbf{I}_{\ell ij} = \mathbf{P}^\top \boldsymbol{\gamma}_{\ell ij}, \qquad \mathbf{Z}_\ell^{\mathrm{pp}} = \mathbf{P}\mathbf{Z}_\ell^{\mathrm{s}}\mathbf{P}^\top.$$

Every full current satisfying $\mathbf{1}^\top \mathbf{I}_{\ell ij} = 0$ has a unique reduced current $\boldsymbol{\gamma}_{\ell ij}$. The coordinate choice may instead be Clarke $\alpha\beta$ or positive/negative sequence; these are bases of the same two-dimensional quotient space. A fixed sequence basis diagonalizes only when the line matrices have the required transposed or circulant structure. It is not a generic decoupling of an untransposed, coupled feeder.

Exactness and the radiality guard

The attached three-wire reduction manuscript proves a useful sufficient condition: a galvanically isolated three-wire section with zero-sum device injections, no line shunts to earth, and at most one ideal earth reference is exact when its active section graph is a tree. The tree condition forces the common-mode current to vanish by induction from the leaves. It is therefore a topological guard on the active member graph, not a synonym for a radial simple projection.

When the section is meshed, multiply grounded, or earth-coupled, write the current as

$$\mathbf{I}_{\ell ij} = \mathbf{P}^\top \boldsymbol{\gamma}_{\ell ij} + \kappa_{\ell ij}\mathbf{1}.$$

The reduced voltage equation then has a residual of the form

$$\mathbf{U}_j^{\mathrm{pp}} = \mathbf{U}_i^{\mathrm{pp}} - \mathbf{Z}_\ell^{\mathrm{pp}}\boldsymbol{\gamma}_{\ell ij} - \mathbf{P}\mathbf{Z}_\ell^{\mathrm{s}}\mathbf{1}\,\kappa_{\ell ij}.$$

The factor $\|\mathbf{P}\mathbf{Z}_\ell^{\mathrm{s}}\mathbf{1}\|$ is a useful line-level sensitivity indicator, but it bounds an equation residual, not the solution or decision error by itself. The missing common-mode coordinate also prevents recovering phase-to-ground voltages without additional information.

Graph-theory trap

The exactness theorem uses a tree of active identified members. A simple graph can be radial while parallel member identity creates a two-edge cycle, and an asset inventory can be meshed while an active state is a tree.

How these transformations fit the book

Transformation	Primary operation	Exactness guard	Typical loss
phase-to-neutral $\mathbf{T}$	terminal-coordinate map plus current recovery	zero or represented shunts, sparse compatible grounding	ground-referenced quantities if recovery is omitted
phase-to-phase $\mathbf{P}$	quotient by common-mode voltage and zero-sum current lift	active member tree, zero-sum injections, limited grounding	common-mode voltage/current and earth-return effects
neutral Kron reduction $\mathcal{K}$	Schur complement of a declared neutral block	invertible neutral block and retained boundary relation	neutral state and source member constraints unless recovered
generic Kron	elimination of internal variables	invertible internal block and declared observations	assets, limits, switching, and decisions unless mapped

The first two rows are coordinate or quotient transformations on conductor spaces. Kron is an elimination of a model block. They may compose, but only after their terminal maps, grounding scope, and recovery contracts have been made explicit. A diagram that shows fewer conductors is not enough evidence to classify the operation.

Executable and research status

The book already has exact conductor-coordinate and transformer-coordinate certificates. The next implementation should add a typed four-wire phase-to-neutral certificate and a three-wire phase-to-phase certificate with positive, mesh, shunt, and grounding guard rejections. The active-state radiality witness now provides the first executable topology guard.

Chapter 39

Degree-two series elimination

Page status: guarded exact series transformation with positive and negative tests; physical line-class closure remains open.

This chapter gives the first executable guarded rewrite. It deliberately separates an exact terminal-behaviour result from the stronger claim that two source assets form one longer homogeneous physical line.

Source and target

Let two multiconductor series elements be oriented as $\ell_1 ib$ and $\ell_2 bj$. The ordered conductor coordinates used by ℓ_1 at the internal junction b need not equal the order used by ℓ_2. Let $\mathbf{P}$ be the permutation satisfying

$$\mathbf{I}_{\ell_2 bj} = \mathbf{P}\mathbf{I}_{\ell_1 ib}, \qquad \mathbf{U}_b^{(2)} = \mathbf{P}\mathbf{U}_b^{(1)}.$$

The outer terminal order at j is relabelled into the coordinates of ℓ_1. The source relations are

$$\mathbf{U}_i - \mathbf{U}_b^{(1)} = \mathbf{Z}_{\ell_1}\mathbf{I}_{\ell_1 ib},$$

$$\mathbf{U}_b^{(2)} - \mathbf{U}_j^{(2)} = \mathbf{Z}_{\ell_2}\mathbf{I}_{\ell_2 bj}.$$

If Kirchhoff conservation at b gives the common series current, multiplying the second relation by $\mathbf{P}^{\mathsf{T}}$ and adding gives

$$\mathbf{U}_i - \mathbf{P}^{\mathsf{T}}\mathbf{U}_j^{(2)} = \left(\mathbf{Z}_{\ell_1} + \mathbf{P}^{\mathsf{T}}\mathbf{Z}_{\ell_2}\mathbf{P}\right)\mathbf{I}_{\ell_1 ib}.$$

Hence the exact behavioural composite has

$$\mathbf{Z}_{\ell_{\mathrm{eq}}} = \mathbf{Z}_{\ell_1} + \mathbf{P}^{\mathsf{T}}\mathbf{Z}_{\ell_2}\mathbf{P}.$$

This is claim `TR-SER-001`. It is a coordinate-aware terminal equivalence, not a license to add matrices whose rows merely happen to have the same position. The displayed formula also assumes that the two voltage-drop relations contain no mutual-impedance terms linking either source element to the other or to an external element. The coupled case is treated separately below.

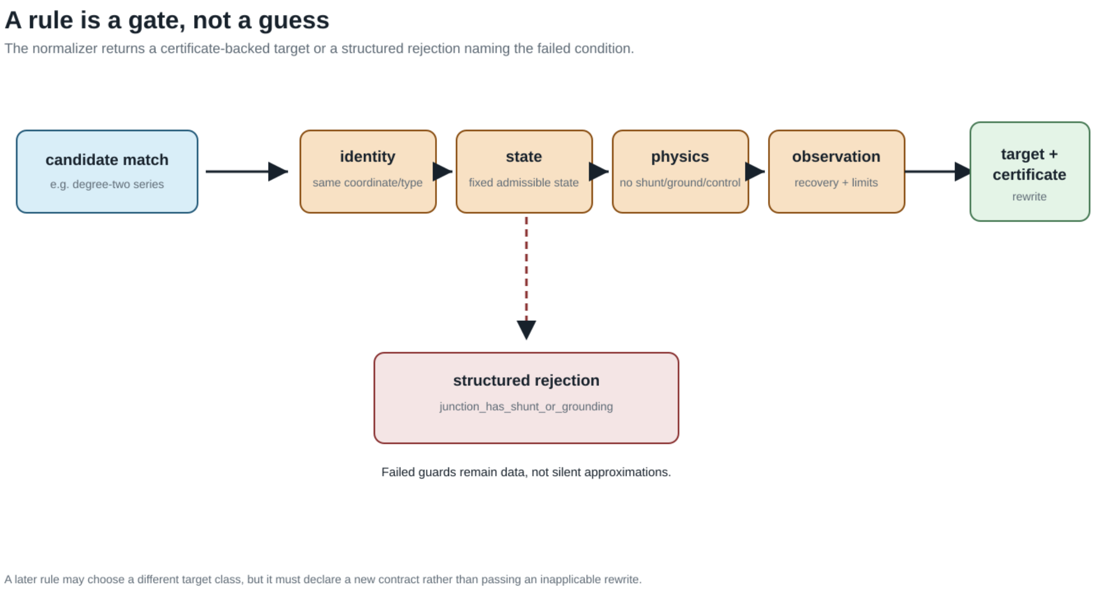

Figure 39.1: The guarded series rewrite: inspect, certify, or reject before lowering.

Guards

The gate is the operational pattern used by the executable rule: a candidate rewrite is accepted only after its structural, constitutive, decision, and provenance preconditions have been checked.

The source type supplies a precondition before the junction guards are even considered: **both factors are series-only multiconductor elements**. A nominal-π or other shunted factor is outside this rule. A shunt carried inside such a factor is not made admissible merely because no separately named shunt object is attached to b.

The implemented rule accepts the rewrite only when all of the following hold:

Guard	Reason
ℓ_1 ends and ℓ_2 begins at b	fixes the declared orientation
conductor labels at b are unique and form the same set	makes $\mathbf{P}$ well defined
no current injection at b	establishes a common series current
no shunt or grounding at b	prevents current from leaving the series path
no measurement, control, or protection boundary at b	keeps elimination within the declared observation contract
neither element is mutually coupled to the other source element	makes the displayed uncoupled impedance sum applicable
neither element is mutually coupled to any external element	prevents loss of a constitutive relation outside the pair

A failed guard returns a structured rejection with the failed condition and the source evidence. In the executable model, mutual coupling is represented by the field $SeriesElement.mutual_couplings[other_element_id]$: an element-pair cross-impedance block keyed by the other asset identity. It is deliberately not a `JunctionContext` flag, because the same pairwise constitutive relation may span a junction or extend beyond it. The guard therefore inspects the source element fields that would invalidate the formula and does not return a best-effort equivalent.

Constraint and recovery maps

For per-conductor current-feasible sets $\mathcal{C}_{\ell_1}$ and $\mathcal{C}_{\ell_2}$, the exact target constraint is

$$\mathcal{C}_{\text{eq}} = \mathcal{C}_{\ell_1} \cap \mathbf{P}^{\mathsf{T}}\mathcal{C}_{\ell_2}.$$

Independent upper current magnitudes therefore become the coordinate-aligned componentwise minimum, not their sum. Source quantities recover as

$$\mathbf{I}_{\ell_1 ib} = \mathbf{I}_{\text{eq}}, \qquad \mathbf{I}_{\ell_2 bj} = \mathbf{P}\mathbf{I}_{\text{eq}},$$

$$\mathbf{U}_b^{(1)} = \mathbf{U}_i - \mathbf{Z}_{\ell_1}\mathbf{I}_{\text{eq}}.$$

The generated target retains both member identities and the eliminated-junction identity in its provenance record.

Why this is not automatically a physical merge

Different construction codes do not invalidate the algebra above. They do invalidate the stronger rewrite into one instance of a homogeneous physical line class. Even equal codes are only a candidate condition: conductor material and geometry, frequency basis, line model, rating semantics, splices, maintenance boundaries, ownership, thermal state, outage state, and other physical facts may still differ.

Graph-theory trap

Degree two is only a structural candidate for elimination. A valid rule must also inspect terminal coordinates, injections, shunts and grounding, observations, constraints, decisions, and the intended target equipment class.

This distinction is claim `TR-SER-002`: closure under behavioural elimination and closure within an equipment class are different questions.

Anti-patterns: algebra is not a type checker

Three tempting rewrites should be shown as refusals or as explicitly typed compositions:

The safe target for the first row is usually a `CompositeSeriesBranch`; for the second and third rows it is a typed multiport retaining the transformer and ground ports. If the target library has no such factor, the transformation is ill-typed even when a matrix calculation can be performed.

This is also why a nominal-π series merge needs more than the displayed Z matrices. Shunt currents make the two segment currents different at the intermediate bus, and cascading the sections generally produces a general two-port rather than a nominal-π section with naively summed parameters.

Rewrite	What can be true	Why the physical merge is unsafe
different line constructions ℓ_1 and ℓ_2 → one line	the pure-series terminal impedance can still be $Z_1 + P^{\mathsf{T}} Z_2 P$	the target may falsely claim one construction, one owner, one thermal state or one rating basis
line + transformer → line	a fixed cascade can have a generic two-port relation	turns ratio, vector group, galvanic boundary, winding limits and controls disappear
external ground + transformer → transformer-only	a fixed nodal admittance can sometimes absorb the branch	neutral-current ownership, earth return, protection and topology dependence disappear

Power-system shorthand

Rejecting every heterogeneous series pair would be too strong. The error is silently asserting membership in a narrower physical line class, or dropping a junction constraint, shunt, grounding branch, control, rating or provenance boundary without recording it.

Mutual-coupling counterexample

Suppose the two series sections are themselves mutually coupled. In coordinates where their drop relations contain cross blocks $\mathbf{Z}_{12}$ and $\mathbf{Z}_{21}$, substituting $\mathbf{I}_2 = \mathbf{P}\mathbf{I}_1$ gives

$$\mathbf{Z}_{\mathrm{eq,coupled}} = \mathbf{Z}_1 + \mathbf{Z}_{12}\mathbf{P} + \mathbf{P}^{\mathsf{T}}\mathbf{Z}_{21} + \mathbf{P}^{\mathsf{T}}\mathbf{Z}_2\mathbf{P}.$$

The two cross terms do not disappear merely because b has zero injection. Coupling between adjacent sections of the same corridor is also not naturally described as *external coupling at the junction*: it is a constitutive relation between the two element identities. The implementation therefore stores mutual-coupling blocks against the other element identity and rejects the local uncoupled rule whenever either an internal-pair or external-pair block exists. A more general coupled-factor elimination could be exact, but it would be a different rule with the full coupled block as its source.

The executable negative witness uses two reciprocal two-conductor sections. It compares the displayed uncoupled sum with the four-term expression above and records an approximately 11.65% relative Frobenius-norm error. This is a counterexample to the insufficient guard, not a claim that mutual coupling always produces an error of that size.

The witness also records the representation boundary explicitly: a junction-only data model cannot express this case, while the pair-keyed `mutual_couplings` field can. A coupled-factor elimination would need the full pair block and a different target interface; silently dropping that field is a semantic loss.

A separate exact rule for a coupled section pair

The cross terms are not merely a reason to weaken the guard. They define a different, narrower rewrite. `TR-SER-003` applies when the two candidate elements are series-only, the junction guards above hold, both pairwise blocks $\mathbf{Z}_{12}$ and $\mathbf{Z}_{21}$ are declared, and neither element has any mutual-coupling block to a third element. No reciprocity assumption is needed by the algebra; if a physical model requires $\mathbf{Z}_{21} = \mathbf{Z}_{12}^{\mathsf{T}}$, that is an additional model-specific guard.

With $\mathbf{P}$ aligning the conductor order at b, the target is the terminal-behaviour composite

$$\mathbf{Z}_{\mathrm{eq,coupled}} = \mathbf{Z}_1 + \mathbf{Z}_{12}\mathbf{P} + \mathbf{P}^{\mathsf{T}}\mathbf{Z}_{21} + \mathbf{P}^{\mathsf{T}}\mathbf{Z}_2\mathbf{P}.$$

The recovery map remains $\mathbf{I}_1 = \mathbf{I}_{\mathrm{eq}}$ and $\mathbf{I}_2 = \mathbf{P}\mathbf{I}_{\mathrm{eq}}$; member current limits therefore map by the same intersection as in the uncoupled rule. The certificate retains the pair identities and declares that physical homogeneous-line closure is *not* asserted. In particular, this rule does not turn a corridor coupling model into two independent line objects, and it rejects any external coupling that would be left dangling after the rewrite. It is exact for the declared external terminal voltage/current relation, not a license to erase the internal coupling semantics.

The executable certificate records this rule alongside the negative witness in `experiments/generated/degree-two-series-cert` The implementation and tests are self-checked; an independent mathematical review of the new rule remains an explicit review item.

Grounding counterexample

If a grounding or shunt admittance $\mathbf{Y}_g$ is attached at b, then

$$\mathbf{I}_{\ell_1 ib} - \mathbf{P}^{\mathsf{T}}\mathbf{I}_{\ell_2 bj} = \mathbf{Y}_g\mathbf{U}_b^{(1)}.$$

The currents are no longer a single common series variable. A Schur complement may still eliminate $\mathbf{U}_b$, but the result is a more general terminal relation and must not be reported as the series rule above. The prototype therefore rejects this application with `junction_has_shunt_or_grounding`.

Executable certificate

The package-independent prototype is implemented in `experiments/transformations/SeriesElimination.jl`. It returns either a target plus preservation certificate or a structured rejection. The executable case uses a two-conductor permutation and heterogeneous construction codes:

```
julia --project=experiments experiments/run_series_elimination.jl
julia --project=experiments experiments/test/runtests.jl
```

Its machine-readable result is `experiments/generated/degree-two-series-certificate.json`. The certificate classifies the result as an exact behavioural reduction, records the permutation and constraint intersection, and explicitly refuses to call the target a homogeneous physical line. It also records the cross-coupled negative witness, its four-term exact expression, the failed element-pair guard, and the relative error made by the uncoupled expression. It conforms to the common interface in Certificate schema and composition, where the separately certified coordinate normalization is composed with this rule.

Chapter 40

Certificate schema and composition

Page status: versioned certificate schema and composition contract; associativity and richer interface compatibility remain open.

A transformation result is useful only if downstream work can determine what it means. Version 1.1.0 of the book's transformation-certificate interface is defined by `schemas/transformation-certificate.schema.json`.

Common interface

Field	Role
`certificate_id`, `rule_id`	stable claim identity and applied rule
`classification`	exact normalization, exact compilation, exact behavioural reduction, inner restriction, outer relaxation, scenario approximation, or mixed
`source`, `target`	model categories and object identities
`interfaces`	typed source/target contracts for states, constraints, decisions, objectives, units, and boundary quantities
`typed_interfaces`	crosswalk from the certificate-local labels to the checked state-space/unit vocabulary
`preconditions`	conditions under which the classification holds
`preserves`, `forgets`	the semantic contract
`constraint_map`	forward transport of declared constraints
`recovery_map`	reconstruction of source quantities from target results
`provenance`	source-to-target lineage and implementation context
`evidence`	derivation data, witnesses, permutations, or computed results
`numerical_evidence`	optional solver status, optimality scope, residual/tolerance, conditioning/backward error, and uncertainty status

JSON-schema validity establishes a structural contract. It does not prove that the reported classification or equations are true. Those remain claims that need derivation, tests, and independent review.

When numerical evidence is reported, `optimality_status` distinguishes a local solve, a branch-scoped comparison, and a globally certified result. A residual and tolerance describe numerical consistency; they do not establish model adequacy. `conditioning` and `backward_error` make numerical fragility visible, while `uncertainty_status` records whether parameter or measurement uncertainty was quantified. Deterministic algebraic certificates may declare solver and optimality as `not_applicable` rather than implying an optimization result.

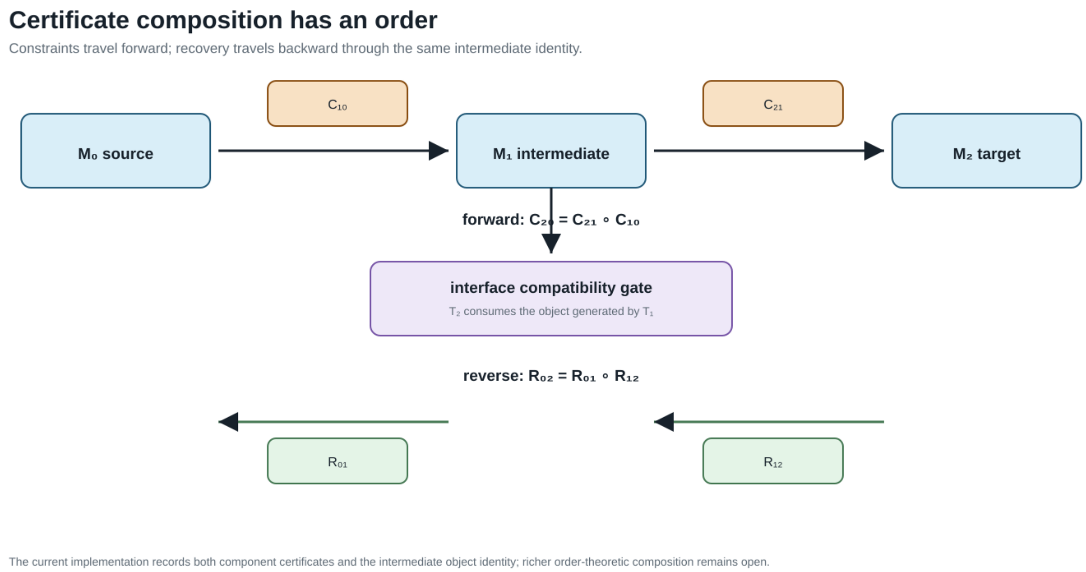

Figure 40.1: Forward and reverse order of certificate composition.

Sequential composition

Let exact transformations be

$$\mathcal{M}_0 \xrightarrow{T_1} \mathcal{M}_1 \xrightarrow{T_2} \mathcal{M}_2.$$

The implemented composition accepts the pair only when an object generated by T_1 is explicitly consumed by T_2. Constraint maps execute in the forward order,

$$C_{20} = C_{21} \circ C_{10},$$

whereas recovery maps execute in reverse order,

$$R_{02} = R_{01} \circ R_{12}.$$

The composite retains the untouched sources of T_2, the original sources of T_1, both component certificate identities, and the intermediate object identity. Non-exact component rules are rejected for now because composing inner, outer, and mixed approximations requires a more explicit order-theoretic calculus.

Composition also carries the six interface contracts and, when present, the typed crosswalk from the first source to the second target. This is a trace, not yet a proof of compatibility: the current rule records both component relations but still uses generated/source object identity as its executable meeting check.

The intermediate identity is the meeting point: constraints compose forward, while recovery reverses the transformation order.

Normalization followed by elimination

The executable example first normalizes ℓ_2bj from (n, a) to (a, n) and then eliminates the zero-injection junction between ℓ_1ib and the normalized element. The composed certificate records:

1. the permutation as the first forward step;
2. series constraint intersection as the second forward step;
3. junction recovery before inverse coordinate recovery; and
4. the unnormalized source identities ℓ_1, ℓ_2, and b.

This is claim `TR-COMP-001`. The `experiments/generated` directory contains both the composed normalization/series certificate and the standalone series certificate.

Version 1.1 boundary

Version 1.1 makes six interface categories mandatory. Each category records a source list, a target list, and the relation between them. Empty lists are allowed and meaningful: for example, a fixed-parameter leakage compilation declares that it introduces no tap or investment decision. The fields are typed by role but their individual entries are still prose rather than a formal state-space or unit algebra. The optional `typed_interfaces` extension now binds the generated certificates to `experiments/generated/state-space-unit-witness.json`: it records the mapped unit families, variable and boundary labels, state-domain IDs, and any unit labels that remain unresolved. The build checker requires this attachment on all sixteen public certificate artifacts, while the schema keeps it optional for older external certificates.

The schema does not yet type uncertainty sets or prove that composed interface relations are compatible. Nor does it establish associativity modulo serialization. Those extensions should be driven by the grounding, switch, and larger OPF case studies. Current Julia generators attach the crosswalk through `TransformationContracts.attach_typed_interfaces`; the Python helper is retained only for migration of older generated artifacts.

Release-oriented semantic matrix

The generated `experiments/generated/semantic-evaluator-matrix.json` binds each of the sixteen public certificate artifacts to the Julia evaluator source, its semantic test file, and the evaluator symbol or rule name. Each row checks that source and target identities, typed unit coverage, the canonical state space witness, evaluator provenance, and nonempty evidence are all present. The package test matrix reads these rows in addition to validating the certificate structure. This is a release traceability contract, not a claim that all evaluators share one numerical backend: the rows intentionally span package-independent algebra, BMOPFTools cross-checks, and JuMP/Ipopt cases.

The release gate additionally runs the public facade, typed state-space tests, and this 16-certificate matrix from a separately instantiated package checkout. The pinned result is recorded in `experiments/generated/clean-package-matrix.j` it is a package-isolation check, not a claim that the full research fixture has no external dependency.

Chapter 41

Kron, Ward, and optimized network equivalents

Page status: scoped reduction definitions, audited literature synthesis, and a package-independent Kron/Ward/scenario comparison; independent mathematical review and source-faithful Opti-KRON implementation remain open.

> **Evidence boundary**
>
> **Scope:** linear and affine boundary reductions, operating-point Ward-style targets, and small scenario-approximation witnesses. **Evidence:** Schur-complement derivations, recovery checks, and package-independent numerical fixtures. **Numerical optimality:** scenario probes and local solves do not establish global nonlinear AC optimality. **Unresolved boundary:** universal physical realizability, uncertainty-robust preservation, and external review.

Three different questions

Kron reduction, Ward equivalents, and Opti-KRON are related, but they do not make the same modelling choice:

Kron:	which linear boundary relation results from elimination?
Ward:	how is the eliminated external system represented at the boundary?
Opti-KRON:	which nodes or clusters should be retained under an error objective?

All three require a declared source model, boundary, injection model, and observation contract. None is inherently an asset-preserving transformation.

Linear Kron reduction

Partition a linear nodal relation into retained boundary variables B and eliminated internal variables I:

$$\begin{bmatrix} \mathbf{i}_B \\ \mathbf{i}_I \end{bmatrix} = \begin{bmatrix} \mathbf{Y}_{BB} & \mathbf{Y}_{BI} \\ \mathbf{Y}_{IB} & \mathbf{Y}_{II} \end{bmatrix} \begin{bmatrix} \mathbf{v}_B \\ \mathbf{v}_I \end{bmatrix}.$$

Proposition. If $\mathbf{Y}_{II}$ is invertible and $\mathbf{i}_I$ is fixed, then eliminating $\mathbf{v}_I$ gives the exact affine boundary relation

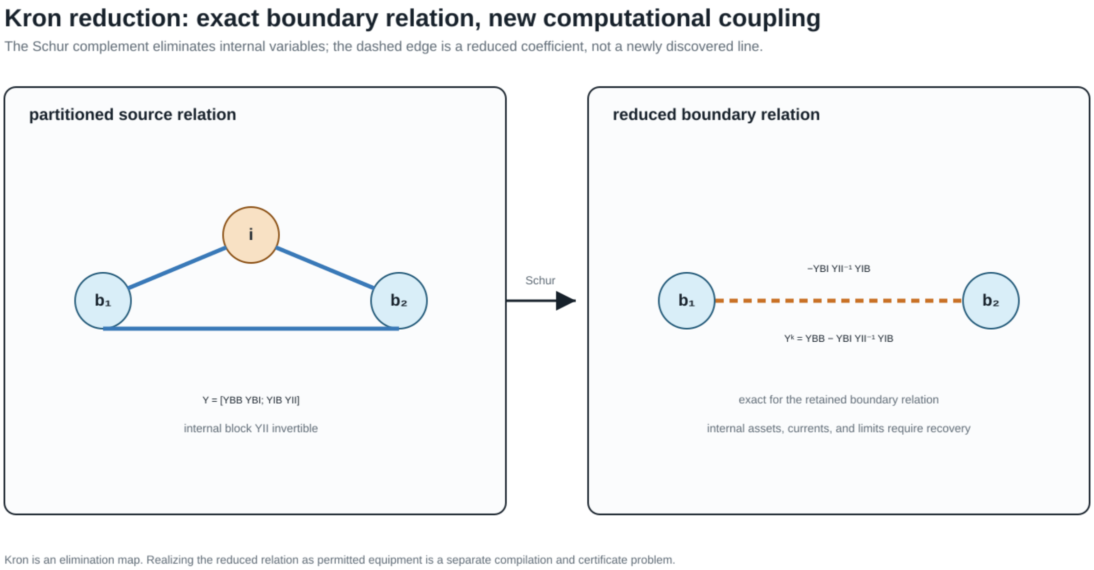

Figure 41.1: Kron reduction creates a boundary fill edge.

$$\mathbf{i}_B = \mathbf{Y}_{\mathrm{K}} \mathbf{v}_B + \mathbf{K}_I \mathbf{i}_I,$$

where

$$\mathbf{Y}_{\mathrm{K}} = \mathbf{Y}_{BB} - \mathbf{Y}_{BI} \mathbf{Y}_{II}^{-1} \mathbf{Y}_{IB}, \qquad \mathbf{K}_I = \mathbf{Y}_{BI} \mathbf{Y}_{II}^{-1}.$$

The eliminated voltages are recovered by

$$\mathbf{v}_I = \mathbf{Y}_{II}^{-1} (\mathbf{i}_I - \mathbf{Y}_{IB} \mathbf{v}_B).$$

Proof. Solve the internal block equation for $\mathbf{v}_I$ and substitute it into the boundary block equation.

For zero internal injection this becomes

$$\mathbf{i}_B = \mathbf{Y}_{\mathrm{K}} \mathbf{v}_B.$$

This is exact equality of a selected linear terminal relation. Schur elimination can create dense coupling among neighbours of eliminated nodes, and the resulting entries need not correspond to physical lines. Dörfler and Bullo analyze the graph and loopy-Laplacian properties of this operation [10]. Closure inside a more restrictive dynamic or device class requires additional physical assumptions [51].

The dashed boundary coupling is a reduced coefficient. It is exact for the retained linear boundary relation, but it is not by itself a physical asset with recoverable currents, limits, or outage identity.

Kron reduction does not by itself preserve:

- eliminated asset identity or topology;
- internal branch currents and limits unless recovery is retained;
- switching, outage, maintenance, or investment decisions;
- protection and failure dependencies;
- sparsity;
- nonlinear constant-power behaviour over arbitrary operating points.

The Schur complement first produces a reduced multiport relation. Realizing that relation as a permitted collection of bus–branch devices is a separate synthesis or compilation problem.

Decision-model consequence

Eliminating a neutral voltage does not eliminate the neutral conductor's current rating. If the source relation gives the neutral branch current as $I_n = A_{nB}v_B + A_{nI}v_I$ and Kron recovery gives $v_I = Y_{II}^{-1}(i_I - Y_{IB}v_B)$, then the reduced decision model must retain the recovered constraint $|A_{nB}v_B + A_{nI}Y_{II}^{-1}(i_I - Y_{IB}v_B)| \le \bar{I}_n$ (or its declared multiconductor norm). The boundary Schur complement alone is not enough for an OPF, hosting-capacity, protection, or contingency problem that observes that neutral limit. The constraint may be omitted only when the neutral rating is genuinely outside the observation set or is proved redundant by a separate certificate.

Typed multiconductor Kron reduction

In the multiconductor case, the symbols B and I denote direct sums of typed terminal-coordinate spaces, not merely lists of scalar buses. Let

$$\mathcal{V}_B = \bigoplus_{k\in B} \mathbb{C}^{n_k}, \qquad \mathcal{V}_I = \bigoplus_{k\in I} \mathbb{C}^{n_k}.$$

Each port voltage is first aligned with its junction by its terminal map. If $\mathbf{N}_q$ maps the ordered junction voltage into the ordered coordinates of port q, then

$$\mathbf{v}_q = \mathbf{N}_q\mathbf{U}_{j(q)}, \qquad \mathbf{I}_{j(q)} \mathrel{+}= \mathbf{N}_q^{\mathsf{H}}\mathbf{i}_q.$$

The conjugate transpose in the current map is the power-dual action: it makes $\mathbf{v}_q^{\mathsf{H}}\mathbf{i}_q$ agree with the corresponding junction power pairing. The assembled nodal blocks in the preceding section are formed only after these terminal maps have been applied. Consequently every product $\mathbf{Y}_{BI}\mathbf{Y}_{II}^{-1}\mathbf{Y}_{IB}$ is a typed map $\mathcal{V}_B \to \mathcal{V}_B$.

Proposition (typed coordinate-covariant Kron reduction). Suppose the assembled linear multiconductor relation has $\mathbf{Y}_{II} : \mathcal{V}_I \to \mathcal{V}_I$ invertible and the internal injection $\mathbf{i}_I$ is fixed data, independent of $\mathbf{v}_I$. Let $\mathbf{T}_B$ and $\mathbf{T}_I$ be invertible coordinate actions such that $\mathbf{T} = \mathrm{diag}(\mathbf{T}_B, \mathbf{T}_I)$ respects the retained/internal partition, with

$$\mathbf{v}_B = \mathbf{T}_B\widetilde{\mathbf{v}}_B, \qquad \mathbf{v}_I = \mathbf{T}_I\widetilde{\mathbf{v}}_I, \qquad \widetilde{\mathbf{i}}_B = \mathbf{T}_B^{\mathsf{H}}\mathbf{i}_B, \qquad \widetilde{\mathbf{i}}_I = \mathbf{T}_I^{\mathsf{H}}\mathbf{i}_I.$$

Then Kron reduction before or after the coordinate change gives the same boundary relation. Specifically,

$$\widetilde{\mathbf{Y}}_{\mathrm{K}} = \mathbf{T}_B^{\mathsf{H}} \mathbf{Y}_{\mathrm{K}} \mathbf{T}_B,$$

and the affine internal-injection term transforms as

$$\widetilde{\mathbf{K}}_I \widetilde{\mathbf{i}}_I = \mathbf{T}_B^{\mathsf{H}} \mathbf{K}_I \mathbf{i}_I.$$

The recovered internal voltage is coordinate consistent:

$$\widetilde{\mathbf{v}}_I = \mathbf{T}_I^{-1} \mathbf{Y}_{II}^{-1} (\mathbf{i}_I - \mathbf{Y}_{IB} \mathbf{v}_B).$$

Proof. The transformed blocks are

$$\widetilde{\mathbf{Y}}_{XY} = \mathbf{T}_X^{\mathsf{H}} \mathbf{Y}_{XY} \mathbf{T}_Y, \qquad X, Y \in \{B, I\}.$$

Using

$$(\mathbf{T}_I^{\mathsf{H}} \mathbf{Y}_{II} \mathbf{T}_I)^{-1} = \mathbf{T}_I^{-1} \mathbf{Y}_{II}^{-1} \mathbf{T}_I^{-\mathsf{H}}$$

in the transformed Schur complement cancels the internal coordinate maps and leaves $\mathbf{T}_B^{\mathsf{H}} \mathbf{Y}_{\mathrm{K}} \mathbf{T}_B$. The same substitution proves the affine and recovery identities.

The proposition covers phase permutations and other invertible port-coordinate actions. Per-port block diagonality within $\mathbf{T}_B$ or $\mathbf{T}_I$ is an additional modelling restriction that keeps the action local to each port; it is not needed by the covariance proof. A dense action within the retained partition and a dense action within the internal partition satisfy the same identities. What is excluded is a coordinate map that mixes retained and internal variables, because then it changes the eliminated/observed partition.

The fixed-injection qualification is load-bearing. If an internal device has a voltage-dependent law, for example $\mathbf{i}_I(\mathbf{v}_I) = \overline{\mathbf{S}_I \oslash \mathbf{v}_I}$, then $\mathbf{K}_I \mathbf{i}_I(\mathbf{v}_I)$ is not a fixed affine term and the proposition does not provide a voltage-independent boundary operator. Such a device may still be eliminated with a nonlinear or state-dependent implicit relation, but that is a different transformation contract.

A sequence-coordinate statement needs its transform convention declared: with the usual non-unitary Fortescue matrix $\mathbf{F}$, applying the same voltage and current coordinates gives the similarity action $\mathbf{F}^{-1}\mathbf{Y}\mathbf{F}$; the power-dual convention above gives the congruence action $\mathbf{F}^{\mathsf{H}}\mathbf{Y}\mathbf{F}$. These coincide only under additional normalization assumptions. Neither convention justifies dropping a conductor, identifying a neutral with earth, or aggregating phases: those maps are not invertible coordinate changes and require separate preservation claims.

Corollary (reciprocity is convention-relative). Kron reduction preserves complex symmetry of a reciprocal nodal matrix in physical coordinates. A real coordinate congruence $T^{\mathsf{T}}\mathbf{Y}T$ also preserves complex symmetry. The power-dual action $T^{\mathsf{H}}\mathbf{Y}T$ preserves it when T is real (or under other explicit compatibility conditions), but not for an arbitrary complex T. Hermitian structure and complex symmetry are distinct properties; the certificate must record which one is required and which coordinate action is being used. Thus "Kron preserves reciprocity" and "this complex coordinate representation remains symmetric" are separate claims.

Executable typed fixture

The first package-independent witness is recorded in `experiments/generated/typed-kron-witness.json` and certified by `experiments/generated/typed-kron-certificate.json`. It has three retained two-conductor ports and one eliminated two-conductor port. The fixture checks the reduced covariance residual, affine-injection covariance, internal-voltage recovery, and source-current limit evaluation after applying complex block-diagonal power-dual coordinate actions. It also checks dense actions within the retained and internal partitions, confirming that per-port block-diagonality is only a locality restriction. The reported residuals are below 2×10^{-15} for the boundary identity and below 7×10^{-17} for internal-state recovery; the fixture records the condition numbers of the internal block and coordinate actions.

The witness evaluates a deliberately different constant-power internal injection at the same recovered $\mathbf{v}_I$. Its current differs materially from the fixed datum, and the affine boundary term changes accordingly. This is a scope diagnostic: it does not disprove nonlinear elimination, but prevents the fixed-injection affine certificate from being reused for a voltage- dependent device.

The same artifact records two target-library outcomes. The general reduced multiport is reciprocal but its off-diagonal conductor blocks are not all individually symmetric, so the direct line–shunt construction is rejected for that restricted library. A separate admissible full-matrix, block-symmetric line–shunt witness is stamped exactly, with residual below 10^{-15}, while a diagonal-only line library is rejected. This is deliberately a positive and negative realizability test, not a claim that every Kron-reduced multiport is a physical bus–branch network.

The same witness now applies a deliberately narrower transformer-library test. In this selected library, each winding block is required to be diagonal in the declared conductor coordinates and the complete relation must remain reciprocal. The coupled target above is rejected because at least one winding block is dense; a companion target obtained by removing those cross-conductor entries is accepted. This is a closure test for a restricted transformer vocabulary, not an identification of turns ratios, leakage parameters, or grounding data.

Direct running-network line witness

The witness family is split by the question it answers rather than presented as one undifferentiated Kron example:

The table is the scope summary; the paragraphs below retain the construction details needed to reproduce each check.

Running-network line and explicit neutral recovery

The canonical running fixture now has a direct typed-Kron check in `experiments/generated/running-network-typed-kron-wit`
Line l_1 is split into two equal four-conductor series sections, with the midpoint retained as the internal block. Eliminating that midpoint reproduces the original i_1–i_2 line primitive to below 10^{-11} and recovers the midpoint voltage $(U_{i_1} + U_{i_2})/2$ to the same tolerance. Terminal order, bus identity, and the four-conductor boundary are retained explicitly.

The same artifact carries a deliberately tight neutral-limit witness. The four-conductor midpoint is eliminated, but the recovered neutral current is $(-0.0430252 + 0.0197760\,\mathrm{i})$ p.u. on both half-sections. A declared limit of 0.0426173 p.u. is therefore violated by this boundary point. The test records that the current recovery is exact and that dropping the neutral constraint would accept a point the source model rejects. This is claim `TR-KRON-NEUTRAL-001`.

The companion `experiments/generated/neutral-kron-independent-reproduction.json` reconstructs the four-conductor impedance and midpoint recovery with a separate standard-library complex solver. It matches both half-section neutral currents, reproduces the exact recovery identity, and retains the deliberately violated limit.

The same witness now includes a midpoint neutral-to-reference shunt probe. The shunt changes the recovered neutral current from the series-only value, adds a reference-current term to the midpoint KCL, and retains a

claims	fixture/question	what varies	what changes	not claimed
TR-KRON-NEUTRAL-001	running four-conductor line midpoint	retained midpoint and neutral limit	recovered neutral current changes feasibility	physical rating generality; nonlinear loads
TR-KRON-NEUTRAL-002	explicit five-conductor earth coordinate	neutral–earth bond and earth terminal	neutral and earth KCL currents remain separate	standards-aligned grounding/protection
TR-KRON-NEUTRAL-003	two explicit grounding points	two bond locations	both bond-current observations survive	one aggregate neutral constraint
TR-KRON-NEUTRAL-004	grounding-impedance sensitivity	two declared impedance pairs	neutral current and feasibility classification	uncertainty quantification
TR-KRON-NEUTRAL-005	one state-dependent bond	endpoint state	frozen nominal map leaves residual; recomputation restores it	global nonlinear theorem
TR-KRON-NEUTRAL-006	two state-dependent bonds	endpoint state and two maps	segment currents and residual change	global continuation/protection
TR-KRON-NEUTRAL-007	finite continuation	five endpoint states	neutral-limit margin changes	adaptive/global continuation
TR-KRON-NEUTRAL-008	local derivative bound	perturbation scale	recomputed Jacobian reduces local error	global error bound
TR-KRON-FIVE-001, -002	five-bus scalar pendant/non-pendant cases	eliminated bus and retained boundary	exact recovery; non-pendant elimination creates fill	general multiconductor closure

separate neutral limit evaluation. Its KCL residual is below 10^{-11}, and the independent reproduction checks the shunted currents as well as the series case.

Explicit earth coordinate and grounding-parameter sensitivity

The next probe makes the earth-return coordinate explicit rather than treating it as an unnamed reference. In TR-KRON-NEUTRAL-002, a synthetic five-conductor (a, b, c, n, e) series midpoint retains an earth terminal e and stamps a midpoint neutral–earth bond. Kron recovery reports separate neutral and earth currents, verifies their KCL equations with opposite bond-current signs, and evaluates the neutral current limit on the recovered half-section. The companion `experiments/generated/explicit-earth-kron-independent-reproduction.json` uses a separate standard-library complex solver. Collapsing e into n would erase the observed bond-current relation.

The same artifact adds a two-grounding-point extension, TR-KRON-NEUTRAL-003. A three-segment five-conductor chain has explicit internal points m_1 and m_2, each with its own neutral–earth bond. The recovered segment currents verify separate neutral and earth KCL at both points, and both bond currents remain observable after the two internal blocks are eliminated. This is the smallest useful warning against replacing a distributed grounding structure by one aggregate neutral constraint.

Finally, TR-KRON-NEUTRAL-004 holds the topology and terminal order fixed while sweeping four declared pairs of grounding impedances. The recovered neutral current changes across the cases, and a fixed 0.028 p.u. neutral limit changes feasibility classification: the structural Kron and KCL checks remain valid, but the decision observation does not. The generated sweep and its standard-library reproduction therefore separate structure preservation from parameter-dependent feasible-set preservation. This is a finite sensitivity probe.

State-dependent grounding bonds

The next boundary is local state dependence. In `TR-KRON-NEUTRAL-005`, the neutral–earth bond current is defined by an illustrative voltage-dependent law $i_{ne} = y_0(1+\alpha_{\mathrm{g}}|V_n - V_e|^2)(V_n - V_e)$. Here α_{g} is a grounding-law coefficient, distinct from the served-load decision α used in the parallel-line cases. After an endpoint state shift, the nominal bond map leaves a nonzero nonlinear residual and a different recovered neutral current; a local Newton solve with the bond recomputed at the shifted state restores the relation and re-evaluates the neutral limit. The independent reproduction uses a separate finite-difference Newton implementation.

The distributed version is also exercised locally in `TR-KRON-NEUTRAL-006`. The three-segment chain has two voltage-dependent neutral–earth bonds. After the same endpoint shift, freezing both nominal bond maps leaves a nonzero chain residual and changes the recovered neutral currents on the three segments; a two-point Newton solve with both maps recomputed restores the local relation. The companion standard-library reproduction checks the midpoint values and residuals.

`TR-KRON-NEUTRAL-007` records a finite endpoint-state continuation of that two-point chain at $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$. Each state is solved with both nonlinear bond maps recomputed; the five nonlinear residuals remain small and the fixed 0.05 p.u. neutral limit changes classification along the path. Reusing the nominal map fails away from $\lambda = 0$. The independent reproduction checks every continuation row. This is a finite local path.

`TR-KRON-NEUTRAL-008` adds a local derivative certificate for the same illustrative voltage-dependent neutral–earth bond law. At a declared base state, the analytic real Jacobian is compared with the frozen nominal coefficient after a state shift. The frozen map leaves a nonzero residual, whereas the recomputed Jacobian gives the smaller local linearisation error; the error decreases as the declared step is reduced. The generated `experiments/generated/nonlinear-grounding-local-bound-witness.json` records the Jacobian, residuals, step scales, and interpretation. This is a local Taylor/conditioning check.

Five-bus scalar reductions

The five-bus companion `experiments/generated/five-bus-typed-kron-witness.json` covers the scalar pendant case directly. Eliminating bus m through its sole incident line u gives the same retained Y-bus as deleting that leaf line from the graph, and the recovered boundary current matches the full nodal relation to machine precision. This is claim `TR-KRON-FIVE-001`.

The same witness also eliminates the non-pendant bus ℓ while retaining i, j, k, m. Boundary-current recovery remains exact, but the Schur complement creates retained support on $j - m$ and $k - m$. These are fill edges: they are couplings in the reduced relation, not automatically new physical lines. The recovered u-branch current is exact as well, but a deliberately tight declared u-limit is violated by the recorded state. This is claim `TR-KRON-FIVE-002` and is the small scalar analogue of the ordering, fill-in, and retained-constraint consequences discussed later in the numerical chapter.

Realizability is a second theorem

The reduced matrix always defines a general linear factor on the retained ports. It need not define a conventional collection of lines. Realizability is therefore relative to a target device library.

> **Power-system shorthand**
>
> A nonzero off-diagonal block in a reduced admittance is a boundary coupling, not automatically a physical line. Any line–shunt realization is a second construction whose asset meaning, limits, states, and provenance must be declared separately.

For a useful baseline, suppose all retained junctions use the same ordered c-conductor coordinates and the target library permits:

1. one reciprocal full-matrix series primitive between any pair of retained junctions; and

2. one full-matrix shunt primitive at every retained junction.

Write the reduced admittance in $c \times c$ blocks $\mathbf{Y}^{\mathrm{K}}_{pq}$.

Proposition (direct line–shunt realization). If every off-diagonal block obeys

$$\mathbf{Y}^{\mathrm{K}}_{pq} = \mathbf{Y}^{\mathrm{K}}_{qp} = (\mathbf{Y}^{\mathrm{K}}_{pq})^{\mathsf{T}},$$

then the reduced relation has the exact complete-graph realization

$$\mathbf{Y}^{\mathrm{s}}_{pq} = -\mathbf{Y}^{\mathrm{K}}_{pq}, \qquad \mathbf{Y}^{\mathrm{sh}}_{p} = \mathbf{Y}^{\mathrm{K}}_{pp} - \sum_{q \neq p} \mathbf{Y}^{\mathrm{s}}_{pq}.$$

Proof. A reciprocal series primitive contributes $-\mathbf{Y}^{\mathrm{s}}_{pq}$ to both off-diagonal blocks and $\mathbf{Y}^{\mathrm{s}}_{pq}$ to both corresponding diagonal blocks. Summing all pairwise stamps reproduces every off-diagonal block. The residual shunt definition then reproduces each diagonal block. Thus the proposition is an exact stamping identity in a library that permits full $c \times c$ reciprocal blocks; it is not a claim that a smaller physical line library is closed under Kron reduction. For N retained junctions the construction uses $N(N-1)/2$ pairwise series blocks plus N shunts.

This is an algebraic realization, not yet a physical line realization. A passive line–shunt library imposes additional conditions such as

$$\mathrm{He}(\mathbf{Y}^{\mathrm{s}}_{pq}) \succeq 0, \qquad \mathrm{He}(\mathbf{Y}^{\mathrm{sh}}_{p}) \succeq 0,$$

along with its reciprocity, frequency, grounding, and parameterization rules. A restricted diagonal or sequence-decoupled library imposes stronger closure conditions. Ideal-transformer terminal maps can realize a broader class of off-diagonal blocks, but then transformer ratios, winding coordinates, grounding, and provenance become part of the target certificate. If none of these libraries closes, retaining $\mathbf{Y}_{\mathrm{K}}$ as one general multiport factor is still exact.

Internal current and limit recovery is yet another layer. If a source branch current has the affine form

$$\mathbf{I}_{\ell} = \mathbf{A}_{\ell B}\mathbf{v}_B + \mathbf{A}_{\ell I}\mathbf{v}_I,$$

substitution of the voltage recovery map gives an exact affine boundary map for $\mathbf{I}_{\ell}$. Keeping that map permits source limits to be checked; discarding it does not make limits on artificial reduced branches equivalent.

Ward equivalents

The affine term

$$\mathbf{K}_I \mathbf{i}_I$$

shows what a network equivalent must do when the eliminated region has nonzero injections. A Ward-type equivalent combines the reduced boundary admittance with equivalent boundary injections, shunts, or sources intended to represent the external system in a power-flow study. Ward's original construction explicitly approximates suppressed loads and generation as constant currents, retains the tie terminals, replaces the eliminated network by a boundary mesh, and places equivalent injections at those terminals [59].

For a linear fixed-current source model, the affine relation above is exact. For a constant-power AC model, however,

$$\mathbf{i}_I(\mathbf{v}_I) = (\mathbf{S}_I \oslash \mathbf{v}_I)^*,$$

where $\oslash$ denotes componentwise division. The internal injection is then voltage dependent, so replacing it by a fixed boundary source is not a global exact reduction of the nonlinear feasible relation. Its validity must instead be tied to an operating point, linearization, iteration, or scenario domain.

The term *extended Ward* is not just another name for that base construction. Monticelli and coauthors build an external equivalent for static security from a single estimated operating state, address boundary-bus designation, and discuss treatment of external shunts and generator-outage studies [60]. A Ward–PV construction instead retains external generator buses after load-node elimination; the reduced Ward–PV model then aggregates selected coherent generator groups [61]. These targets preserve different state and control structure.

The resulting source taxonomy is:

- **classical Ward:** constant-current approximation followed by external-bus elimination and boundary-mesh/injection realization;
- **operating-state extended Ward:** a base-state-calibrated external equivalent with extra boundary, shunt, and contingency treatment;
- **Ward–PV:** retention of selected generator/PV structure before any subsequent coherent aggregation;
- **nonlinear or iterative Ward-type methods:** later constructions that update or fit the boundary source model over operating points and must state their own domain.

These are historically related, not mathematically interchangeable. In particular, the exact affine result for fixed currents does not make a base-state-calibrated AC equivalent globally exact for constant-power or voltage-controlled devices.

Opti-KRON

Opti-KRON adds a structural selection problem around a Kron-based electrical reduction. An assignment matrix maps original nodes to retained supernodes; the selection trades the degree of reduction against reproduction error for declared voltage observations and operating scenarios. The three-phase work also constrains phase availability and connectivity [53]. A related extension identifies nodes to restore so that the final reduced network recovers radiality [54].

It is useful to factor the method conceptually as

$$\text{optimized structural assignment} + \text{Kron-based electrical reduction} + \text{scenario observation metric}.$$

The Schur-complement step can be exact for its retained linear boundary relation while the supernode representation of eliminated voltages is approximate. The combined method is therefore not classified simply as *exact Kron reduction*. Its certificate must record at least:

- the retained-node and assignment decision spaces;
- phase and connectivity guards;
- the operating scenarios used to evaluate voltage error;
- the voltage observation norm and bound;
- any radiality restoration;
- the source injections and controls represented in those scenarios;
- whether source constraints and decisions can be recovered.

Low voltage error over a scenario set does not alone prove equality of AC OPF feasible sets, active limits, discrete decisions, or objective values. Those are additional observation families requiring their own evidence.

Decision-model consequence

Scenario voltage accuracy answers one observation question. It is not a surrogate theorem for feasibility, active-limit, objective, or discrete decision accuracy.

Classification in the book's transformation language

The classification depends on the complete construction, not on whether its name contains *Kron*:

- **Zero-injection linear Kron:** from a linear nodal or multiport relation to its boundary relation. This is exact behavioural reduction; physical realization and internal constraints are not automatic.
- **Fixed-current affine Kron:** from a linear relation with fixed internal injections to an affine boundary relation. This is exact behavioural reduction for that source model, not for arbitrary voltage-dependent injections.
- **Ward equivalent:** from an external-system study model to a boundary network with equivalent injections. It may be exact, local, or approximate depending on the injection model; there is no assumed universal definition across Ward variants.
- **Opti-KRON:** from scenario data and a candidate topology to a selected reduced network. This is mixed structural optimization and scenario approximation; general decision equivalence is not automatic.
- **Radiality restoration:** from a nonradial reduced topology to one with selected nodes restored. This is structural postprocessing under its stated rule, not electrical or decision equivalence by itself.

These distinctions place each method relative to a preservation contract before relating it to the book's local rewrite rules.

Relation to local transformations

The guarded degree-two series rule is a special zero-injection elimination for which the reduced relation remains inside a declared series-element family. A star–mesh transformation is another local Schur-complement realization. Parallel primitive summation is different: it combines factors sharing a boundary but eliminates no bus. Redundant-limit removal is different again: it is exact presolve on the constraints while the physical members remain.

This separation prevents every operation involving a smaller network from being called Kron reduction.

Open research boundary

Executable comparison: exact, operating-point, and scenario-selected

The comparison artifact `experiments/generated/kron-ward-scenario-comparison.json` uses four declared scenarios and a shared observation contract for three targets. It records boundary voltage and current, recovered internal voltage and current, the source-current constraint margin, and the scenario objective together with the selected structural decision. This prevents a sparse candidate from looking successful merely because its boundary-current error was reported without the state, constraint, or decision quantities that the study actually uses.

Target	Construction	Result in the fixture
exact Kron	recompute the affine boundary relation for each fixed internal injection	exact for every declared scenario
operating-point Ward	retain the exact reduced admittance but freeze the boundary injection at the base scenario	exact at the base point; relative current errors of about 1.5–3.3% off base
Opti-KRON-style target	select full, banded, or diagonal retained couplings using an explicit scenario error plus sparsity penalty	selects the banded target; this is scenario approximation, not decision equivalence

The selection is intentionally small and transparent. It demonstrates the classification boundary rather than reproducing a particular published Opti-KRON implementation: the target candidates and penalty are declared in the artifact, and the selected target is judged on the same observation family as the alternatives. The result is claim `TR-KRON-002` and does not establish global AC feasibility, objective, or control preservation.

The fixture now also exposes the boundary-support distinction directly. An **extended Ward support target** retains the same base-calibrated reduced admittance and fixed base injection, but supplies the explicit support term

$$\Delta \mathbf{i}_B^{\text{support}} = \mathbf{K}_I(\mathbf{i}_I - \mathbf{i}_I^{\text{base}}).$$

For the declared fixed-current linear fixture this support term makes the target exact at every scenario, and its off-base norm is nonzero. That result does not make the construction a globally exact AC Ward equivalent: it records the additional boundary quantity that must be available, and the source model under which it is valid. The generated comparison records these rows under `extended_ward_rows` alongside the operating-point rows.

Certified approximation chain

The next artifact composes the approximation vocabulary into a decision test. In the same one-state fixture, the Ward target freezes the internal injection at the base scenario. For a scenario injection mismatch $\delta i_I = i_I^{\text{base}} - i_I$, the exact linear maps give

$$\delta i_I \longmapsto Y_{II}^{-1}\delta i_I \longmapsto K_I \delta i_I \longmapsto m = L - \|\widehat{i}_B\|,$$

where the middle term bounds recovered-state error, the next term bounds boundary-current error, and m is the approximate current-limit margin. A declared error bound e yields the same three-way test used in the numerical consequences chapter: $m > e$ is certified feasible, $m < -e$ is certified violated, and $|m| \le e$ is ambiguous.

Scenario	Approximate margin	Error bound	Classification
base	-0.09535	0	certified violated
high-load	0.02203	0.03430	ambiguous
low-voltage	0.12026	0.03028	certified feasible
internal-outage proxy	-0.07692	0.07498	certified violated

The generated witness `experiments/generated/certified-approximation-witness.json` reports this chain for all four scenarios:

This is claim `TR-KRON-003`. The normwise bound is exact for the declared one-state linear fixture, so it demonstrates composition of the machinery, not a general nonlinear or uncertainty-aware certification theorem. In particular, the high-load row is intentionally ambiguous even though the nominal Ward point satisfies the limit: the error interval crosses the decision boundary.

Scoped nonlinear AC probe

The generated `nonlinear-ward-witness.json` takes one deliberately small step toward the AC case. Its eliminated state has a constant-power injection, so the internal current is $\mathbf{i}_I(\mathbf{v}_I) = \overline{\mathbf{S}/\mathbf{v}_I}$ and the exact state is found with a damped Newton solve. The Ward target still freezes the base internal current. For each scenario the witness reports the nonlinear residual at the Ward state, a local inverse-Jacobian estimate, the direct boundary-current error, and the resulting local decision classification:

Scenario	Local result	What it demonstrates
base	locally certified feasible	calibration is exact at the base point
small shift	locally certified feasible	the local estimate dominates the observed boundary-current error
large shift	local-bound ambiguous	nonlinear residual and the error interval grow away from calibration

This is an exploratory numerical witness, not a theorem. The inverse-Jacobian estimate is local, depends on the chosen Newton solution, and does not certify global AC feasibility, bifurcation behaviour, parameter uncertainty, or KKT preservation. A solver-exported Jacobian/KKT comparison remains a separate roadmap item.

The current evidence leaves four implementation questions open:

1. executable realizability tests for selected line, shunt, transformer, and general-factor libraries;
2. recovery and constraint maps for internal currents, powers, and losses;
3. an executable small example comparing exact Kron, a Ward operating-point equivalent, and an Opti-KRON-style scenario approximation;
4. a decision experiment measuring feasibility, active constraints, and objective error in addition to voltage error;
5. extension of the certified-approximation chain to nonlinear AC residuals, uncertain parameters, and an independently derived error analysis beyond this scoped probe.

Chapter 42

Conductor-coordinate normalization

Page status: guarded exact coordinate transformation with an executable certificate; independent mathematical review remains open.

Conductor order is representation, not physics. It is nevertheless part of the meaning of every vector, matrix, and componentwise constraint. A rewrite that changes an order without changing all dependent objects is therefore a model error, not a harmless formatting operation.

Vocabulary bridge

Normalization in this chapter is an invertible rewrite into a declared conductor order. It is not per-unit conversion, schema normalization, or ML feature scaling. Those operations require their own maps, units, inverses where applicable, and preservation claims.

Rule

Consider an element ℓij with source conductor order $\gamma_\ell = (n, a)$ and a requested order $\widehat{\gamma}_\ell = (a, n)$. Let $\mathbf{P}_\ell$ be the unique permutation for which

$$\widehat{\mathbf{x}}_{\ell ij} = \mathbf{P}_\ell \mathbf{x}_{\ell ij}.$$

Intrinsic element data retain only the element index. In particular, the coordinate-normalized impedance is

$$\widehat{\mathbf{Z}}_\ell = \mathbf{P}_\ell \mathbf{Z}_\ell \mathbf{P}_\ell^\mathsf{T},$$

and a vector of componentwise limits becomes

$$\widehat{\mathbf{I}}_\ell^{\max} = \mathbf{P}_\ell \mathbf{I}_\ell^{\max}.$$

The terminal map at j is reordered by preserving each original from–to conductor pairing. Thus a coordinate position can move, but conductor identity cannot silently change.

Exactness and inverse

A permutation matrix satisfies $\mathbf{P}_\ell^{-1} = \mathbf{P}_\ell^{\mathsf{T}}$. Hence every normalized state and every transformed intrinsic matrix has a unique recovery:

$$\mathbf{x}_{\ell ij} = \mathbf{P}_\ell^{\mathsf{T}} \widehat{\mathbf{x}}_{\ell ij}, \qquad \mathbf{Z}_\ell = \mathbf{P}_\ell^{\mathsf{T}} \widehat{\mathbf{Z}}_\ell \mathbf{P}_\ell.$$

The rule is therefore an exact normalization (`TR-COORD-001`). It forgets no declared source semantics. This statement depends on permuting every indexed quantity consistently; permuting only $\mathbf{Z}_\ell$ or only a terminal list is not the same rule.

Guards and rejection

The executable rule requires:

- unique conductor labels in both orders;
- equal source and requested conductor sets;
- equal arity; and
- preservation of the original paired terminal map.

A missing, duplicated, or new conductor label produces a structured rejection. The implementation does not guess whether, for example, `g` and `n` are interchangeable.

Executable example

For the order reversal used by the running series example,

$$\mathbf{P}_\ell = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

The rule moves limits $(80, 110)$ A in order (n, a) to $(110, 80)$ A in order (a, n) and permutes both axes of $\mathbf{Z}_\ell$. Run:

```
julia experiments/test/coordinate_normalization.jl
julia --project=experiments experiments/run_coordinate_series_composition.jl
```

The machine-readable result is `experiments/generated/coordinate-normalization-certificate.json`. The next transformer chapter applies the shared coordinate action to a winding's full terminal-to-coil relation. The series chapter then composes line-coordinate normalization with degree-two elimination.

Chapter 43

Transformer-winding coordinate normalization

Page status: guarded exact winding-coordinate normalization with executable round trips; compact serialization closure remains open.

A winding is a port of a multiwinding device, not an independent two-terminal transformer. Its terminal coordinates may nevertheless need normalization before factor assembly, comparison, or compilation. This chapter applies the same typed coordinate action used for lines while retaining transformer and winding identity.

Winding factor

Let transformer x have winding k at bus i. Its ordered terminal voltage vector is $\mathbf{U}_{xki}$. A connection-incidence matrix $\mathbf{A}_{xk}$ maps terminal voltages to stable coil coordinates:

$$\mathbf{V}_{xk}^{\mathrm{coil}} = \mathbf{A}_{xk}\mathbf{U}_{xki}.$$

For a grounded-wye winding, each row of $\mathbf{A}_{xk}$ subtracts the neutral voltage from one phase voltage. For a delta winding, each row takes a declared line-to-line difference. The row order identifies the coils and is distinct from the terminal-coordinate order.

Terminal-coordinate action

Let

$$\widehat{\mathbf{U}}_{xki} = \mathbf{P}_{xk}\mathbf{U}_{xki}$$

for a permutation $\mathbf{P}_{xk}$. Coil coordinates are retained, so the normalized incidence matrix must be

$$\widehat{\mathbf{A}}_{xk} = \mathbf{A}_{xk}\mathbf{P}_{xk}^{\mathsf{T}}.$$

Then

$$\widehat{\mathbf{A}}_{xk}\widehat{\mathbf{U}}_{xki} = \mathbf{A}_{xk}\mathbf{P}_{xk}^{\mathsf{T}}\mathbf{P}_{xk}\mathbf{U}_{xki} = \mathbf{A}_{xk}\mathbf{U}_{xki}.$$

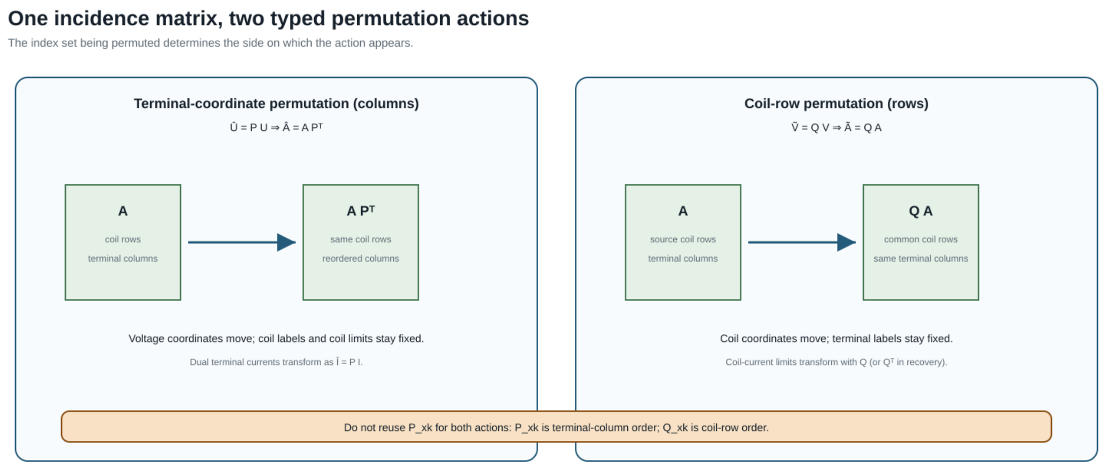

Figure 43.1: Terminal-column and coil-row permutation actions on the same winding incidence matrix.

The terminal-current map is the dual of the voltage map:

$$\widehat{\mathbf{I}}_{xki} = \mathbf{P}_{xk}\mathbf{I}_{xki}, \qquad \mathbf{I}_{xki} = \mathbf{A}_{xk}^{\mathsf{T}}\mathbf{I}_{xk}^{\mathrm{coil}}.$$

The transpose (rather than an adjoint) is correct because the incidence matrix is real. These two maps give complex-power invariance between source and target terminal coordinates, and between terminal and coil coordinates.

Coil-indexed limits are unchanged because their coordinates have not been permuted. A terminal-current limit is different: when it is declared in the source model, it must be reordered by the same P action. If coil coordinates were also reordered, they would require a second coordinate action on the rows and on every coil-indexed constraint.

Why delta needs the full relation

Changing only a delta winding's `terminal_map` can change which terminal pair forms each coil. A cyclic order, vector-group convention, or `delta_roll` field is not a substitute for transforming the actual relation unless the serialization semantics prove that they are equivalent.

The executable example takes winding 3 of running transformer x_1, whose source terminal order is (a, b, c) and whose connection is delta. It requests (c, a, b), transforms the terminal-to-coil matrix, and verifies for a complex voltage witness that source and target coil voltages agree. Grounded-wye winding 1 is tested separately.

This exact typed-factor normalization is claim `TR-XFMR-001`. It does not yet claim that the normalized factor can be written back into every transformer's compact BMOPFTools fields; that serialization question is a separate guarded compilation problem.

Executable rule

The reusable coordinate action, winding factor, and tests are package independent:

```
julia experiments/test/transformer_winding_normalization.jl
julia --project=experiments experiments/run_transformer_winding_normalization.jl
```

The generated transformer-winding certificate records the source fixture winding, both terminal orders, the permutation, both incidence matrices, the dual current map, terminal-versus-coil limit semantics, and executable complex-power checks. It deliberately does not claim all*declared*source_semantics: only the listed identities and constraints are certified.

The next step, Multiwinding terminal leakage assembly, aligns these coil-row identities and composes all winding connection factors with the reference-invariant leakage relation.

Chapter 44

Multiwinding leakage reference compilation

Page status: exact reference-compilation construction with executable round-trip evidence; independent transformer review remains open.

Pairwise short-circuit data are a compact source description of transformer leakage, but they are not yet the matrix relation required by a multiconductor network model. This chapter gives an exact compilation for an arbitrary number of windings. It keeps winding identity, ratios, and limits explicit and does not assume that the general result is a diagonal star.

Source contract

Let transformer x have ordered windings

$$\mathcal{K}_x = \{1, \ldots, n_x\},$$

The source provides nominal coil voltages v_k^{nom}, winding resistances r_k, current limits $\bar{i}_k$, and a short-circuit reactance x_{ij}^{sc} for every unordered pair $1 \le i < j \le n_x$. It must identify the winding s whose voltage base is used for those reactances.

The compilation may independently select any winding $r \in \mathcal{K}_x$ as its internal reference. Define

$$N_k^{(r)} = \frac{v_k^{\mathrm{nom}}}{v_r^{\mathrm{nom}}}, \qquad r_k^{(r)} = \frac{r_k}{(N_k^{(r)})^2}, \qquad x_{ij}^{\mathrm{sc},(r)} = x_{ij}^{\mathrm{sc},(s)} \left(\frac{v_r^{\mathrm{nom}}}{v_s^{\mathrm{nom}}} \right)^2,$$

and the pairwise impedance referred to the selected winding,

$$z_{ij}^{\mathrm{sc},(r)} = r_i^{(r)} + r_j^{(r)} + \mathrm{j} x_{ij}^{\mathrm{sc},(r)}.$$

The pair indices describe winding tests, not fictitious independent two-winding transformers. Per-winding limits remain attached to k.

Exact reference coordinates

Let p_r enumerate the windings in $\mathcal{K}_x \setminus \{r\}$ without changing their relative order. For nonreference windings i and j, form

$$\left[\mathbf{Z}_x^{\mathrm{B},(r)}\right]_{p_r(i),p_r(j)} = \frac{1}{2}\left(z_{ri}^{\mathrm{sc},(r)} + z_{rj}^{\mathrm{sc},(r)} - z_{ij}^{\mathrm{sc},(r)}\right),$$

using $z_{ii}^{\mathrm{sc},(r)} = 0$ on the diagonal. This produces the full $(n_x - 1) \times (n_x - 1)$ reference impedance matrix. Its off-diagonal entries are generally nonzero and are essential when $n_x > 3$.

The construction is invertible on the declared pairwise data:

$$z_{rj}^{\mathrm{sc},(r)} = \left[\mathbf{Z}_x^{\mathrm{B},(r)}\right]_{p_r(j),p_r(j)},$$

and, for $i, j \neq r$,

$$z_{ij}^{\mathrm{sc},(r)} = \left[\mathbf{Z}_x^{\mathrm{B},(r)}\right]_{p_r(i),p_r(i)} + \left[\mathbf{Z}_x^{\mathrm{B},(r)}\right]_{p_r(j),p_r(j)} - 2\left[\mathbf{Z}_x^{\mathrm{B},(r)}\right]_{p_r(i),p_r(j)}.$$

Thus the compilation forgets none of the pairwise tests. It changes their coordinates and exposes a matrix factor suitable for network assembly.

External winding admittance

Let $\mathbf{C}_x^{(r)} \in \mathbb{R}^{(n_x-1)\times n_x}$ have row $p_r(i)$ equal to $\mathbf{e}_r^\mathsf{T} - \mathbf{e}_i^\mathsf{T}$, and let $\mathbf{D}_x^{(r)} = \mathrm{diag}(N_1^{(r)}, \ldots, N_{n_x}^{(r)})$. When $\mathbf{Z}_x^{\mathrm{B},(r)}$ is nonsingular, the leakage admittance in the external winding coordinates is

$$\mathbf{Y}_x^{\mathrm{w}} = (\mathbf{D}_x^{(r)})^{-1}(\mathbf{C}_x^{(r)})^\mathsf{T}(\mathbf{Z}_x^{\mathrm{B},(r)})^{-1}\mathbf{C}_x^{(r)}(\mathbf{D}_x^{(r)})^{-1}.$$

This is a compilation of the leakage relation. Wye/delta terminal-to-coil incidence, terminal ordering, tap decisions, and network interconnection are separate factors and must be composed explicitly.

Reference-choice invariance

Changing r changes the impedance base, reference differences, and entries of $\mathbf{Z}_x^{\mathrm{B},(r)}$. It does not change the external relation in winding-own voltage and current coordinates:

$$\mathbf{Y}_x^{\mathrm{w},(r)} = \mathbf{Y}_x^{\mathrm{w},(q)}, \qquad r, q \in \mathcal{K}_x.$$

This equality is the appropriate invariant. Comparing the internal Z_B matrices directly would be a coordinate error because they use different impedance bases and reference-difference rows. The executable rule compiles every possible r and compares the resulting external admittances. For the running BMOPFTools fixture, the schema convention makes winding 1 the source impedance reference s; the generated certificate records that interpretation separately from the selected compilation reference r.

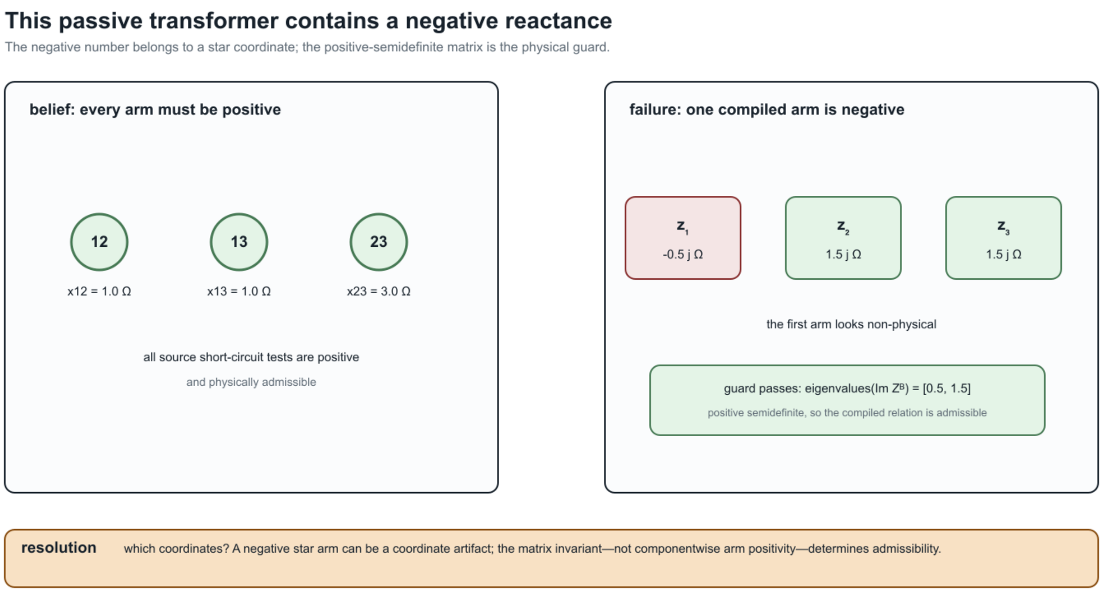

Figure 44.1: A negative star-arm reactance can be a valid coordinate representation.

Three windings are the special case

For $n_x = 3$, the familiar star/T arms are

$$z_1 = \tfrac{1}{2}(z_{12}^{\mathrm{sc}} + z_{13}^{\mathrm{sc}} - z_{23}^{\mathrm{sc}}), \quad z_2 = \tfrac{1}{2}(z_{12}^{\mathrm{sc}} + z_{23}^{\mathrm{sc}} - z_{13}^{\mathrm{sc}}), \quad z_3 = \tfrac{1}{2}(z_{13}^{\mathrm{sc}} + z_{23}^{\mathrm{sc}} - z_{12}^{\mathrm{sc}}).$$

An individual arm reactance can be negative without invalidating the pairwise test set. The implemented physical guard therefore checks positive semidefiniteness of the symmetric matrix $\mathrm{Im}(\mathbf{Z}_x^{\mathrm{B},(r)})$; it does not impose a componentwise nonnegative-arm rule. The executable tests include a valid negative-arm case and a nearby non-PSD rejection.

The reader-facing witness makes the distinction concrete: three positive pairwise tests $(1.0, 1.0, 3.0)\ \Omega$ compile to a star with $\mathrm{Im}(z_1) = -0.5\ \Omega$ and $\lambda(\mathrm{Im}(Z_B)) = (0.5, 1.5)$. The negative entry is a coordinate result, not a negative physical test. The generated evidence is `negative_star_arm_witness` in `experiments/generated/multiwinding-leakage-compilation-certificate.json`.

Running transformer

For fixture transformer x_1, all three referred winding resistances are $0.38875225\ \Omega$. The compiled matrix is

$$\mathbf{Z}_{x_1}^{\mathrm{B}} = \begin{bmatrix} 0.7775045 + \mathrm{j}4.665027 & 0.38875225 + \mathrm{j}3.110018 \\ 0.38875225 + \mathrm{j}3.110018 & 0.7775045 + \mathrm{j}5.4425315 \end{bmatrix}\ \Omega.$$

Its reactance eigenvalues are approximately 1.91956 and 8.18800, and the round trip recovers the three source reactances exactly to floating-point tolerance. The corresponding star-arm reactances are 3.110018, 1.555009, and $2.3325135\ \Omega$.

Repeating the compilation with each of the three windings as r gives distinct matrices on the corresponding voltage bases. After mapping back to winding-own coordinates, the largest entrywise difference among their external admittance matrices is $2.84 \times 10^{-14}\ \mathrm{S}$. The certificate records this fixture-level invariance check.

The generated certificate `TR-XFMR-002` classifies this as an `exact_compilation`. Its typed interfaces state that winding limits and external coil quantities remain at the boundary, while reference-coordinate states are introduced internally. The package-independent implementation and positive, negative, incomplete-data, and four-winding tests live under `experiments/transformations` and `experiments/test`.

Decision-model boundary

The exactness claim concerns the fixed-parameter leakage relation and the declared pairwise source data. A target optimization model is decision-exact only if it also retains winding current limits, connection factors, tap or phase-shift decisions, thermal states, and any objective terms indexed by the original windings. The certificate retains the fixture's per-winding current limits by identity; it does not claim to compile an adjustable tap model.

Multiwinding terminal leakage assembly performs the next exact composition: it combines this coil-coordinate relation with grounded-wye and delta connection factors while retaining a lifted coil-current constraint map.

Chapter 45

Multiwinding terminal leakage assembly

Page status: guarded exact terminal assembly with executable certificates; tap-dependent and excitation extensions remain open.

The pairwise-leakage compilation produces a relation among winding coil coordinates. A network model, however, connects transformer terminals to buses. This chapter composes the leakage relation with the winding connection factors while keeping coil currents available for operational constraints.

The result is an exact terminal-level **leakage factor**, not yet a complete transformer model. Magnetizing and core-loss branches, vector-group phase shifts not represented by the connection incidence, and adjustable taps remain separate factors.

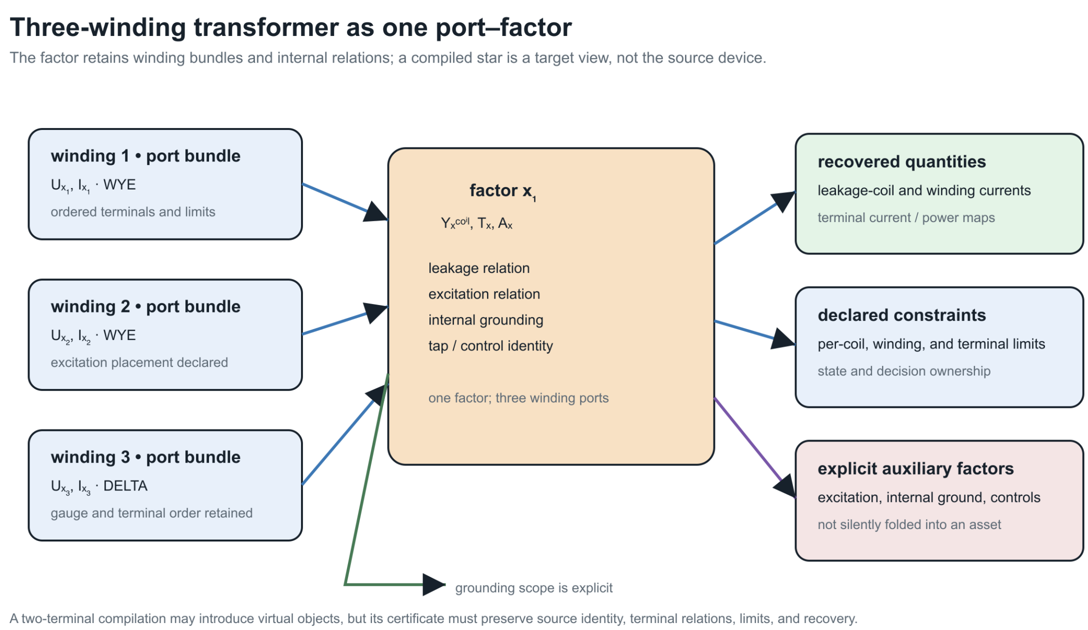

Figure 45.1: The running transformer retains two WYE windings and an open DELTA tertiary; the terminal assembly is a multi-port factor, not an ordinary line.

Typed winding factors

For transformer x, winding k at bus i has ordered terminal voltage $\mathbf{U}_{xki}$ and labelled coil coordinates $\mathcal{C}_{xk}$. Its connection incidence satisfies

$$\mathbf{V}_{xk}^{\mathrm{coil}} = \mathbf{A}_{xk}\mathbf{U}_{xki}.$$

The row labels of $\mathbf{A}_{xk}$ are part of the type. Two rows occupying the same array position do not thereby represent the same magnetic coordinate. The executable rule requires every winding to declare the same set $\mathcal{C}_x$ and constructs the coil-row permutation $\mathbf{Q}_{xk}$ from its source order to a common order. It then uses

$$\widetilde{\mathbf{A}}_{xk} = \mathbf{Q}_{xk}\mathbf{A}_{xk}, \qquad \widetilde{\boldsymbol{\imath}}_{xk}^{\max} = \mathbf{Q}_{xk}\boldsymbol{\imath}_{xk}^{\max}.$$

This is an output-coordinate action on the terminal-to-coil operator. It is distinct from permuting the input terminal coordinates, which right-multiplies $\mathbf{A}_{xk}$ by an inverse permutation. We use $\mathbf{Q}_{xk}$ for the coil-row action throughout this chapter so it cannot be confused with the terminal-coordinate permutation $\mathbf{P}_{xk}$ in the preceding chapter.

Block assembly

Order windings by k and form

$$\mathbf{A}_x = \operatorname{blkdiag}\left(\widetilde{\mathbf{A}}_{x1}, \ldots, \widetilde{\mathbf{A}}_{xn_x}\right).$$

Let $\mathbf{Y}_x^{\mathrm{w}}$ be the reference-invariant winding admittance from `TR-XFMR-002`, and let $n_c = |\mathcal{C}_x|$. With winding-major, coil-minor stacking, the repeated magnetic-coordinate relation is

$$\mathbf{Y}_x^{\mathrm{coil}} = \mathbf{Y}_x^{\mathrm{w}} \otimes \mathbf{I}_{n_c}.$$

This Kronecker form is a declared model assumption: the same winding leakage matrix acts independently on every common coil coordinate. A transformer with inter-coordinate magnetic coupling requires a full block or tensor relation in place of $\mathbf{Y}_x^{\mathrm{w}} \otimes \mathbf{I}_{n_c}$.

The component equations are

$$\mathbf{V}_x^{\mathrm{coil}} = \mathbf{A}_x \mathbf{U}_x, \qquad \mathbf{I}_x^{\mathrm{coil}} = \mathbf{Y}_x^{\mathrm{coil}} \mathbf{V}_x^{\mathrm{coil}}, \qquad \mathbf{I}_x = \mathbf{A}_x^{\mathsf{T}} \mathbf{I}_x^{\mathrm{coil}}.$$

Consequently,

$$\boxed{\mathbf{Y}_x^{\mathrm{terminal}} = \mathbf{A}_x^{\mathsf{T}} (\mathbf{Y}_x^{\mathrm{w}} \otimes \mathbf{I}_{n_c}) \mathbf{A}_x}.$$

Because the incidence is real, the assembly also preserves complex power:

$$\mathbf{U}_x^{\mathsf{H}} \mathbf{I}_x = (\mathbf{A}_x \mathbf{U}_x)^{\mathsf{H}} \mathbf{I}_x^{\mathrm{coil}}.$$

The numerical certificate evaluates this identity for a deterministic complex voltage witness; its residual is $1.62 \times 10^{-13}\ \mathrm{VA}$.

Why the current map must remain

A bare $\mathbf{Y}_x^{\mathrm{terminal}}$ preserves the unconstrained terminal relation. It does not by itself expose the current in winding k and coil c. The exact decision-aware target therefore retains

$$\mathbf{I}_x^{\mathrm{coil}} = \mathbf{Y}_x^{\mathrm{coil}} \mathbf{A}_x \mathbf{U}_x$$

and enforces every source constraint

$$\left| \left[\mathbf{Q}_{xk}^{\mathsf{T}} \mathbf{I}_{xk}^{\mathrm{coil}} \right]_c \right| \le \bar{i}_{xkc}.$$

Thus eliminating coil-current variables is optional, but eliminating their recovery map is not exact for a decision problem containing winding limits. This is the same preservation distinction exposed earlier by heterogeneous parallel lines.

Coordinate and reference covariance

Three independent changes are tested:

1. changing the internal leakage reference leaves $\mathbf{Y}_x^{\mathrm{w}}$ and hence the terminal factor unchanged;
2. reordering a winding's coil rows and labels is removed by $\mathbf{Q}_{xk}$; and
3. if terminal coordinates change by $\widehat{\mathbf{U}}_x = \mathbf{P}_x \mathbf{U}_x$, then

$$\widehat{\mathbf{Y}}_x^{\mathrm{terminal}} = \mathbf{P}_x \mathbf{Y}_x^{\mathrm{terminal}} \mathbf{P}_x^{\mathsf{T}}.$$

The rule also accepts winding factors in arbitrary serialization order; stable winding positions determine the assembly order.

Running transformer

Fixture transformer x_1 has two grounded-wye four-terminal windings and one delta three-terminal winding. Each has three labelled coil coordinates. The compiled matrices therefore have dimensions

Object	Dimension
terminal-to-coil incidence $\mathbf{A}_{x_1}$	9×11
coil admittance $\mathbf{Y}_{x_1}^{\mathrm{coil}}$	9×9
terminal admittance $\mathbf{Y}_{x_1}^{\mathrm{terminal}}$	11×11
lifted coil-current map	9×11

For the recorded witness, terminal-current recovery has residual 9.10×10^{-14} A. Recompiling the leakage relation with winding 2 rather than winding 1 as its internal reference changes the terminal matrix by at most 8.53×10^{-14} S.

The focused test suite also reads the canonical running-network $x1$ contract directly and rebuilds this WYE/WYE/DELTA assembly from its declared winding maps, short-circuit data, and current limits. This is a fixture smoke test, not nameplate validation: the contract itself labels the illustrative extensions and does not turn the executable model into measured equipment data.

The companion conductor-terminal lift is now direct for the serialized multiwinding contract. It creates one ordered port for each winding, preserves the neutral terminal on the two WYE windings, leaves the DELTA winding as a three-terminal port, and records the internal grounding and excitation shunt as separate observations. This is a structural incidence result; it does not replace the terminal leakage assembly or claim that the reduced port graph alone preserves winding-current constraints.

The companion typed-Kron probe makes the reduction precondition visible. If the DELTA terminal block is selected for elimination from this serialized terminal admittance, its three-by-three internal block is singular because no terminal grounding for that port is declared. The reduction therefore refuses the Schur complement without a pseudoinverse. The DELTA winding's current-limit observation remains in the source contract and is not silently removed.

The package-independent implementation rejects mismatched coil-label sets, inconsistent winding positions, differing transformer identities, nonfinite matrices, and disagreement between winding and leakage current limits. Its machine-readable result is certificate `TR-XFMR-003`.

Model boundary

The compilation assumes one common labelled set of magnetic coil coordinates and fixed connection incidence, with leakage repeated independently across those coordinates. It does not infer phase correspondence from array position, duplicate pairwise tests into independent transformers, or replace coil-current limits by terminal-current magnitudes. A complete transformer factor must later compose this leakage block with any declared phase shift, tap, magnetizing, core-loss, grounding, thermal, and decision relations. The fixed-linear transformer factor completion performs that composition for fixed linear components; adjustable controls remain parameterized decision factors.

Chapter 46

Fixed-linear transformer factor completion

Page status: guarded fixed-linear transformer completion with an executable anatomy witness; broader rating and control models remain open.

The leakage factor is only one part of a transformer model. A linear phasor model may also contain fixed ideal voltage transfers, a no-load excitation branch, and transformer-internal neutral grounding. These components can be assembled exactly, but only after their placement, coordinates, and control status have been declared.

This chapter completes the fixed-linear factor. Here *complete* means complete for the declared linear steady-state components. Saturation, frequency dependence, thermal state, protection, and adjustable controls are not thereby eliminated or approximated.

The figure is generated by `experiments/render_transformer_anatomy.py`. It shows the source factor and its typed interfaces; it is not a physical winding cross-section or a claim that the device decomposes into

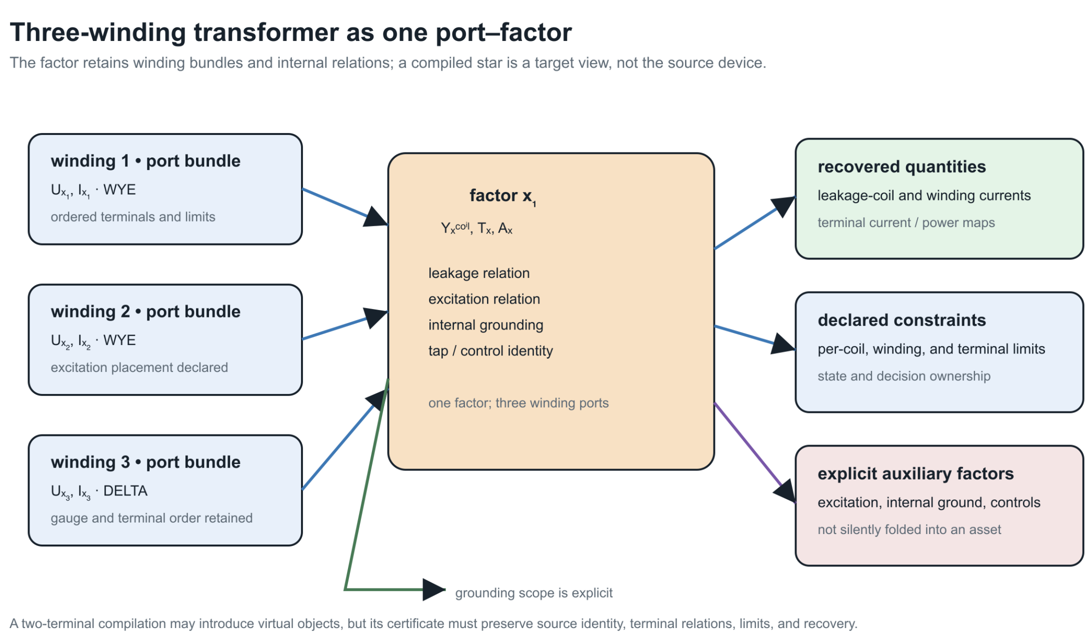

Figure 46.1: Three-winding transformer port-factor anatomy with recovered quantities, constraints, auxiliary factors, and explicit grounding.

independent two-winding elements.

Serialization contract

For transformer x, winding k at bus i retains:

- its stable winding identity and position k;
- ordered terminal labels $\mathbf{N}_{xk}$ and labelled coil coordinates $\mathcal{C}_{xk}$;
- the connection incidence $\mathbf{A}_{xk}$;
- the source impedance-test reference and selected leakage-compilation reference;
- a declared voltage-transfer mode, coefficient convention, and decision identity when adjustable;
- the placement and coordinates of every excitation or grounding factor.

The executable JSON contract uses the versioned convention

$$V_{xkc}^{\mathrm{leak}} = a_{xkc} V_{xkc}^{\mathrm{coil}}, \qquad c \in \mathcal{C}_{xk}.$$

Thus a_{xkc} is not an unqualified `tap` field whose direction must be guessed. An adapter may map a package's tap convention into a_{xkc}, but it must record that map. The running contract is stored separately from the canonical BMOPFTools fixture because its excitation and grounding values are illustrative, not nameplate data.

Power-dual voltage transfer

Let $\mathbf{A}_x$ be the aligned block connection matrix from `TR-XFMR-003` and let

$$\mathbf{T}_x = \mathrm{blkdiag}\left(\mathrm{diag}(\mathbf{a}_{x1}), \ldots, \mathrm{diag}(\mathbf{a}_{xn_x})\right).$$

The connected-coil and leakage-coordinate voltages are

$$\mathbf{V}_x^{\mathrm{coil}} = \mathbf{A}_x \mathbf{U}_x, \qquad \mathbf{V}_x^{\mathrm{leak}} = \mathbf{T}_x \mathbf{A}_x \mathbf{U}_x.$$

A voltage map does not determine the same current map. Complex-power preservation requires the conjugate-transpose dual:

$$\mathbf{I}_x^{\mathrm{w,leak}} = \mathbf{T}_x^{\mathrm{H}} \mathbf{I}_x^{\mathrm{leak}}.$$

Indeed,

$$(\mathbf{V}_x^{\mathrm{coil}})^{\mathrm{H}} \mathbf{I}_x^{\mathrm{w,leak}} = (\mathbf{T}_x \mathbf{V}_x^{\mathrm{coil}})^{\mathrm{H}} \mathbf{I}_x^{\mathrm{leak}}.$$

This distinction is invisible for a real unit coefficient. It is essential for an off-nominal magnitude or an explicitly declared phase-shifting transfer. Connection-induced vector-group displacement should remain in the real winding incidences—such as the declared delta roll—rather than being duplicated in $\mathbf{T}_x$.

With $\mathbf{Y}_x^{\mathrm{coil}}$ from the leakage assembly, define

$$\begin{aligned}\mathbf{B}_x &= \mathbf{T}_x\mathbf{A}_x,\\ \mathbf{I}_x^{\mathrm{leak}} &= \mathbf{Y}_x^{\mathrm{coil}}\mathbf{B}_x\mathbf{U}_x.\end{aligned}$$

The leakage contribution at transformer terminals is therefore

$$\mathbf{Y}_x^{\mathrm{series}} = \mathbf{B}_x^{\mathrm{H}}\mathbf{Y}_x^{\mathrm{coil}}\mathbf{B}_x.$$

A complex phase-shifting transfer need not produce a complex-symmetric nodal matrix. The correctness condition here is the declared voltage/current dual and its complex-power identity, not symmetry copied from the real-incidence special case.

Excitation and core loss

Let winding k_0 carry the explicitly placed excitation branch. The real selection-incidence matrix $\mathbf{S}_x$ maps terminal voltages to its aligned coil voltages, and $\mathbf{Y}_x^0$ is its labelled coil admittance:

$$\mathbf{V}_x^0 = \mathbf{S}_x\mathbf{U}_x, \qquad \mathbf{I}_x^0 = \mathbf{Y}_x^0\mathbf{V}_x^0.$$

The implementation accepts a full reciprocal matrix $\mathbf{Y}_x^0$ rather than assuming diagonal phases. Its Hermitian part must be positive semidefinite within tolerance, so the shunt cannot generate real power. A diagonal $G_x^0 + \mathrm{j}B_x^0$ is the common core-loss and magnetising special case.

Placement is part of the contract. The executable example follows the BMOPFTools/OpenDSS convention of placing the branch across winding 2's coils, but the book-level compiler does not infer that placement from an array position.

Transformer-internal grounding

For each declared transformer-internal grounding branch g, let y_g be its passive terminal-to-earth admittance and $\mathbf{e}_g$ select the qualified terminal (x, k, N). Then

$$\mathbf{Y}_x^{\mathrm{ground}} = \sum_g y_g\mathbf{e}_g\mathbf{e}_g^{\mathsf{T}}.$$

This factor is distinct from a neutral conductor, a voltage reference, and an external bus grounding. The compiler rejects an object whose scope is `external_bus`: absorbing it would change asset ownership and could invalidate later grounding, protection, or topology decisions.

Completed fixed-linear factor

The terminal current is the sum of the three retained contributions,

$$\mathbf{I}_x = \mathbf{B}_x^{\mathrm{H}}\mathbf{I}_x^{\mathrm{leak}} + \mathbf{S}_x^{\intercal}\mathbf{I}_x^{0} + \mathbf{Y}_x^{\mathrm{ground}}\mathbf{U}_x,$$

and hence

$$\boxed{\mathbf{Y}_x^{\mathrm{complete}} = \mathbf{B}_x^{\mathrm{H}}\mathbf{Y}_x^{\mathrm{coil}}\mathbf{B}_x + \mathbf{S}_x^{\intercal}\mathbf{Y}_x^{0}\mathbf{S}_x + \mathbf{Y}_x^{\mathrm{ground}}}.$$

The target retains maps for leakage-coil current, winding-side leakage current, excitation current, and internal-ground current. A leakage-path coil limit is therefore enforced on

$$\left|[\mathbf{T}_x^{\mathrm{H}}\mathbf{Y}_x^{\mathrm{coil}}\mathbf{T}_x\mathbf{A}_x\mathbf{U}_x]_{xkc}\right| \le \bar{i}_{xkc}^{\mathrm{leak}}.$$

If a source instead defines a rating on total terminal current or apparent power, that semantic must be declared separately and applied to the corresponding recovered sum. The compiler does not silently reinterpret a leakage-path rating.

Adjustable taps are a different target

For a tap decision t_x or discrete position d_x, the relation is

$$\mathbf{V}_x^{\mathrm{leak}} = \mathbf{T}_x(t_x, d_x)\mathbf{A}_x\mathbf{U}_x, \qquad \mathbf{I}_x^{\mathrm{w,leak}} = \mathbf{T}_x(t_x, d_x)^{\mathrm{H}}\mathbf{I}_x^{\mathrm{leak}}.$$

This is a parameterized factor in the decision model, not one fixed admittance matrix. Replacing t_x by its start value changes the feasible set and loses the tap decision. The static compiler therefore returns the structured guard `adjustable_winding_transfer_requires_factorized_decision_model` for both continuous and discrete modes. The serialization still retains the mode and decision identity so a later optimization compiler can implement the correct factor.

Executable evidence

The illustrative completion of the running WYE/WYE/DELTA transformer has 11 external terminals, nine leakage-coil coordinates, a three-coordinate excitation branch on winding 2, and one internal neutral-ground branch. It records:

- component-current recovery residual 9.38×10^{-14} A;
- complex-power residual 1.13×10^{-13} VA; and
- difference 2.96×10^{-17} S from the independently constructed `BMOPFTools.nwinding_yprim` after removing the separately added internal-ground branch, which that implementation does not stamp for the n-winding subtype.

The tests also use nonuniform real tap coefficients and a complex fixed phase transfer, reorder labelled transfer coordinates, and reject active shunts, missing grounding terminals, external bus grounding, and adjustable controls. The machine-readable result is certificate `TR-XFMR-004`.

Model boundary

`TR-XFMR-004` is exact for the declared fixed linear phasor factor. It is not a claim about saturation, harmonics, frequency-dependent core behavior, thermal state, inrush, protection, mechanical tap dynamics, or uncertainty. Those are additional factors and states. Parameterized transformer tap decisions provides the continuous/discrete continuation that preserves feasible sets and decision identities rather than evaluating one fixed snapshot.

Chapter 47

Parameterized transformer tap decisions

Page status: exact parameterized tap-factor construction with solver-backed evidence; broader control domains remain open.

A fixed transformer primitive is a snapshot, not a decision model. If a tap is continuous or discrete, substituting its start value before optimization removes feasible operating points, the decision identity, and any objective or constraint that depends on the tap.

This chapter compiles an adjustable winding transfer into an exact parameterized factor. It does not select a tap and does not claim that one fixed admittance represents all positions.

Typed tap domain

For transformer x and adjustable winding k, let t_{xk} be a stable decision identity. The compact contract declares one of

$$\mathcal{D}_{xk}^{\mathrm{cont}} = [\underline{t}_{xk}, \overline{t}_{xk}], \qquad \mathcal{D}_{xk}^{\mathrm{disc}} = \{t_{xk}^{(1)}, \dots, t_{xk}^{(m)}\}.$$

It also records the start value, but the start is initialization data—not a replacement for the domain. Values must be finite and positive; discrete positions must be unique and ordered. Every adjustable winding has a unique decision identifier such as `tap/x1/winding/2`.

The first executable parameterization is a scalar ganged tap applied to all labelled coils on one winding:

$$a_{xkc}(t_{xk}) = t_{xk} a_{xkc}^{0}, \qquad c \in \mathcal{C}_{xk}.$$

The base coefficient a_{xkc}^{0} may itself be complex and need not be equal across coils. The scalar tap parameterization is a declared model family, not an inference from a field named `tap`. Independent per-phase controls require distinct decision identities and a richer domain.

A family of exact fixed-linear factors

Stack all retained decisions as $\mathbf{t}_x$. The power-dual transfer from the previous chapter becomes

$$\mathbf{B}_x(\mathbf{t}_x) = \mathbf{T}_x(\mathbf{t}_x)\mathbf{A}_x.$$

At every admissible decision $\mathbf{t}_x \in \mathcal{D}_x$, the terminal factor is

$$\mathbf{Y}_x(\mathbf{t}_x) = \mathbf{B}_x(\mathbf{t}_x)^{\mathrm{H}}\mathbf{Y}_x^{\mathrm{coil}}\mathbf{B}_x(\mathbf{t}_x) + \mathbf{S}_x^{\mathsf{T}}\mathbf{Y}_x^{0}\mathbf{S}_x + \mathbf{Y}_x^{\mathrm{ground}}.$$

The corresponding winding-side leakage current is retained as

$$\mathbf{I}_x^{\mathrm{w,leak}}(\mathbf{t}_x) = \mathbf{T}_x(\mathbf{t}_x)^{\mathrm{H}}\mathbf{Y}_x^{\mathrm{coil}}\mathbf{T}_x(\mathbf{t}_x)\mathbf{A}_x\mathbf{U}_x.$$

The compiler stores this parameterized relation and the decision domain. Its evaluator accepts a complete admissible decision assignment, substitutes that same value into the transfer, and invokes the certified fixed-linear compilation `TR-XFMR-004`. Missing, additional, continuous out-of-range, or unlisted discrete values return structured rejections.

Exactness for decision problems

Let z contain network states and other controls. For constraints and an objective that use the declared terminal and recovered component quantities, the source feasible set has the form

$$\mathcal{F}_x = \left\{(z, \mathbf{t}_x) : \mathbf{t}_x \in \mathcal{D}_x, \quad g_x(z, \mathbf{t}_x) = 0, \quad h_x(z, \mathbf{t}_x) \le 0\right\}.$$

`TR-XFMR-005` maps the decision by identity,

$$\widehat{\mathbf{t}}_x = \mathbf{t}_x,$$

and uses the same pointwise component relations. Hence it preserves the declared feasible set and any objective $f(z, \mathbf{t}_x)$ expressed through those interfaces. This is an exact compilation because the tap remains a variable; it is not a claim that the resulting optimization problem is convex or easy.

By contrast, evaluating the start value $\mathbf{t}_x^0$ produces only the slice

$$\mathcal{F}_x^{\mathrm{frozen}} = \{(z, \mathbf{t}_x) \in \mathcal{F}_x : \mathbf{t}_x = \mathbf{t}_x^0\}.$$

Unless the domain is already a singleton or the decision is provably irrelevant to the requested result, this is an inner restriction rather than an exact compilation.

Discrete decision witness

The illustrative contract layers a discrete winding-2 tap $\{0.95, 1.00, 1.05\}$ on the running WYE/WYE/DELTA transformer. It does not modify the canonical fixture or claim these are nameplate settings. At a fixed balanced boundary-voltage witness, winding 2 is at 0.97 of its nominal voltage. The decision problem minimizes the maximum winding-2 leakage current subject to the original $2200\ \mathrm{A}$ winding limit.

Tap $t_{x_1,2}$	Maximum winding-2 current (A)	Feasible
0.95	4732.320	no
1.00	1903.716	yes
1.05	1232.656	yes

Both the direct source evaluation and parameterized target retain feasible positions $\{1.00, 1.05\}$ and select 1.05. Freezing the factor at its 1.00 start value gives objective $1903.716\ \mathrm{A}$ rather than $1232.656\ \mathrm{A}$, a gap of $671.060\ \mathrm{A}$. It therefore loses the optimal decision even though the frozen point itself is feasible.

Across all three positions, the recorded source/target terminal-admittance difference is zero to machine precision. The largest component complex-power residual is $4.88 \times 10^{-9}\ \mathrm{VA}$ at the SI-scaled witness.

Continuous and coordinate tests

The same compiler accepts a continuous interval. The executable test uses $[0.94, 1.06]$, evaluates an interior value 1.013, and rejects 1.061. Reordering a winding's labelled base coefficients changes only their stored coordinates: alignment produces the same terminal factor at the same retained tap value.

Malformed domains are rejected before a start snapshot is constructed. The guards cover missing or duplicate decision identities, unsupported coefficient parameterizations, nonpositive or unsorted positions, invalid continuous bounds, and start values outside the declared domain.

Model boundary

The current rule covers scalar ganged magnitude taps whose fixed excitation and internal-grounding factors do not depend on tap. Phase-angle controls, independent phase taps, mechanically coupled devices, tap-dependent leakage, deadbands, switching costs, operation-count limits, and automatic control logic need additional typed relations. A solver adapter must encode the retained continuous or discrete factor without changing its domain or current recovery semantics.

The machine-readable result is certificate `TR-XFMR-005`. The transformer tap AC decision case now couples this factor to network voltage, neutral-KCL, power-balance, and recovered-current constraints rather than holding the boundary-voltage witness fixed.

Chapter 48

Guarded normalization rules

Page status: candidate rule catalogue; individual rules require explicit certificates.

This chapter begins a catalogue of candidate rules. Each rule distinguishes physical normalization from more general behavioral reduction.

The catalogue's central operational rule is refusal with a structured reason when a guard fails; a candidate is never silently approximated.

Canonical conductor coordinates

Before comparing or composing multiconductor elements, normalize:

- terminal orientation;
- conductor labels and ordering;
- phase and neutral identity;
- voltage and current reference directions;
- units and base quantities;
- matrix permutation under terminal relabeling.

For a conductor permutation matrix P, a series impedance matrix transforms as

$$Z' = PZP^{\mathsf{T}}.$$

Equality of raw matrices without equality of conductor coordinates is not a valid equivalence test.

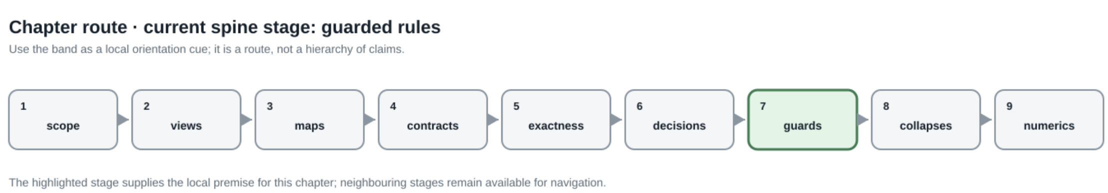

Figure 48.1: Chapter route with the guarded-rules stage highlighted.

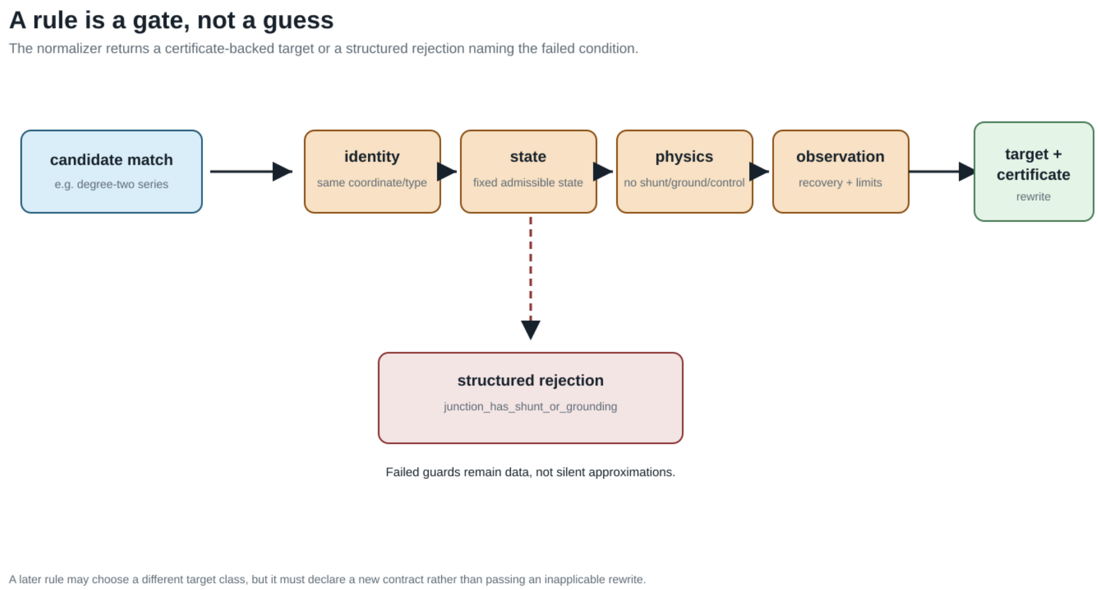

Figure 48.2: A guarded transformation rule as a gate.

Degree-two series elimination

The catalogue points to Degree-two series elimination for the complete coordinate-aware rule, executable guards, mutual-coupling counterexample, grounding counterexample, recovery map, and nominal-π closure warning. In particular, the target impedance is $Z_1 + P^\top Z_2 P$, not an unqualifiedZ_1+Z_2‘: the second section must first be pulled back into the first section's conductor coordinates.

This catalogue entry deliberately records only the rule-family distinction: behavioural series elimination may be exact in a generic two-port library, whereas merging two assets into one homogeneous physical line requires a separate closure proof for construction, ratings, ownership, grounding, charging, and outage semantics.

Constraint transformation for series elements

When exactly the same conductor current traverses both series elements and the current-feasible sets are $\mathcal{C}_1$ and $\mathcal{C}_2$, the exact equivalent set is

$$\mathcal{C}_{\mathrm{eq}} = \mathcal{C}_1 \cap \mathcal{C}_2.$$

For independent upper current magnitudes this becomes the componentwise minimum. It does not automatically handle terminal-MVA limits, dynamic thermal ratings, emergency durations, direction dependence, or temperature states. Those constraints should either be lifted through recovered internal variables or preserved as residual source constraints.

Parallel recognition and aggregation

Parallel assets should first be recognized as a **bundle** without destroying membership. For linear branches,

$$i_{\text{total}} = \sum_{e} Y_e \Delta v,$$

but each branch still has

$$i_e = Y_e \Delta v.$$

An aggregate electrical factor can coexist with the bundle if it stores the individual flow-recovery map and constraints. Destructive replacement is valid only for an observation and decision contract that cannot distinguish bundle members.

This is stricter than the practical `MergeParallel` operation offered by OpenDSS [56]. OpenDSS's reduction tools are important evidence of engineering demand, but also show why application-level procedures should not silently define the canonical data semantics.

Switch contraction

An ideal closed switch can identify its endpoint junctions for a fixed state. The rule must be rejected or qualified when:

- the switch is lossy or has nonzero impedance;
- its status is a decision variable;
- switch flow is measured or limited;
- protection or operational procedures refer to the switch;
- alternative states must be studied from the same model.

The quotient should retain a membership map from each derived bus to all source connectivity nodes and switches.

Multiwinding transformer compilation

A multiwinding transformer is naturally a multiport factor. A two-port-only target formulation may compile it into ideal two-winding transformers, internal buses, impedance branches and shunts. This is a semantics-preserving realization only if it preserves:

- winding voltage/current conventions;
- vector group, polarity and phase displacement;
- short-circuit impedance data consistency;
- magnetizing and no-load losses;
- taps and their control ownership;
- per-winding ratings and terminal constraints.

Round-trip tests should verify that the compiled network's terminal relation matches the source device and that source-level decisions map uniquely to the virtual objects.

Rewrite-system questions

Local rules can overlap. Merging two line segments before compiling grounding, for example, may produce a different intermediate graph than compiling the ground first. A scientific normalization system must study:

- termination of repeated rewrites;
- critical pairs and confluence;
- uniqueness only up to typed isomorphism;
- canonical selection when several electrically equivalent minimal networks exist;
- preservation of provenance under rewrite composition.

Electrical network equivalence already shows that a unique minimal graph need not exist: series, parallel and Y–Δ transformations can relate distinct critical networks with the same boundary response. The inverse-network literature provides the appropriate caution [21], while algebraic graph transformation provides tools for typed rules, critical pairs and confluence [26].

Part VI

Cases and consequences

Chapter 49

Multiconductor parallel AC decision case

Page status: guarded multiconductor decision case with executable redundancy certificates; general state-dependent classification remains open.

This is the multiconductor specialization of the canonical parallel-member failure in the first scalar counterexample. That chapter owns the general warning: an aggregate terminal relation does not automatically preserve member-constrained decisions. The present chapter adds coupling, complex voltages, and AC power balance without restating that warning as a separate modelling claim.

The scalar parallel example shows the feasible-set mechanism with almost no algebra. This case retains complex conductor voltages, mutual impedance, phase-to-neutral power, voltage bounds, and nonlinear AC power balance. Its purpose is to test whether the same representation failure changes an AC decision optimum.

The grid explains why the later cases are not repetitions of the scalar example. Coupling, explicit neutrals, end shunts, two-end observations, and control decisions are added deliberately; the evidence obligation grows with the model rather than being hidden behind a larger diagram.

Source model

Two buses i and j have ordered conductor set (a, n). The sending voltage is fixed at

$$\mathbf{U}_i = (1, 0)^\intercal \text{ p.u.}$$

and two parallel members satisfy

$$\mathbf{I}_{\ell ij} = \mathbf{Y}_\ell(\mathbf{U}_i - \mathbf{U}_j), \qquad \ell \in \{1, 2\}.$$

The full, coupled series impedances are

$$\mathbf{Z}_{\ell_1} = \begin{bmatrix} 0.04 + 0.08\text{j} & 0.01 + 0.02\text{j} \\ 0.01 + 0.02\text{j} & 0.04 + 0.08\text{j} \end{bmatrix}, \qquad \mathbf{Z}_{\ell_2} = 10\mathbf{Z}_{\ell_1}.$$

Every member and conductor has limit $|I_{\ell ij,c}| \leq 0.6$ p.u. Receiving-end conservation requires

$$\sum_\ell I_{\ell ij,a} + \sum_\ell I_{\ell ij,n} = 0.$$

The worked cases are an escalation, not repetition

Each row adds a modelling distinction or a stronger evidence obligation.

Parallel-member decision cases

case	coupling	non-proportionality	neutral	shunts	end structure	check
scalar DC	—	—	—	—	—	analytic
scalar AC π	—	—	—	yes	one-end	analytic
proportional multiconductor	yes	no	explicit	—	one-end	certificate
non-proportional four-wire	yes	yes	explicit	—	two-end	independent
four-wire nominal-π	yes	yes	explicit	yes	two-end	independent

Transformer compilation and control cases

case	electrical scope	control domain	forgotten/auxiliary detail	evidence
leakage relation	pairwise	fixed	—	—
reference compilation	all windings	fixed	recovery	round-trip
terminal assembly	WYE/DELTA	fixed	ground	certificate
factor completion	multi-port	fixed	auxiliary	cross-check
tap decisions	multi-port	discrete	controls	solver-backed

The progression makes the research programme visible: conductor coupling and decision observations grow together with the guard and reproduction burden.

yes/explicit = present; partial = scoped or one-end; — = deliberately absent in that case.

Figure 49.1: Escalation grid for the parallel-member and transformer worked cases.

The decision $\alpha \geq 0$ scales a constant-power direction across phase and neutral:

$$S_j = \alpha(1 + 0.2\mathrm{j}) = (U_{j,a} - U_{j,n}) \left(\sum_{\ell} I_{\ell ij,a} \right)^* .$$

The phase-to-neutral voltage magnitude is restricted to $[0.70, 1.05]$ p.u., and the objective maximizes α.

Four formulations

The **source** formulation retains each $\mathbf{I}_{\ell ij}$ as a variable and enforces every member limit. The **naive aggregate** uses

$$\mathbf{Y}_{\mathrm{eq}} = \mathbf{Y}_{\ell_1} + \mathbf{Y}_{\ell_2}$$

and assigns each aggregate conductor the summed limit 1.2 p.u. The **exact lifted aggregate** uses the same aggregate terminal relation but recovers

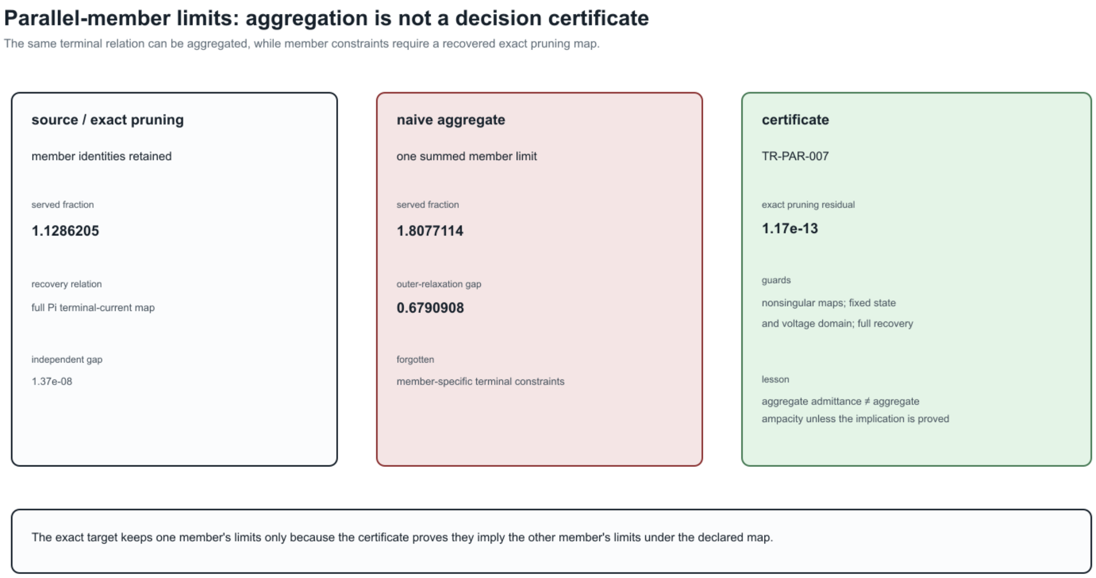

Figure 49.2: Parallel-member aggregation preserves a terminal relation, but exact decision pruning needs a recovered member-current map and a proved implication.

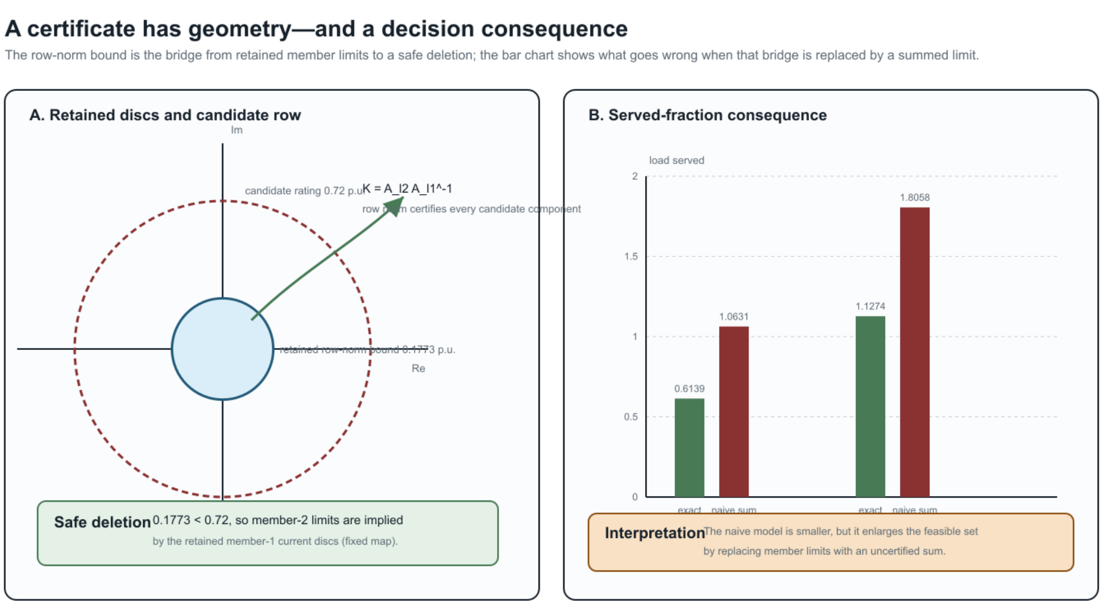

Figure 49.3: The parallel redundancy certificate maps retained current discs to a candidate rating and contrasts the certified decision with the naive aggregate.

$$\mathbf{I}_{\ell ij} = \mathbf{Y}_{\ell}(\mathbf{U}_i - \mathbf{U}_j)$$

inside the target model and applies the original 0.6 p.u. limits.

The **exact pruned aggregate** first observes that $\mathbf{Z}_{\ell_2} = 10\mathbf{Z}_{\ell_1}$, and hence

$$\mathbf{Y}_{\ell_2} = 0.1\mathbf{Y}_{\ell_1}, \qquad \mathbf{I}_{\ell_2 ij} = 0.1\mathbf{I}_{\ell_1 ij}.$$

Because the members have equal componentwise limits, every ℓ_2 current circle is implied by the corresponding ℓ_1 circle. The formulation therefore keeps both recovery maps but enforces only the certified nonredundant ℓ_1 limits. This is the multiconductor proportional special case of the constraint-pruning idea; it does not assume that the general scalar quadratic test in [3] automatically extends to arbitrary matrix-valued conductor models.

General linear-current containment test

The proportional proof is now implemented as a special case of a reusable linear-current certificate. Let A_r map the stacked complex endpoint voltages to a retained current group, and let A_c define a candidate constraint. For any complex matrix A, define its realification

$$\mathcal{R}(A) = \begin{bmatrix} \Re(A) & -\Im(A) \\ \Im(A) & \Re(A) \end{bmatrix}$$

and the normalized quadratic form

$$Q(A, I^{\max}) = \frac{\mathcal{R}(A)^{\mathsf{T}}\mathcal{R}(A)}{(I^{\max})^2}.$$

The retained constraint implies the candidate constraint exactly when

$$Q(A_r, I_r^{\max}) - Q(A_c, I_c^{\max}) \succeq 0.$$

To see this, write the constraints as $x^{\mathsf{T}}Q_r x \leq 1$ and $x^{\mathsf{T}}Q_c x \leq 1$. Positive-semidefinite dominance gives the forward implication immediately. Conversely, homogeneity lets any x with $x^{\mathsf{T}}Q_r x > 0$ be scaled to the retained boundary; directions in the nullspace can be scaled without bound and therefore must also lie in the candidate nullspace. Thus implication requires $x^{\mathsf{T}}Q_c x \leq x^{\mathsf{T}}Q_r x$ for every x. The argument includes singular cylinders, not only bounded ellipsoids.

For componentwise multiconductor limits, the implementation applies this test to every aligned conductor at both terminal ends. A non-proportional test uses different row factors, 0.2 and 0.4, so the member admittance matrices are not scalar multiples even though every candidate current circle is certifiably implied. A second test is safe at ij and unsafe at ji and is correctly rejected. This establishes claim `TR-PAR-005`.

The test is necessary and sufficient for each individual centered Euclidean norm implication. The current member-level algorithm is only a pairwise certificate: it does not yet detect a constraint implied jointly by several other limits, nor does it cover affine offsets, non-Euclidean thermal regions, or decision-dependent line, tap, outage, and switching states.

Formulation	Served fraction	Receiving voltage magnitude	Largest recovered member current	Variables / constraints
source members	0.6138908	0.9485579	0.6000000	13 / 19
naive aggregate	1.0630833	0.9034471	1.0909091	5 / 9
exact lifted	0.6138908	0.9485579	0.6000000	5 / 11
exact pruned	0.6138908	0.9485579	0.6000000	5 / 9

Results

The naive target serves about 73% more load than the source by violating the stronger member's current limit. The exact lifted formulation reproduces the source optimum while using the aggregate current relation and eight fewer real current variables in this implementation (four fewer complex member currents). This is claim `TR-PAR-004`. The exact pruned formulation has the same variable and constraint counts as the naive target but the source optimum: model size alone therefore does not establish fidelity.

Solver-independent check

For the chosen proportional matrices, a phase-to-neutral current sees loop impedances

$$z_\ell = Z_{\ell,aa} + Z_{\ell,nn} - Z_{\ell,an} - Z_{\ell,na}.$$

The equivalent loop impedance is $z = 0.05454545 + 0.10909091\mathrm{j}$ p.u. If C is the limiting total-current magnitude, $s = 1 + 0.2\mathrm{j}$, and v is the receiving voltage magnitude, then

$$1 = v^2 + 2Cv\frac{\Re(z)\Re(s) + \Im(z)\Im(s)}{|s|} + |z|^2C^2, \qquad \alpha = \frac{Cv}{|s|}.$$

The source member limit gives $C = 0.66$ p.u.; the summed aggregate gives $C = 1.2$ p.u. The served-power derivative on the high-voltage branch remains positive at both limits, so each current cap is binding. The positive quadratic roots reproduce both Ipopt objectives to better than 10^{-7}. These are values on the traced high-voltage branch: Ipopt supplies local solves here, not a global-optimality certificate. The exact pruning conclusion also relies on the deliberately exact proportionality $\mathbf{Z}_{\ell_2} = 10\mathbf{Z}_{\ell_1}$; near-proportional data require the general quadratic-containment test.

Scope and reproducibility

This is a deliberately minimal nonlinear AC case, not a three-phase benchmark. It includes conductor coupling and an explicit return path, but uses proportional member matrices and one scalable load direction so that a closed form check remains possible. Its proportional current map supplies a complete redundancy proof for this example. The non-proportional three-phase four-wire case is the next extension: it breaks proportionality inside the solved decision problem, adds all three phases plus neutral, certifies joint componentwise implication, and cross-checks the line primitives with BMOPFTools.

Run:

```
julia --project=experiments experiments/run_multiconductor_parallel_ac.jl
julia --project=experiments experiments/test/multiconductor_parallel_ac.jl
```

The generated AC certificate contains all four solutions, the proportional cross-check, the two-end quadratic-containment certificate, recovered member currents, model sizes, residuals, and closed-form differences. An automated independent re-derivation reproduces the reported figures and the binding constraint interpretation; it is not human peer review. The generic checker is in `experiments/transformations/MulticonductorFlowLimitRedundan`

Chapter 50

Non-proportional three-phase four-wire parallel case

Page status: guarded AC decision case with self-checked and independently reproduced numerical evidence; broader global claims remain open.

This case reuses the canonical member-versus-aggregate distinction from the first scalar counterexample; its new content is the non-proportional four-wire current map and joint quadratic containment certificate.

The shared certificate geometry and decision-gap plate are introduced in the multiconductor parallel case; this chapter reuses that visual contract while changing the member data and solving the non-proportional four-wire decision problem.

The preceding phase–neutral example deliberately used proportional member matrices. This case removes that simplification while retaining the book's general baseline: ordered (a, b, c, n) conductors, full mutual coupling, an explicit neutral return, unbalanced constant-power loading, individual member limits, and a decision objective.

Reciprocal non-proportional members

Two series-only members $\ell_1 ij$ and $\ell_2 ij$ share buses i and j. Their full impedance matrices are complex symmetric and have positive resistive and reactive diagonals. The second matrix is not a scalar multiple of the first: after fitting the best complex scalar ρ, the infinity-norm residual is

$$\|\mathbf{Y}_{\ell_2} - \rho\mathbf{Y}_{\ell_1}\|_\infty = 0.3663, \qquad \frac{\|\mathbf{Y}_{\ell_2} - \rho\mathbf{Y}_{\ell_1}\|_\infty}{\|\mathbf{Y}_{\ell_2}\|_\infty} = 0.1219.$$

Both members follow

$$\mathbf{I}_{\ell ij} = \mathbf{Y}_\ell(\mathbf{U}_i - \mathbf{U}_j),$$

with component limits $|I_{\ell ij,c}| \leq 0.72$ p.u. at both ends. A balanced four-wire slack supplies an unbalanced wye load with phase directions

$$(s_a, s_b, s_c) = (0.70 + 0.14\mathrm{j},\ 0.55 + 0.12\mathrm{j},\ 0.42 + 0.09\mathrm{j}),$$

all multiplied by the served-load decision α. Each phase-to-neutral voltage magnitude lies in $[0.88, 1.05]$ p.u.; neutral KCL is explicit.

Joint componentwise redundancy certificate

Because $\mathbf{Y}_{\ell_1}$ is nonsingular, the common voltage drop can be eliminated between the member laws:

$$\mathbf{I}_{\ell_2 ij} = \mathbf{K}\mathbf{I}_{\ell_1 ij}, \qquad \mathbf{K} = \mathbf{Y}_{\ell_2}\mathbf{Y}_{\ell_1}^{-1}.$$

For retained component discs $|I_{\ell_1 ij,k}| \le I^{\max}_{\ell_1,k}$, the exact worst-case magnitude of candidate component c is

$$\max |I_{\ell_2 ij,c}| = \sum_k |K_{ck}| I^{\max}_{\ell_1,k}.$$

The equality is constructive: the independent complex currents can choose phases that align every term in row c. Nonsingularity ensures that every retained current vector corresponds to a voltage drop. Thus the row-norm test is necessary and sufficient for each candidate component to be implied jointly by all retained component limits.

For (a, b, c, n), the certified worst cases are respectively $(0.1773, 0.1710, 0.1647, 0.1636)$ p.u., all well below 0.72 p.u. Since a series-only reverse-end current changes only sign, the same proof covers ℓji. The target can remove all four ℓ_2 component constraints while retaining its line law, identity, recovered currents, and possible use by other observations.

This extends the one-constraint PSD result in `TR-PAR-005`. It does not yet cover singular retained maps, shunt currents, limits implied jointly by several different retained members, or topology- and decision-dependent parameters. The four-wire nominal-pi case next adds distinct from/to shunt currents and certifies the full stacked terminal map.

Decision results

Formulation	Served fraction	Phase-a voltage	Largest ℓ_1 loading	Largest ℓ_2 loading (fraction of 0.72 p.u. rating)	Variables / constraints
source	1.1274329	0.9394441	1.0000000	0.1898951	9 / 23
exact lifted	1.1274329	0.9394441	1.0000000	0.1898951	9 / 23
exact pruned	1.1274329	0.9394441	1.0000000	0.1898951	9 / 19
naive summed-limit aggregate	1.8058181	0.8952127	1.6807715	0.3192597	9 / 19

The exact-pruned target removes four constraints and agrees with the source to 1.7×10^{-14} in objective value. The naive target has the same model size but serves 60% more of the load direction by violating the binding ℓ_1 phase-a constraint. Again, size does not determine fidelity. The certified worst-case currents are absolute p.u. magnitudes; the loading columns are current divided by the 0.72 p.u. member rating.

The unbalanced solution has neutral voltage 0.02796 p.u.; this is not a balanced transmission case with a cosmetic fourth coordinate. All four conductors participate in the coupled member recovery and redundancy proof.

Independent checks

Three checks use different seams:

1. JuMP and Ipopt solve the source, lifted, pruned, and naive nonlinear models.
2. A LinearAlgebra-only finite-difference Newton continuation and bisection reproduces the source boundary at 1.1274329171, within 1.4×10^{-8} of Ipopt, with power-flow residual below 5×10^{-16}.
3. BMOPFTools' public `line_yprim` reconstructs each ordered four-wire primitive from the stored impedance entries and matches the direct admittance blocks to 10^{-12}.

Together these establish claim `TR-PAR-006`; they do not constitute a global optimality proof for an arbitrary nonconvex AC OPF.

Run:

```
julia --project=experiments experiments/run_four_wire_parallel_ac.jl
julia --project=experiments experiments/test/four_wire_parallel_ac.jl
```

The generated certificate is `experiments/generated/four-wire-parallel-ac-certificate.json`.

Chapter 51

Four-wire nominal-pi parallel case

Page status: guarded nominal-π decision case with executable certificates; singular shunted refusal and state/voltage-dependent recomputation are explicit, while global extensions remain open.

This is the nominal-π extension of the canonical parallel-member failure in the first scalar counterexample. Only the additional shunt-current observations and both-end limit contract are new here.

The shared certificate geometry and decision-gap plate are introduced in the multiconductor parallel case; the nominal-π chapter then adds the full two-end primitive and its refusal and recomputation guards.

The series-only four-wire case uses the same current magnitude at opposite member ends. A nominal-π line removes that shortcut: shunt current depends on the local terminal voltage, and $\mathbf{I}_{\ell ij}$ is generally not $-\mathbf{I}_{\ell ji}$. Redundancy must therefore be certified on the full two-end primitive, as in the scalar setting of [3].

Full terminal-current map

For series admittance $\mathbf{Y}_\ell$ and end shunts $\mathbf{Y}^{\mathrm{sh}}_{\ell,i}$ and $\mathbf{Y}^{\mathrm{sh}}_{\ell,j}$, define

$$\begin{bmatrix}\mathbf{I}_{\ell ij}\\ \mathbf{I}_{\ell ji}\end{bmatrix} = \underbrace{\begin{bmatrix}\mathbf{Y}_\ell + \mathbf{Y}^{\mathrm{sh}}_{\ell,i} & -\mathbf{Y}_\ell\\ -\mathbf{Y}_\ell & \mathbf{Y}_\ell + \mathbf{Y}^{\mathrm{sh}}_{\ell,j}\end{bmatrix}}_{\mathbf{A}_\ell} \begin{bmatrix}\mathbf{U}_i\\ \mathbf{U}_j\end{bmatrix}.$$

The experiment adds unequal diagonal charging blocks at the two ends of both reciprocal, non-proportional (a, b, c, n) members from `TR-PAR-006`. The retained primitive $\mathbf{A}_{\ell_1}$ is nonsingular, with condition number 2922.9. Hence

$$\begin{bmatrix}\mathbf{I}_{\ell_2 ij}\\ \mathbf{I}_{\ell_2 ji}\end{bmatrix} = \mathbf{A}_{\ell_2}\mathbf{A}_{\ell_1}^{-1} \begin{bmatrix}\mathbf{I}_{\ell_1 ij}\\ \mathbf{I}_{\ell_1 ji}\end{bmatrix}.$$

Applying the complex-polydisc row norm to this eight-dimensional recovery map gives an exact worst-case candidate magnitude no larger than 0.17793 p.u. for any of the eight ij or ji components, below the 0.72 p.u. rating. All member-2 terminal limits are therefore jointly implied by the full set of member-1 terminal limits.

The invertibility guard matters. The series-only full primitive is singular because a common endpoint-voltage shift produces no series current; that case uses the reduced voltage-drop coordinate of `TR-PAR-006`. Nominal-π shunts make absolute terminal voltage observable, so silently applying the series recovery would omit charging current.

The certificate records the retained-map condition number, a normalized backward error for the recovery solve, and a relative margin for every deleted limit. It rejects a map that is too ill-conditioned or a limit whose margin is numerically ambiguous. The result is exact for the nominal matrices and centered complex discs; uncertainty in those matrices requires a separate robustness analysis.

For reproducibility, the implementation rejects a retained map when its 2-norm condition number exceeds $\kappa_{\max} = 10^8$; it also rejects a candidate row when its relative margin is below $\varepsilon_{\mathrm{margin}} = 10^{-8}$. These are numerical certification guards, not physical operating limits. The present fixture has $\kappa_2(\mathbf{A}_{\ell_1}) = 2922.9$, comfortably inside the declared conditioning bound.

Guarded extensions: singular, jointly retained, and state-dependent

The companion witness `experiments/generated/guarded-parallel-reduction-witness.json` makes three boundaries executable:

Situation	Witnessed treatment	Classification
series-only full terminal map	reject the singular full $[\mathbf{I}_{ij}; \mathbf{I}_{ji}]$ recovery map, then use the endpoint-voltage-drop coordinate	guarded exact reduction in the reduced coordinate
candidate constrained by several retained members	sum the exact support contributions $\sum_k \lvert K_{ck}\rvert \bar{I}_k$ over all retained discs	jointly implied when the declared candidate rating contains the sum
state-dependent admittance/control	recompute the recovery map at each declared state; the base map is not reused off-state	decision-conditioned map required

`TR-PAR-JOINT-001` records the scope of the middle row: the support sum is an exact linear-disc implication when every retained limit is imposed together. The generated witness now uses three retained discs. It is not yet a full nonlinear AC result for several retained members.

Jointly retained constraints (`TR-PAR-JOINT-001`, `TR-PAR-AC-JOINT-001`)

A separate three-member AC probe crosses this linear certificate into the nonlinear network model. It sets $I_{\ell_3} = 0.10 I_{\ell_1} + 0.10 I_{\ell_2}$, assigns member 3 a 0.15 p.u. component limit, and obtains a joint support bound of 0.144 p.u. The source and exact-pruned formulations both solve at served fraction 1.2401762 (gap below 7×10^{-14}). This is claim `TR-PAR-AC-JOINT-001`: a fixed-map, locally solved AC witness, not a global nonlinear optimality theorem.

The same certificate includes an independent finite-difference damped-Newton continuation and bisection check of the source boundary. It brackets the boundary within 6.0×10^{-9} served-fraction units, with power-flow residual below 1.1×10^{-15}; its boundary differs from the Ipopt source solve by about 1.3×10^{-8}. This is an independent numerical reproduction of the declared branch, not a proof of global AC optimality.

The same three-member certificate now includes `TR-PAR-STATE-001`, a finite four-state admittance envelope. At the base, higher-admittance, lower-admittance, and phase-selective unbalanced states, the member maps, joint support certificate, and source and exact-pruned AC formulations are rebuilt. Pruning remains exact in all four local solves, while the optimal served value changes across states. The state rows also carry independent boundary checks. This is finite state-dependent evidence, not a global control-policy or nonlinear optimality guarantee. The companion `three-member-state-envelope-independent-reproduction.json` reconstructs all four state boundaries with a separate standard-library Newton/bisection implementation.

This does not certify a singular shunted primitive or arbitrary AC control policy. It records the guard and the correct next representation, rather than treating a pseudoinverse or a frozen nominal map as an exact rewrite.

Singular-map refusal (`TR-PAR-SINGULAR-001`)

The generated nominal-π certificate now includes singular-map refusal probes. One constructs a rank-deficient neutral series map with zero endpoint shunts; a second retains a nonzero from-end neutral shunt while leaving the to-end neutral shunt absent. Both verify that the full two-end recovery matrix is singular after realification and that the redundancy evaluator refuses it. The fallback is an endpoint-voltage or factor-coordinate formulation; this is a refusal witness, not a pseudoinverse-based reduction.

For the series-only singular fixture, the certificate also demonstrates the valid reduced-coordinate fallback. Writing $\Delta\mathbf{U} = \mathbf{U}_i - \mathbf{U}_j$ gives $\mathbf{I}_{\ell_1} = \mathbf{Y}_{\ell_1}\Delta\mathbf{U}$ and $\mathbf{I}_{\ell_2} = \mathbf{Y}_{\ell_2}\Delta\mathbf{U}$; the declared neutral rows are zero and are retained as an explicit invariant. The reduced recovery map is therefore exact on the endpoint-voltage-drop coordinate even though the full two-end terminal map remains rank deficient. This is a scoped series-only result: it does not use a pseudoinverse and does not extend to singular shunted maps or a global nonlinear AC theorem. This scoped executable result is claim `TR-PAR-SINGULAR-001`.

State-conditioned maps (`TR-PAR-STATE-001`)

The same certificate includes scoped state-conditioned map probes: changing the endpoint shunts changes the nominal-π primitive, so the base-state map is rejected off-state and the shifted map is recomputed. A voltage-dependent shunt probe makes the stronger point that the map changes with terminal voltage, not merely with a separately declared state. These records provide the local guard needed for a control/state-dependent study; they do not provide a global robust AC bound or an optimizer-independent equivalence theorem.

The certificate now also solves the two declared shunt states with their own maps. The served-fraction boundary changes from 1.1286205497 in the base state to 1.1285736287 after the shunt shift. Re-running the exact-pruned formulation at the shifted state agrees with its shifted source formulation to 4.5×10^{-14}. Thus the probe demonstrates a state-conditioned full-AC decision calculation and exact pruning after recomputation; it does not justify reusing the base map or claim a global control policy theorem.

The generated evidence now evaluates a finite three-state envelope: the base state, the shifted state above, and a reverse shunt shift. Each state rebuilds its nominal-π map, rechecks the joint limit certificate, and solves both the source and exact-pruned AC formulations. All three states are locally solved, with maximum source/pruned objective gap below 1.2×10^{-13}. This is finite declared-state evidence, not a bound over a continuous control or uncertainty set.

The finite envelope is summarized once here; the map and both local nonlinear solves are rebuilt for every row.

Declared state	From/to shunt scales	Map certified	Source served fraction	Pruned served fraction	Objective gap
base	1.00 / 1.00	yes	1.1286205497	1.1286205497	1.2×10^{-13}
shifted_shunt	1.35 / 0.70	yes	1.1285736287	1.1285736287	4.5×10^{-14}
re-verse_shift	0.75 / 1.25	yes	1.1286586199	1.1286586199	1.1×10^{-13}

AC decision comparison

The load directions, explicit neutral KCL, phase-to-neutral voltage bounds, and served-load objective are retained from the preceding four-wire case. Receiving-bus power uses the negative ji terminal current, including the receiving-end shunt.

The exact-pruned target deletes eight constraints and agrees with the source to 1.2×10^{-13} in objective value. The same-size naive formulation serves substantially more load by violating the binding member-1 limit. The certified worst-case currents are absolute p.u. magnitudes; the loading columns are current divided by the corresponding 0.72 p.u. rating.

Formulation	Served fraction	Phase-a voltage	Largest ℓ_1 loading	Largest ℓ_2 loading (fraction of 0.72 p.u. rating)	Variables / constraints
source	1.1286205	0.9404444	1.0000000	0.1898792	9 / 31
exact lifted	1.1286205	0.9404444	1.0000000	0.1898792	9 / 31
exact pruned	1.1286205	0.9404444	1.0000000	0.1898792	9 / 23
naive summed-limit aggregate	1.8077114	0.8961512	1.6807809	0.3192479	9 / 23

Independent checks and scope

BMOPFTools' `line_yprim` reconstructs both complete nominal-π primitives from their series and from/to shunt entries and matches the direct matrices to 10^{-12}. A separate finite-difference Newton continuation and bisection reproduces the source boundary at 1.1286205634, within 1.4×10^{-8} of Ipopt, with residual below 4×10^{-15}.

This establishes claim `TR-PAR-007` for fixed nominal-π members whose retained full primitive is nonsingular. The accompanying refusal and recomputation probes expose, but do not solve, singular shunted maps, voltage-dependent shunts, tap or switching states, or implication by constraints distributed across several retained members.

Run:

```
julia --project=experiments experiments/run_pi_four_wire_parallel_ac.jl
julia --project=experiments experiments/test/pi_four_wire_parallel_ac.jl
```

The generated certificate is `experiments/generated/pi-four-wire-parallel-ac-certificate.json`.

Chapter 52

Four-wire impedance-model ladder

Page status: deterministic four-wire transformation witness; authored overhead and cable reproductions remain a follow-on case-study tranche.

This case turns a familiar engineering workflow into an auditable sequence of typed transformations. The source is a coupled four-wire conductor primitive with ordered terminals a, b, c, n, a series matrix $\mathbf{Z}_\ell$, and an explicit shunt matrix. The target views include a shunt-free series model, a neutral-reduced model, a phase-to-neutral model, sequence coordinates, a diagonal sequence approximation, and a positive-sequence view.

The source record retains frequency, length, units, ground convention, matrix ordering, and the fact that the fixture is deterministic rather than derived from a particular geometry solver. This is the minimum data needed to compare the views without mistaking a derived matrix for a primitive asset property.

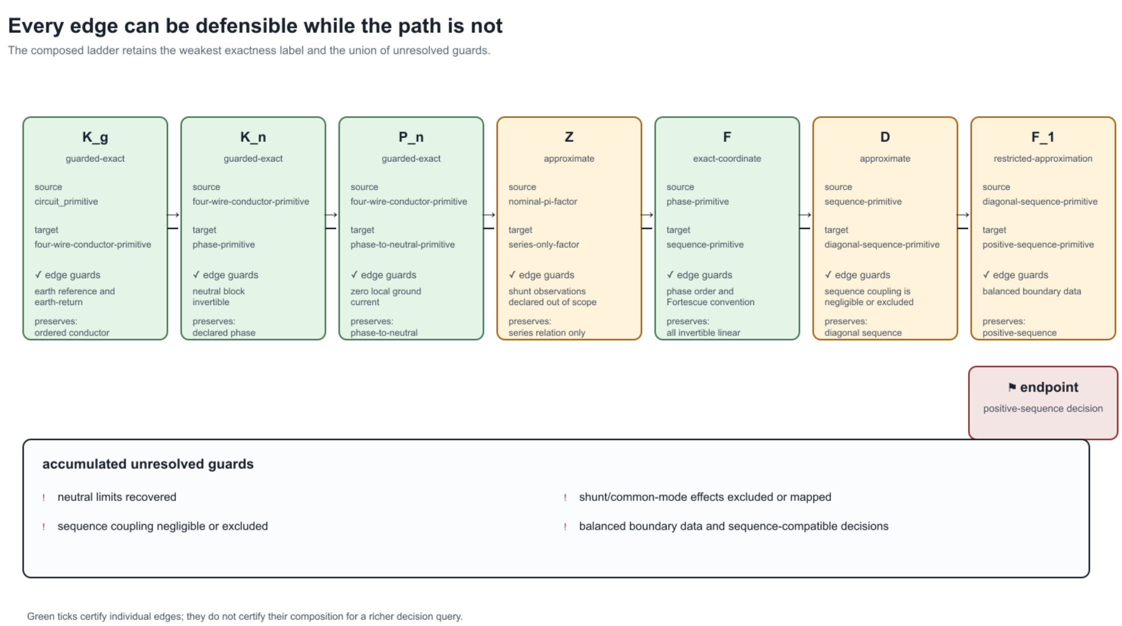

Figure 52.1: The transformation ladder is locally defensible but globally guarded: each edge is checked, while the endpoint retains unresolved decision guards.

The green checks in this plate certify edge-local contracts only. The red endpoint is intentional: composition carries forward the weakest exactness label and the union of unresolved guards. A positive-sequence decision model is therefore not certified merely because each preceding coordinate or elimination step has a valid derivation.

The path

four-wire primitive $\xrightarrow{K_n}$ Kron phase view $\xrightarrow{F}$ full sequence view $\xrightarrow{D}$ diagonal sequence view $\xrightarrow{F_1}$ positive-sequence view.

The phase-to-neutral map is also evaluated directly. With T_v mapping conductor voltages to phase-to-neutral voltages and T_i lifting phase currents under the zero-ground-current assumption,

$$\mathbf{v}_\ell^{pn} = T_v \mathbf{v}_\ell, \qquad \mathbf{i}_\ell = T_i \mathbf{i}_\ell^p, \qquad I_{\ell,n} = -(I_{\ell,a} + I_{\ell,b} + I_{\ell,c}).$$

The resulting phase-to-neutral relation is exact for the declared current map, but it does not recover common-mode voltage. This is a different contract from neutral Kron reduction, which requires an invertible neutral block and an explicit grounding/neutral-voltage assumption.

What the fixture checks

The generated `experiments/generated/four-wire-impedance-model-ladder.json` witness checks:

- the source matrix is complex symmetric but not Hermitian;
- the neutral block is invertible for the declared Kron rule;
- neutral current and phase-to-neutral voltage are recoverable under the declared map;
- the Fortescue transform is invertible;
- the deliberately non-circulant matrix produces visible sequence mixing;
- shunt deletion is visibly a model change; and
- every path edge carries explicit guards, preserved layers, forgotten facts, and risk tags.

The witness additionally composes the main path and the phase-to-neutral branch. Composition is accepted only when adjacent target and source types match; its result retains the weakest exactness label and the union of undischarged guards. The test suite includes a mismatched-path negative case.

The fixture therefore does not claim that a positive-sequence model is always wrong. It shows the narrower and more useful result: sequence coordinates are available by an invertible change of basis, while sequence decoupling and positive-sequence decision models require additional symmetry and query guards.

Decision consequences

The same bus–branch graph can support materially different decision models. Dropping shunts changes charging and loss observations. Eliminating the neutral can remove neutral-voltage observations and requires a recovered neutral-current limit. Deleting sequence coupling can admit a target that cannot reproduce an unbalanced source state. Retaining only positive sequence removes zero- and negative-sequence, phase-specific, and neutral decisions from the target unless they are separately certified.

This is why the case belongs beside the transformation register and the geometry-to-impedance fidelity ladder rather than in a catalogue of impedance labels. The question is not “which impedance model is standard?” but “which transformation path is admissible for this study?”

Reproduction

```
julia --project=experiments experiments/run_four_wire_impedance_model_ladder.jl
julia --project=experiments experiments/test/four_wire_impedance_model_ladder.jl
```

The current fixture is intentionally small. The next tranche can import the authored overhead-line and cable matrices, then add geometry/Carson provenance, balanced and unbalanced load rows, neutral-grounding variants, voltage and loss comparisons, and an OpenDSS or BMOPFTools cross-check.

Chapter 53

Transformer tap AC decision case

Page status: solver-backed transformer-control case with independent local reproduction; broader control and global-optimality claims remain open.

The fixed-voltage witness in the previous chapter proves that evaluating a tap start value can remove the best decision. This case takes the next step: the tap-dependent 11-terminal transformer factor is stamped into nonlinear network voltage and power-balance equations and solved with JuMP and Ipopt.

Its result is deliberately different from the preceding fixed-boundary witness: that witness minimises a transformer-local leakage-current metric at a prescribed boundary voltage and selects 1.05. Here the tap is embedded in a network decision that maximises served load subject to AC voltage, KCL, and recovered leakage-current constraints, and the boundary voltages are variables. The network objective therefore selects 0.95. The examples are complementary objective/boundary-condition cases, not contradictory tap recommendations.

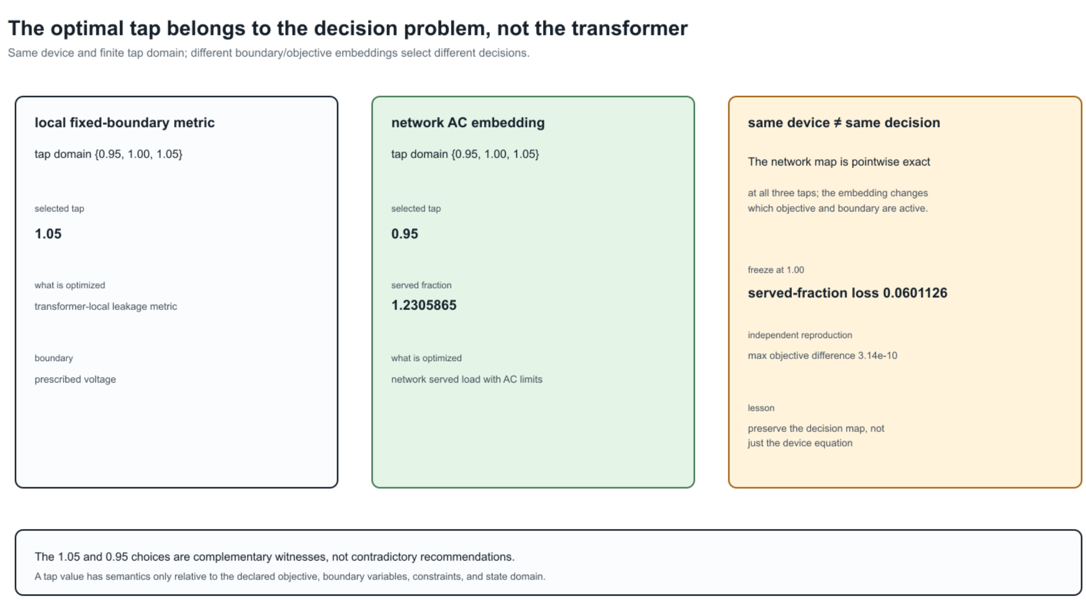

Figure 53.1: The optimal tap belongs to the decision problem, not the transformer: the same finite tap domain gives different choices under different embeddings.

The example deliberately retains the running transformer's WYE/WYE/DELTA structure. It is not a balanced positive-sequence transformer surrogate.

Network embedding

Transformer x_1 has winding set $\mathcal{K}_{x_1} = \{1, 2, 3\}$. Winding 1 is a balanced three-phase slack with an explicit neutral voltage. Winding 2 is a four-terminal WYE load bus. Its phase-to-neutral voltage is

$$V_{x_1,2,p} = U_{x_1,2,p} - U_{x_1,2,n}, \qquad p \in \{a, b, c\}.$$

The scalar decision $t_{x_1,2}$ retains the contract domain $\{0.95, 1.00, 1.05\}$. At a selected value, `TR-XFMR-005` evaluates the complete terminal and lifted leakage-current maps

$$\mathbf{I}^{\mathrm{term}}_{x_1} = \mathbf{Y}_{x_1}(t_{x_1,2})\mathbf{U}_{x_1}, \qquad \mathbf{I}^{\mathrm{w,leak}}_{x_1} = \mathbf{M}^{\mathrm{leak}}_{x_1}(t_{x_1,2})\mathbf{U}_{x_1}.$$

Currents are positive into the transformer factor. A balanced constant-power load has per-phase direction $0.50 + 0.10\mathrm{j}$ MVA and served fraction $\alpha \geq 0$. Phase power balance is therefore

$$-V_{x_1,2,p}\left(I^{\mathrm{term}}_{x_1,2,p}\right)^* = \alpha(0.50 + 0.10\mathrm{j})\ \mathrm{MVA}.$$

The winding-2 neutral is not silently grounded or eliminated. Its KCL equation is

$$\sum_{c \in \{a,b,c,n\}} I^{\mathrm{term}}_{x_1,2,c} = 0.$$

All three phase-to-neutral magnitudes remain in $[0.90, 1.05]$ p.u., and all nine original leakage-path limits are recovered and enforced:

$$\left|I^{\mathrm{w,leak}}_{x_1,k,c}\right| \leq I^{\max}_{x_1,k,c}, \qquad k \in \mathcal{K}_{x_1}.$$

Open delta tertiary and its gauge

Winding 3 is an open delta. Its three terminal currents are zero, but the terminal-to-ground voltage common mode is not physically observable by the delta incidence. The implementation imposes two independent complex KCL rows; the third follows from their sum. It then fixes

$$\sum_{p \in \{a,b,c\}} U_{x_1,3,p} = 0$$

only as a voltage-reference gauge. This does not add a grounding branch or change any line-to-line voltage. Separating the coordinate gauge from physical grounding is essential in a multiconductor model.

Exact discrete enumeration

The finite-domain problem is solved by enumeration. For each declared tap, the direct source contract and the parameterized target each produce a 15-variable, 30-constraint continuous nonlinear subproblem. The target does not relax, interpolate, round, or freeze the tap. Selection occurs only after all three tap-conditioned subproblems have been solved.

Tap $t_{x_1,2}$	Served fraction α	Served MW	Winding-2 voltage (p.u.)	Largest leakage loading
0.95	1.2305865	1.845880	1.0291923	1.0000000
1.00	1.1704739	1.755711	0.9789174	1.0000000
1.05	1.1159504	1.673926	0.9333170	1.0000000

The direct source and parameterized target agree at every tap to the reported solver tolerance and both select $t_{x_1,2} = 0.95$. The binding constraints are the three $2200\ \mathrm{A}$ winding-2 leakage-current limits; the voltage bounds remain slack.

Freezing the tap at its 1.00 start reduces the served fraction by 0.0601126. With three 0.50 MW phase directions, that is a served-load loss of 0.090169 MW. This is a network decision error, not merely a change in a transformer-local current metric.

What the certificate establishes

Certificate `TR-XFMR-006` has two distinct layers of evidence:

1. algebraically, direct source evaluation and parameterized factor evaluation stamp the same terminal admittance and leakage-current recovery map at each retained tap;
2. numerically, the corresponding nonlinear AC subproblems return the same states, constraint activity, objectives, and selected tap to the recorded tolerances.

The first layer establishes exact compilation of the network equations. The second is reproducible evidence for this nonconvex example; an Ipopt local termination status is not a general global-optimality proof.

Maximum recorded residuals are below 10^{-7} A for winding-2 neutral KCL and open-tertiary KCL, and below 10^{-8} MVA for phase power balance. The delta common-mode gauge residual is below 10^{-8} p.u.

Independent numerical reproduction

`TR-XFMR-007` reproduces the decision with a separate numerical engine. After receiving the same certified transformer matrices and case data, this engine uses only linear-algebra operations; it does not construct a JuMP model or call an external optimizer.

For fixed tap t and served fraction α, the seven unknown complex terminal voltages become fourteen normalized rectangular coordinates $\mathbf{x}$. The independent residual stacks

$$\rho(z) = \begin{bmatrix} \Re(z) \\ \Im(z) \end{bmatrix}, \qquad \mathbf{r}_t(\mathbf{x}, \alpha) = \begin{bmatrix} \rho(\Delta S_a) \\ \rho(\Delta S_b) \\ \rho(\Delta S_c) \\ \rho(\Delta I_n) \\ \rho(I_{x_1,3,a}) \\ \rho(I_{x_1,3,b}) \\ \rho(\Delta U_3^0) \end{bmatrix} \in \mathbb{R}^{14},$$

where ΔS_p is phase-power mismatch, ΔI_n is winding-2 neutral KCL mismatch, and ΔU_3^0 is the open-delta common-mode gauge residual. Nonzero residual blocks are scaled by fixed MVA, current, or voltage bases; the scaling is invertible and does not change their zero set.

A central finite-difference Jacobian and damped Newton iteration solve $\mathbf{r}_t = 0$. Continuation begins at the direct linear no-load solution and increases α along the high-voltage branch. The search must observe a feasible point followed by a converged infeasible point before it may bisect an upper boundary. This guard is tested explicitly: a truncated scan and a case with no feasible voltage interval both return structured rejections.

Tap	Independent α	JuMP/Ipopt α	Difference
0.95	1.2305865268	1.2305865271	-2.90×10^{-10}
1.00	1.1704738807	1.1704738810	-3.14×10^{-10}
1.05	1.1159503676	1.1159503679	-2.74×10^{-10}

Both methods select $t_{x1,2} = 0.95$. Across all positions, the largest secondary-voltage difference is 1.94×10^{-12} p.u. and the largest leakage-current difference is 5.18×10^{-7} A. The independent boundary states have scaled equality residuals below 5.6×10^{-11}.

This reproduction changes the nonlinear algorithm and optimization machinery, but it deliberately shares the certified input matrices and case assembly. It is therefore an independent numerical check, not an independent data-model or transformer-primitive implementation. Continuation also certifies only the traced high-voltage branch under the recorded bracketing assumptions; it is not a general global-optimality method for nonconvex AC problems.

Run:

```
julia --project=experiments experiments/run_transformer_tap_ac_decision.jl
julia --project=experiments experiments/test/transformer_tap_ac_decision.jl
julia --project=experiments experiments/run_transformer_tap_ac_independent_reproduction.jl
julia --project=experiments experiments/test/transformer_tap_ac_independent_reproduction.jl
```

Boundary and next controls

This first solver-backed case uses one ganged scalar magnitude tap, a finite domain, a balanced load direction, and tap-independent leakage, excitation, and internal grounding. It does not yet cover phase-angle regulation, independent phase taps, mechanically coupled tap decisions, automatic deadbands, or tap-dependent loss parameters in this full 11-terminal network.

The scoped companion artifact `experiments/generated/transformer-control-family-witness.json` now checks the control-domain boundary without overstating solver evidence. It shows that scalar magnitude, phase-angle, independent-phase, mechanically coupled, and automatic-deadband controls can all compile pointwise when the same typed control map is retained. It also shows that a tap-dependent loss parameter must be evaluated at the retained tap: freezing it at the base tap leaves a nonzero off-tap residual. Each declared map also has a small JuMP/Ipopt feasibility probe, establishing executable solver-backed control-domain evidence. The phase-angle and tap-dependent-loss maps additionally run through a two-bus AC served-current network probe, while independent-phase and mechanically coupled maps run through a three-phase uncoupled probe. The report therefore crosses the control/network boundary without presenting these fixtures as a full neutral-coupled unbalanced network OPF. Richer multiwinding domains remain open; the four-wire probe now records mutual impedance, neutral displacement, and return-current KCL explicitly, but does not replace a full multiwinding network case.

The same certificate now includes a two-scenario switching-cost ledger. It enumerates all (3^2=9) ordered tap pairs, evaluates the two locally solved AC scenario objectives, subtracts the declared switching cost, and records the selected pair. This is branch-complete for the declared finite pair domain; it is not a global certificate for the continuous nonconvex subproblems. The certificate also sweeps five switching-cost values. For this fixture the selected pair remains ((0.95,0.95)) throughout the tested range, which is a reported stability result—not an assumption that the policy is cost-invariant in other scenarios. It also records the positive intersections of the affine branch objectives, so potential policy changes can be inspected analytically before choosing a cost sweep.

The companion `TR-XFMR-008` ledger keeps the same 11-terminal transformer but changes the second scenario by phase: its three constant-power directions are multiplied by the explicitly recorded vector $(1.08, 0.91, 1.04)$. All nine ordered tap pairs are still enumerated, and the scenario directions remain attached to their phase identities. This is the useful distinction between a phase-selective unbalanced scenario and a scalar stress factor: the former cannot be represented faithfully by silently rescaling one aggregate load. The result is finite, local solver-backed evidence; it does not claim global optimality or a general unbalanced multiwinding theorem.

The finite path extension `TR-XFMR-009` evaluates three phase-selective scenarios, using the explicitly recorded scales $(1, 1, 1)$, $(1.02, 0.98, 1.01)$, and $(0.99, 1.03, 0.98)$, and enumerates all $3^3 = 27$ ordered tap triples. Its objective is the sum of the three locally solved served fractions minus a declared cost on tap movement between consecutive scenarios. This makes the temporal/control semantics explicit without pretending that a two-scenario ledger is a general multi-period OPF. The branch ledger is complete for this finite path domain; continuous global optimality, operation-count limits, and richer topology decisions remain open. The separate finite-difference reproduction also traces the nine scenario/tap boundaries and selects the same 27-branch path, with a recorded maximum net-objective difference below 10^{-8}.

`TR-XFMR-010` adds an explicit operation policy: at most one tap movement is allowed across the two scenario transitions. The ledger still enumerates all 27 triples before filtering, leaving 15 admissible branches. This ordering matters—filtering first would hide the distinction between the full decision domain and the policy-constrained feasible set.

Chapter 54

BIM/BFM parallel lines: an expressiveness audit

Page status: literature-informed formulation case; the equations illustrate scope boundaries and are not a new executable certificate.

The notes by Geth and Liu provide a compact warning about expressive notation. They study BIM and BFM second-order-cone formulations for parallel lines and Π-sections with ideal transformers and shunts [62]. The case belongs in this book because it is not only about convex relaxation: it shows how a missing branch index changes what a proof can state.

This is a notation-capability plate, not a new numerical certificate. It makes the scope boundary visible before the equations are interpreted as a theorem.

Variable signatures determine which BIM/BFM questions are expressible

A shared bus-pair variable and member-indexed variables describe different relaxation spaces.

question	shared W_ij	member-indexed W_lij / S_lij
member current limit	✗	✓
independent outage/state	✗	✓
recovered branch current	✗	✓
member measurement	✗	✓
common voltage-drop consistency	**implicit**	**explicit**

Missing relation: Z_l* S_lij = Z_k* S_kij. Aggregate balance can pass while no common voltage drop exists.

Notation capability plate, not a new numerical certificate: the chapter's equations define the scope boundary.

Figure 54.1: A variable-signature capability matrix shows which member-level questions a shared BIM/BFM bus-pair variable can and cannot express.

The branch identity is part of the variable signature

Consider two parallel branches ℓij and kij. Their impedances are Z_ℓ and Z_k and their series flows are $S_{\ell ij}$ and S_{kij}. A BIM representation can use one bus-pair cross-product $W_{ij} = U_i U_j^*$ and write

$$S_{\ell ij} = Y_\ell^*(W_i - W_{ij}), \qquad S_{kij} = Y_k^*(W_i - W_{ij}).$$

The shared W_{ij} expresses that both members see the same endpoint voltage product. In a BFM representation, each branch may instead retain its own lifted current L_ℓ and L_k. The two variable spaces have different coordinates and different relaxation geometry. They are not equivalent merely because both are described as "the branch-flow model for the same network."

For parallel members, the missing consistency relation can be written as

$$Z_\ell^* S_{\ell ij} = Z_k^* S_{kij}.$$

Without it, independently chosen branch flows can satisfy the balance equations while failing to arise from one common voltage drop. Adding the relation is a formulation repair, not a graph transformation.

Decision-model consequence

A theorem whose signature contains only W_{ij} cannot quantify over member-specific current limits, outages, measurements or recovered branch currents. A theorem using $W_{\ell ij}$ can name those quantities, but it has changed the relaxation and must state that change explicitly.

Terminal power is not series power

For a nominal-Π member, distinguish the series current from the current seen at each terminal:

$$I_{\ell ij}^{\mathrm{tot}} = \frac{I_{\ell ij}^{s} + I_{\ell ij}^{\mathrm{sh}}}{T_{\ell ij}^*}, \qquad I_{\ell ji}^{\mathrm{tot}} = I_{\ell ji}^{s} + I_{\ell ji}^{\mathrm{sh}}.$$

Consequently,

$$S_{\ell ij}^{\mathrm{tot}} = U_i (I_{\ell ij}^{\mathrm{tot}})^*, \qquad S_{\ell ij}^{s} = \frac{U_i}{T_{\ell ij}} (I_{\ell ij}^{s})^*.$$

An apparent-power limit placed at the terminal therefore constrains the total current, including shunt current and the ideal-transformer scaling. A limit on the series impedance current is a different observation. In a lossy branch, there is no single conserved scalar called "the flow on the edge."

Implied current limits and relaxations

If a terminal apparent-power limit is $|S^{\mathrm{tot}}| \le S^{\max}$ and $|U_i| \ge U_i^{\min}$, then

$$|I_{\ell ij}^{\mathrm{tot}}| \le \frac{S^{\max}}{U_i^{\min}}.$$

This is a valid implied current bound. In the exact AC model it cannot tighten the feasible set beyond the original apparent-power limit, but in a relaxation it can bind first and strengthen the relaxation. The distinction is important: an implied constraint is not a new nameplate rating, and a relaxation proof is not automatically a physical-network proof.

Four common overclaims

Tempting statement	What is actually established
“BIM and BFM are equivalent for this parallel network.”	Only after the variable correspondence and parallel-consistency constraints are stated; otherwise the relaxations may be incomparable.
“The branch has a current limit.”	Which current: series, sending terminal, receiving terminal, conductor total, or a recovered winding current?
“We can use one edge for the two lines.”	The aggregate terminal relation may be preserved, but member identity and member limits require a recovery map or a certified projection.
“The proof covers the power-flow model.”	It may cover only a fixed-parameter SOC relaxation, a selected projection, or a numerical test family.

The paper’s two-bus examples make these differences visible without a large network. They complement the book’s multiconductor parallel cases: the latter focus on terminal-current recovery and feasible-set preservation, while this case focuses on variable signatures, Π-section bound semantics and the boundary between physical and relaxation-level equivalence.

House notation for the book

The book will use the BMOPFTools-style convention consistently:

- ℓij is a stored oriented arc, not a claim about operating power flow;
- ℓ identifies the physical/model member;
- symmetric intrinsic parameters such as Z_ℓ carry only the member index;
- directional quantities such as $I_{\ell ij}$ and $S_{\ell ij}$ carry the full arc triple;
- total-terminal and series quantities receive distinct superscripts;
- a shared bus-pair quantity such as W_{ij} is used only when the theorem declares the sharing relation explicitly.

This is an expressive-notational rule: the symbols needed to state a constraint must remain visible in the theorem's variable signature.

Chapter 55

Australian Carson reproduction

Page status: source-backed Carson/OpenDSSDirect reproduction; overhead and underground reference matrices remain independently compared outputs, and the CS1035 construction mapping is explicitly unresolved.

This case is a provenance check, not a transcription of the matrices already stored in the source repository. We lift the construction data that is available in the `ImpedanceModels.jl` line-library history, regenerate the multiconductor primitive with BMOPFTools' Carson implementation, and then solve a small four-wire OpenDSS circuit with OpenDSSDirect.jl.

What is lifted

The reproducible input record is `experiments/data/australian_source_inputs.toml`. It records the source commit, the 50 Hz modified-Carson setting, and the construction fields:

- the Pluto overhead wire diameter, AC resistance, four conductor positions, and 0.2 km length;
- the available UGHV 185 Al PILC four-wire construction fixture, its wire diameter and AC resistance, its four positions, and 0.1 km length.

The generated line code is assembled in memory from those fields. No `rmatrix`, `xmatrix`, or `cmatrix` from an existing `.dss` case is used to construct the model. Each OpenDSS load explicitly sets `vminpu=0` and `vmaxpu=2`, so the reproduction does not inherit a hidden load-model voltage clamp.

The companion register `experiments/data/australian_source_audit.toml` keeps provenance judgements separate from those inputs. Each case records whether a field is lifted, derived, inferred, or unresolved, together with machine-readable finding codes. In particular, the overhead 60 Hz/order alignment is an inference from the probe, not a statement found in the source file; the underground `CS1035` mapping remains unresolved.

The generated artifact also carries the versioned `power-network-impedance` contract from `experiments/data/impedance_contr`
It requires ordered terminals, series and shunt blocks, units, ampacity limits, grounding assumptions, source lineage, named derived views, and findings. The validator is deliberately small: it checks that a record is structurally safe to exchange, while the audit register carries the domain-specific judgements that no generic schema can infer.

What is compared, but not used

The files under `data/opendss/Australian_overhead` and `data/opendss/Australian_underground` contain derived reference matrices. We parse those matrices after the new solve and report the maximum complex-series error in the generated artifact `experiments/generated/australian-carson-reproduction.json`. The artifact marks these matrices with `matrix_used_as_input = false`.

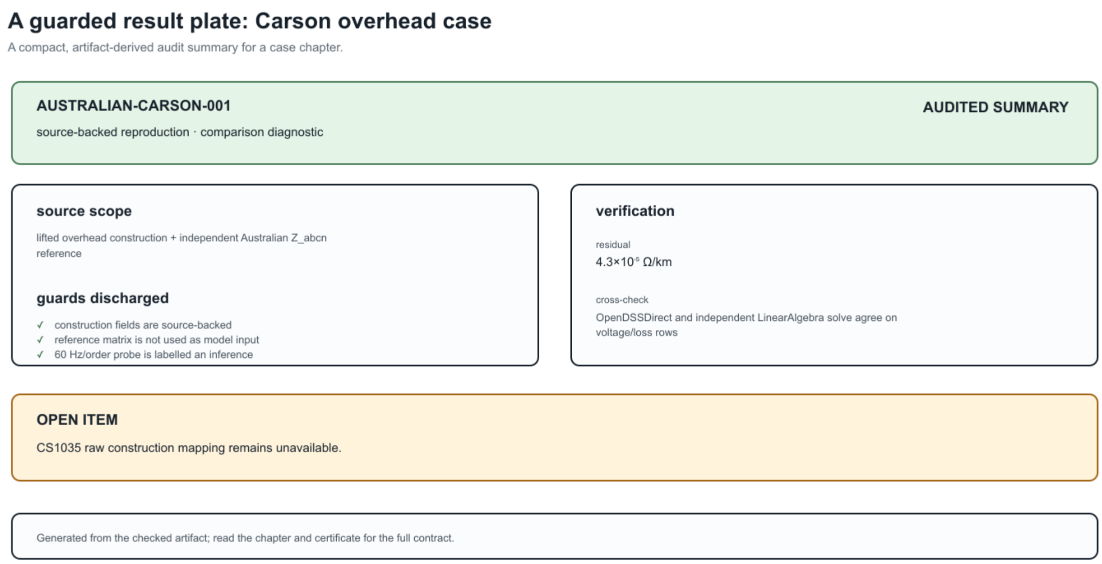

Figure 55.1: A generated guarded-result plate makes the Carson claim, discharged guards, residual, cross-check, and open item visible without replacing the certificate.

The overhead source geometry in the line-library history regenerates a series matrix with approximately $R_{ll} = 0.21735\,\Omega/\mathrm{km}$ and $X_{ll} = 0.73853\,\Omega/\mathrm{km}$. The Australian `Z_{abcn}` reference file instead has approximately $0.22722 + j0.87933\,\Omega/\mathrm{km}$ on its diagonal. A frequency/order probe resolves this apparent mismatch: at 60 Hz, and with the source conductor order $[4, 1, 2, 3]$ rather than the lifted primitive order $[1, 2, 3, 4]$, the maximum complex-series error falls to about $4.3 \times 10^{-5}\,\Omega/\mathrm{km}$. The generated artifact retains the source-backed 50 Hz result as its primary comparison and records this 60 Hz/order diagnosis separately; the reference file itself does not declare that frequency in the available source data.

The underground `CS1035` file is handled more cautiously. Its active 4-by-4 matrix is available for comparison, but the source repository does not provide the raw cable construction mapping for that case. Changing the Carson probe from 50 Hz to 60 Hz does not reduce the mismatch (the maximum error increases slightly), so frequency is not a sufficient explanation. We therefore label the UGHV construction as an explicit fixture, keep the CS1035 matrix independent, and leave the mapping as an open data task.

The source `StandardLinecodes.dss` also warns that the underground geometry's negative physical heights cannot be passed directly to OpenDSS; the reproduction uses the documented positive reference-plane convention and records that choice in the construction code rather than silently treating it as a measured cable depth.

Reproduce

From the repository root:

```
julia --project=experiments experiments/run_australian_carson_reproduction.jl
```

The JSON records the construction inputs, generated series and capacitance matrices, OpenDSS commands, convergence diagnostics, bus voltages, losses, the independent-reference comparison, and a status for each cross-check. A second LinearAlgebra-only constant-power solve agrees with OpenDSS on all rows when voltage

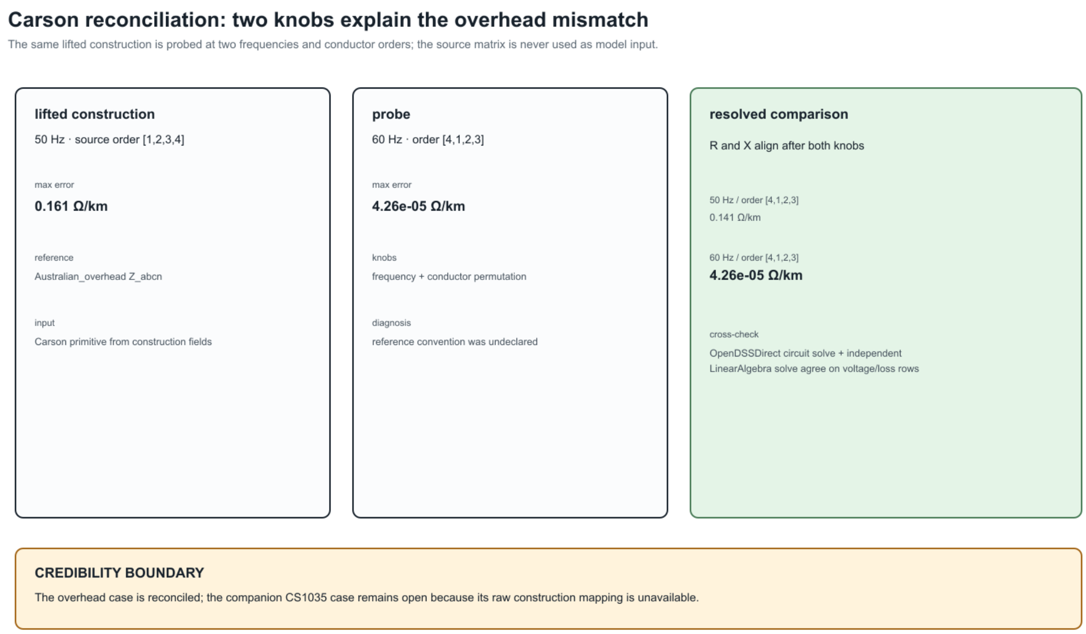

Figure 55.2: The overhead mismatch is reconciled by frequency and conductor-order probes, while the independent CS1035 matrix remains outside the reproduced model.

and line losses are compared (the voltage error is below 10^{-6} V and the line-loss error below 10^{-3} VA in the current fixture). OpenDSS `Circuit.Losses()` reports the line-element loss but not the separately modelled grounding-reactor loss; the artifact therefore keeps both line-loss and total-loss comparisons. The embedded OpenDSS engine is left at its stable 60 Hz default; the Carson primitive follows the 50 Hz source-library setting. This distinction is explicit in the artifact rather than hidden behind a text-level `set frequency` command, which is unstable in the current OpenDSSDirect release.

The underground fixture also includes low and high grounding-impedance rows. They keep the construction and load row fixed while changing only the grounding factor, making the neutral-voltage and grounding-loss sensitivity observable rather than an implicit assumption. Both rows remain within the same independent line-loss cross-check contract.

The result is therefore useful in two ways: it verifies that the available construction data can be regenerated end to end, and it identifies exactly which published cases still need their original construction provenance before they can be claimed as faithful reproductions.

Reconciliation and the unresolved companion case

The two knobs are shown together because either one alone leaves a substantial residual. They are diagnostic probes, not retroactive source metadata: the artifact records the 60 Hz/order alignment as an inference.

This is a deliberate credibility boundary. The available fixture fields and the independent matrix are useful, but without the raw conductor/screen geometry, earth-return convention, frequency, and ordering, the construction map cannot be reconstructed faithfully.

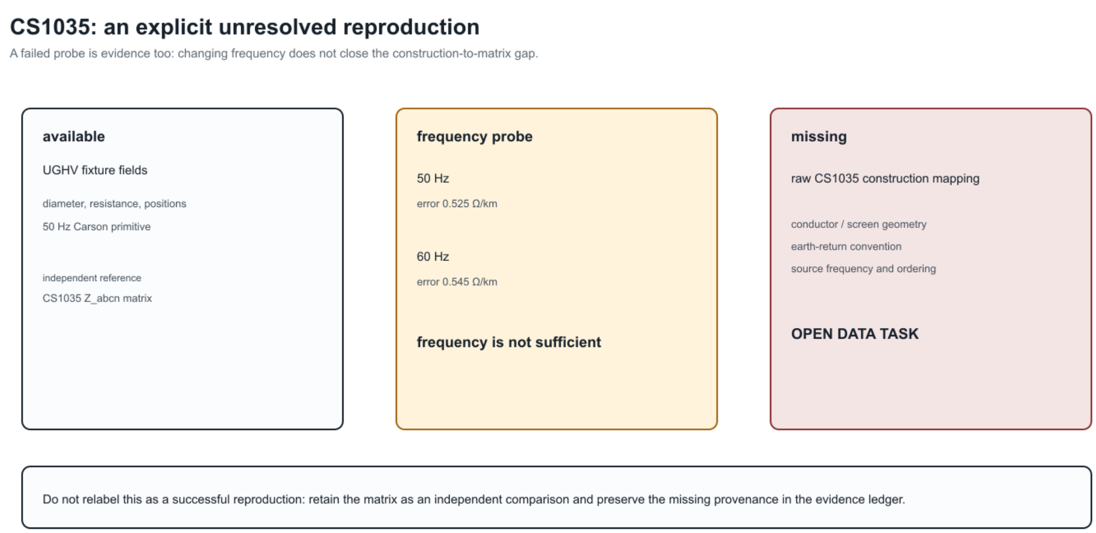

Figure 55.3: CS1035 is retained as an explicit unresolved data task rather than being labelled a successful reproduction.

Part VII

Research record

Chapter 56

Literature map

Page status: research record; coverage is provisional and not an exhaustive systematic review. Current matrix counts, checksums, search coverage, and coding status are published in the generated review protocol and evidence status page.

The current seed matrix spans several communities. No included record in that matrix supplies the full combination of typed physical assets, multiconductor terminal structure, exact/approximate behavioral maps, decision-constraint preservation, provenance, and executable normalization rules.

Circuit and graph theory

A landscape of graph models

The literature does not converge on one graph class because different models answer different questions. Simple undirected graphs support connectivity, cuts, and generic algorithms; identified multigraphs retain parallel-member identity; oriented incidence graphs supply sign conventions for conservation equations; and hypergraph or factor representations retain relations whose arity exceeds two. Port-based and compositional circuit work gives a rigorous language for multi-terminal behaviour [49], while port-Hamiltonian graph models emphasize interconnection, energy, and passivity [63]. Typed graph-transformation theory adds matching, negative conditions, and rewrite composition [26].

Power-system information models add a different axis: equipment, terminals, connectivity nodes, topological nodes, ownership, protection, and provenance. CIM/CGMES and engineering compilers such as PowerModelsDistribution therefore provide typed data and state-processing views rather than a single electrical graph [32, 47, 50, 55]. Equation and sparsity graphs then project a chosen formulation onto variables, constraints, or nonzero blocks. These models are alternatives or companions, not successive rungs of one universal refinement ladder.

The book selects a linked asset/dependency model and hierarchical port–factor electrical model as its source pair because they jointly retain the identities, terminal structure, behavioural relations, limits, states, and provenance needed by the declared multiconductor decision problems. Simple graphs, oriented multigraphs, nodal-support graphs, and tableau/MNA systems remain important derived views or formulation targets. This is a scoped canonicality claim—canonical for the book's source contract—not a claim that these are the only valid graph models or that the literature has a unique standard.

Kron reduction gives the foundational boundary-variable elimination through a Schur complement. Dörfler and Bullo analyze the resulting topology, algebra, spectrum, effective resistance, and sensitivity for loopy Laplacians [10]. This is the right reference for exact linear terminal reduction, but its retained graph is an equivalent network rather than a physical asset model.

Caliskan and Tabuada extend Kron ideas to generalized electrical networks in the time domain and identify homogeneity conditions under which compatible network structure survives [51]. This is one of the closest theoretical precedents for construction-aware closure rules.

Circular planar resistor-network theory studies response matrices, local electrical transformations and recoverability. Curtis and Morrow provide a book-length treatment [21]. This literature teaches that boundary equivalence, internal identifiability, minimality, and a unique normal form are different questions.

Baez and Fong separate circuit syntax from external behavior through a compositional black-box construction [49]. Port-Hamiltonian systems similarly emphasize energy, ports, interconnection and passivity [63]. These frameworks are strong foundations for multiport composition but do not by themselves encode utility asset semantics or OPF decision constraints.

Multiphase topology and nodal assembly

Gan and Low make an important distinction between macro- and scalar-level topology: a radial multiphase feeder has an equivalent scalar support graph with a clique associated with each densely coupled line [22]. Their BIM/BFM treatment likewise identifies each bus–phase pair with a scalar coordinate while continuing to exploit radiality at the bus level [23]. This is direct precedent for keeping bus-level radiality separate from conductor-expanded matrix cycles.

Kettner and Paolone assemble the compound nodal admittance matrix from a polyphase branch incidence matrix and block-diagonal primitive admittances, then derive rank conditions relevant to Kron reduction [19]. Coppo, Bignucolo and Turri expose nested primitive, winding, and connection maps for general multiphase transformers [20]. Together these works support the book's factor-stamping view. They do not make the inverse decomposition unique: asset identity, limits, states, and primitive lineage still have to be retained outside the assembled nodal operator.

A software-facing companion makes the same bridge explicit for four-wire unbalanced power flow: matrix-valued series and shunt data feed a nodal current-injection method, while a radial backward–forward sweep uses an element-wise impedance view [1]. The primary three-phase distribution methods of Cheng and Shirmohammadi and of Zimmerman and Chiang provide complementary precedents for unbalanced feeder equations and radial solution structure [64, 65]. These sources establish method-specific modelling and solution results, not a universal four-wire, grounding, or decision- preservation contract.

Mutually coupled corridors and multi-voltage lines

The classical building-block literature does not require mutually coupled branches to share endpoints. Wortman, Allen and Grigsby construct a multiport steady-state network model from linear-graph incidence and joint component blocks, including mutual coupling, neutral and static conductors, and finite earth conductivity [37]. Standard power-system analysis texts then interpret the nodal stamp of two coupled scalar branches as an equivalent uncoupled terminal lattice [38]. This supports an exact equation-level lowering, not identification of the lattice edges with physical lines.

Kersting develops phase impedance and shunt admittance models for physically parallel distribution lines on common poles, rights of way, or trenches [45]. Yan and Saha address the more revealing cross-voltage case: an 11 kV three-wire circuit and a 415 V four-wire circuit sharing poles, with the joint phase-coordinate relation assembled into an unbalanced current- injection load flow [44]. Dziendziel, Kocot and Kubek develop multi-circuit, multi-voltage HVAC line models and state the product-voltage base required for mutual per-unit quantities under a common power base [39].

Protection work emphasizes a narrower but operationally important projection: zero-sequence mutual coupling can dominate ground-fault and distance-element behaviour, and partial shared corridors require model section boundaries at the coupling start and end points [40]. CIM, PowSyBl, and PowerWorld accordingly retain explicit relations between two line identities and their coupled intervals [41–43]. These records are implementation precedent for section-to-section coupling, while their usual short-circuit and scalar zero-sequence scope is narrower than a full phase-domain source primitive.

The book's coupled multi-voltage corridor case synthesizes these strands as a typed path from line assets and section-coupling records to one joint factor, then to a guarded nodal stamp or equivalent lattice with current and provenance recovery.

Ground-return impedance and earth modelling

Carson's original ground-return treatment derives overhead-wire propagation under a homogeneous half-space idealization [66]. It is an important physical anchor for earth-return impedance, but it does not identify the earth conductor, grounding topology, protection state, or uncertainty semantics of a modern multiconductor network. The book therefore treats Carson-style impedance as one possible reduced-earth factor and keeps explicit earth conductors, bonds, and grounding observations as separate model objects.

Circuit formulations beyond nodal admittance

The circuit literature supplies several established equation targets rather than one universally correct matrix. Classical nodal analysis is compact when each retained element contributes a voltage-to-current relation. Modified nodal analysis augments node voltages with selected branch currents, making ideal voltage sources and current-controlled elements explicit [13]. Sparse tableau formulations retain branch- and device-level variables and equations, with sparsity and elimination treated as separate design choices [24].

This distinction is directly relevant to power-network models. Sparse-tableau OPF and node–breaker work keeps multi-port elements, breaker actions, and member-level constraints in the formulation instead of rebuilding a different fixed Y_{bus} matrix for every state [25]. The lesson is not that tableau is always preferable: it is that a nodal admittance target is a guarded lowering for a declared variable set and query family, not a universal representation of a power network.

The book therefore treats nodal support, MNA/tableau systems, branch-current models, hybrid port parameters, and general port–factor relations as related but non-identical formulation families. A formulation may be equivalent after regular elimination for one boundary-voltage observation while failing to preserve switching decisions, asset identity, grounding paths, current limits, or multi-terminal behaviour. The new circuit formulations and lowering boundary chapter records these guards as part of the representation choice rather than as an implementation detail.

Power-system network reduction

The cited power-system reduction literature emphasizes external-system equivalents, Kron/Ward/REI methods, coherency, and dynamic model reduction. "Structure preserving" has several meanings.

Ward's original construction treats suppressed loads and generation as approximately constant current, retains tie terminals, and realizes the external network as a boundary mesh with equivalent terminal injections [59]. The operating-state extended Ward construction addresses boundary-bus designation, external shunts, and contingency use from a single estimated state [60]. Ward–PV instead retains external generator buses after load-node elimination and may then aggregate coherent generator groups [61]. These sources support a family taxonomy, not one universal `Ward` operator.

Grudzien and coauthors use topology-guided reductions of lines, trees and triangular subgraphs while preserving power-flow behavior for their model class [67]. Sistermanns and coauthors seek to avoid artificial entities and maintain physical correspondence for selected transmission-grid features [68]. These works are directly relevant, although they do not establish a general typed asset-preservation framework.

The line-limit-preserving equivalent work demonstrates that thermal transfer constraints require special treatment beyond ordinary network equivalencing [4]. This should be regarded as a prototype of a broader theory of decision-set preservation.

Molzahn studies a complementary operation for parallel scalar AC lines: delete a flow-limit constraint only after a positive-semidefinite quadratic containment test proves that another parallel member implies it at both line ends [3]. This is especially important for the book's taxonomy. It is exact presolve on an unchanged physical model, not parallel asset aggregation. The result gives a concrete decision-feasible-set certificate and a useful transmission special case, while leaving open the multiconductor, state-dependent, and asset-decision generalizations. In the reported PEGASE cases, the method identified between 203 and 650 redundant parallel-line limits, with MIPS OPF runtime reductions of 2.0% to 5.7%; these are useful presolve results, not evidence for replacing the lines.

Distribution feeder reduction

Pecenak and coauthors address multiphase unbalance, mutual coupling and spatial variation while targeting accuracy at selected critical buses [52]. Opti-KRON work targets voltage reproduction, phase connectivity [53] and restored radiality [54]. This literature is more sensitive than classical transmission equivalents to phase availability and feeder topology, but explicit neutral grounding, physical line codes, parallel-asset decisions, protection and provenance remain underdeveloped.

OpenDSS implements practical reduction options including unloaded-intermediate bus elimination, short-line merging, parallel-line merging and lateral aggregation [56]. These procedures are important test cases for formalization: their operational usefulness is clear, while their preservation domains and failure cases deserve explicit certificates.

Topology processing and information models

CIM's connectivity/topological-node distinction and software such as PowSyBl document state-dependent topology processing [32, 33]. Its limitation for this agenda is not correctness but scope: connectivity quotienting is only one of the transformations needed. An automated topology-processor study makes the EMS boundary concrete: breaker and substation configuration data are converted into network topology for state estimation, with implementation and timing evaluated on test systems [69]. That is prior evidence for a topology- processing function, not evidence that the book's typed quotient preserves all electrical, asset, or decision semantics. The evidence rows for CGMES and PowSyBl describe this same underlying connectivity-node-to-topological-node map, while remaining separate records: one is a standards/building-process source and the other is an implementation and API source. They are therefore related evidence, not duplicate citations.

PowerModelsDistribution provides a concrete engineering-to-mathematical compiler and supports conductor terminals, grounding and multiwinding transformers [50, 55]. It offers an excellent implementation case study for provenance-aware compilation.

Graph transformation and model-driven engineering

Typed algebraic graph transformation supplies formal rule matching, negative application conditions, rewrite composition, critical-pair analysis and confluence [26]. These ideas are well-developed in formal software and systems model-transformation literature but have not been deeply integrated with power-system network equivalents. The W3C PROV-DM recommendation supplies a separate provenance vocabulary for entities, activities, agents, and derivations [70]. The book combines these as conceptual anchors for transformation traces and source-to-target lineage; it does not claim that either source already defines the proposed electrical certificate calculus.

Literature-position table

The table records the boundary between prior work, the book's synthesis, and repository evidence. “Repository demonstrates” refers to the checked fixtures and generated artifacts in this project; it is not independent validation of the cited literature or external review of the book's assumptions.

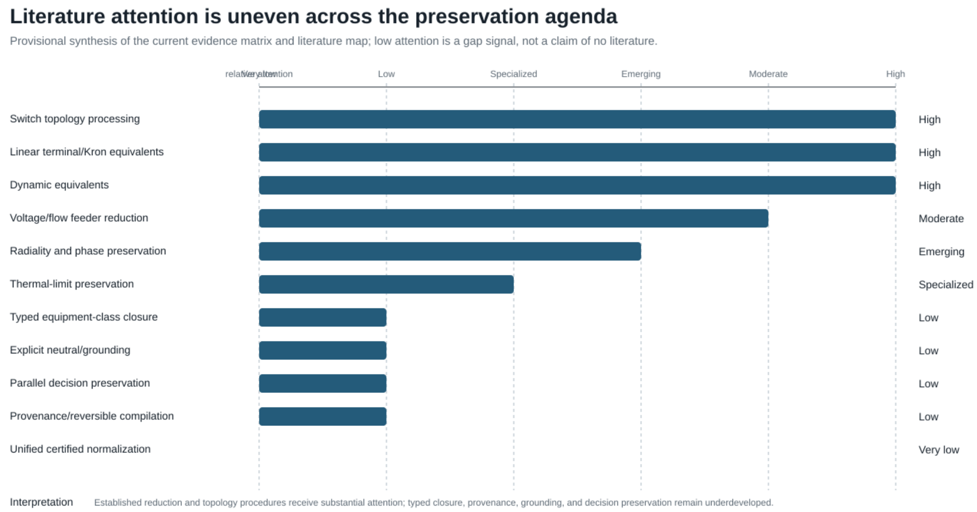

Figure 56.1: Provisional literature attention and structured gaps across the preservation agenda.

Source-type legend

The literature map keeps source authority separate from the book's own evidence. The same topic may therefore appear in several rows without those rows being interchangeable.

Source class	What it can establish here	What it cannot establish here
Peer-reviewed primary paper	A published method, theorem, derivation, or reported case under its stated assumptions	That the book's broader model, implementation, or claims are equivalent to the paper's scope
Standard or information-model specification	Normative vocabulary, profile structure, exchange requirements, or documented semantics	Electrical behaviour, solver convergence, or preservation of omitted asset/decision detail
Official software documentation	Intended API, data model, topology operation, or documented implementation behaviour	A peer-reviewed theorem, universal semantics, or independent validation of the current version
Author-derived result or repository reproduction	A traceable computation, fixture result, or independent reimplementation when its method is declared	External review, source-faithful utility validation, or global optimality unless separately demonstrated
Project proposal or synthesis	The book's definitions, taxonomies, research questions, and proposed certificates	Prior-art status or evidence that the proposed calculus is already established

Current assessment

This table is a provisional assessment to be replaced by a systematic review with a reproducible search protocol and coded evidence matrix.

The figure is a synthesis aid, not a bibliometric score or prevalence claim. Its attention labels are the book's provisional assessment of the coded seed matrix; they identify where the cited record is more developed for this agenda and where the book's proposed typed-closure, grounding, decision-preservation, provenance, and certified-normalization questions remain weakly represented.

Chapter or foundation	Literature establishes	Book synthesizes	Repository demonstrates	Open boundary
When the general model collapses	Symmetrical coordinates and sequence analysis [48]	Exact positive-sequence restriction requires operator, grounding, decision, and observation invariance	Circulant/non-circulant sequence witnesses and balanced transmission fixture	Unbalanced controls, phase-specific limits, and external mathematical review
Circuit coordinate transformations	Ground-return physics [66] and multiphase power-flow methods [64, 65]	Phase-to-neutral and phase-to-phase maps are distinct from neutral deletion and Kron elimination	Four-wire ladder, grounding guards, and coordinate-action tests	Typed four-wire/three-wire certificates and standards-aligned validation
Kron, Ward, and optimized equivalents	Kron, Ward, and feeder-reduction families [10, 52, 59]	Boundary behavior, calibration-point approximation, decision preservation, and physical realizability are separate objects	Typed Kron covariance, Ward/scenario probes, and recovery witnesses	Source-faithful Opti-KRON validation and global nonlinear claims
Node–breaker and topology processing	Connectivity/topological-node models and EMS topology processors [32, 33, 69]	Topology processing is a state-conditioned quotient with provenance, not a complete electrical reduction	Active-state connectivity, two-level topology, and graph-surgery witnesses	Full state-estimation/protection integration and utility-model validation
Data model crosswalk and provenance	CIM/CGMES information models, engineering compilers, graph transformation, and PROV [26, 47, 55, 70]	Source-to-canonical maps must expose losses, inferences, identities, and recovery obligations	Version-pinned crosswalks, certificates, provenance manifests, and schema checks	Round-trip adapters, profile conformance, and independent domain review
Parallel-member and decision cases	Scalar exact constraint-pruning theory [3] and multiphase OPF formulations [22, 23]	Preserving a feasible set is not the same as aggregating assets or preserving topology	Multiconductor, four-wire, nominal-π, and joint implication witnesses	Broader contingencies, protection, uncertainty, and external case validation

Current evidence-matrix snapshot

The versioned matrix in `review/evidence-matrix.csv` is a single-coded snapshot; its generated status page reports the current included/excluded counts. The records cover parallel-line constraint pruning, graph-aware Kron/power-flow reductions, multiphase OPF formulations, structure-preserving feeder and transmission reduction, CGMES topology processing, and circuit/graph formulation precedents. The rows intentionally preserve different exactness labels and limitations; they are not merged into a single claim about “network reduction.”

Topic	Attention	Main limitation relative to this agenda
Switch topology processing	High	state quotient, not general physical normalization
Linear terminal/Kron equivalents	High	artificial edges; limits and assets usually omitted
Dynamic equivalents	High	preservation is input–output/dynamic, not asset semantic
Voltage/flow-oriented feeder reduction	Moderate	scenario/application dependent
Radiality and phase preservation	Emerging	limited device and neutral detail
Thermal-limit preservation	Specialized	not generalized to arbitrary decision constraints
Typed equipment-class closure	Low	few formal rewrite guards
Explicit neutral/grounding preservation	Low	often eliminated or treated as an implementation detail
Parallel-asset decision preservation	Low	exact scalar constraint pruning exists, but aggregation and multiconductor/state-dependent cases remain open
Provenance and reversible compilation	Low	implemented inconsistently, weakly theorized
Unified certified normalization	Very low	central research gap

The 2026-08-14 seed search and its limitations are archived in `review/search-runs/2026-08-14-seed-batch.md`; the targeted 2026-08-16 multiphase/practical-reduction pass is archived in `review/search-runs/2026-08-16-multiphase-and-` Database exports, duplicate resolution beyond the coded seed, full-text screening, and independent double-coding remain open tasks.

Chapter 57

Representation implementation record

Page status: research and software record; implementation coverage is evidence about the current repository, not a claim that the architecture is complete.

The mathematical architecture is defined normatively in Formal representation frameworks. This page records the engineering status of the small executable facade and its fixture coverage so that package maturity is not confused with representation theory.

Public transformation API boundary

The reusable facade in `experiments/src/GraphModelsForPowerNetworks.jl` currently exports a small, dependency-light surface: identified multigraph primitives, incidence and cycle-space queries, simple projection, typed linear Kron operations, coordinate and recovery maps, unit conversion, boundary projections, and structural certificate validation. Solver-backed AC cases, generated figures, literature adapters, and BMOPFTools integration remain experimental evidence, not package-stability promises.

The generated `experiments/generated/public-api-manifest.json` is the checked record of that boundary. Promotion to a standalone package should wait until the state-space and unit types, conversion contracts, and recovery semantics are versioned. Otherwise a representation change could silently alter the meaning of a reduction while leaving the Julia method names unchanged.

Fixture coverage and evidence status

`experiments/generated/fixture-coverage-matrix.json` keeps coverage separate from API existence. It distinguishes direct fixture evidence, related evidence, and explicit not_yet_tested rows for the running network, five-bus cycle-space example, and multiwinding transformer. A certificate on a synthetic fixture is therefore not silently promoted to validation of a canonical network fixture.

The multiwinding fixture records two cycle views: its native factor–port incidence is a star with cycle rank zero, while a derived clique compilation has cycle rank one. The latter is a view choice, not a new electrical loop. Likewise, active radiality for the isolated transformer contract remains not_yet_tested because no switching or energized-state domain is declared.

The matrix distinguishes typed map families as well. The running-network state-space/unit witness is direct evidence for its declaration and conversion contract; the multiwinding entry is related evidence because it is not a separate state-space instantiation. Winding-terminal normalization is direct for the serialized $x1$ transformer contract but only related to the running network as a whole. Parameterized transformer control is tracked separately from fixed transformer compilation and has direct evidence in the retained-tap AC certificate.

Current research boundary

These records establish a reproducible implementation inventory. They do not establish categorical composition, universal factor evaluation, complete schema coverage, or package API stability. Those remain research tasks and are kept separate from the normative definitions and the generated transformation certificates.

Chapter 58

Review protocol and evidence status

Page status: generated scoping-review snapshot and evidence-status record.

This page is generated from `review/snapshot-manifest.json` and the canonical evidence matrix. It publishes the review state without implying that a single-coded seed corpus is an independently validated systematic review.

Published snapshot

field	value
protocol version	`0.1.0`
snapshot date	`2026-08-25`
matrix records	`37`
matrix SHA-256	`3cbb795c0cd6063da9da5979056c04c294ebe6ae8bc7545f1b0a708dc39ee9f8`
deduplication rows	`37`
deduplication SHA-256	`75143b8c286bbba4bd674c11fb205d594a1b7385783d634de0d11dcf79477b86`
independent human double-coding	no

Screening and coding counts

dimension	counts
screening status	include: **36**; exclude: **1**; uncertain: **0**
coding status	`single_coded`: **37**
exactness labels	`exact`: **14**; `not_reported`: **4**; `outer`: **1**; `scenario_approximate`: **8**; `unclassified`: **10**

The current snapshot is therefore a **single-coded seed snapshot**, not a double-coded corpus. The 2026-08-15 second-coding log recommends eight rows for promotion after slot repairs and identifies six substantive coding conflicts; those recommendations remain pending and are not silently reflected in the canonical matrix.

Search coverage

The protocol names the following information sources:

- IEEE Xplore

- Scopus
- Web of Science
- Engineering Village/Compendex
- Inspec
- MathSciNet or zbMATH where available
- arXiv
- relevant standards and official software documentation

The dated search runs included in this snapshot are:

- `2026-08-14-seed-batch.md`
- `2026-08-15-formulation-landscape.md`
- `2026-08-15-information-model-citation-chase.md`
- `2026-08-16-multiphase-and-practical-reductions.md`
- `2026-08-25-coupled-multivoltage-corridors.md`

The search-run files record query families, available platforms, and limitations for each run. The protocol also requires backward and forward citation chasing, duplicate resolution, and a saved-search rerun before a tagged release; these are not claimed complete merely because a row is in the matrix.

Evidence-status interpretation

A populated matrix row means that one source-to-target result was coded at a declared scope. It does not mean that the source is a proof of the book's architecture, that an exactness label applies to a feasible set, or that a software conversion preserves provenance and limits. The `exactness_object`, `recovery_map`, `constraint_map`, and `provenance_map` fields must be read together.

The second-coding and independent-technical-review documents are retained as review evidence, but they are not substitutes for an independent human double-coding pass. The canonical matrix remains the source of published counts; dated snapshots and reconciliation notes explain proposed changes without rewriting history.

Reproducibility inputs

The manifest records 9 hashed protocol, coding, bibliography, and search-run inputs. Run `scripts/check_review_snapshot.py` to verify the matrix, deduplication register, record identifiers, and input hashes against this published snapshot.

The protocol's promised PRISMA-style flow is represented here as a scoped count table rather than a clinical-review claim: the available evidence is a seed search with one explicit exclusion, and the screening/coding pipeline is still being expanded.

Chapter 59

Search runs

Page status: generated search-run record.

Protocol: 0.1.0 **Run date:** 2026-08-14 **Purpose:** small seed expansion for the scoping-review matrix; not an exhaustive database search and not double-coded.

Queries

1. `power system network reduction Kron Ward equivalent multiconductor paper`
2. `power system parallel line redundant flow limits optimization paper`
3. `CIM CGMES TopologicalNode connectivity topology processing official documentation`

The web search run returned the following primary or official records used for seed coding:

- Structure- & Physics-Preserving Reductions of Power Grid Models
- Structure-preserving Optimal Kron-based Reduction of Radial Distribution Networks
- CGM Building Process Implementation Guide v2.0
- Identifying Redundant Flow Limits on Parallel Lines

Matrix effects

The run added `EV-0002` through `EV-0004` to `review/evidence-matrix.csv`; `EV-0001` was already present. All four rows are `single_coded` and pass `scripts/check_evidence_matrix.py`. The records are deliberately coded with scoped exactness and limitations rather than being promoted to established claims.

Limitations

- No IEEE Xplore, Scopus, Web of Science, Compendex, or MathSciNet export was available in this run.
- Deduplication, citation chasing, full-text screening, and second-coder review remain outstanding.
- Search-result snippets were used only to locate primary/official records; the matrix notes identify where full-text coding is still incomplete.

Page status: generated search-run record.

Protocol: 0.2.0 **Run date:** 2026-08-15 **Purpose:** expand the scoping-review seed matrix from network-reduction papers to graph, port, and circuit-formulation precedents. This is still a seed run, not an exhaustive systematic review or a double-coded release.

Queries

The following portable queries were executed as web searches and then checked against the primary record or publisher/official landing page where available:

1. `"power network" graph representation port factor hypergraph multigraph`
2. `"modified nodal" network analysis voltage source branch current`
3. `"sparse tableau" power flow node breaker optimal power flow`
4. `"port-Hamiltonian" graph interconnection electrical network`
5. `typed graph transformation" electrical network rewrite provenance`

The search strings are preserved in `review/search-strings.md`; database-specific translations, export files, deduplication, and citation chasing remain future work.

Records added

The run added six formulation/framework rows to `review/evidence-matrix.csv`:

- `EV-0005` — Ho, Ruehli, and Brennan, modified nodal analysis;
- `EV-0006` — Hachtel, Brayton, and Gustavson, sparse tableau;
- `EV-0007` — Park, Holzer, and DeMarco, sparse-tableau node–breaker OPF;
- `EV-0008` — Baez and Fong, compositional passive linear networks;
- `EV-0009` — van der Schaft and Maschke, port-Hamiltonian systems on graphs;
- `EV-0010` — Ehrig et al., typed algebraic graph transformation.

All rows are `include` and `single_coded`. They are coded as scoped precedents, not as evidence that one formulation dominates the others or that the book's source architecture is uniquely canonical.

Matrix effects

The evidence matrix now contains ten included single-coded records: four network-reduction/topology seeds, three circuit-formulation records, and three graph/port/transformation-framework records. `scripts/check_evidence_matrix.py` passes the controlled vocabulary and schema checks.

Limitations and next actions

- No Scopus, Web of Science, Compendex, MathSciNet, or IEEE Xplore export was available in this run.
- Search-result pages were used only to locate records; full-text coding should be independently repeated for every included row.
- Deduplication, backward/forward citation chasing, and a second coder remain open.
- The next evidence pass should add information-model and engineering-compiler records, then test whether the graph-family taxonomy is missing a recurring power-system representation.

Page status: generated search-run record.

Protocol: 0.2.0 **Run date:** 2026-08-15 **Purpose:** backward and targeted citation chasing from the CGMES topology seed and the representation crosswalk, adding official software and information-model records to the evidence map.

Targeted searches

1. `PowSyBl topology views node breaker bus breaker official documentation`
2. `PowerModelsDistribution engineering data model conversion mathematical model`
3. `OpenDSS circuit reduction parallel lines intermediate buses official`
4. `MATPOWER data model developer manual bus branch official`

The records were checked against the official project or standards documentation rather than search-result snippets.

Records added

- `EV-0011` — PowSyBl topology views;
- `EV-0012` — PowerModelsDistribution engineering-to-mathematical conversion;
- `EV-0013` — OpenDSS circuit-reduction procedures;
- `EV-0014` — MATPOWER developer data model.

All records are `include` and `single_coded`. They are coded as engineering or software precedents with explicit scope limitations; none is treated as a proof of universal preservation.

Matrix effects and limitations

The matrix now contains fourteen included single-coded records. The additional records improve coverage of deployed topology processing and solver-oriented data models, but they do not replace independent full-text screening or database-level deduplication. A second coder and a DOI/title duplicate pass remain required before a release snapshot can be called double-coded.

Page status: generated search-run record.

Protocol: 0.2.0 **Run date:** 2026-08-16 **Purpose:** close the next scoped coverage gap identified in the roadmap: the multiphase OPF formulation precedents and practical feeder/transmission reduction records that sit between abstract Kron theory and software topology processing.

Records added

The pass added seven included, single-coded records to the canonical matrix:

- `EV-0022` and `EV-0023` — Gan–Low chordal and multiphase convex/linear formulation records;
- `EV-0024` — Caliskan–Tabuada generalized Kron reduction;
- `EV-0025` — Coppo–Bignucolo–Turri multiphase transformer construction;
- `EV-0026` — Pecenak et al. multiphase feeder reduction;

- EV-0027 — Sistermanns et al. feature- and structure-preserving transmission reduction;
- EV-0028 — the journal Opti-KRON record, kept distinct from the radiality preprint already coded as EV-0003.

Coding decisions

The Gan–Low records are coded as formulation/relaxation precedents rather than as physical asset reductions. The chordal result is an outer feasible-set relaxation for its declared class; the companion record contains multiple approximation regimes and is therefore left unclassified rather than given a single exactness label. The practical feeder and transmission records are coded as scenario-approximate observation studies. None is treated as a universal multiconductor, neutral-preserving, or decision-preserving equivalence.

Limits

This is targeted citation chasing from the verified bibliography, not a database export or independent second coding. The new rows remain single_coded; the matrix still cannot support PRISMA-style flow counts or claims of exhaustive coverage. As a screening-control exercise, the same pass records Nanopass2005 as EV-0029 with exclude / wrong_domain: it is retained as a compiler analogy in the bibliography but has no explicit power-network mapping.

Page status: generated search-run record.

Protocol: 0.2.0 **Run date:** 2026-08-25 **Purpose:** identify the electrical-modeling, protection, and data-model precedents needed to discuss parallel line sections at different voltage levels that share a mutual-impedance primitive.

Targeted searches

The pass combined title/subject searches with backward citation chasing from the direct multi-voltage papers. Query families included:

1. `mutual coupling parallel lines different voltage levels phase impedance`;
2. `multi-circuit multi-voltage overhead line mutual impedance per unit`;
3. `primitive admittance equivalent lattice mutually coupled branches`;
4. `partial mutual coupling line section protection out of service grounded`;
5. `CIM MutualCoupling line segment terminal`;
6. `PowSyBl line coupling extension`; and
7. `PowerWorld mutual impedance record coupled line fractions`.

DOI metadata was checked for journal and conference sources. The software and information-model records were checked against official, version-pinned where available, documentation rather than search-result snippets.

Records added

Eight included, single-coded records were added:

- `EV-0030` — Wortman–Allen–Grigsby, a foundational unbalanced multiport building-block construction;
- `EV-0031` — Kersting, a joint model for physically parallel distribution circuits;
- `EV-0032` — Yan–Saha, the direct 11 kV/415 V phase-coordinate case;
- `EV-0033` — Dziendziel–Kocot–Kubek, geometry and per-unit treatment for multi-circuit multi-voltage HVAC lines;
- `EV-0034` — Tziouvaras–Altuve–Calero, protection consequences of partial overlap, orientation, switching, and grounded out-of-service circuits;
- `EV-0035` — IEC CIM `MutualCoupling` as a first-class relation;
- `EV-0036` — PowSyBl line couplings over identified line intervals; and
- `EV-0037` — PowerWorld's oriented, partial-section mutual-impedance record.

Grainger–Stevenson's textbook construction was added to the bibliography as the classical source for converting an invertible reciprocal joint primitive to an ordinary equivalent lattice. It is not coded as a separate evidence row because this tranche uses the primary building-block paper for the general equation-assembly result and the book supplies its own derivation and executable certificate for the lattice identity.

Synthesis decision

The literature supports three distinct statements that the chapter keeps separate:

1. co-located circuits need not be graph-parallel or share nominal voltage;
2. their coupled sections belong to one joint phase-coordinate primitive; and
3. an invertible fixed-linear primitive can be stamped directly or represented by a generated ordinary-edge lattice at the equation level.

The source line assets and the coupling relation therefore remain canonical. Generated cross-voltage edges are a lowering target with provenance, not new conductors or a claim of galvanic connection.

Limits

This was a targeted search and citation chase, not a fresh export from every database named in the protocol. All new rows remain `single_coded`. The pass does not establish geometry-to-impedance equivalence across all earth, frequency, transposition, bundling, and shunt models. Nor does fixed-linear terminal equivalence prove preservation of relay behavior, thermal limits, switching decisions, or nonlinear feasible sets. Those scopes require separate contracts and evidence.

Chapter 60

Research agenda

Page status: proposal and open-work register.

The current paper-sized dissemination cuts are recorded in the repository's review track. They reuse this agenda's claim and evidence boundaries rather than creating a second roadmap.

Research objective

Develop a theory, reference architecture, and executable toolkit for **typed, asset-, constraint-, and provenance-preserving transformations of multiconductor power-network models**.

The agenda should deliberately connect formal results to utility-relevant applications. A transformation is valuable when it enables faster or more reliable computation without silently invalidating the decisions being made.

Workstream A: model categories and semantics

1. Define the typed hierarchical port–factor category.
2. Define linked asset/property semantics and stable identities.
3. Specify conductor, phase, neutral, ground, orientation and reference-frame types.
4. Formalize observation contracts and relative expressiveness $\succeq_Q$.
5. Separate physical equivalence, terminal behavioral equivalence, feasible-set equivalence, and approximation.

Candidate result A1. A representation theorem showing that ordinary bus–branch multigraphs, conductor-expanded graphs, hypergraphs/factor graphs, and common component compilations embed into the port–factor kernel.

Status: partial — `ARCH-BLOCK-001`, `ARCH-LOWER-001`, `ARCH-PORT-001`, and the representation taxonomy establish finite typed embeddings and lowering examples; a general representation theorem remains open.

Candidate result A2. Conditions under which a projection between model categories is faithful, conservative, or admits a reconstruction functor on a restricted subcategory.

Status: partial — `ARCH-LENS-001`, `ARCH-RECOVERY-002`, and the representation-map query-sufficiency analysis classify scoped faithful, set-identifiable, and non-identifiable cases; a general reconstruction result is open.

Workstream B: normalization calculus

1. Specify typed rewrite rules and negative application conditions.
2. Prove semantic preservation of conductor permutation, ideal-switch contraction, homogeneous line concatenation, grounding extraction, and transformer compilation.
3. Characterize critical pairs among rules.
4. Determine whether useful subsets terminate and are confluent up to typed isomorphism.
5. Define normal forms by purpose rather than one universal form.

Candidate result B1. Necessary and sufficient conditions for degree-two multiconductor bus elimination to remain inside a selected line model class.

Status: partial — `TR-SER-001` and `TR-SER-002` discharge the guarded uncoupled behavioural rule and show why homogeneous line-class closure is separate; `TR-SER-003` now gives a distinct exact rule for a complete mutually coupled section pair, while necessary-and-sufficient closure conditions for broader line libraries remain open.

Candidate result B2. A closure classification for series composition of series-only, nominal-π, exact distributed-parameter, frequency-dependent, and thermally coupled line models.

Status: partial — the guarded-normalization catalogue and the degree-two series chapter cover series-only, nominal-π, and distributed-parameter warnings; frequency-dependent and thermally coupled closure are open.

Candidate result B3. A non-existence result showing that no single simple edge with conventional scalar or per-conductor limits can exactly represent the feasible set of general heterogeneous parallel branches.

Status: partial — the parallel decision cases and `TR-PAR-001`/`TR-PAR-002` give scalar counterexamples and explicit outer-relaxation witnesses; a formal non-existence theorem for the general heterogeneous multiconductor class is open.

Candidate result B4. Necessary and sufficient redundancy certificates for multiconductor parallel-member constraint sets, extending scalar quadratic containment to coupled phase, neutral, ground, and terminal-direction models, with explicit guards for topology and control states.

Current partial result. Claim `TR-PAR-005` gives a necessary-and-sufficient PSD test for each individual centered linear-current norm implication and a two-end componentwise certificate. Claim `TR-PAR-006` adds an exact complex polydisc row-norm test when all component limits of one nonsingular series member jointly imply another member's limits, and exercises it in a reciprocal non-proportional four-wire AC decision case. Claim `TR-PAR-007` generalizes the same support-function argument to an invertible stacked terminal-current map and exercises distinct from/to shunts in a nominal-π case. Singular shunted maps, implication by several different members, non-Euclidean regions, and state-conditioned models remain open parts of B4.

Workstream C: decision-preserving reduction

1. Treat equations and feasible sets together.
2. Develop recovery maps for internal voltages, currents, losses and thermal states.
3. Classify exact, inner, outer and scenario-approximate constraint maps.
4. Treat certified removal of implied constraints as exact presolve, retaining the asset laws, identities, recovery maps, and all nonredundant constraints.
5. Study preservation for OPF, security-constrained OPF, reconfiguration, expansion planning, dynamic operating envelopes and state estimation.

6. Quantify when reduced models change optimal decisions rather than merely state-variable errors.

Candidate result C1. A general lifting theorem: if eliminated variables are uniquely recoverable and all source constraints are composed with that recovery map, optimization over boundary variables is exactly equivalent.

Status: partial — `PRESERVE-001`, the recovery-map chapter, and the Kron, parallel, and transformer certificates establish the statement for declared finite linear and decision cases; a general theorem over nonlinear and mixed discrete models remains open.

Candidate result C2. Complexity or representability bounds for projecting branch-wise thermal constraints onto boundary variables.

Status: open — current work provides exact recovery and support-function certificates, but no general complexity or representability bound.

Workstream D: approximate but certified models

1. Define application-specific observation norms.
2. Develop scenario and uncertainty-domain error certificates.
3. Preserve radiality, phase availability, grounding modes and selected physical corridors.
4. Compare Kron, clustering, aggregation, sparsification and learned surrogates under the same contract.
5. Measure errors in feasibility, optimal objective, active constraints and decisions—not voltage alone.

Workstream E: implementation and interoperability

Develop a Julia reference implementation with:

- immutable source identities and explicit generated-object identities;
- typed ports, factors and hierarchy;
- a transformation registry with machine-readable certificates;
- rule tracing and reversible provenance;
- adapters for CIM/CGMES, OpenDSS, PowerModelsDistribution and selected Julia optimization models;
- generated multigraph, simple-graph and sparse-matrix views;
- property-based tests and adversarial counterexamples.

Graph representation should be independent of any one solver. Mathematical models should consume generated views and expose the mapping back to stable source entities.

Workstream F: empirical corpus

Build a deliberately difficult test corpus containing:

- heterogeneous parallel lines with distinct ratings and decisions;
- four-wire feeders with multi-grounded and impedance-grounded neutrals;
- phase discontinuities and conductor permutations;
- same-code and mixed-code degree-two line chains;
- nominal-π versus distributed line concatenation;
- physically parallel and endpoint-parallel circuits with full, sequence-only, partial-overlap, different-voltage, open, and grounded coupling states;
- multiwinding and autotransformer/regulator models;
- lossy and controllable switches;
- measurements and protection zones at otherwise eliminable nodes.

Each case should include an expected preservation/failure certificate. Small symbolic cases are as important as large benchmarks because they expose exact semantic errors.

High-value applications

The first demonstrations should target decisions for which structural loss has obvious consequences:

1. **Parallel-line OPF and contingency analysis:** show incorrect feasible regions from naïve aggregation.
2. **Four-wire state estimation:** show the effect of grounding-aware versus topology-only normalization.
3. **Distribution model cleaning:** safely merge genuine line subdivisions while retaining construction and provenance.
4. **Multiwinding transformer compilation:** prove terminal equivalence and source-level constraint recovery.
5. **Feeder reduction for hosting capacity or operating envelopes:** compare voltage accuracy with decision accuracy.

Longer-term formalization

A formal methods track could encode the core semantics and selected rewrite proofs in Lean. The initial targets should be finite-dimensional linear relations, incidence conservation, conductor permutations, series composition, parallel feasible sets, and Schur-complement recovery. This should follow a stable mathematical specification rather than precede it.

Part VIII

Reference

Chapter 61

Notation and modelling conventions

Page status: maintained modelling convention; not a standalone empirical claim.

This page fixes the index discipline and physical conventions used throughout the book. The starting point is the notation of the BMOPFTools model specification, generalized only where the book must discuss representations that BMOPFTools does not directly implement.

Symbols describe semantic ownership

An index appears on a quantity when it identifies the object or oriented attachment that owns that quantity. Indices are not added merely because an equation is evaluated at an endpoint.

Identifiers need not be consecutive integers. A data model may use stable strings while the mathematical model uses the corresponding symbols. When an implementation needs an ordinary array, introduce an explicit enumeration map, such as

$$\kappa_{\mathcal{B}} : \mathcal{B} \xrightarrow{\sim} \{1, \ldots, |\mathcal{B}|\}.$$

The semantic object is the label-indexed family $(Y_{ij})_{i,j\in\mathcal{B}}$; the stored array is $[Y]_{\kappa_{\mathcal{B}}(i),\kappa_{\mathcal{B}}(j)}$. This book names the semantic indices before displaying their integer coordinate realization. Integer positions are not asset identities, and changing the enumeration must not change a claim about the network.

For multiconductor models, bold $\mathbf{Y}^{\mathrm{N}}_{ij}$ denotes a block map between terminal spaces. It becomes a scalar matrix entry only after the terminal coordinates and bus enumeration have both been declared. By contrast, $\mathbf{Y}_{\ell}$ is intrinsic data owned by element ℓ and does not acquire endpoint indices merely because it appears in a nodal assembly.

Graph objects and graph-derived counts

The normative finite undirected multigraph is

$$G = (V, E, \mathcal{F}, s, \mathrm{ed}),$$

with two flags in each edge fibre $\mathrm{ed}^{-1}(e)$. The complete definition, including graph loops and matrix conventions, is maintained in Multigraphs for expert modelers. Power-system chapters may specialize V to buses $\mathcal{B}$ and E to identified lines $\mathcal{L}$, but they must not infer that specialization from an unqualified word such as *network* or *graph*.

Symbol	Represents	Typical index
$\mathcal{B}$	buses or connectivity objects	i, j
$\mathcal{L}$	lines	ℓ
$\mathcal{X}$	transformers	x
$\mathcal{W}$	switches	w
$\mathcal{D}$	demands	d
$\mathcal{G}$	generators	g
$\mathcal{H}$	shunts and grounding impedances	h
$\mathcal{F}$	edge ends or flags in the normative multigraph object	f
$\mathcal{P}$	phases	p, q
$\mathcal{N}$	terminal names	p, q
$\mathbf{A}$	oriented incidence matrix	declared bus/edge coordinates
$\mathbf{A}_\ell$	full two-end primitive of element ℓ	nominal-π terminal-current map
A_r, A_c	retained and candidate current row maps	redundancy certificates
$\mathbf{A}_\phi, \mathbf{A}_x$	factor-incidence maps	conductor/factor assembly
$\mathbf{F}$	Fortescue phase-to-sequence transform	$abc \leftrightarrow 012$
C_+	restriction to the positive-sequence subspace	phase state $\to$ positive sequence
E_+	embedding of a positive-sequence state	positive sequence $\to$ phase state
σ	declared equipment/topology state; use σ_e for an element state and G^σ for its resolved graph	active/open/closed state, not a nominal-π subscript
π	projection or quotient map when decorated (for example π_σ); nominal-π is a model-family label	π_σ for connectivity quotient; π section for a circuit model
α	scalar decision or model coefficient; qualify the role, such as served-load α or ZIP α_Z	α in an optimization case; $\alpha_Z, \alpha_I, \alpha_P$ in a load law
Λ	asset-to-electrical relation, generally many-to-many rather than a function	$(a, e) \in \Lambda$
ρ	a declared realification map, sequence-mixing residual, or fitted proportionality scalar; qualify each use	$\rho(z)$, ρ_+ or $\rho^\star$
κ	enumeration map when subscripted by a semantic set, or a condition-number diagnostic	$\kappa_{\mathcal{B}}$ versus $\kappa(\mathbf{A})$
Θ	admissible family of typed factors or parameters; Θ_ϕ denotes a factor family	model-class or factor-parameter set
η	backward-error, residual, or scenario-validity diagnostic; qualify the source	$\eta(\widehat{x})$ or η_{asm}

For the loopless bus–branch specialization, $\partial(\ell)$ denotes the derived unordered endpoint pair and

$$\mathrm{simp} : \mathcal{L}^\circ \to E_{\mathrm{s}}$$

denotes the simple endpoint projection on non-loop members. The symbols p and q remain available for phase and terminal indices; graph maps use descriptive operators. For arbitrary-arity incidence structures, $\mathrm{rel} : \mathcal{F} \to R$ maps a flag to its owning relation. The state-conditioned connectivity-node quotient remains π_σ.

The following names are not interchangeable:

- d_{inc} is incidence degree and counts the two flags of a graph loop twice;
- d_{nbr} is distinct-neighbour degree in a loopless simple projection;

- incident-member count is an engineering count on identified equipment and equals d_{inc} only in the declared loopless specialization;
- terminal count belongs to a port model; and
- row or block nonzero count belongs to a compiled matrix-support graph.

Likewise, B denotes a generic signed graph-incidence matrix in the expert reference chapter, while $\mathbf{A}$ remains the preferred power-system incidence symbol in the engineering chapters. Both use -1 at the selected tail and $+1$ at the head unless a reproduced source declares another convention.

Oriented element triples

A two-terminal branch is identified independently of its orientation. Its physical incidence is the unordered endpoint pair $\partial\ell = \{i, j\}$. For a line ℓ whose stored reference orientation is from bus i to bus j, write

$$\ell ij \in \mathcal{T}^{L\rightarrow} \subseteq \mathcal{L} \times \mathcal{B} \times \mathcal{B}.$$

The opposite terminal arc and the bidirected terminal-arc set are

$$\mathcal{T}^{L\leftarrow} = \{\, \ell ji \mid \ell ij \in \mathcal{T}^{L\rightarrow} \,\}, \qquad \mathcal{T}^{L} = \mathcal{T}^{L\rightarrow} \cup \mathcal{T}^{L\leftarrow}.$$

The triple retains the identity of parallel branches. The arrow is a coordinate and terminal-order convention; it does not assert the operating direction of current or active power. Transformer and switch topology sets $\mathcal{T}^{X}$ and $\mathcal{T}^{W}$ use the same pattern when the device is genuinely two-terminal.

Quantities belonging to an oriented terminal or arc use the triple index:

$$\mathbf{I}_{\ell ij}, \qquad \mathbf{S}_{\ell ij}, \qquad \mathbf{Y}^{\mathrm{sh}}_{\ell ij}.$$

The two terminal quantities need not be negatives: terminal currents can include local shunts, and the two shunt half-sections may be asymmetric. A stored-orientation reversal swaps endpoint records; only a declared internal series-current coordinate has the simple antisymmetry relation. The full distinction is developed in Orientation, terminal quantities, and power transfer.

Element-intrinsic quantities

A physical parameter that does not change when the branch orientation is reversed carries only the element index. For a multiconductor line,

$$\mathbf{Z}^{\mathrm{s}}_{\ell} = \mathbf{R}_{\ell} + \mathrm{j}\mathbf{X}_{\ell}$$

is the series impedance of line ℓ. It is not written $\mathbf{Z}_{\ell ij}$ unless the model actually assigns different directional constitutive data. Likewise, length is L_{ℓ} and an element-level construction or provenance record belongs to ℓ.

This distinction is central to the book:

- ℓ identifies the physical or model element;
- ℓij identifies one oriented attachment of that element;
- ℓji identifies the opposite attachment;
- terminal maps determine how the element's ordered conductor coordinates meet the endpoint buses.

Bus terminals and terminal maps

Each bus i declares an ordered terminal vector

$$\mathbf{N}_i = [N_{i,1}, \dots, N_{i,n_i}],$$

for example $[a, b, c, n]$. Its complex terminal-to-ground voltages form

$$\mathbf{U}_i \in \mathbb{C}^{n_i}.$$

An element does not assume that two buses have identical terminal order or even identical terminal sets. The ordered terminal maps $\mathbf{N}_{\ell i}$ and $\mathbf{N}_{\ell j}$ select the bus terminals to which the ordered conductors of line ℓ attach. Conductor permutations and phase discontinuities are therefore explicit mappings, not hidden matrix conventions.

For the declared forward orientation, a nominal series relation has the form

$$\mathbf{U}_j[\mathbf{N}_{\ell j}] = \mathbf{U}_i[\mathbf{N}_{\ell i}] - \mathbf{Z}^{\mathrm{s}}_{\ell}\mathbf{I}^{\mathrm{s}}_{\ell ij}.$$

The matrix $\mathbf{Z}^{\mathrm{s}}_{\ell}$ is full unless a more restrictive model is declared.

Current and power signs

Terminal currents are defined relative to the terminal at which they enter an element. In the bus balance, the corresponding contribution is given the sign required by the declared KCL convention. Every chapter must state that convention before using an unqualified current direction.

For this book, the preferred physical statements use complex terminal vectors. Rectangular real and imaginary parts are an implementation realization. Per-conductor complex power at an oriented terminal is

$$\mathbf{S}_{\ell ij} = \mathbf{U}_i[\mathbf{N}_{\ell i}] \circ \mathbf{I}^{*}_{\ell ij},$$

where $\circ$ is the Hadamard product and $(\cdot)^*$ is element-wise conjugation.

The pair $(\mathbf{S}_{\ell ij}, \mathbf{S}_{\ell ji})$ consists of terminal complex-power injections into the element. It is not generally one conserved edge flow: a series impedance absorbs power even when its end currents are opposite, and internal nominal-π shunts also make the terminal currents non-antisymmetric. An unqualified phrase such as *power from i to j* must therefore identify both the terminal sign convention and the operating quantity being reported.

Multi-terminal and multiwinding devices

A genuinely multi-terminal device is not assigned an artificial pair of endpoint buses. A transformer x has an ordered winding or port set $\mathcal{K}_x$. Its attachment map is

$$\beta_x : \mathcal{K}_x \to \mathcal{B}, \qquad \beta_x(k) = i,$$

and winding k has its own terminal map $\mathbf{N}_{xk}$, connection, reference voltage, current variables, limits, and provenance. Pairwise short-circuit data may be indexed by winding pairs, while a reconstructed winding-star or primitive admittance remains owned by x.

Only an explicit compilation may replace x by two-terminal elements and virtual internal buses. The generated arcs then receive their own identities and triples, together with a map back to (x, k) and to any generated internal object.

In this book, *winding* counts a transformer port or voltage-level connection unless a passage explicitly refers to physical coils.

Nodal attachments

A nodal element uses an element–bus pair, such as

$$di \in \mathcal{C}^D, \qquad gi \in \mathcal{C}^G, \qquad hi \in \mathcal{C}^H.$$

Its ordered terminal or connection map is recorded separately. A load attached at bus i is not therefore assumed to be three-phase, wye-connected, or phase-to-neutral.

Compound nodal operator

The scalar-to-matrix transition is a change in coordinate space, not a change of graph vocabulary. A scalar branch law uses $i_{\ell ij} = y_\ell(v_i - v_j)$; the multiconductor analogue uses ordered vectors and a matrix primitive. The translation bridge and its diagram-reading checklist are given in How to read power-network diagrams and equations.

The assembled linear current–voltage operator on a declared ordered set of retained junction coordinates is

$$\mathbf{I} = \mathbf{Y}^{\mathrm{N}}\mathbf{U}.$$

The superscript N means *nodal* and does not imply a scalar, positive-sequence, or simple-graph model. Conventional $\mathbf{Y}_{\mathrm{bus}}$ and Ybus are retained as aliases when reproducing software interfaces or established scalar bus-matrix language, but every use must state the coordinate order and model class. A reduced image is distinguished as $\widehat{\mathbf{Y}}$ or $\mathbf{Y}_{\mathrm{K}}$; it is not silently overwritten onto the source operator.

Ground, neutral, and reference

The following are distinct:

- a voltage reference used to remove gauge freedom;

- the physical earth or an earth-return model;
- a neutral conductor;
- a perfectly grounded terminal;
- a grounding impedance or admittance;
- the elimination of a neutral variable by a declared reduction.

No chapter may use *ground* as an unqualified synonym for all of them. Perfectly grounded terminals have fixed voltage. Impedance grounding is an explicit shunt or factor. A neutral conductor remains a network conductor unless an admissible transformation eliminates it.

Parameters, variables, and sets

The mathematical prose distinguishes:

- ordinary italic lower-case symbols for scalars;
- bold symbols for vectors and matrices;
- calligraphic symbols for sets;
- roman subscripts or superscripts for descriptive roles such as sh or ref;
- a superscript $*$ for complex conjugation and H for conjugate transpose.

The HTML may use colour to distinguish real and complex quantities or parameters and variables, following BMOPFTools. No definition may depend on colour: typography, prose, and context must remain complete in monochrome PDF output.

The same rule applies to generated figures. Titles, labels, line styles, panel placement, and captions carry the interpretation; colour only provides a secondary visual cue. In matrix-pattern figures, filled cells mean nonzeros and the panel title identifies the matrix or formulation. In the running network multiview, x_1^* is labelled as a compiled star so its orange styling is not the sole indication of compilation.

Units and coordinate systems

Foundational physical equations use SI units. A per-unit system, sequence transform, conductor permutation, or real rectangular realization is a declared coordinate transformation. Its bases, ordering, orientation, and inverse or recovery map belong in the transformation record.

Equality of raw arrays is not evidence of physical equality when their terminal order, units, reference direction, or base differ.

Decision notation

A model M defines both equations and an admissible set $\mathcal{F}_M$. A decision problem is written abstractly as

$$\min_{z \in \mathcal{F}_M} \; f_M(z),$$

where z may include network states, continuous controls, and discrete decisions. A transformation is not decision preserving merely because it reproduces selected voltages. Its contract must state the relationship between source and target feasible sets, objectives, active constraints, and optimal decisions.

Local deviations

A chapter may introduce specialized notation when that notation makes a derivation substantially clearer. It must provide a local symbol table and an explicit map to this contract. Silent changes of current direction, terminal order, impedance base, or winding meaning are not permitted.

Chapter 62

Terminology

Page status: maintained glossary and translation aid.

This page records the preferred vocabulary used in definitions, claims, and certificates. Readers approaching from power engineering, software and data, mathematical modelling, graph theory, or graph machine learning should begin with One network, five languages.

The canonical editorial policy and its three usage statuses are defined in House policy: familiar words with explicit qualifiers. This page applies that policy as a lookup surface. Community terms are not declared synonyms merely because they occupy the same row: the vocabulary relation and the permitted editorial usage are separate classifications. For the exhaustive generated views, use the community-to-book and book-to-community indexes. Those indexes are generated from the controlled registry; this page remains the maintained definition surface.

Cross-community collision index

The sections below give the compact house definitions. Detailed counterexamples belong to Translation traps; rigorous graph objects belong to Representation frameworks.

Terms to use carefully

Structure preserving

Always qualify the structure: Laplacian, sparsity, radiality, phase connectivity, port-Hamiltonian form, equipment class, asset correspondence, limits, or dynamic equations. The unqualified phrase is too ambiguous.

Equivalent

Always state the interface, operating/model domain, observations, constraints, and whether the claim is exact, conservative, relaxed, or approximate.

Bus

Distinguish at least:

- physical busbar or busbar section;
- connectivity node;
- state-dependent topological node;

Familiar term or cluster	Required question in this book	Common unsafe inference
bus, node, vertex, junction, terminal, port	What object type and representation?	every node is one physical bus
line, branch, edge, arc, relation, nonzero	What owns identity and what records attachment or coupling?	every graph edge is one physical branch
factor	Behavioural relation, factor-graph node, power factor, or matrix factor?	these uses are interchangeable
graph, topology, adjacency	Asset, active-connectivity, factor-incidence, equation, support, or message graph?	one topology answers every query
directed, oriented, upstream, downstream	Stored order, operating sign, rooted-tree relation, causality, or admissibility?	an arrow predicts physical transfer
flow, current, power, message	Which terminal or internal quantity and which conservation law?	one antisymmetric scalar lives on every edge
parallel	Same endpoints, same terminals, shared physical route, homogeneous equipment, or parallel computation?	members can be merged without a contract
cycle, loop, mesh, radial	In which graph and active state?	a graph cycle is a circulating physical flow
state	Electrical variables, equipment status, scenario, estimator state, or ML hidden state?	one state object contains all of them
rating, limit, constraint	Source datum, operational policy, or mathematical encoding?	one scalar constraint reproduces the equipment limit
equivalent, exact, preserving	Which interface and preservation object?	equal terminal equations imply equal decisions
projection, compilation, elimination, aggregation, reduction, pooling	Which objects are changed, forgotten, solved, or recoverable?	every operation merely makes a graph smaller
normalization	Per-unit, coordinate, schema, physical-model, or feature normalization?	all normalizations preserve the same semantics
loss	Electrical dissipation, information loss, or training objective?	a shared word implies a shared quantity
ground, neutral, reference	Physical earth, conductor, impedance, or gauge choice?	all are one zero-voltage node
phase, conductor, sequence, coordinate	Asset label, physical path, terminal slot, phase-domain quantity, or transformed coordinate?	one label fixes every coordinate and transformation
label, index, coordinate, array position	Semantic object identity or storage location?	an integer row/column is the physical bus, line, or cycle identity

- mathematical nodal variable group;
- reporting or planning bus.

Nodal-admittance graph

State whether this means block support, scalar conductor-coordinate support, or a retained stamp decomposition. The first two are simple support graphs of an operator. They do not recover the identified asset multigraph that was assembled into that operator.

Line

Distinguish:

- physical circuit or cable system;
- homogeneous construction segment;
- model section introduced by discretization;
- mathematical two-port branch;
- behavioral equivalent with no single physical counterpart.

Parallel

Distinguish physically parallel line sections, topological parallel edges, terminal or electrical parallelism, and operational interchangeability. *Physically parallel* follows line-modeling usage for spatially co-located sections; it is not the transfer meaning of *corridor*. Electromagnetic mutual coupling is a separate relation and can exist between sections at different voltage levels and with different endpoint pairs.

Mutual coupling

A mutual-coupling record is a relation over two oriented line sections. Its connected components are coupling groups, and each group is assembled into one joint primitive. These source objects remain distinct from any generated weighted-lattice realization of the primitive.

Flow

Distinguish a commodity-flow variable, an internal series current, a terminal current, a terminal complex power, and an observed operating-point transfer. A lossy AC branch generally has a pair of terminal powers rather than one conserved edge-flow scalar.

Directed

State whether direction means stored orientation, terminal-current sign, positive power reference, observed operating transfer, causal dependency, or one-way admissibility. These meanings do not imply one another.

Connected and energized

Graph connectivity is a structural predicate. Energization additionally depends on active state, admissible sources, terminal connectivity, and the electrical semantics of the study.

Ground, neutral, and reference

Distinguish physical earth or an earth-return model, a neutral conductor, a grounding impedance or connection, and a voltage reference used to remove gauge freedom. A neutral voltage may be eliminated from a reduced coordinate system while its recovered current, rating, grounding path, and protection meaning remain part of the source feasible set.

Phase, conductor, sequence, and coordinate

State whether *phase* names an asset label, a physical conductor, a bus terminal, a phase-domain component, or a reporting aggregate. A terminal coordinate is an ordered interface slot; a sequence component is a transformed coordinate. Conductor permutations, phase discontinuities, and sequence transforms therefore require declared maps rather than matching labels alone.

Term	Meaning in this book	Definition or route
Asset	A physical or organizational entity with stable identity and lifecycle facts	Frameworks
Port	A typed interface at which variables interact with a component or subsystem	Frameworks
Junction	An interconnection object imposing equality/conservation structure	Frameworks
Factor	A constitutive, control, limit, measurement, or decision relation over ports	Frameworks
Hierarchy	Explicit containment with declared subsystem boundary ports	Frameworks
Projection	A map that forgets distinctions without solving governing equations	Transformation register
Compilation	Replacement of a high-level component by a realization in a target vocabulary	Transformation register
Normalization	A semantics-preserving rewrite into a selected canonical physical/model form	Transformation register
Behavioral reduction	Elimination of hidden variables preserving a declared external relation	Projection, compilation, and reduction
Approximate reduction	Reduction with a stated observation domain, metric and error	Projection, compilation, and reduction
Preservation contract	Precise statement of retained observations, constraints, assumptions and recovery	Contracts
Limit disposition	Six-way classification of a source limit as retained, mapped, conservative, relaxed, scenario/uncertain, or forgotten	Rating semantics
Observed-set class	Feasible-set relation classified as exact, inner/conservative, outer/relaxed, or scenario-approximate	Preservation contracts
Recovery map	Map from retained/reduced variables to eliminated source quantities	Contracts
Provenance	Traceable correspondence among source, generated and reduced objects	Source architecture
Morphism	A map preserving the declared structure within one representation framework	Representation maps
Isomorphism	A reversible morphism; a change of names or coordinates rather than a quotient or compilation	Representation maps
Query factorization	Evidence that a source query can be answered from a target because it factors through the declared transformation	Contracts
Computational dependency graph	A graph whose vertices and edges encode declared variables, equations, operations, or dependencies rather than physical equipment	Frameworks
Message graph	The compiled graph on which a graph-learning architecture exchanges messages; its nodes and edges need not be physical buses and branches	Vocabulary bridge
Feature	An attribute supplied to a statistical or learned model; physical meaning exists only through its source map, units, coordinate convention, and state association	Vocabulary bridge
Pooling	A many-to-one learned or algorithmic aggregation whose sufficiency is relative to a downstream query and retained side information	Transformation register
Electrical state	The declared continuous electrical variables at an operating point, distinct from equipment status, scenario labels, estimator metadata, and learned hidden state	Load models
Equipment state	A discrete or continuous device condition such as switch status or tap position, with source identity and admissible transitions retained	Topology processing
Neutral conductor	An explicit conductor with voltage, current, connectivity, ratings, and provenance rather than a synonym for earth or reference	Earth and neutral
Voltage reference	A gauge choice used to select a voltage representative; not automatically physical earth or a neutral conductor	Earth and neutral

Chapter 63

Cross-community vocabulary indexes

Page status: generated bidirectional vocabulary index; not a standards crosswalk.

This page is generated from `vocabulary/vocabulary.toml`. It does not declare community terms to be synonyms. Each row records a dominant relation class, an editorial usage status, a diagnostic question, and a route to the maintained definition. The conceptual introduction remains One network, five languages; the compact house definitions remain on the Terminology page. This page answers “how does a community phrase map to the house vocabulary?”; for claim, chapter, and verification retrieval, use the knowledge-base indexes.

The five target communities are indexed separately. Circuit theory is included after them as shared technical vocabulary rather than as a sixth audience route.

Relation and usage keys

The **relation class** describes the map between a community phrase and the house vocabulary. The **usage status** describes how the phrase may be used in this book. These axes are orthogonal.

Registry value	Reader-facing meaning
`exact_alias`	exact alias
`scoped_alias`	scoped alias
`broader_narrower`	broader / narrower
`representation_dependent`	representation dependent
`false_friend`	false friend
`preferred_house_term`	preferred house term
`accepted_qualified_shorthand`	accepted qualified shorthand
`unsafe_unqualified_term`	unsafe unqualified term

Community-to-book index

Power engineering

Software and network data

Mathematical modelling and optimization

Mathematical graph theory

Graph machine learning

Circuit theory (shared technical vocabulary)

Book-to-community index

Use this view when a house term is familiar but the language likely to appear in another community is not. Blank cells are meaningful: the registry does not invent an alias merely to complete a row.

Scope and review boundary

This first registry contains 18 collision concepts. It is a controlled book vocabulary, not an empirical claim that every practitioner uses the recorded terms identically. Community review remains required. Changes should update the registry first and regenerate this page rather than editing the generated tables.

Familiar terms	House terms	Relation / usage	Diagnostic question	Unsafe inference	Route
bus; node; terminal	asset; junction; port; terminal; vertex	representation dependent; unsafe unqualified term	What object type and representation does the word name?	Every node is one physical bus.	definition
line; branch; circuit	asset; line; oriented attachment; support edge	representation dependent; unsafe unqualified term	What owns identity, and what records attachment or algebraic coupling?	Every edge is one physical branch.	definition
device model; power factor	factor; behavioural relation	false friend; unsafe unqualified term	Does factor mean a behavioural relation, factor-graph node, power factor, or matrix factorization?	The shared word denotes the same mathematical object.	definition
network graph; topology; one-line	named graph; typed incidence; active topology; support graph	representation dependent; unsafe unqualified term	Which asset, connectivity, factor, equation, support, or message graph is meant?	One topology answers every physical and computational query.	definition
from bus; to bus; upstream; downstream	stored orientation; terminal sign; operating transfer; rooted-tree relation	scoped alias; accepted qualified shorthand	Is direction a stored order, sign convention, operating result, rooted hierarchy, causal relation, or admissibility rule?	An arrow predicts physical transfer or permanent upstream/downstream status.	definition
power flow; line flow; branch current	series current; terminal current; terminal power; operating transfer; message	false friend; unsafe unqualified term	Which terminal or internal quantity is meant, and which conservation law applies?	One antisymmetric conserved scalar lives on every edge.	definition
parallel lines; physically parallel lines; double circuit; shared right of way	physical parallelism; parallel fibre; terminal parallelism; electrical aggregability	broader / narrower; accepted qualified shorthand	Do members share endpoints, terminals, a physical route, boundary variables, or operating interchangeability?	Parallel members can be merged without a preservation contract.	definition
mutual impedance; mutually coupled lines; joint line model	mutual-coupling record; coupling group; joint primitive	representation dependent; preferred house term	Is this the pairwise source relation, its connected group, the assembled primitive, or a generated realization?	Mutual coupling is an ordinary parallel branch or every generated lattice edge is a physical conductor.	definition
radial feeder; mesh; loop	simple cycle; line-identity cycle;	representation dependent	In which graph and active state is the cycle, tree, mesh, or	A graph cycle is a circulating physical flow, or radiality is	definition

Familiar terms	House terms	Relation / usage	Diagnostic question	Unsafe inference	Route
`equipment`; `terminal`; `connectivity node`; `topological node`	`asset`; `junction`; `port`; `terminal`; `vertex`	representation dependent; unsafe unqualified term	What object type and representation does the word name?	Every node is one physical bus.	definition
`conducting equipment`; `branch record`; `terminal pair`	`asset`; `line`; `oriented attachment`; `support edge`	representation dependent; unsafe unqualified term	What owns identity, and what records attachment or algebraic coupling?	Every edge is one physical branch.	definition
`component model`; `equipment model`	`factor`; `behavioural relation`	false friend; unsafe unqualified term	Does factor mean a behavioural relation, factor-graph node, power factor, or matrix factorization?	The shared word denotes the same mathematical object.	definition
`node-breaker topology`; `bus-branch topology`; `connectivity graph`	`named graph`; `typed incidence`; `active topology`; `support graph`	representation dependent; unsafe unqualified term	Which asset, connectivity, factor, equation, support, or message graph is meant?	One topology answers every physical and computational query.	definition
`terminal 1`; `terminal 2`; `from`; `to`	`stored orientation`; `terminal sign`; `operating transfer`; `rooted-tree relation`	scoped alias; accepted qualified shorthand	Is direction a stored order, sign convention, operating result, rooted hierarchy, causal relation, or admissibility rule?	An arrow predicts physical transfer or permanent upstream/downstream status.	definition
`from-end flow`; `to-end flow`; `measurement`	`series current`; `terminal current`; `terminal power`; `operating transfer`; `message`	false friend; unsafe unqualified term	Which terminal or internal quantity is meant, and which conservation law applies?	One antisymmetric conserved scalar lives on every edge.	definition
`duplicate endpoint pair`; `parallel equipment records`	`physical parallelism`; `parallel fibre`; `terminal parallelism`; `electrical aggregability`	broader / narrower; accepted qualified shorthand	Do members share endpoints, terminals, a physical route, boundary variables, or operating interchangeability?	Parallel members can be merged without a preservation contract.	definition
`MutualCoupling`; `line coupling`; `mutual impedance record`	`mutual-coupling record`; `coupling group`; `joint primitive`	representation dependent; preferred house term	Is this the pairwise source relation, its connected group, the assembled primitive, or a generated realization?	Mutual coupling is an ordinary parallel branch or every generated lattice edge is a physical conductor.	definition
`radial topology`; `island`; `loop`	`simple cycle`; `line-identity cycle`;	representation dependent	In which graph and active state is the cycle, tree, mesh, or	A graph cycle is a circulating physical flow, or radiality is	defini-

Familiar terms	House terms	Relation / usage	Diagnostic question	Unsafe inference	Route
node; variable block; index	asset; junction; port; terminal; vertex	representation dependent; unsafe unqualified term	What object type and representation does the word name?	Every node is one physical bus.	definition
branch; arc; relation	asset; line; oriented attachment; support edge	representation dependent; unsafe unqualified term	What owns identity, and what records attachment or algebraic coupling?	Every edge is one physical branch.	definition
constraint relation; factor	factor; behavioural relation	false friend; unsafe unqualified term	Does factor mean a behavioural relation, factor-graph node, power factor, or matrix factorization?	The shared word denotes the same mathematical object.	definition
equation graph; dependency graph; sparsity pattern	named graph; typed incidence; active topology; support graph	representation dependent; unsafe unqualified term	Which asset, connectivity, factor, equation, support, or message graph is meant?	One topology answers every physical and computational query.	definition
positive direction; oriented arc; causal dependency	stored orientation; terminal sign; operating transfer; rooted-tree relation	scoped alias; accepted qualified shorthand	Is direction a stored order, sign convention, operating result, rooted hierarchy, causal relation, or admissibility rule?	An arrow predicts physical transfer or permanent upstream/downstream status.	definition
flow variable; injection; transfer	series current; terminal current; terminal power; operating transfer; message	false friend; unsafe unqualified term	Which terminal or internal quantity is meant, and which conservation law applies?	One antisymmetric conserved scalar lives on every edge.	definition
parallel branches; aggregate branch	physical parallelism; parallel fibre; terminal parallelism; electrical aggregability	broader / narrower; accepted qualified shorthand	Do members share endpoints, terminals, a physical route, boundary variables, or operating interchangeability?	Parallel members can be merged without a preservation contract.	definition
coupling relation; block impedance matrix; connected component	mutual-coupling record; coupling group; joint primitive	representation dependent; preferred house term	Is this the pairwise source relation, its connected group, the assembled primitive, or a generated realization?	Mutual coupling is an ordinary parallel branch or every generated lattice edge is a physical conductor.	definition
cycle basis; tree formulation;	simple cycle; line-identity cycle;	representation depen-	In which graph and active state is the cycle, tree, mesh, or	A graph cycle is a circulating physical flow, or radiality is	defini-

Familiar terms	House terms	Relation / usage	Diagnostic question	Unsafe inference	Route
vertex; node	asset; junction; port; terminal; vertex	representation dependent; unsafe unqualified term	What object type and representation does the word name?	Every node is one physical bus.	definition
edge; arc	asset; line; oriented attachment; support edge	representation dependent; unsafe unqualified term	What owns identity, and what records attachment or algebraic coupling?	Every edge is one physical branch.	definition
factor node; hyperedge	factor; behavioural relation	false friend; unsafe unqualified term	Does factor mean a behavioural relation, factor-graph node, power factor, or matrix factorization?	The shared word denotes the same mathematical object.	definition
graph; adjacency; incidence structure	named graph; typed incidence; active topology; support graph	representation dependent; unsafe unqualified term	Which asset, connectivity, factor, equation, support, or message graph is meant?	One topology answers every physical and computational query.	definition
orientation; directed edge; rooted tree	stored orientation; terminal sign; operating transfer; rooted-tree relation	scoped alias; accepted qualified shorthand	Is direction a stored order, sign convention, operating result, rooted hierarchy, causal relation, or admissibility rule?	An arrow predicts physical transfer or permanent upstream/downstream status.	definition
network flow; circulation	series current; terminal current; terminal power; operating transfer; message	false friend; unsafe unqualified term	Which terminal or internal quantity is meant, and which conservation law applies?	One antisymmetric conserved scalar lives on every edge.	definition
parallel edges; edge multiplicity	physical parallelism; parallel fibre; terminal parallelism; electrical aggregability	broader / narrower; accepted qualified shorthand	Do members share endpoints, terminals, a physical route, boundary variables, or operating interchangeability?	Parallel members can be merged without a preservation contract.	definition
relation graph; connected component; weighted realization	mutual-coupling record; coupling group; joint primitive	representation dependent; preferred house term	Is this the pairwise source relation, its connected group, the assembled primitive, or a generated realization?	Mutual coupling is an ordinary parallel branch or every generated lattice edge is a physical conductor.	definition
cycle; forest; bridge;	simple cycle; line-identity cycle;	representation dependent	In which graph and active state is the cycle, tree, mesh, or	A graph cycle is a circulating physical flow, or radiality is	definition

Familiar terms	House terms	Relation / usage	Diagnostic question	Unsafe inference	Route
node; node type	asset; junction; port; terminal; vertex	representation dependent; unsafe unqualified term	What object type and representation does the word name?	Every node is one physical bus.	definition
edge; edge type; relation	asset; line; oriented attachment; support edge	representation dependent; unsafe unqualified term	What owns identity, and what records attachment or algebraic coupling?	Every edge is one physical branch.	definition
factor node; relation type	factor; behavioural relation	false friend; unsafe unqualified term	Does factor mean a behavioural relation, factor-graph node, power factor, or matrix factorization?	The shared word denotes the same mathematical object.	definition
input graph; message graph; adjacency	named graph; typed incidence; active topology; support graph	representation dependent; unsafe unqualified term	Which asset, connectivity, factor, equation, support, or message graph is meant?	One topology answers every physical and computational query.	definition
message direction; directed relation	stored orientation; terminal sign; operating transfer; rooted-tree relation	scoped alias; accepted qualified shorthand	Is direction a stored order, sign convention, operating result, rooted hierarchy, causal relation, or admissibility rule?	An arrow predicts physical transfer or permanent upstream/downstream status.	definition
message; aggregation	series current; terminal current; terminal power; operating transfer; message	false friend; unsafe unqualified term	Which terminal or internal quantity is meant, and which conservation law applies?	One antisymmetric conserved scalar lives on every edge.	definition
multi-edge; duplicate relation	physical parallelism; parallel fibre; terminal parallelism; electrical aggregability	broader / narrower; accepted qualified shorthand	Do members share endpoints, terminals, a physical route, boundary variables, or operating interchangeability?	Parallel members can be merged without a preservation contract.	definition
relation edge; heterogeneous coupling edge	mutual-coupling record; coupling group; joint primitive	representation dependent; preferred house term	Is this the pairwise source relation, its connected group, the assembled primitive, or a generated realization?	Mutual coupling is an ordinary parallel branch or every generated lattice edge is a physical conductor.	definition
graph cycle; tree	simple cycle; line-identity cycle;	represen-tation depen-	In which graph and active state is the cycle, tree, mesh, or	A graph cycle is a circulating physical flow, or radiality is	def-i-ni-

Familiar terms	House terms	Relation / usage	Diagnostic question	Unsafe inference	Route
node; terminal; port	asset; junction; port; terminal; vertex	representation dependent; unsafe unqualified term	What object type and representation does the word name?	Every node is one physical bus.	definition
branch; one-port; two-port	asset; line; oriented attachment; support edge	representation dependent; unsafe unqualified term	What owns identity, and what records attachment or algebraic coupling?	Every edge is one physical branch.	definition
constitutive relation; multiport	factor; behavioural relation	false friend; unsafe unqualified term	Does factor mean a behavioural relation, factor-graph node, power factor, or matrix factorization?	The shared word denotes the same mathematical object.	definition
circuit graph; network topology; incidence	named graph; typed incidence; active topology; support graph	representation dependent; unsafe unqualified term	Which asset, connectivity, factor, equation, support, or message graph is meant?	One topology answers every physical and computational query.	definition
reference direction; passive sign convention	stored orientation; terminal sign; operating transfer; rooted-tree relation	scoped alias; accepted qualified shorthand	Is direction a stored order, sign convention, operating result, rooted hierarchy, causal relation, or admissibility rule?	An arrow predicts physical transfer or permanent upstream/downstream status.	definition
branch current; terminal current; complex power	series current; terminal current; terminal power; operating transfer; message	false friend; unsafe unqualified term	Which terminal or internal quantity is meant, and which conservation law applies?	One antisymmetric conserved scalar lives on every edge.	definition
parallel connection; common terminal pair	physical parallelism; parallel fibre; terminal parallelism; electrical aggregability	broader / narrower; accepted qualified shorthand	Do members share endpoints, terminals, a physical route, boundary variables, or operating interchangeability?	Parallel members can be merged without a preservation contract.	definition
mutually coupled branches; primitive impedance matrix; equivalent lattice	mutual-coupling record; coupling group; joint primitive	representation dependent; preferred house term	Is this the pairwise source relation, its connected group, the assembled primitive, or a generated realization?	Mutual coupling is an ordinary parallel branch or every generated lattice edge is a physical conductor.	definition
loop; mesh; loop current	simple cycle; line-identity	representation	In which graph and active state is the	A graph cycle is a circulating physical	def- i-

House-term cluster	Power engineering	Software / data	Mathematical modelling	Graph theory	Graph ML	Circuit theory	Relation	Route
asset; junction; port; terminal; vertex	bus; node; terminal	equipment; terminal; connectivity node; topological node	node; variable block; index	vertex; node	node; node type	node; terminal; port	representation dependent	definition
asset; line; oriented attachment; support edge	line; branch; circuit	conducting equipment; branch record; terminal pair	branch; arc; relation	edge; arc	edge; edge type; relation	branch; one-port; two-port	representation dependent	definition
factor; behavioural relation	device model; power factor	component model; equipment model	constraint relation; factor	factor node; hyperedge	factor node; relation type	constitutive relation; multiport	false friend	definition
named graph; typed incidence; active topology; support graph	network graph; topology; one-line	node-breaker topology; bus-branch topology; connectivity graph	equation graph; dependency graph; sparsity pattern	graph; adjacency; incidence structure	input graph; message graph; adjacency	circuit graph; network topology; incidence	representation dependent	definition
stored orientation; terminal sign; operating transfer; rooted-tree relation	from bus; to bus; upstream; downstream	terminal 1; terminal 2; from; to	positive direction; oriented arc; causal dependency	orientation; directed edge; rooted tree	message direction; directed relation	reference direction; passive sign convention	scoped alias	definition
series current; terminal current; terminal power; operating transfer; message	power flow; line flow; branch current	from-end flow; to-end flow; measurement	flow variable; injection; transfer	network flow; circulation	message; aggregation	branch current; terminal current; complex power	false friend	definition
physical parallelism; parallel fibre; terminal parallelism; electrical aggregability	parallel lines; physically parallel lines; double circuit; shared right of	duplicate endpoint pair; parallel equipment records	parallel branches; aggregate branch	parallel edges; edge multiplicity	multi-edge; duplicate relation	parallel connection; common terminal pair	broader / narrower	definition

Chapter 64

Evidence map and verification summary

Page status: generated reference navigation and evidence-gap summary.

This page is generated from `claims/claims.toml`. It is a retrieval aid and gap analysis, not a replacement for the claims ledger or the evidence matrix. Empty cells are intentionally visible.

<!– generated-evidence-summary:start –> The largest named gaps in this snapshot are measurement (no coded claims at any verification state), state and numerical-structure claims without independent implementations, and external review (zero claims across all nine preservation dimensions). These are gaps in the current evidence record, not evidence that the corresponding literature or engineering practice is empty. The verification summary reports 100 self-checked, 5 independently implemented, and 0 externally reviewed claims out of 105; self-checking and independent implementation are useful but are not external peer review. <!– generated-evidence-summary:end –>

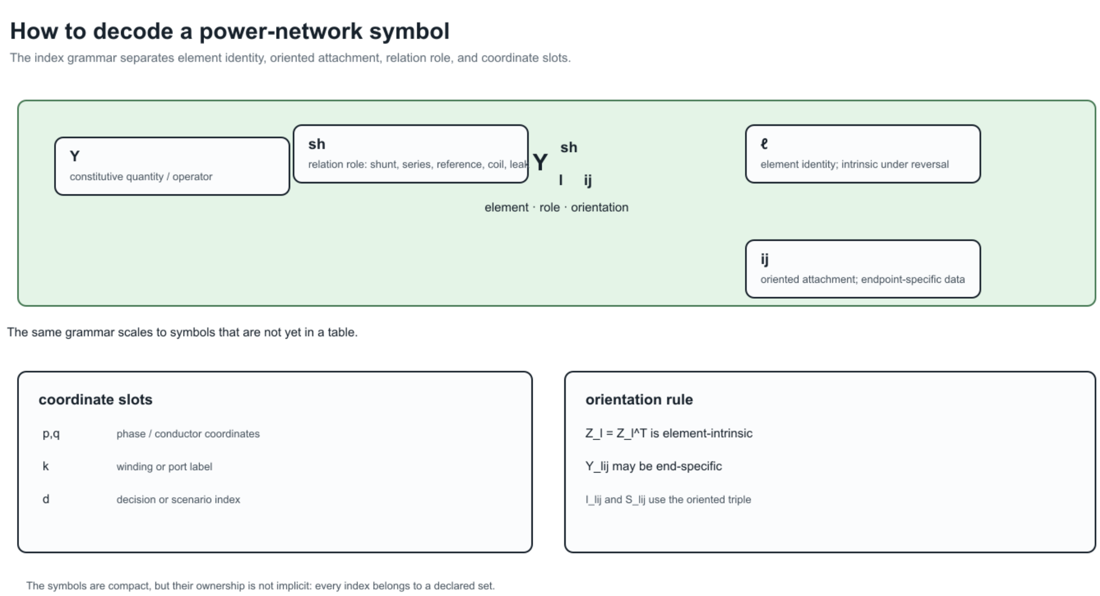

Figure 64.1: Symbol anatomy for the book's index grammar.

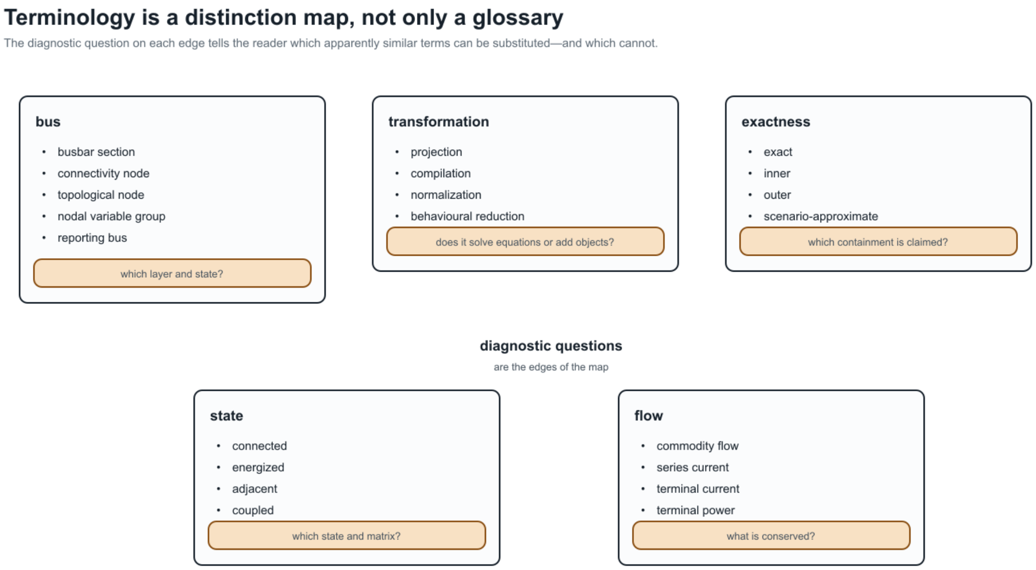

Figure 64.2: Terminology distinction map.

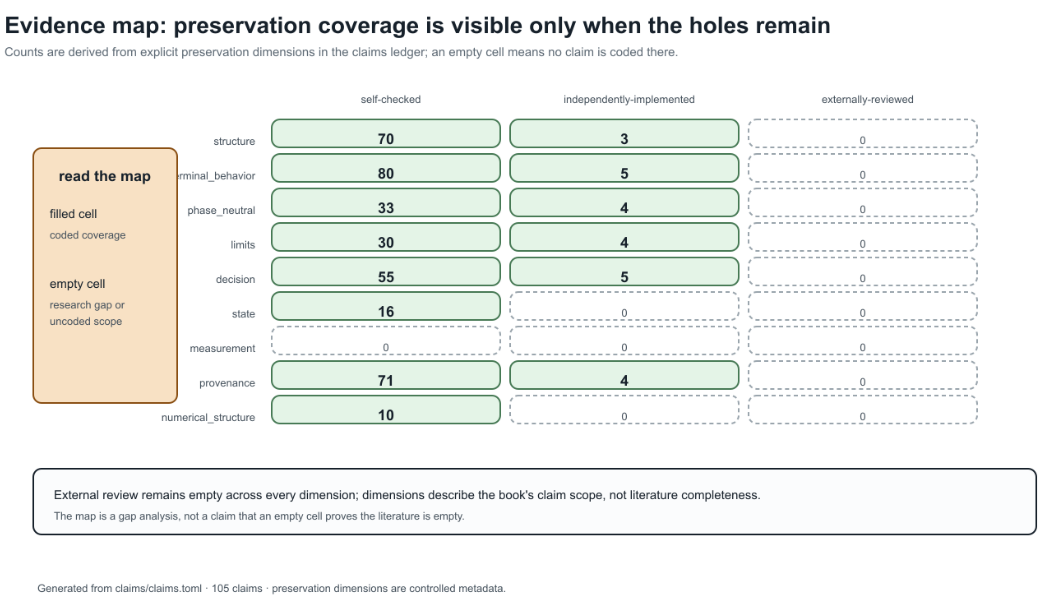

Figure 64.3: Evidence coverage by preservation dimension and verification state.

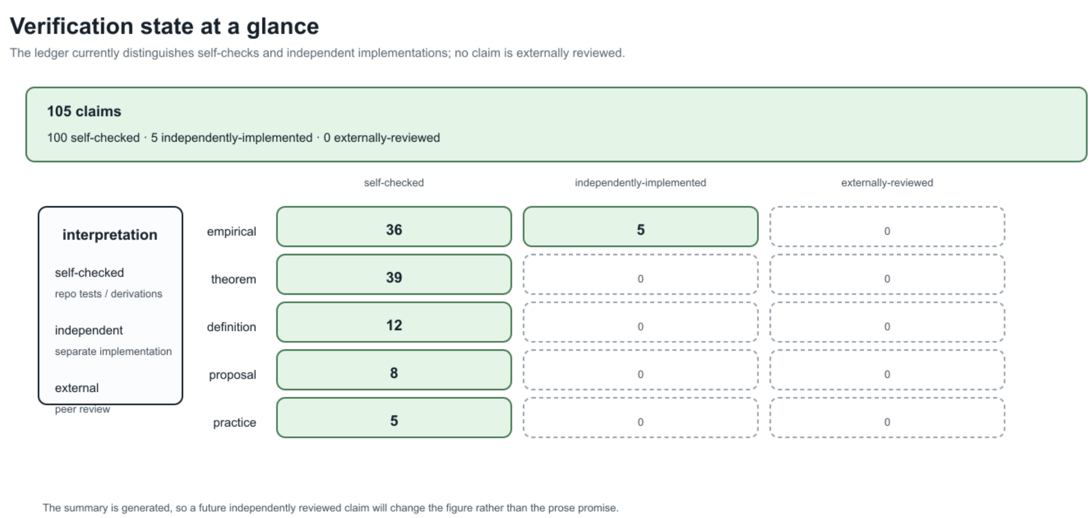

Figure 64.4: Verification state by claim type.

Chapter 65

Knowledge-base indexes

<!– generated-from claims/claims.toml sha256:7665586ed6a8a607a3d601f703193adb55a49ad64ed8a50cfa5557d5cbd92aeb –> This page is generated from `claims/claims.toml` and the JSON artifacts under `experiments/generated/`. It is the HTML knowledge base's retrieval layer; the curated PDF route does not attempt to reproduce these indexes as a linear chapter sequence. For a compact visual summary of notation, terminology distinctions, coverage gaps, and verification state, see the evidence map and verification summary.

Indexed claims: 105 **Indexed chapters:** 44

Claims by type

`definition` (12)

`empirical` (41)

`practice` (5)

`proposal` (8)

`theorem` (39)

Claims by verification state

Unresolved issues

Facet indexes

These retrieval facets are provisional and path-derived. They are navigation aids, not additional verification labels; explicit facet fields can replace them when the claims schema is normalised.

`decision-cases` (30)

`general` (1)

`graph-and-topology` (23)

`numerical-evidence` (5)

`physical-modelling` (9)

`representation` (55)

`software-and-data` (23)

`study-and-literature` (2)

`transformations` (45)

Generated artifacts

This file is regenerated during the documentation build.

Claim	Chapter	Verification
FORMULATION-Y-SPLIT-001 — A load or generator may be represented as a factor attached to a source network while a declared study formulation places a constant-admittance, Norton, or other linearized part in the nodal operator and retains the remainder as an injection, control, limit, or decision relation; the resulting nodal matrix is therefore mode-, state-, and linearization-qualified rather than a unique graph of the source system.	Circuit formulations and the lowering boundary	self-checked
GRAPH-LOOPY-001 — Network-reduction and microgrid-stability literature uses loopy-Laplacian self-loop terms for grounded or differential-conductance diagonal effects; that matrix-level usage is distinct from the book's ordinary graph-loop convention with a zero signed-incidence column.	Multigraphs for expert modelers	self-checked
GRAPH-MULTI-001 — The book's finite undirected multigraph is an identified edge-and-flag object in which every edge owns exactly two distinct flags; loops have two flags incident to one vertex, parallel members retain distinct edge identities, and attributes remain typed maps rather than being inferred from adjacency.	Multigraphs for expert modelers	self-checked
GRAPH-NPORT-001 — Allowing a relation to own an arbitrary finite flag fibre generalizes the two-uniform multigraph to an incidence structure, while an expert mathematical model additionally needs typed flag spaces, relation roles or ordering, and a constitutive relation to represent an n-port factor rather than merely its hypergraph incidence.	Multigraphs for expert modelers	self-checked
GRAPH-SELF-LOOP-001 — In the book's loopless bus–branch circuit specialization, ordinary edges connect distinct retained circuit nodes; a graph self-loop may remain in the source multigraph or arise from a topology quotient, but it is not interchangeable with an electrical circuit loop or mesh, a grounded shunt, or a diagonal self-admittance term.	Multigraphs for expert modelers	self-checked
GROUND-SCOPE-001 — Reference, neutral, earth-return, and grounding-asset semantics are distinct model objects; reductions involving them must declare an earth-return class, grounding points, retained observations, and recovery data.	Earth, neutral, and reference model classes	self-checked
NUMERICAL-001 — Representation and reduction choices have numerical consequences that must be reported separately from electrical preservation: coordinate scaling changes conditioning without changing an invertible solution set, Jacobian dependency graphs need not equal physical graphs, Schur elimination can create fill-in, and decision certificates require residual/error estimates and margins.	Numerical consequences of representation and reduction	self-checked
PRESERVE-001 — Equivalence of two power-network models is indexed by a declared observation family and admissible input set; equality of an unconstrained terminal relation alone does not establish equality of constrained feasible observable sets.	Preservation contracts	self-checked
RATING-001 — A power-network rating must identify its constrained asset or terminal, measured quantity and feasible region, duration, ambient/scenario validity, and ownership/provenance before a transformation can claim to preserve it.	Rating and limit semantics	self-checked
THESIS-001 — Representation adequacy is evaluated relative to declared observations, constraints, and decisions.	Scope and thesis	self-checked
TOPOLOGY-001 — For a fixed switch state, topological nodes are the connected components of the closed-switch connectivity graph; compiling them into bus–branch buses is a state-conditioned quotient that requires provenance and does not preserve switching decisions	Node–breaker, bus–breaker, and topology processing	self-checked

Claim	Chapter	Verification
`ARCH-BLOCK-001` — For a declared two-bus four-conductor linear factor, the vector-edge, port-factor, block nodal, scalar-support, and realified-coordinate views are related representations of one assembled relation; coordinate expansion and realification do not create physical assets, while factor identity remains separate provenance.	How to read power-network diagrams and equations	`self-checked`
`ARCH-CONDUCTOR-002` — The five-bus scalar line-identity fixture lifts to fourteen scalar endpoint ports, five terminal junctions, and seven two-port line factors; the lift retains the q/r parallel fibre and its extra cycle dimension while adding no multiconductor, switch, or transformer semantics.	Formal representation frameworks	`self-checked`
`ARCH-FIVEBUS-XFMR-001` — In the recorded five-bus structural extension, one three-port transformer has acyclic local factor-incidence and star realizations, while eliminating the virtual star point generically yields a terminal clique with cycle rank one; the embedded factor-incidence and star member ranks coincide at five despite carrying different semantics, while the clique member rank is six, without implying additional physical transformer loops.	Five buses through a multi-port lowering	`self-checked`
`ARCH-PORT-001` — A minimal executable port–factor bundle instantiated from the running network validates typed port-to-junction and port-to-factor incidence, a three-port multiwinding factor, grounding as an explicit factor, and a many-to-many asset/electrical relation Λ.	Formal representation frameworks	`self-checked`
`ARCH-PORT-002` — The five-bus identified scalar multigraph has a direct structural port–factor lift with five bus junctions, seven two-port scalar line factors, fourteen endpoint ports, and one asset-to-factor relation per identified line; parallel members q and r remain distinct factors despite sharing the same bus pair.	Formal representation frameworks	`self-checked`
`AU-CARSON-001` — The Australian Carson reproduction regenerates overhead and underground multiconductor primitives from lifted construction inputs, compares them with independent OpenDSS reference matrices, and identifies the CS1035 construction mapping as unresolved rather than presenting it as a faithful reproduction.	Australian construction inputs: Carson and OpenDSSDirect	`independently-implemented`
`COLLAPSE-002` — The generated Fortescue witness diagonalizes a circulant three-phase impedance matrix and preserves the positive-sequence subspace, while a non-circulant perturbation produces sequence mixing and a positive-subspace residual.	When the general model collapses	`self-checked`
`FIXTURE-001` — Running-network fixture v0.1.0 passes the current BMOPFTools JSON schema and conformance checks without errors or warnings.	Executable running network	`self-checked`
`FIXTURE-002` — The v0.1.0 continuous PF and OPF instances terminate locally solved in the recorded environment.	Executable running network	`self-checked`
`GROUND-SCOPE-002` — On the recorded two-conductor fixture, floating, finite-impedance, and ideal customer-end grounding relations share the same simple bus–branch graph but change neutral voltage, ground-current allocation, and the associated observations.	Earth, neutral, and reference model classes	`self-checked`
`GROUND-SCOPE-003` — In the scoped E_2 witness, an explicit earth conductor with a finite neutral-to-earth bond has distinct in-service, earth-conductor-outage, and phase-to-earth-fault states; the outage changes earth-current availability and the fault crosses the declared protection-current threshold while the simple bus graph remains fixed.	Earth, neutral, and reference model classes	`self-checked`
`GROUND-SCOPE-004` — In the explicit-earth witness, a declared	Earth, neutral, and	`self-checked`

Claim	Chapter	Verification
`DATA-XWALK-001` — CIM/CGMES, PowerModelsDistribution, OpenDSS, and MATPOWER provide distinct partial correspondences to the book's asset, terminal, topology, factor, state, and rating objects; successful import is not by itself semantic or decision equivalence.	Data-model crosswalk	`self-checked`
`PRACTICE-ADAPTER-001` — A safe source-to-canonical adapter should publish stable identities, terminal maps, state and control treatment, factor and rating mappings, generated-object provenance, unsupported fields, validation findings, and declared recovery checks before downstream graph transformations are trusted.	From source data to a canonical network model	`self-checked`
`PRACTICE-ARCH-001` — The representation implementation record keeps public API maturity, fixture coverage, direct-versus-related evidence, and not-yet-tested rows separate from the normative mathematical representation definitions.	Representation implementation record	`self-checked`
`PRACTICE-IMPEDANCE-001` — A safe impedance adapter should retain conductor order and terminal maps, units and frequency, geometry or linecode provenance, earth-return assumptions, matrix diagnostics, shunt placement, and the limits and decisions that use the resulting coordinates.	From conductor geometry to impedance fidelity	`self-checked`
`VOCAB-BRIDGE-001` — Load-bearing statements use a preferred house term or qualified shorthand that declares the relevant representation, quantity, state, and preservation object; an unqualified term is unsafe when it can change the claim.	One network, five languages	`self-checked`

Claim	Chapter	Verification
`ARCH-DEGENERACY-001` — Duplicate ideal switches with identical terminal sets and state domains have a well-defined electrical connectivity quotient but an unresolved asset-attribution relation; the model should retain both identities and emit a diagnostic rather than invent protection, maintenance, or failure ownership.	From source graphs to views and graph surgery	`self-checked`
`ARCH-DEGENERACY-002` — The book proposes that missing grounding/reference declarations and rank-deficient active-state maps be treated as model-quality diagnostics; a compiler should refuse to infer a reference or invert a singular map without an additional declaration or restricted coordinate query.	From source graphs to views and graph surgery	`self-checked`
`ARCH-LENS-001` — A layer–lens matrix can attach concrete data, optimization, sparse-matrix, and graph-learning interfaces to different construction stages without treating software packages as representation levels; each attachment must declare preserved and omitted semantics.	Maps between representation frameworks	`self-checked`
`ARCH-SURGERY-001` — The book proposes that state-conditioned graph surgery return state-indexed graphs or graph families with diagnostics and provenance; for many unknown switches, a three-valued certain-connected/certain-separated/undetermined summary should be available instead of silently collapsing to one active graph.	From source graphs to views and graph surgery	`self-checked`
`ARCH-SURGERY-002` — The book proposes that an n-terminal surgery retain port-coordinate identity and return port-specific active and isolated sets; it cannot be inferred by replacing an n-port factor with implicit pairwise edges.	From source graphs to views and graph surgery	`self-checked`
`ARCH-VIEW-001` — The book proposes that a power-network visualisation declare its object level, preserved and forgotten semantics, identity fibres, and reverse-map status; single-line, multi-line, port-factor, node-breaker, nodal-support, and reduced views are distinct typed projections.	From source graphs to views and graph surgery	`self-checked`
`COUPLED-CORRIDOR-001` — Physically parallel line sections need not be parallel in the bus multigraph; a source model can retain stable line assets plus oriented section-to-section coupling records, with each connected coupling group compiling into one joint electrical factor before inversion or nodal stamping.	A coupled multi-voltage corridor	`self-checked`
`TRANSFORM-CATALOG-001` — The guarded-normalization catalogue treats coordinate normalization, series elimination, parallel bundling, switch contraction, multiwinding compilation, and rooted-tree views as distinct rule families whose acceptance depends on declared closure, recovery, constraint, and provenance guards.	Guarded normalization rules	`self-checked`

Claim	Chapter	Verification
`ARCH-CHORDAL-001` — For a simple bus-level tree with a common m-coordinate block at every bus and a structurally dense two-terminal stamp on every tree edge, the scalar structural-support graph is chordal and leaf-bus block elimination is a zero-fill perfect elimination ordering.	Two topology levels and the nodal projection	`self-checked`
`ARCH-LOWER-001` — A typed lowering from an identity-bearing n-port source graph to ordinary-edge incidence objects may preserve a declared equation relation while forgetting factor identity unless source fibres and provenance are retained.	From source graphs to views and graph surgery	`self-checked`
`ARCH-LOWER-002` — A four-winding transformer factor can be compiled pointwise into a complete terminal equation operator while retaining a full non-diagonal reference impedance, mixed winding connection maps, connection-specific shunts, internal grounding, recovery maps, and an explicit finite tap/phase decision domain; this does not by itself provide a source-faithful ordinary-edge realization.	Five buses through a multi-port lowering	`self-checked`
`ARCH-NODAL-001` — Assembly of typed linear factor stamps into a compound nodal operator is not generally injective: distinct admissible parallel-factor decompositions can produce an identical nodal operator and identical normalized assembly residuals.	Two topology levels and the nodal projection	`self-checked`
`ARCH-RECOVERY-001` — Source recovery from a compound nodal operator is class-dependent: support-separated single-factor classes can be identifiable, over-parameterized classes can be set-identifiable at the terminal-primitive level, and parallel multiplicity or eliminated internal coordinates can be non-identifiable; a recovery interface must report the status and ambiguity rather than infer asset identity.	Two topology levels and the nodal projection	`self-checked`
`ARCH-RECOVERY-002` — Auxiliary observations and declarations lift source-recovery ambiguity only through their joint observation map: catalog bounds may produce a compact but non-singleton feasible set, whereas member-current measurements, explicit grounding attribution, or a declared transformer state can make the restricted map injective in a scoped model class.	Two topology levels and the nodal projection	`self-checked`
`ARCH-RECOVERY-003` — For a matrix-valued multiconductor factor observed through member-current snapshots, full primitive recovery requires voltage snapshots spanning the retained conductor space and coverage of every current coordinate; single-snapshot or phase-selective observations retain reciprocal ambiguity even when the assembled nodal operator is known.	Two topology levels and the nodal projection	`self-checked`
`ARCH-RECOVERY-004` — For noisy full-rank multiconductor voltage/current snapshots, the pseudoinverse source estimate has a deterministic Frobenius error bound proportional to the noise radius and the voltage-snapshot pseudoinverse norm; nearly dependent excitation therefore enlarges the certified uncertainty set even when the observation map is full rank.	Two topology levels and the nodal projection	`self-checked`
`ARCH-SUPPORT-001` — Block and scalar nonzero-support graphs of a declared compound nodal operator are simple graphs by construction, while the identified factor-stamp decomposition is separate data and may be a multigraph.	Two topology levels and the nodal projection	`self-checked`
`COLLAPSE-001` — Under compatible three-phase terminals, cyclic (circulant) series and shunt matrices, balanced boundary data, sequence-compatible grounding, two-terminal factor closure, phase-symmetric decisions, and positive-sequence observations, the general phase-domain relation restricts exactly to the positive-sequence scalar network.	When the general model collapses	`self-checked`

Verification	Claims
`self-checked`	100
`independently-implemented`	5
`externally-reviewed`	0

Claim	Issue
ARCH-BLOCK-001	Extend to mixed terminal counts, multi-terminal factors, and numerical-zero policies.
ARCH-CHORDAL-001	Characterize chordality and minimal fill under missing phases, sparse coupling blocks, parallel factors, multi-terminal devices, and meshed bus graphs.
ARCH-CONDUCTOR-002	Compare the scalar terminal lift with a full conductor-terminal factor evaluator and state-conditioned topology maps.
ARCH-DEGENERACY-001	Specify utility-data remediation policies for duplicate or indistinguishable switch assets.
ARCH-DEGENERACY-002	Connect diagnostics to standards-aware import remediation and quantify restricted-coordinate recovery policies.
ARCH-FIVEBUS-XFMR-001	Independently review the layer interfaces and extend the witness to evaluated four- and n-winding factors, non-diagonal reference matrices, connection-specific shunts, controls, and decision-preserving edge realizations.
ARCH-LENS-001	Exercise the matrix against version-pinned external imports and independently review the cross-community API terminology.
ARCH-LOWER-001	Add a full evaluated multiwinding factor lowerer and independently review equation-preservation conditions.
ARCH-LOWER-002	Independently review the four-winding equation-preservation conditions and extend the decision family to richer controls and non-diagonal coupled conductor blocks.
ARCH-NODAL-001	Classify identifiable and bounded ambiguity families under realistic line, transformer, shunt, catalog, measurement, and state constraints.
ARCH-PORT-001	Lift the data witness to evaluated factor relations and independently review the architecture against a non-synthetic asset model.
ARCH-PORT-002	Extend the lift to evaluated factor relations and compare its boundary semantics with the simple quotient and line-identity cycle basis.
ARCH-RECOVERY-001	Classify realistic catalog, measurement, grounding, transformer, and state-dependent recovery classes beyond the four finite witnesses.
ARCH-RECOVERY-002	Extend the augmented-observation criterion to multiconductor current measurements, nonlinear grounding, transformer controls, and partial observability.
ARCH-RECOVERY-003	Extend the rank and partial-observation criterion to noisy measurements, nonlinear operating-point data, coupled transformer ports, and experimental design.
ARCH-RECOVERY-004	Extend deterministic bounds to noisy partial observations, structured/passive uncertainty sets, nonlinear operating points, and experiment design.
ARCH-SUPPORT-001	Extend the support/stamp distinction to frequency-coupled, rectangular realified, Jacobian, and multi-terminal compiled operator families.
ARCH-SURGERY-001	Extend the surgery contract to energized islands, protection states, n-terminal factors, and optimization decisions.
ARCH-SURGERY-002	Extend port-selective surgery to coupled conductor bundles, grounding factors, and protection/energization states.
ARCH-VIEW-001	Add independent technical review of the house visual grammar and compare additional utility and manufacturer diagram conventions.
AU-CARSON-001	Recover the raw CS1035 conductor, screen, earth-return, frequency, and ordering provenance before claiming a faithful reconstruction.
COLLAPSE-001	Extend the network witness to controls, phase-specific limits, contingencies, and an independent mathematical review before making a global decision-equivalence claim.
COLLAPSE-002	Extend the witness to controls, phase-specific limits, contingencies, and independent mathematical review.
COUPLED-CORRIDOR-001	Obtain information-model and protection review; define a machine-readable coupling-group schema and test partial-overlap import round trips.
COUPLED-CORRIDOR-002	Independently review the sign/orientation convention; extend the result to block-valued full-pi and singular tableau targets; and separately test limit, state, protection, and optimization-decision preservation.
DATA-XWALK-001	The running-fixture contract is checked against pinned documentation profiles; external package imports and file-level round-trip provenance/rating checks remain open.
FIXTURE-001	Add an independent fixture reviewer.
FIXTURE-002	Re-run with an independent solver where possible.
FORMULATION-NODAL-001	Add independent review and extend the witness to controlled, dynamic, and multi-terminal power-network factors.

Claim	Chapter	Type
`AU-CARSON-001` — The Australian Carson reproduction regenerates overhead and underground multiconductor primitives from lifted construction inputs, compares them with independent OpenDSS reference matrices, and identifies the CS1035 construction mapping as unresolved rather than presenting it as a faithful reproduction.	Australian construction inputs: Carson and OpenDSSDirect	`empirical`
`COUPLED-CORRIDOR-001` — Physically parallel line sections need not be parallel in the bus multigraph; a source model can retain stable line assets plus oriented section-to-section coupling records, with each connected coupling group compiling into one joint electrical factor before inversion or nodal stamping.	A coupled multi-voltage corridor	`proposal`
`COUPLED-CORRIDOR-002` — For two fixed linear reciprocal scalar series sections with a nonsingular joint impedance matrix, A*Gamma^T Z*Gamma^{-1} A*Gamma has an exact six-edge weighted-lattice realization on the four source terminals; source section currents are recovered by Z*Gamma^{-1} A_Gamma u, while the generated cross edges carry no asset or galvanic interpretation.	A coupled multi-voltage corridor	`theorem`
`FIXTURE-001` — Running-network fixture v0.1.0 passes the current BMOPFTools JSON schema and conformance checks without errors or warnings.	Executable running network	`empirical`
`FIXTURE-002` — The v0.1.0 continuous PF and OPF instances terminate locally solved in the recorded environment.	Executable running network	`empirical`
`IMPEDANCE-LADDER-001` — In the deterministic four-wire impedance ladder fixture, phase-to-neutral current and voltage recovery are exact under the declared zero-ground-current map, while the deliberately non-circulant reduced matrix has visible sequence mixing; shunt deletion and positive-sequence use therefore require explicit decision-domain guards.	Four-wire impedance-model ladder	`empirical`
`LOAD-CONNECTION-001` — On the recorded balanced three-phase terminal fixture, explicit wye phase-to-neutral and delta phase-to-phase connection maps share the same bus and graph but produce different load-voltage observations: unit-magnitude wye voltages and sqrt(3)-magnitude delta voltages.	Load models and decision dependence	`empirical`
`LOAD-CONTINUATION-001` — On the recorded scalar two-bus continuation probe, the damped CP branch first fails to converge at demand scale 1.8 after a converged scale 1.7, while CI, CZ, and the declared ZIP branch remain converged through scale 3.0; this is an iteration-scoped branch diagnostic, not a global collapse theorem.	Load models and decision dependence	`empirical`
`LOAD-DECISION-001` — On the recorded two-bus fixture, CP, CI, CZ, and a normalized ZIP load law share the same bus–branch graph but produce distinct high-voltage solutions and decision margins: CP violates both the declared voltage and current limits, while CI, CZ, and ZIP satisfy both.	Load models and decision dependence	`empirical`
`TR-PAR-001` — Summed admittance preserves the unconstrained terminal relation of parallel linear branches.	A first failure: heterogeneous parallel branches	`theorem`
`TR-PAR-002` — Using the sum of member current ratings can create an outer relaxation of the member-constrained feasible set.	A first failure: heterogeneous parallel branches	`theorem`
`TR-PAR-003` — In the recorded two-bus maximum-served-load problem, the naive summed-rating aggregate serves 200 MW while the source and exact lifted formulations each serve 110 MW.	A first failure: heterogeneous parallel branches	`empirical`
`TR-PAR-004` — In the recorded two-conductor AC maximum-served-load case, the source, exact lifted, and certified exact-pruned formulations have objective 0.6138908, while a summed-limit aggregate has objective 1.0630833 and violates a 0.6 p.u. member limit.	Multiconductor parallel AC decision case	`empirical`

Claim	Chapter	Type
`VOCAB-BRIDGE-001` — Load-bearing statements use a preferred house term or qualified shorthand that declares the relevant representation, quantity, state, and preservation object; an unqualified term is unsafe when it can change the claim.	One network, five languages	`practice`

Claim	Chapter	Type
`ARCH-FIVEBUS-XFMR-001` — In the recorded five-bus structural extension, one three-port transformer has acyclic local factor-incidence and star realizations, while eliminating the virtual star point generically yields a terminal clique with cycle rank one; the embedded factor-incidence and star member ranks coincide at five despite carrying different semantics, while the clique member rank is six, without implying additional physical transformer loops.	Five buses through a multi-port lowering	`empirical`
`ARCH-LOWER-002` — A four-winding transformer factor can be compiled pointwise into a complete terminal equation operator while retaining a full non-diagonal reference impedance, mixed winding connection maps, connection-specific shunts, internal grounding, recovery maps, and an explicit finite tap/phase decision domain; this does not by itself provide a source-faithful ordinary-edge realization.	Five buses through a multi-port lowering	`theorem`
`GRAPH-CYCLE-001` — The recorded connected five-bus bus–branch multigraph has seven identified lines, incidence rank four, and cycle-space dimension three; collapsing its parallel q/r pair to a simple edge reduces the cycle-space dimension to two, whereas the spanning-tree-plus-chords representation retains all three source dimensions.	A five-bus multigraph: identities, cycles, and tree coordinates	`theorem`
`GRAPH-LOOPY-001` — Network-reduction and microgrid-stability literature uses loopy-Laplacian self-loop terms for grounded or differential-conductance diagonal effects; that matrix-level usage is distinct from the book's ordinary graph-loop convention with a zero signed-incidence column.	Multigraphs for expert modelers	`definition`
`GRAPH-MATRIX-001` — Under the declared convention A*uv equals non-loop edge multiplicity off diagonal, A*vv equals twice the graph-loop count, and D contains incidence degrees, so D-A equals B B transpose; graph-loop columns vanish from signed incidence while a grounded shunt contributes a distinct diagonal constitutive term.	Multigraphs for expert modelers	`theorem`
`GRAPH-MULTI-001` — The book's finite undirected multigraph is an identified edge-and-flag object in which every edge owns exactly two distinct flags; loops have two flags incident to one vertex, parallel members retain distinct edge identities, and attributes remain typed maps rather than being inferred from adjacency.	Multigraphs for expert modelers	`definition`
`GRAPH-NPORT-001` — Allowing a relation to own an arbitrary finite flag fibre generalizes the two-uniform multigraph to an incidence structure, while an expert mathematical model additionally needs typed flag spaces, relation roles or ordering, and a constitutive relation to represent an n-port factor rather than merely its hypergraph incidence.	Multigraphs for expert modelers	`definition`
`GRAPH-PI-COLLAPSE-001` — For a fixed linear two-terminal pi factor with series admittance Y*s and endpoint shunts Y*a and Y*b, identifying both terminals through the common attachment map T*pi=[1,1]^T gives T*pi^T Y*pi T*pi=Y*a+Y_b, so the series contribution cancels and the exact nodal image is a one-terminal constant-admittance shunt under the declared coordinate and reference assumptions.	Multigraphs for expert modelers	`theorem`
`GRAPH-SELF-LOOP-001` — In the book's loopless bus–branch circuit specialization, ordinary edges connect distinct retained circuit nodes; a graph self-loop may remain in the source multigraph or arise from a topology quotient, but it is not interchangeable with an electrical circuit loop or mesh, a grounded shunt, or a diagonal self-admittance term.	Multigraphs for expert modelers	`definition`
`TR-GRAPH-001` — For a loopless identified multigraph and its simple endpoint projection, the multigraph cycle rank exceeds the simple-graph cycle rank by the sum over edge fibres of fibre size minus one; the lost dimensions are line-identity cycles supported on	Cycles, parallelism, and radial structure	`theorem`

Claim	Chapter	Type
`FIXTURE-001` — Running-network fixture v0.1.0 passes the current BMOPFTools JSON schema and conformance checks without errors or warnings.	Executable running network	`empirical`
`FIXTURE-002` — The v0.1.0 continuous PF and OPF instances terminate locally solved in the recorded environment.	Executable running network	`empirical`
`NUMERICAL-001` — Representation and reduction choices have numerical consequences that must be reported separately from electrical preservation: coordinate scaling changes conditioning without changing an invertible solution set, Jacobian dependency graphs need not equal physical graphs, Schur elimination can create fill-in, and decision certificates require residual/error estimates and margins.	Numerical consequences of representation and reduction	`definition`
`NUMERICAL-002` — For the pinned running-network fixture, BMOPFTools exports a 20-by-20 passive Ybus with 166 nonzeros; the constant-Z linearized Ybus agrees with it, and realification produces a 40-by-40 current-voltage matrix with 664 nonzeros. The complex matrices have numerical rank 18 at the declared tolerance, with rank-aware effective 2-norm condition about 6.50e8 (1.13e7 after equilibration); the realified embedding preserves support and dimension but is not complex-transpose-symmetric.	Numerical consequences of representation and reduction	`empirical`
`NUMERICAL-003` — In the pinned nonlinear two-bus parallel-member witness, retaining two explicit member-current laws produces a 6-by-7 residual Jacobian and 13-by-13 KKT pattern, while the summed-current aggregate produces a 4-by-5 Jacobian and 9-by-9 KKT pattern; symbolic fill changes with elimination order in both formulations.	Numerical consequences of representation and reduction	`empirical`

Claim	Chapter	Type
`GROUND-SCOPE-001` — Reference, neutral, earth-return, and grounding-asset semantics are distinct model objects; reductions involving them must declare an earth-return class, grounding points, retained observations, and recovery data.	Earth, neutral, and reference model classes	`definition`
`GROUND-SCOPE-002` — On the recorded two-conductor fixture, floating, finite-impedance, and ideal customer-end grounding relations share the same simple bus–branch graph but change neutral voltage, ground-current allocation, and the associated observations.	Earth, neutral, and reference model classes	`empirical`
`GROUND-SCOPE-003` — In the scoped E_2 witness, an explicit earth conductor with a finite neutral-to-earth bond has distinct in-service, earth-conductor-outage, and phase-to-earth-fault states; the outage changes earth-current availability and the fault crosses the declared protection-current threshold while the simple bus graph remains fixed.	Earth, neutral, and reference model classes	`empirical`
`GROUND-SCOPE-004` — In the explicit-earth witness, a declared inverse-time relay curve maps the CT-scaled phase-earth fault current to a 0.2466 s operation, while the neutral-earth fault remains below pickup; a separate declared CT-saturation cap changes the phase-fault trip decision.	Earth, neutral, and reference model classes	`empirical`
`RATING-001` — A power-network rating must identify its constrained asset or terminal, measured quantity and feasible region, duration, ambient/scenario validity, and ownership/provenance before a transformation can claim to preserve it.	Rating and limit semantics	`definition`
`TR-GRAPH-001` — For a loopless identified multigraph and its simple endpoint projection, the multigraph cycle rank exceeds the simple-graph cycle rank by the sum over edge fibres of fibre size minus one; the lost dimensions are line-identity cycles supported on parallel fibres.	Cycles, parallelism, and radial structure	`theorem`
`TR-GRAPH-002` — An identified line is a multigraph bridge exactly when its simple endpoint edge is a bridge and its parallel fibre is a singleton; consequently the identified multigraph is a forest exactly when its simple projection is a forest and every edge fibre is a singleton.	Cycles, parallelism, and radial structure	`theorem`
`TR-GRAPH-ACTIVE-001` — For the five-bus fixture, the inventory has identified-member cycle rank 3 and simple-projection cycle rank 2, while the declared spanning tree is radial at both levels; active radiality is therefore a state-specific property, not an inventory-only label.	Cycles, parallelism, and radial structure	`empirical`
`TR-NEG-001` — The executable anti-pattern witness rejects or classifies four tempting compositions: a heterogeneous series composite is not a homogeneous physical line, external grounding is not absorbed into a transformer, a three-port transformer is not a two-terminal line, and aggregate BIM/BFM branch balance does not imply member voltage compatibility.	Translation traps: graphs, circuits, and power-system language	`empirical`

Claim	Chapter	Type
ARCH-BLOCK-001 — For a declared two-bus four-conductor linear factor, the vector-edge, port-factor, block nodal, scalar-support, and realified-coordinate views are related representations of one assembled relation; coordinate expansion and realification do not create physical assets, while factor identity remains separate provenance.	How to read power-network diagrams and equations	empirical
ARCH-CHORDAL-001 — For a simple bus-level tree with a common m-coordinate block at every bus and a structurally dense two-terminal stamp on every tree edge, the scalar structural-support graph is chordal and leaf-bus block elimination is a zero-fill perfect elimination ordering.	Two topology levels and the nodal projection	theorem
ARCH-CONDUCTOR-002 — The five-bus scalar line-identity fixture lifts to fourteen scalar endpoint ports, five terminal junctions, and seven two-port line factors; the lift retains the q/r parallel fibre and its extra cycle dimension while adding no multiconductor, switch, or transformer semantics.	Formal representation frameworks	empirical
ARCH-DEGENERACY-001 — Duplicate ideal switches with identical terminal sets and state domains have a well-defined electrical connectivity quotient but an unresolved asset-attribution relation; the model should retain both identities and emit a diagnostic rather than invent protection, maintenance, or failure ownership.	From source graphs to views and graph surgery	proposal
ARCH-DEGENERACY-002 — The book proposes that missing grounding/reference declarations and rank-deficient active-state maps be treated as model-quality diagnostics; a compiler should refuse to infer a reference or invert a singular map without an additional declaration or restricted coordinate query.	From source graphs to views and graph surgery	proposal
ARCH-FIVEBUS-XFMR-001 — In the recorded five-bus structural extension, one three-port transformer has acyclic local factor-incidence and star realizations, while eliminating the virtual star point generically yields a terminal clique with cycle rank one; the embedded factor-incidence and star member ranks coincide at five despite carrying different semantics, while the clique member rank is six, without implying additional physical transformer loops.	Five buses through a multi-port lowering	empirical
ARCH-LENS-001 — A layer–lens matrix can attach concrete data, optimization, sparse-matrix, and graph-learning interfaces to different construction stages without treating software packages as representation levels; each attachment must declare preserved and omitted semantics.	Maps between representation frameworks	proposal
ARCH-LOWER-001 — A typed lowering from an identity-bearing n-port source graph to ordinary-edge incidence objects may preserve a declared equation relation while forgetting factor identity unless source fibres and provenance are retained.	From source graphs to views and graph surgery	theorem
ARCH-LOWER-002 — A four-winding transformer factor can be compiled pointwise into a complete terminal equation operator while retaining a full non-diagonal reference impedance, mixed winding connection maps, connection-specific shunts, internal grounding, recovery maps, and an explicit finite tap/phase decision domain; this does not by itself provide a source-faithful ordinary-edge realization.	Five buses through a multi-port lowering	theorem
ARCH-NODAL-001 — Assembly of typed linear factor stamps into a compound nodal operator is not generally injective: distinct admissible parallel-factor decompositions can produce an identical nodal operator and identical normalized assembly residuals.	Two topology levels and the nodal projection	theorem
ARCH-PORT-001 — A minimal executable port–factor bundle instantiated from the running network validates typed port-to-junction and port-to-factor incidence, a three-port multiwinding factor, grounding as an explicit factor, and a	Formal representation frameworks	empirical

Claim	Chapter	Type
ARCH-BLOCK-001 — For a declared two-bus four-conductor linear factor, the vector-edge, port-factor, block nodal, scalar-support, and realified-coordinate views are related representations of one assembled relation; coordinate expansion and realification do not create physical assets, while factor identity remains separate provenance.	How to read power-network diagrams and equations	empirical
ARCH-CHORDAL-001 — For a simple bus-level tree with a common m-coordinate block at every bus and a structurally dense two-terminal stamp on every tree edge, the scalar structural-support graph is chordal and leaf-bus block elimination is a zero-fill perfect elimination ordering.	Two topology levels and the nodal projection	theorem
ARCH-CONDUCTOR-002 — The five-bus scalar line-identity fixture lifts to fourteen scalar endpoint ports, five terminal junctions, and seven two-port line factors; the lift retains the q/r parallel fibre and its extra cycle dimension while adding no multiconductor, switch, or transformer semantics.	Formal representation frameworks	empirical
ARCH-DEGENERACY-001 — Duplicate ideal switches with identical terminal sets and state domains have a well-defined electrical connectivity quotient but an unresolved asset-attribution relation; the model should retain both identities and emit a diagnostic rather than invent protection, maintenance, or failure ownership.	From source graphs to views and graph surgery	proposal
ARCH-DEGENERACY-002 — The book proposes that missing grounding/reference declarations and rank-deficient active-state maps be treated as model-quality diagnostics; a compiler should refuse to infer a reference or invert a singular map without an additional declaration or restricted coordinate query.	From source graphs to views and graph surgery	proposal
ARCH-FIVEBUS-XFMR-001 — In the recorded five-bus structural extension, one three-port transformer has acyclic local factor-incidence and star realizations, while eliminating the virtual star point generically yields a terminal clique with cycle rank one; the embedded factor-incidence and star member ranks coincide at five despite carrying different semantics, while the clique member rank is six, without implying additional physical transformer loops.	Five buses through a multi-port lowering	empirical
ARCH-LENS-001 — A layer–lens matrix can attach concrete data, optimization, sparse-matrix, and graph-learning interfaces to different construction stages without treating software packages as representation levels; each attachment must declare preserved and omitted semantics.	Maps between representation frameworks	proposal
ARCH-LOWER-001 — A typed lowering from an identity-bearing n-port source graph to ordinary-edge incidence objects may preserve a declared equation relation while forgetting factor identity unless source fibres and provenance are retained.	From source graphs to views and graph surgery	theorem
ARCH-LOWER-002 — A four-winding transformer factor can be compiled pointwise into a complete terminal equation operator while retaining a full non-diagonal reference impedance, mixed winding connection maps, connection-specific shunts, internal grounding, recovery maps, and an explicit finite tap/phase decision domain; this does not by itself provide a source-faithful ordinary-edge realization.	Five buses through a multi-port lowering	theorem
ARCH-NODAL-001 — Assembly of typed linear factor stamps into a compound nodal operator is not generally injective: distinct admissible parallel-factor decompositions can produce an identical nodal operator and identical normalized assembly residuals.	Two topology levels and the nodal projection	theorem
ARCH-PORT-001 — A minimal executable port–factor bundle instantiated from the running network validates typed port-to-junction and port-to-factor incidence, a three-port multiwinding factor, grounding as an explicit factor, and a many-to-many asset/electrical relation Λ.	Formal representation frameworks	empirical
ARCH-PORT-002 — The five-bus identified scalar multigraph has a direct structural port–factor lift with five bus junctions, seven two-port scalar line factors, fourteen endpoint ports, and one asset-to-factor relation per	Formal representation frameworks	empirical

Claim	Chapter	Type
`LIT-PAR-001` — For fixed scalar AC pi-line models on common endpoints, a parallel member's current- or apparent-power limit at one terminal is redundant when its normalized terminal-voltage quadratic feasible set contains that of another member; applying the test at both terminals certifies removal of both directional limits without aggregating the line models.	A first failure: heterogeneous parallel branches	`theorem`
`PRACTICE-ARCH-001` — The representation implementation record keeps public API maturity, fixture coverage, direct-versus-related evidence, and not-yet-tested rows separate from the normative mathematical representation definitions.	Representation implementation record	`practice`

Claim	Chapter	Type
TR-COMP-001 — Two exact certified transformations compose when the first target is consumed by the second source; constraint maps apply forward and recovery maps apply in reverse order.	Certificate schema and composition	theorem
TR-COORD-001 — A simultaneous permutation of conductor coordinates, terminal pairing, element matrices, and componentwise limits is an exact normalization with an inverse permutation.	Conductor-coordinate normalization	theorem
TR-GRAPH-001 — For a loopless identified multigraph and its simple endpoint projection, the multigraph cycle rank exceeds the simple-graph cycle rank by the sum over edge fibres of fibre size minus one; the lost dimensions are line-identity cycles supported on parallel fibres.	Cycles, parallelism, and radial structure	theorem
TR-GRAPH-002 — An identified line is a multigraph bridge exactly when its simple endpoint edge is a bridge and its parallel fibre is a singleton; consequently the identified multigraph is a forest exactly when its simple projection is a forest and every edge fibre is a singleton.	Cycles, parallelism, and radial structure	theorem
TR-GRAPH-ACTIVE-001 — For the five-bus fixture, the inventory has identified-member cycle rank 3 and simple-projection cycle rank 2, while the declared spanning tree is radial at both levels; active radiality is therefore a state-specific property, not an inventory-only label.	Cycles, parallelism, and radial structure	empirical
TR-GRAPH-SIMPLIFY-001 — The loopless simple endpoint projection preserves the vertex set, adjacency, connected components, distinct-neighbour sets, and unweighted vertex distances, but it does not preserve identified edge count, incidence degree, cycle-space dimension, bridges, edge connectivity, spanning-tree multiplicity, or member-level state and provenance.	Multigraphs for expert modelers	theorem
TR-KRON-001 — Typed multiconductor Kron reduction commutes with invertible coordinate actions that preserve the retained/internal partition when currents transform by the power-dual action; per-port block diagonality is an optional locality restriction, and the affine statement requires fixed internal injections.	Kron, Ward, and optimized network equivalents	theorem
TR-KRON-002 — In the declared linear scenario fixture, exact Kron reproduces each fixed-injection boundary relation, an operating-point Ward-style equivalent is exact only at its calibration point, and an explicit scenario objective can select a sparser non-exact target.	Kron, Ward, and optimized network equivalents	empirical
TR-KRON-003 — In the declared one-state linear Ward scenario fixture, an internal-injection residual propagates through a recovered-state bound and a boundary-current bound to classify the approximate source-limit decision as certified feasible, ambiguous, or certified violated; the bound is exact for this fixture but is not a general nonlinear error theorem.	Kron, Ward, and optimized network equivalents	empirical
TR-KRON-FIVE-001 — In the five-bus scalar fixture, eliminating the pendant bus m through line u by typed Kron reduction reproduces the retained boundary Y-bus obtained by direct deletion of the leaf line, with exact boundary-current recovery for the recorded voltage state.	Kron, Ward, and optimized network equivalents	empirical
TR-KRON-FIVE-002 — In the same five-bus fixture, eliminating the non-pendant bus l by typed Kron reduction preserves the recorded boundary current relation, creates Schur-complement fill edges j-m and k-m among the retained buses, and exactly recovers the u-branch current whose deliberately tight declared limit is violated by the recorded state.	Kron, Ward, and optimized network equivalents	empirical
TR-KRON-NEUTRAL-001 — In the running four-conductor midpoint Kron fixture, the eliminated neutral half-section current is exactly recoverable from the retained boundary solution and midpoint recovery; a neutral-current limit must therefore remain in the reduced	Kron, Ward, and optimized network equivalents	empirical

Artifact	Evidence summary
active-radiality-witness.json	generated evidence
artifact-manifest.json	generated evidence
australian-carson-reproduction.json	generated evidence
balanced-transmission-independent-reproduction.json	COLLAPSE-001 — generated evidence
balanced-transmission-witness.json	COLLAPSE-001 — generated evidence
block-structure-bridge-witness.json	ARCH-BLOCK-001 — two-bus four-conductor fixed linear factor with full complex series matrix and endpoint shunts
certified-approximation-witness.json	TR-KRON-003 — kron*ward*scenario*fixture*v0.1.0
circuit-formulation-witness.json	FORMULATION-NODAL-001 — ideal voltage source, floating linear network, and aligned parallel members with source-level current limits
clean-package-matrix.json	PKG-CLEAN-001 — generated evidence
compiled-views-surgery-witness.json	ARCH-VIEWS-SURGERY-001 — finite typed source graph with one three-port factor, duplicate ideal switches, four-wire phase-only switching, and one state-conditioned zone surgery
conductor-terminal-lift-witness.json	ARCH-CONDUCTOR-001 — running-network v0.1.0 conductor-terminal incidence with line, switch, and three-winding factor compilation
connection-map-independent-reproduction.json	LOAD-CONNECTION-001 — generated evidence
coordinate-normalization-certificate.json	TR-COORD-001 — generated evidence
coordinate-series-composition-certificate.json	TR-COMP-001 — generated evidence
coupled-corridor-lattice-witness.json	COUPLED-CORRIDOR-002 — two reciprocal fixed-linear scalar series sections on four retained terminal-voltage coordinates
data-model-crosswalk-witness.json	DATA-XWALK-001 — data/running-network/v0.1.0.json
degree-two-series-certificate.json	TR-SER-001 — generated evidence
explicit-earth-independent-reproduction.json	GROUND-SCOPE-004 — generated evidence
explicit-earth-kron-independent-reproduction.json	TR-KRON-NEUTRAL-002 — generated evidence
explicit-earth-kron-witness.json	TR-KRON-NEUTRAL-002 — synthetic five-conductor linear series midpoint with ordered (a,b,c,n,e) terminals and a fixed midpoint neutral-earth bond
five-bus-active-radiality-witness.json	TR-GRAPH-ACTIVE-001 — experiments/generated/five-bus-cycle-space-analysis.json
five-bus-conductor-terminal-lift-witness.json	ARCH-CONDUCTOR-002 — five-bus cycle-space scalar line identities lifted to scalar terminal junctions and two-port factors
five-bus-cycle-space-analysis.json	GRAPH-CYCLE-001 — connected loopless scalar series bus-branch multigraph
five-bus-figure-manifest.json	generated evidence
five-bus-port-factor-witness.json	ARCH-PORT-002 — experiments/generated/five-bus-cycle-space-analysis.json
five-bus-transformer-lowering-witness.json	ARCH-FIVEBUS-XFMR-001 — five-bus scalar line topology plus a structural three-port transformer extension attached at j, l, and m
five-bus-typed-kron-witness.json	TR-KRON-FIVE-001 — experiments/generated/five-bus-cycle-space-analysis.json
fixture-coverage-matrix.json	PKG-FIXTURE-001 — generated evidence
four-winding-lowering-witness.json	ARCH-LOWER-002 — fixed-frequency four-winding factor with a full non-diagonal reference matrix, mixed WYE/DELTA ports, connection-specific shunts, internal grounding, and finite pointwise tap/phase states
four-wire-impedance-model-ladder.json	IMPEDANCE-LADDER-001 — deterministic four-wire matrix fixture
four-wire-parallel-ac-certificate.json	TR-PAR-006 — generated evidence
grounding-impedance-sweep-independent-reproduction.json	TR-KRON-NEUTRAL-004 — generated evidence
grounding-impedance-sweep-witness.json	TR-KRON-NEUTRAL-004 — generated evidence
guarded-parallel-reduction-witness.json	TR-PAR-GUARDED-001 — series-only singular terminal map, jointly retained current discs, and state-dependent admittance maps
hierarchy-boundary-witness.json	ARCH-BOUNDARY-001 — running-network hierarchy, typed boundary refinement, open-system gluing, and state-conditioned switch maps
kron-ward-scenario-comparison.json	TR-KRON-002 — kron*ward*scenario*fixture*v0.1.0

Chapter 66

Chapter status

<!– generated-from claims/claims.toml sha256:7665586ed6a8a607a3d601f703193adb55a49ad64ed8a50cfa5557d5cbd92aeb –> This page is generated from the claims ledger. It makes the evidence state visible without requiring readers to inspect TOML or generated JSON files. Claim absence means the page is tracked as explanatory, definitional, or proposed material rather than silently treated as a verified empirical result.

This file is regenerated during the documentation build.

Chapter	Page status	Claims	Claim types	Verification	Open issue
Australian construction inputs: Carson and OpenDSSDirect	source-backed Carson/OpenDSSDirect reproduction; overhead and underground reference matrices remain independently compared outputs, and the CS1035 construction mapping is explicitly unresolved.	1	empirical	`independently-reproduced`	Recover the raw CS1035 conductor, screen, earth-return, frequency, and ordering provenance before claiming a faithful reconstruction.
BIM/BFM parallel lines: an expressiveness audit	literature-informed formulation case; the equations illustrate scope boundaries and are not a new executable certificate.	0	—	`untracked`	—
A coupled multi-voltage corridor	literature-backed representation and exact fixed-linear lowering case with an executable scalar certificate; a geometry-derived multi-voltage certificate and decision-preservation tests remain follow-on work.	2	proposal, theorem	`self-checked`	Obtain information-model and protection review; define a machine-readable coupling-group schema and test partial-overlap import round trips.; Independently review the sign/orientation convention; extend the result to block-valued full-pi and singular tableau targets; and separately test limit, state, protection, and optimization-decision preservation.
Executable running network	executable fixture, local solver evidence, and a derived state-conditioned radiality witness; claims are versioned and verification-scoped.	2	empirical	`self-checked`	Add an independent fixture reviewer.; Re-run with an independent solver where possible.
Four-wire impedance-model ladder	deterministic four-wire transformation witness; authored overhead and cable reproductions remain a follow-on case-study tranche.	1	empirical	`self-checked`	Reproduce authored overhead-line and underground-cable cases with geometry or linecode provenance, balanced and unbalanced load rows, grounding variants, and an external solver cross-check.
Non-proportional three-phase four-wire parallel case	guarded AC decision case with self-checked and independently reproduced numerical evidence; broader global claims remain open.	1	theorem	`self-checked`	TR-PAR-007 covers nonsingular nominal-pi primitives; the companion certificate now refuses singular and singular-shunted recovery maps. Extend from that refusal boundary to exact singular reductions, several retained members, state-dependent topology and controls, and obtain an independent global optimality bound where required.

Chapter 67

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Page status: bibliography and source register.

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